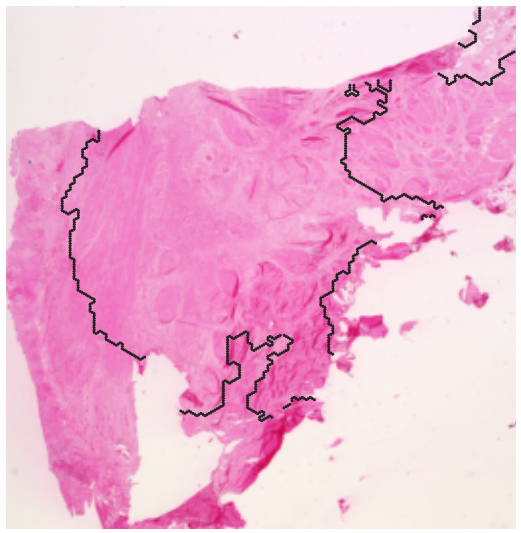
GSE238145_B10 | Tissue context

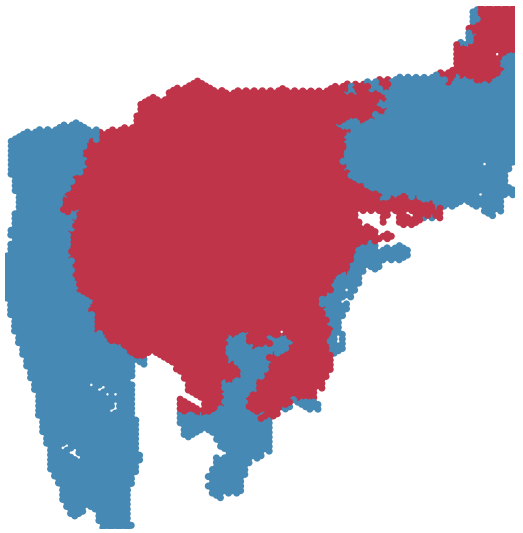
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



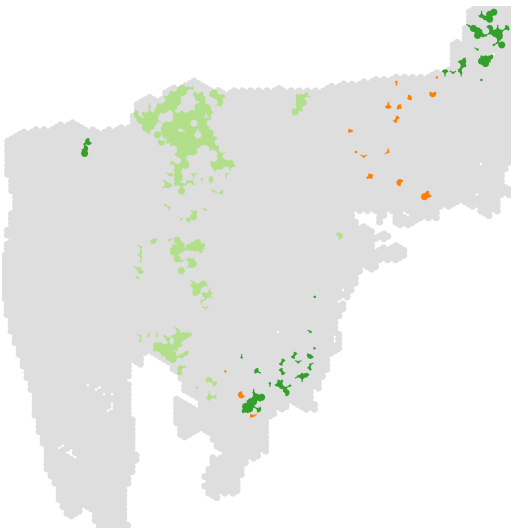
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



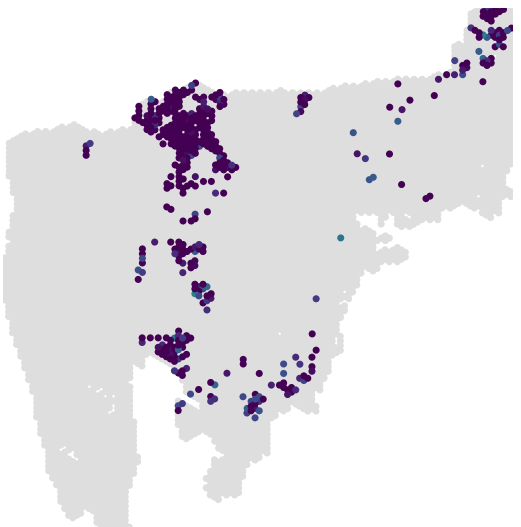
2 3 5

Epithelial support

0 1

Fraction of all 30

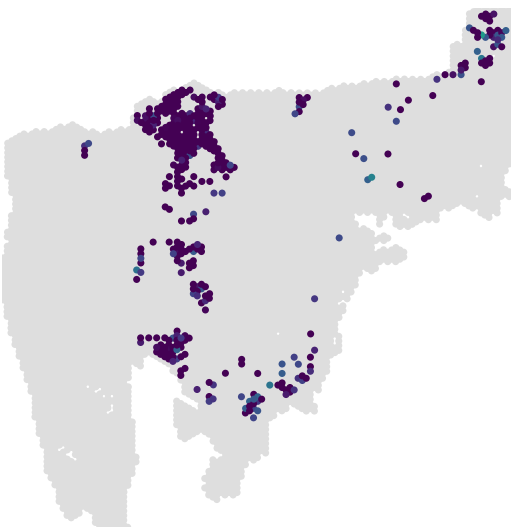
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



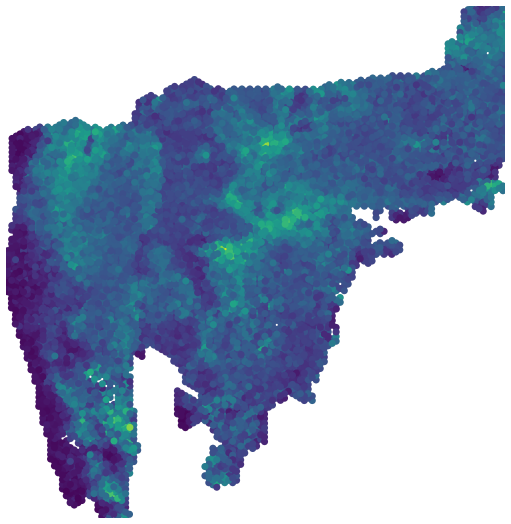
0 0.14

Matched-model AUCell

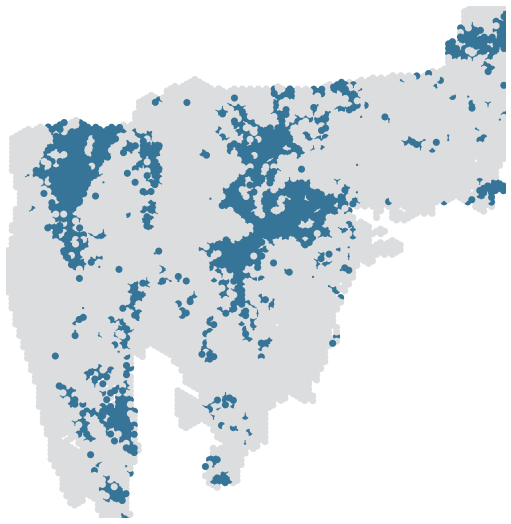
GSE238145_B10 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

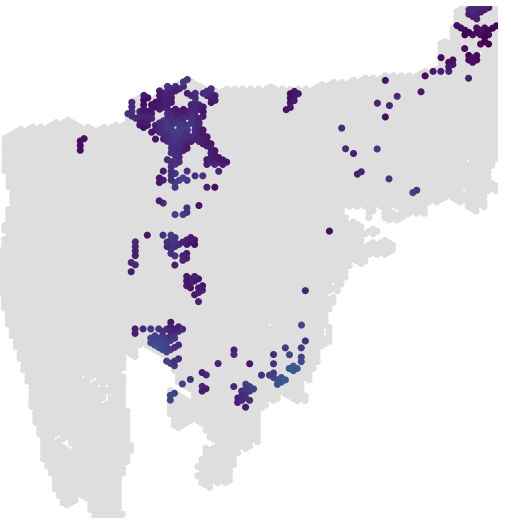
CAF composition

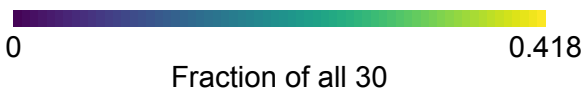


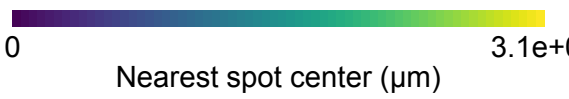
CAF P80 targets



CAF distance

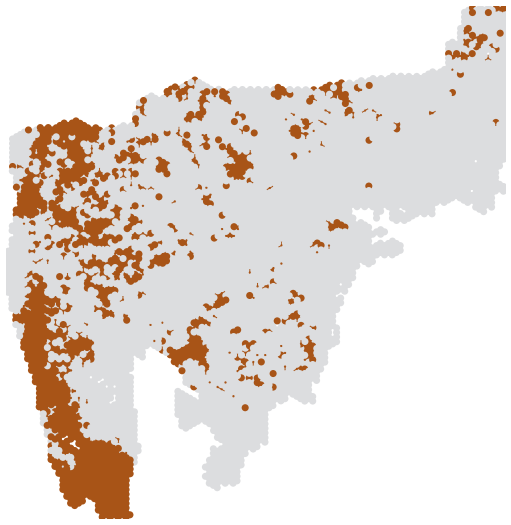




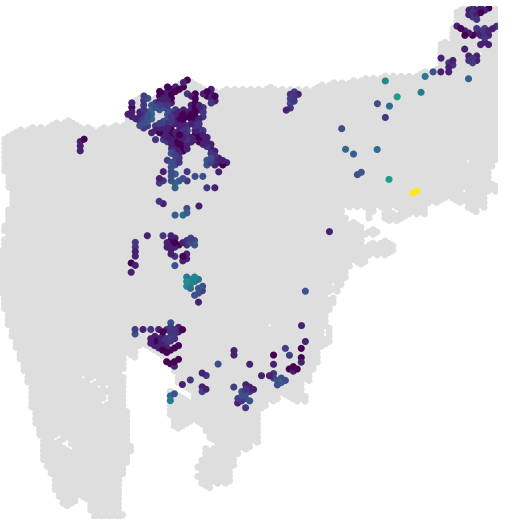


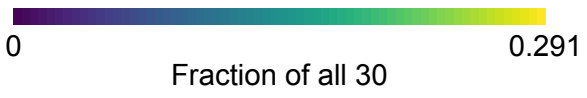
TAM composition

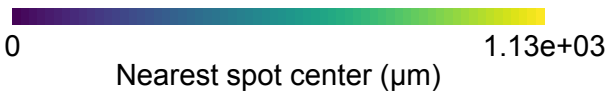
TAM P80 targets



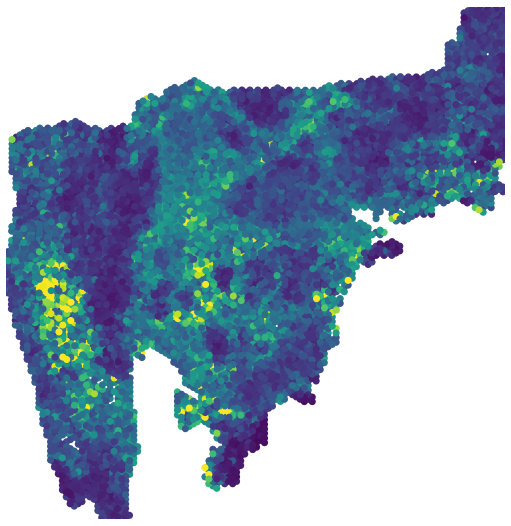
TAM distance



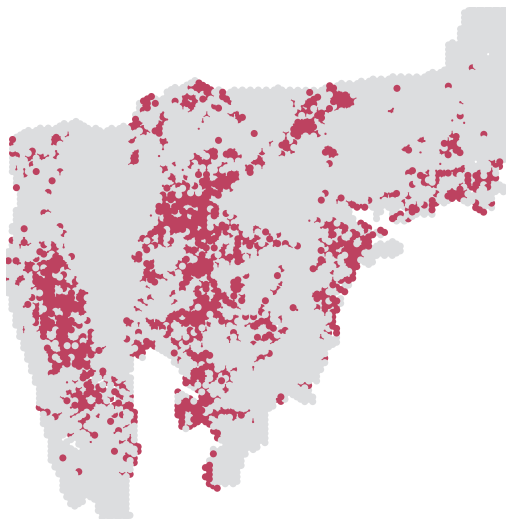




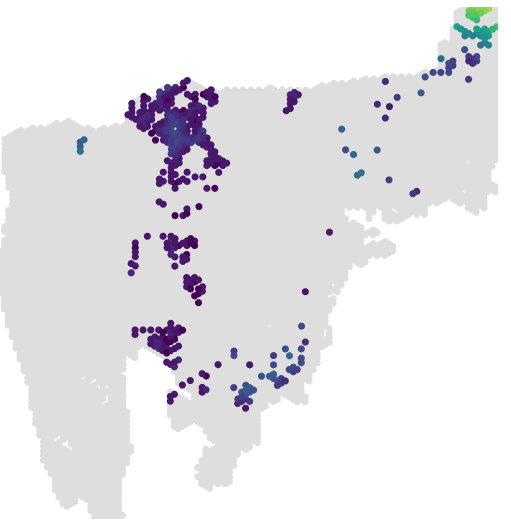
Treg composition

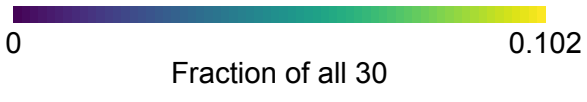


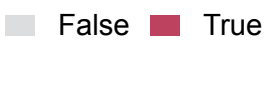
Treg P80 targets

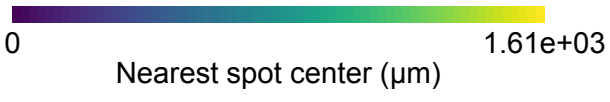


Treg distance



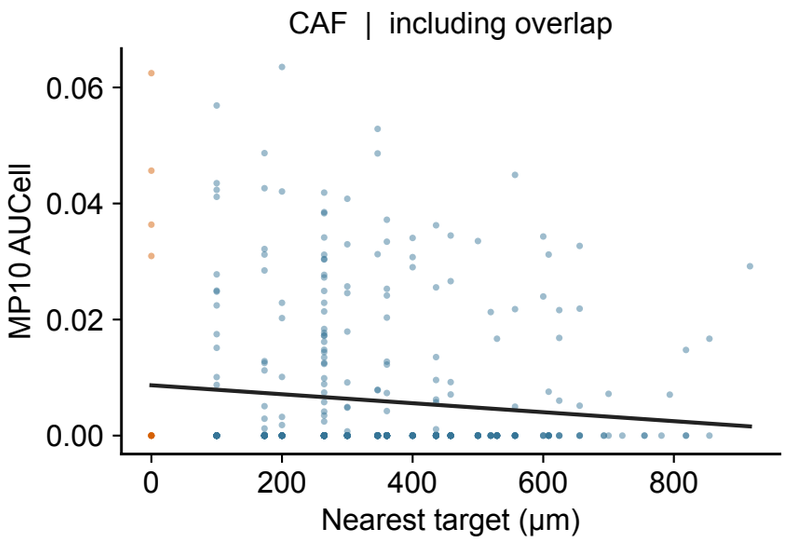




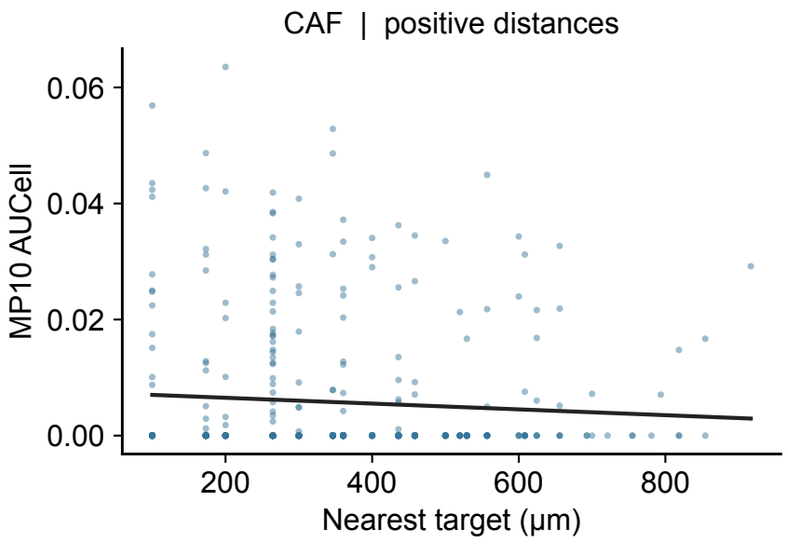


GSE238145_B10 | MP10 versus nearest target

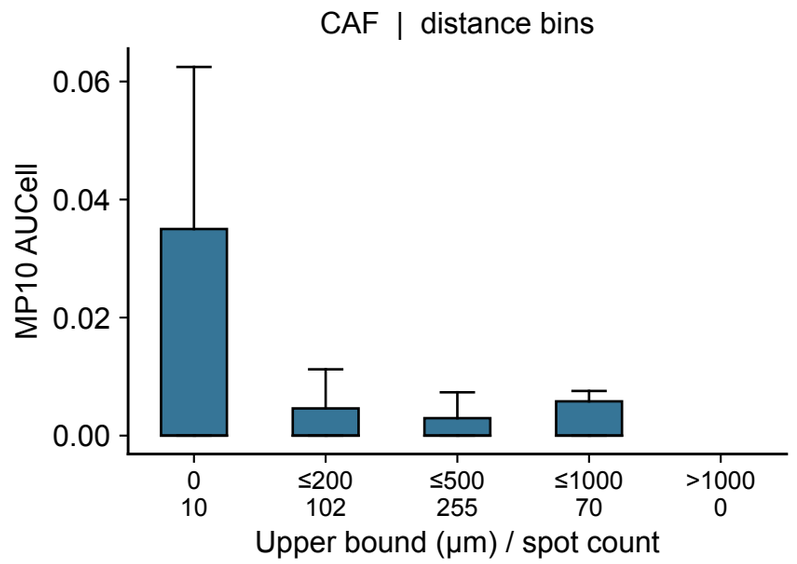
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



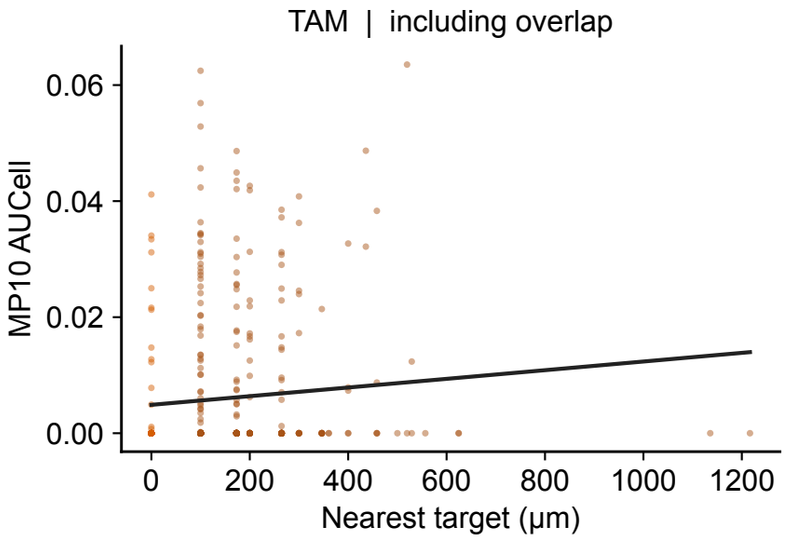
r -0.11 ρ -0.08
n 437/437 queries • 1480 targets



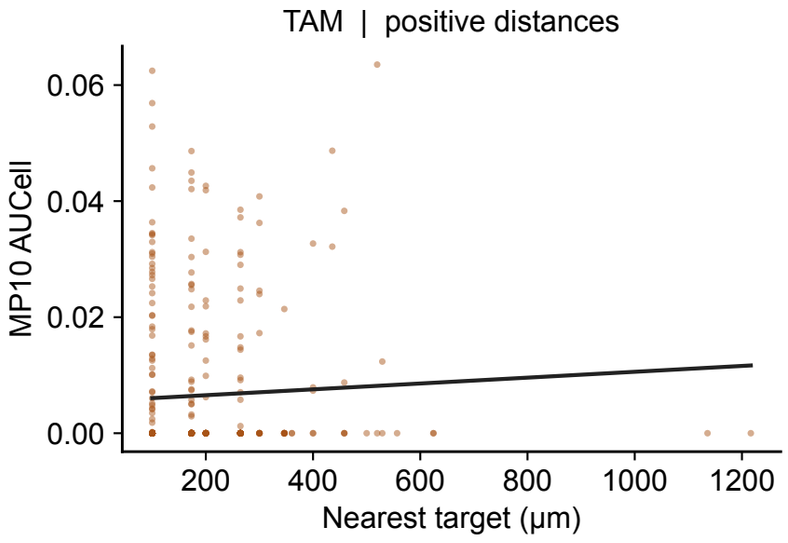
r -0.07 ρ -0.07
n 427/437 queries • 1480 targets



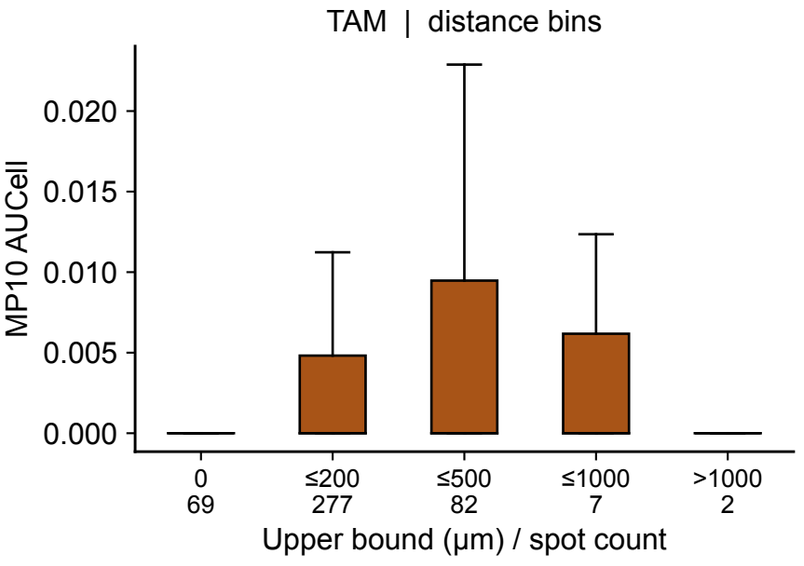
Orange = zero distance
Line = within-section linear fit



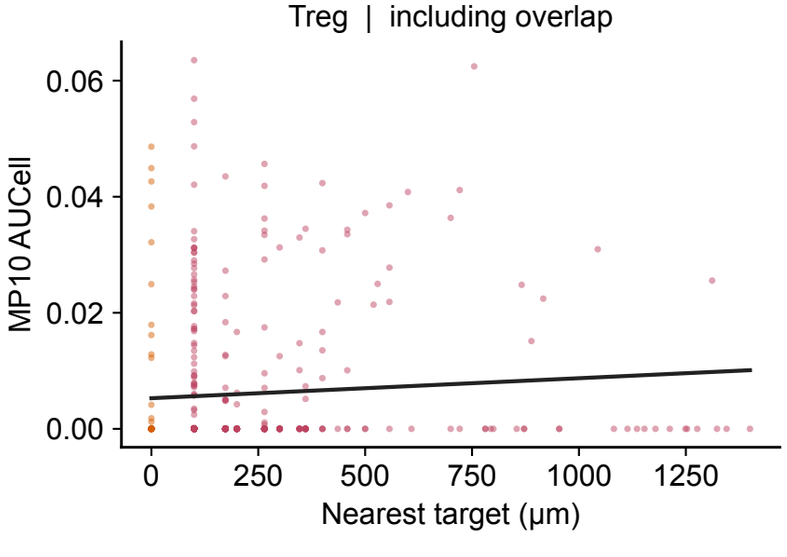
r +0.08 ρ +0.08
n 437/437 queries • 1480 targets



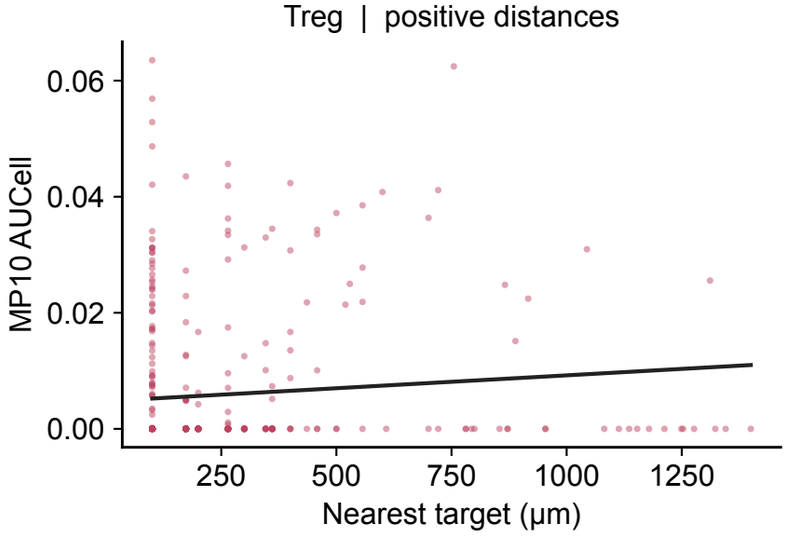
r +0.05 ρ +0.04
n 368/437 queries • 1480 targets



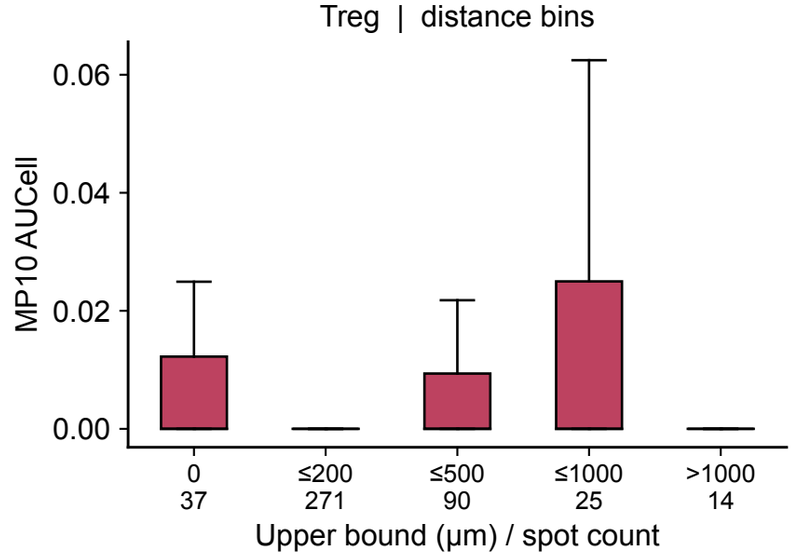
Orange = zero distance
Line = within-section linear fit



r +0.07 ρ +0.01
n 437/437 queries • 1480 targets



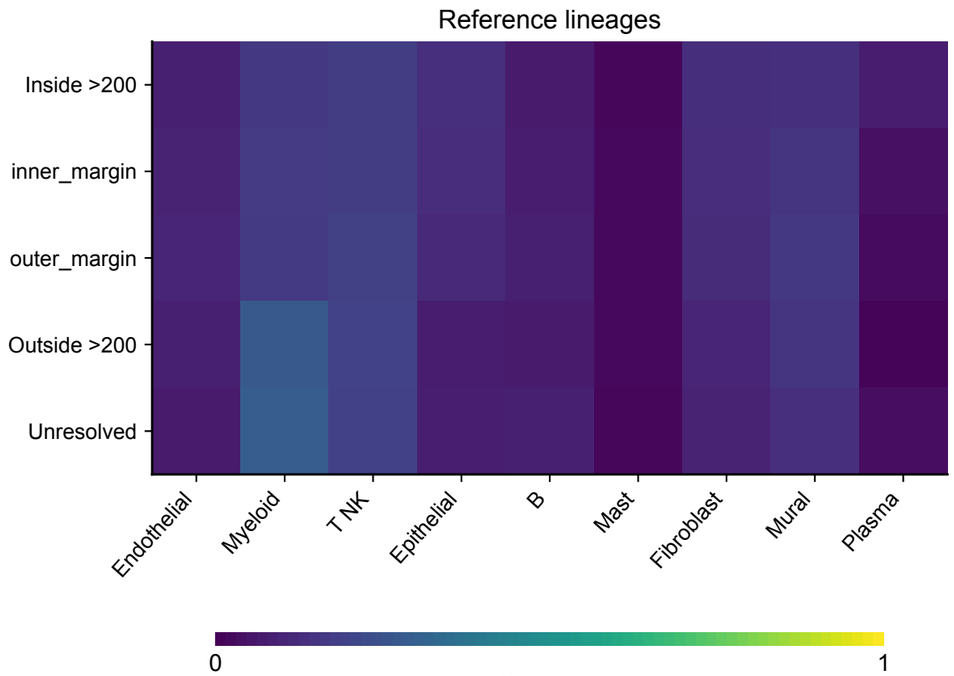
r +0.09 ρ +0.05
n 400/437 queries • 1480 targets

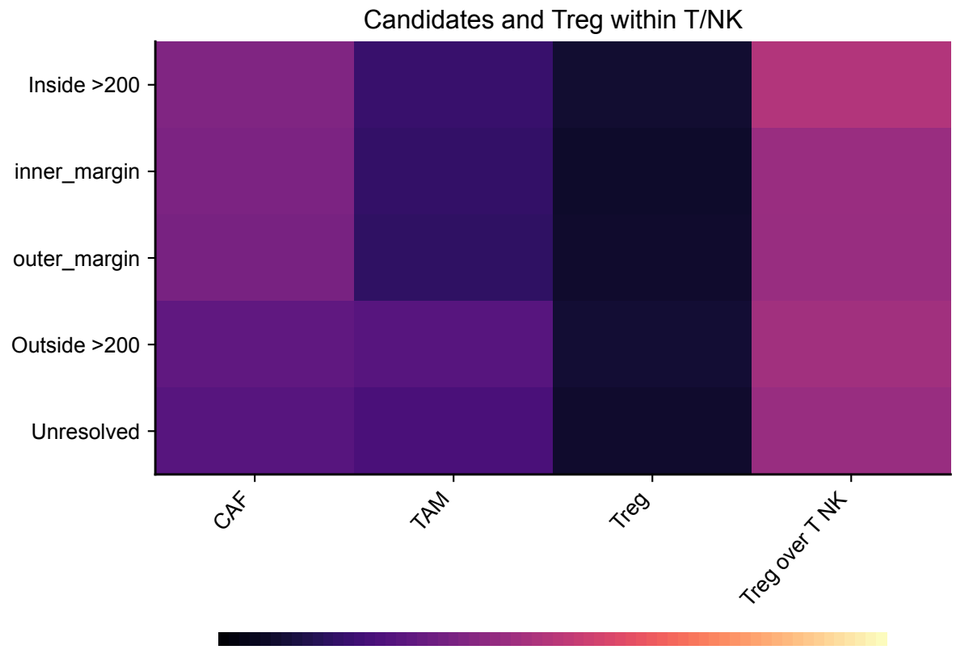


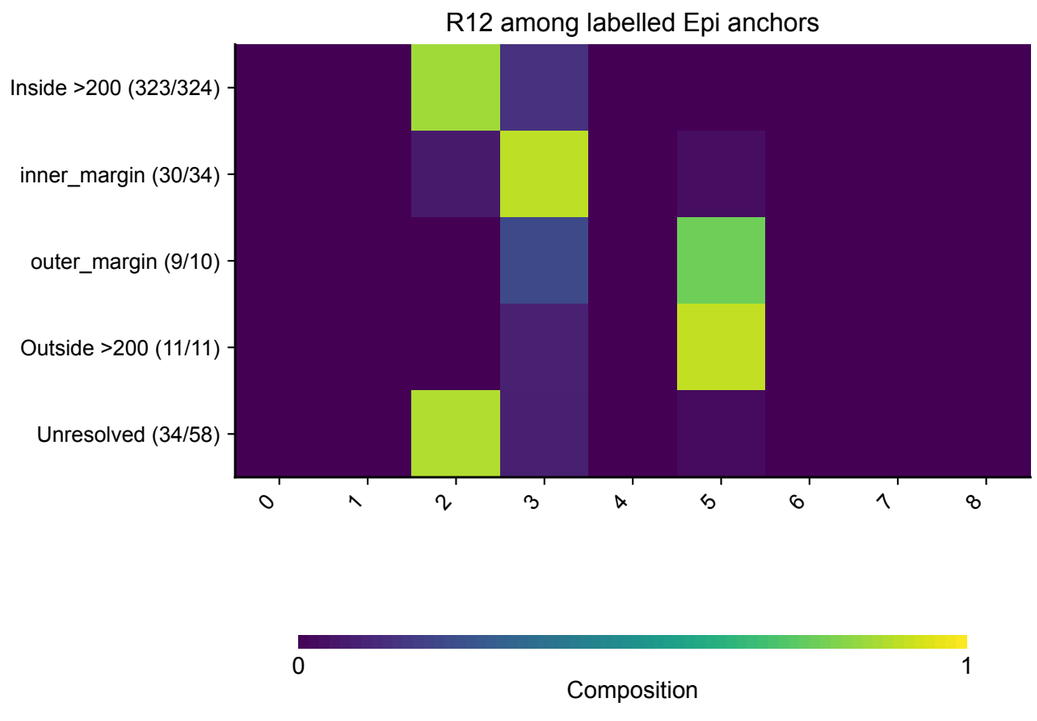
Orange = zero distance
Line = within-section linear fit

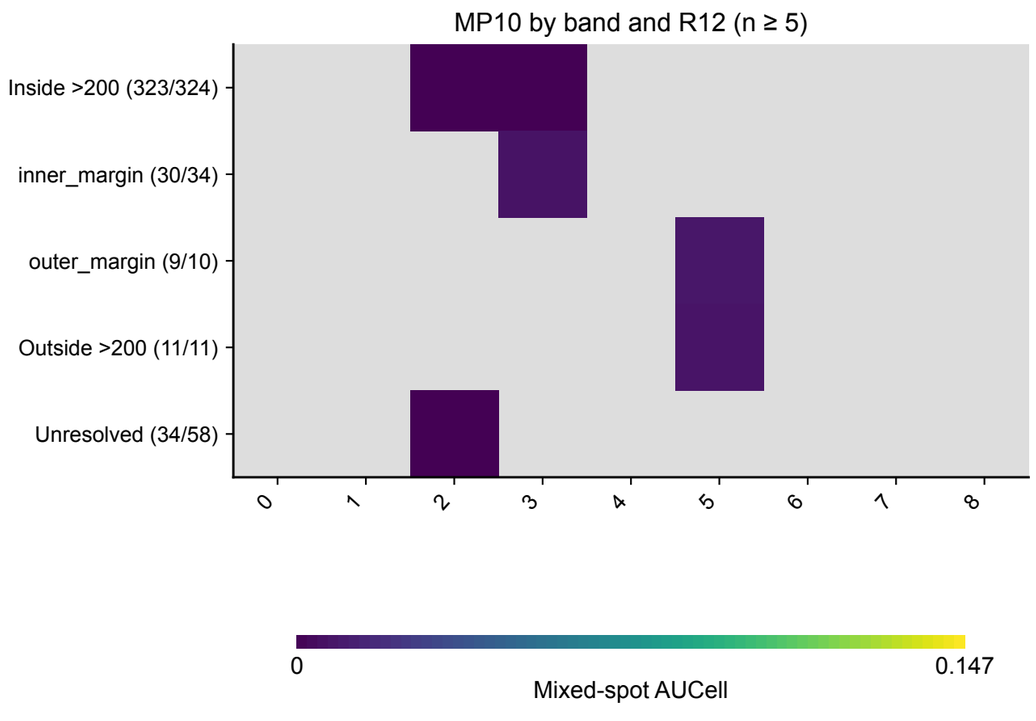
GSE238145_B10 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable









GSE238145_B10 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets

0 0.152

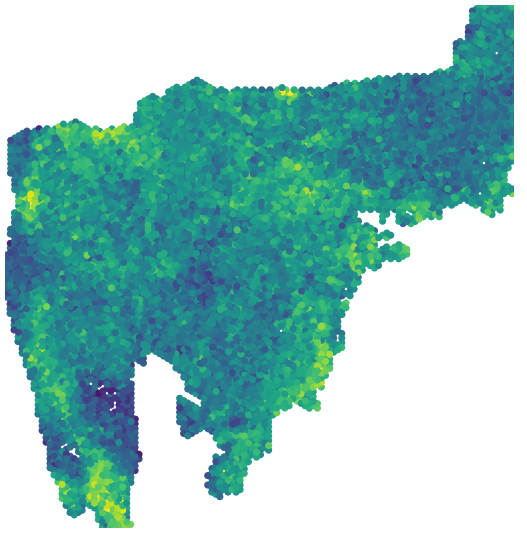
UCell score

G2M Checkpoint

0 0.134

UCell score

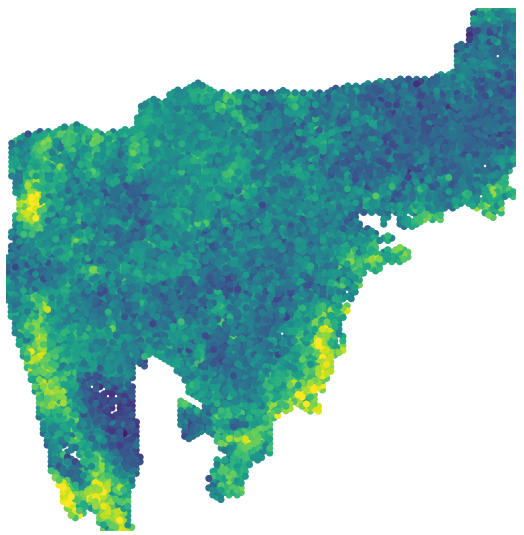
Hypoxia



0 0.211

UCell score

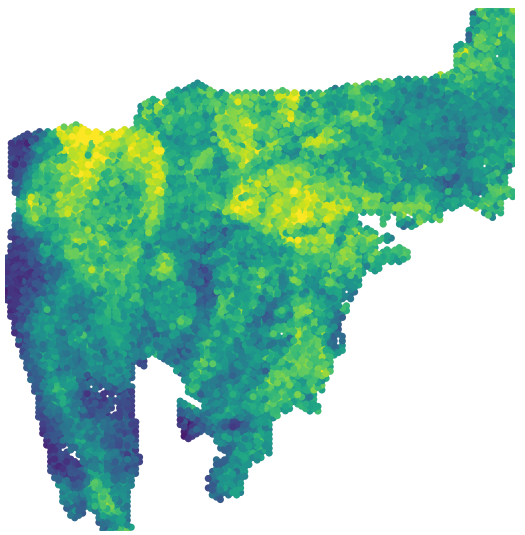
Tnfa Signaling Via Nfkb



0 0.275

UCell score

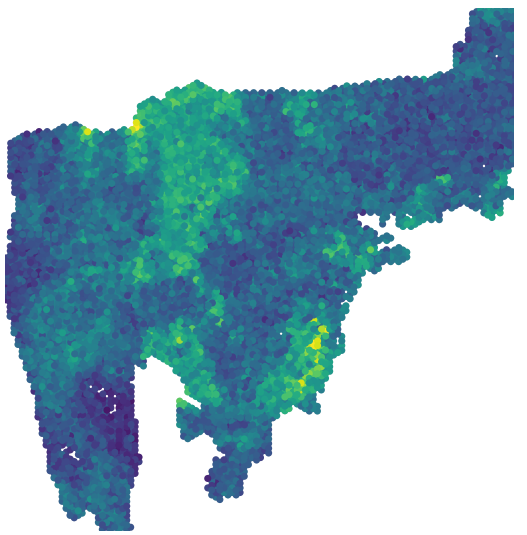
Emt



0 0.386

UCell score

Interferon Gamma Response



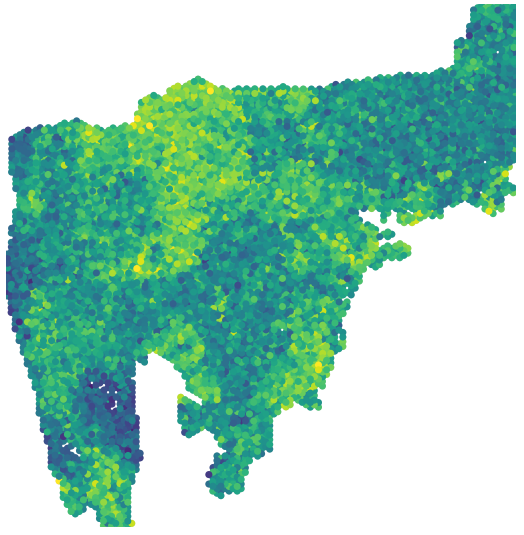
0 0.275

UCell score

GSE238145_B10 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

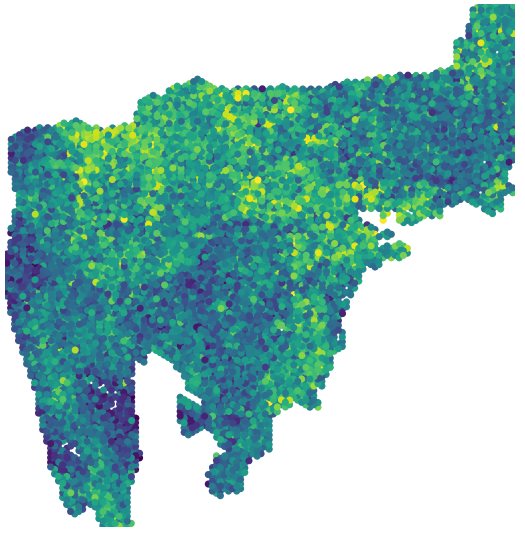
Il2 Stat5 Signaling



0 0.144

UCell score

Tgf Beta Signaling



0 0.232

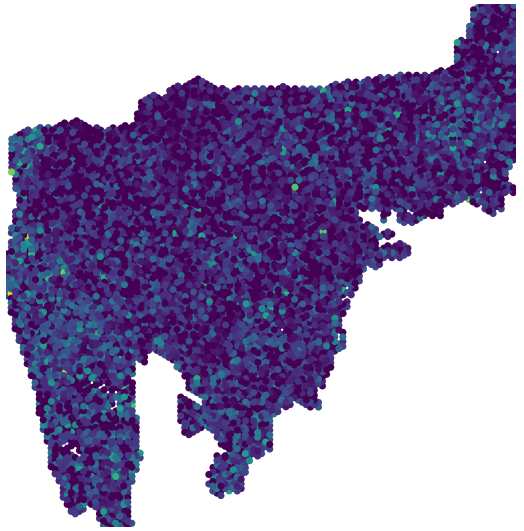
UCell score

Cholesterol Homeostasis

0 0.21

UCell score

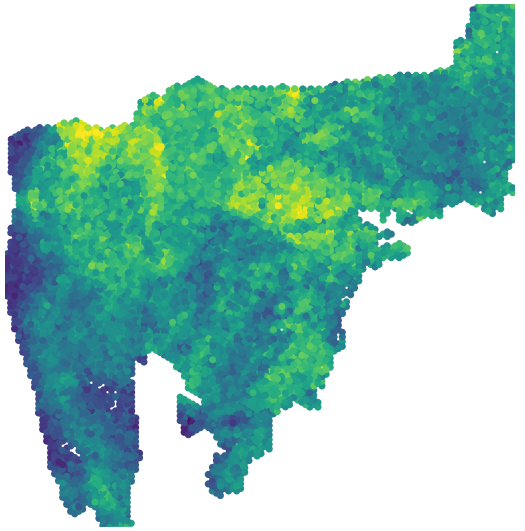
Cholesterol Biosynthesis



0 0.166

UCell score

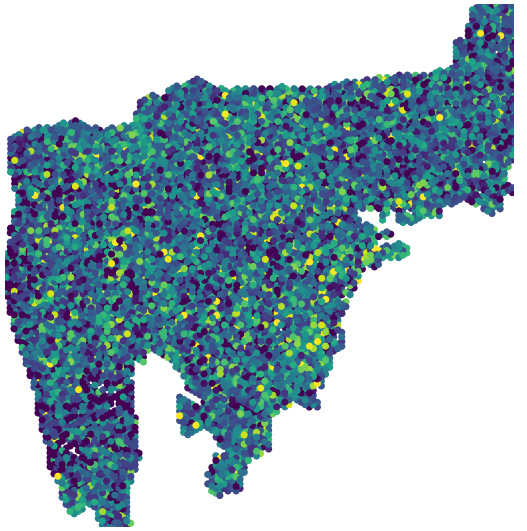
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



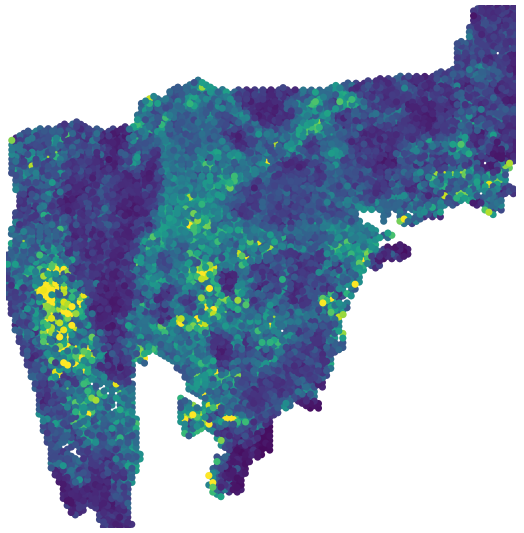
0 0.151

UCell score

GSE238145_B10 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

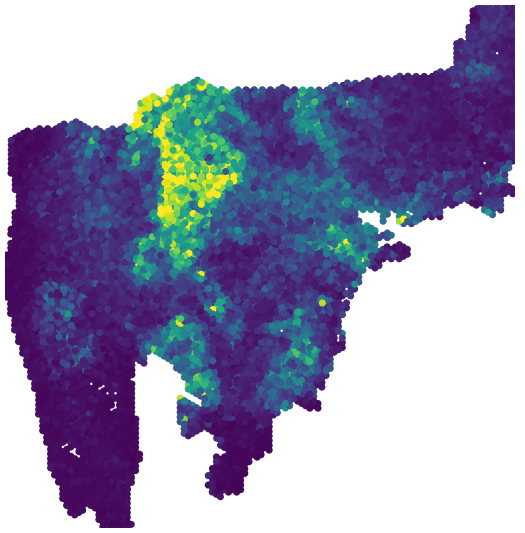
Treg composition



0 0.102

Fraction of all 30

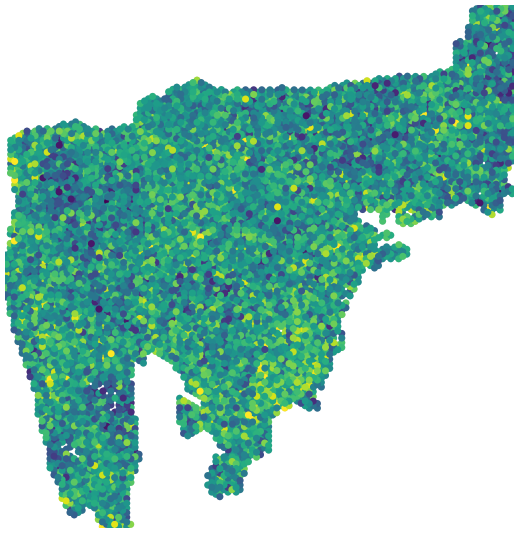
Treg estimated abundance



0 2.85

Posterior mean estimate

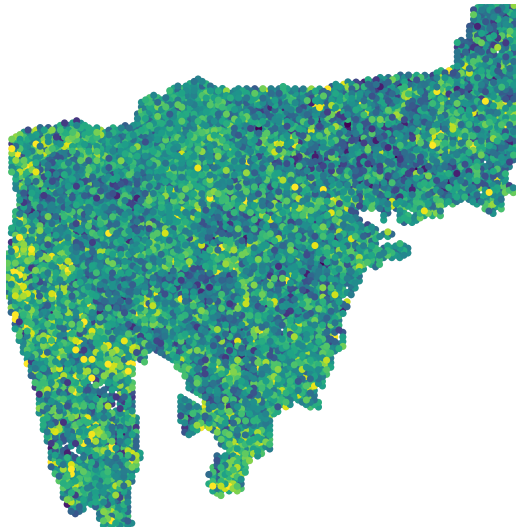
Treg Resting Vs Activated Dn



0 0.0778

UCell score

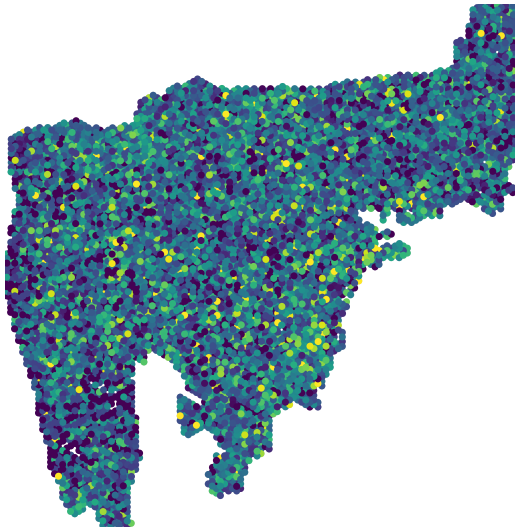
Treg Resting Vs Activated Up



0 0.0609

UCell score

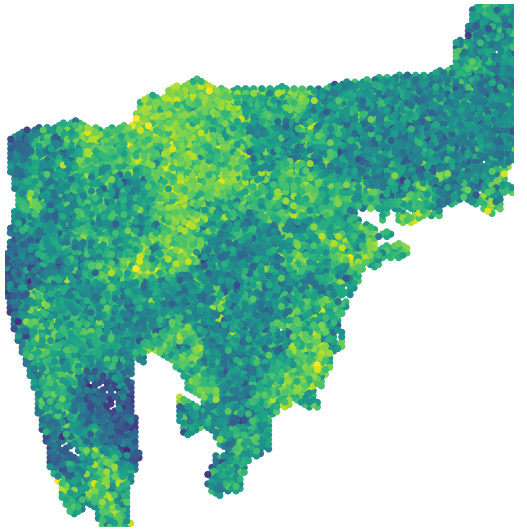
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

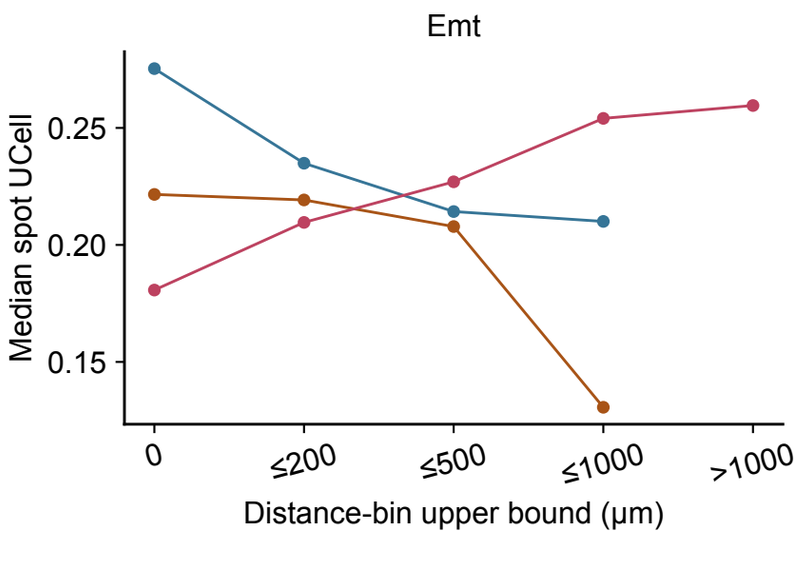


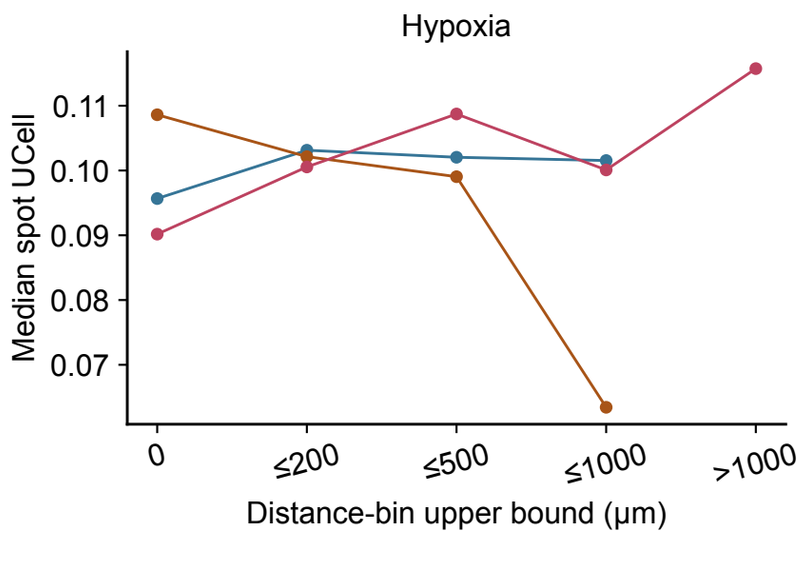
0 0.144

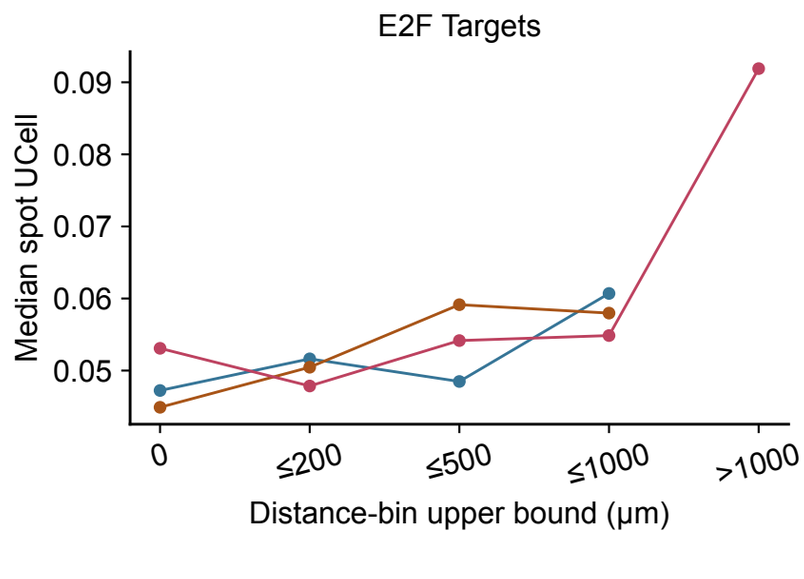
UCell score

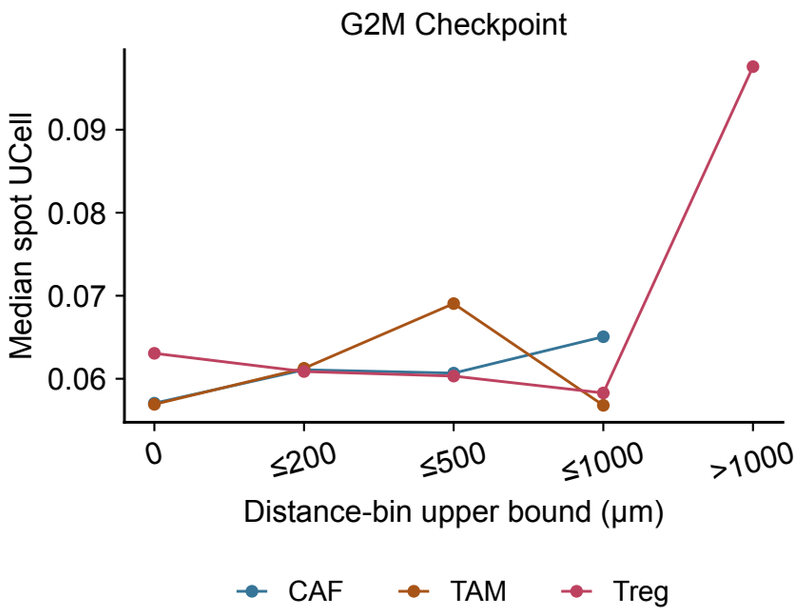
GSE238145_B10 | Programs by target distance

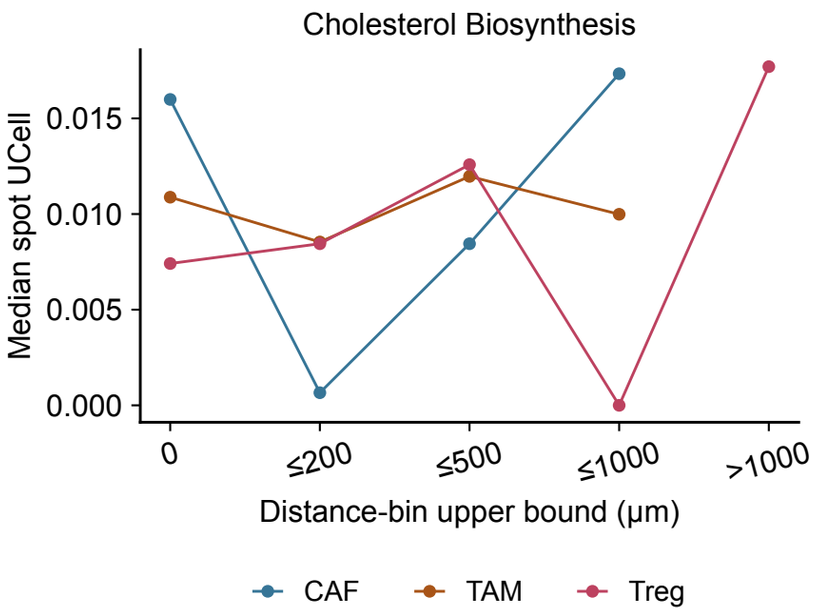
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

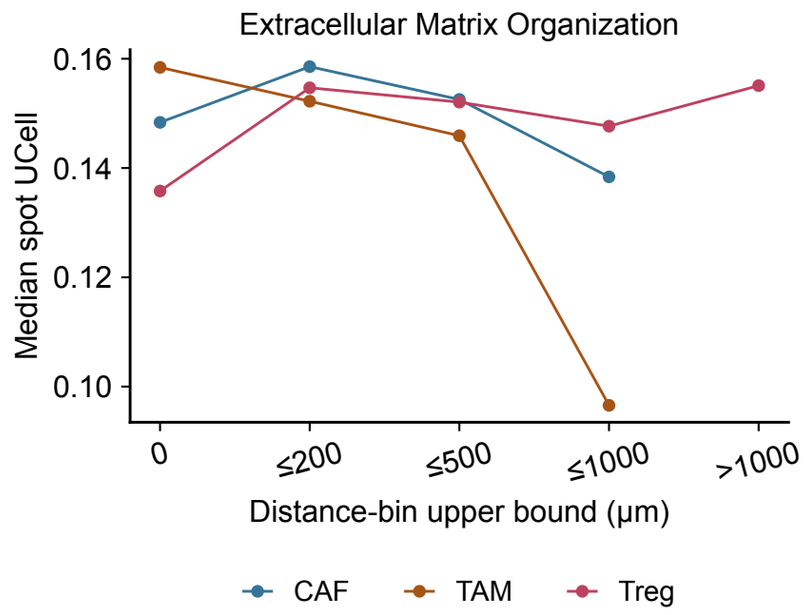






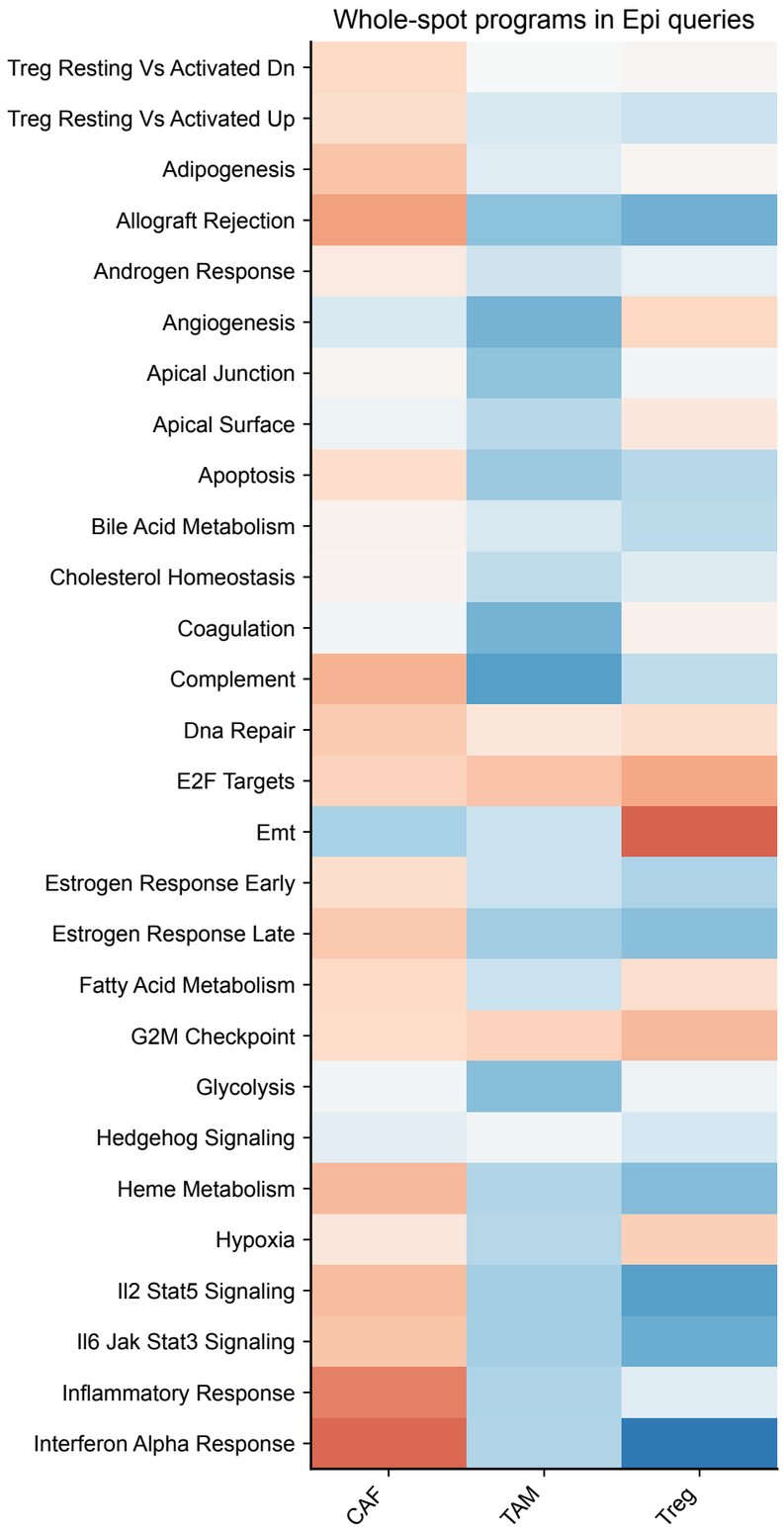


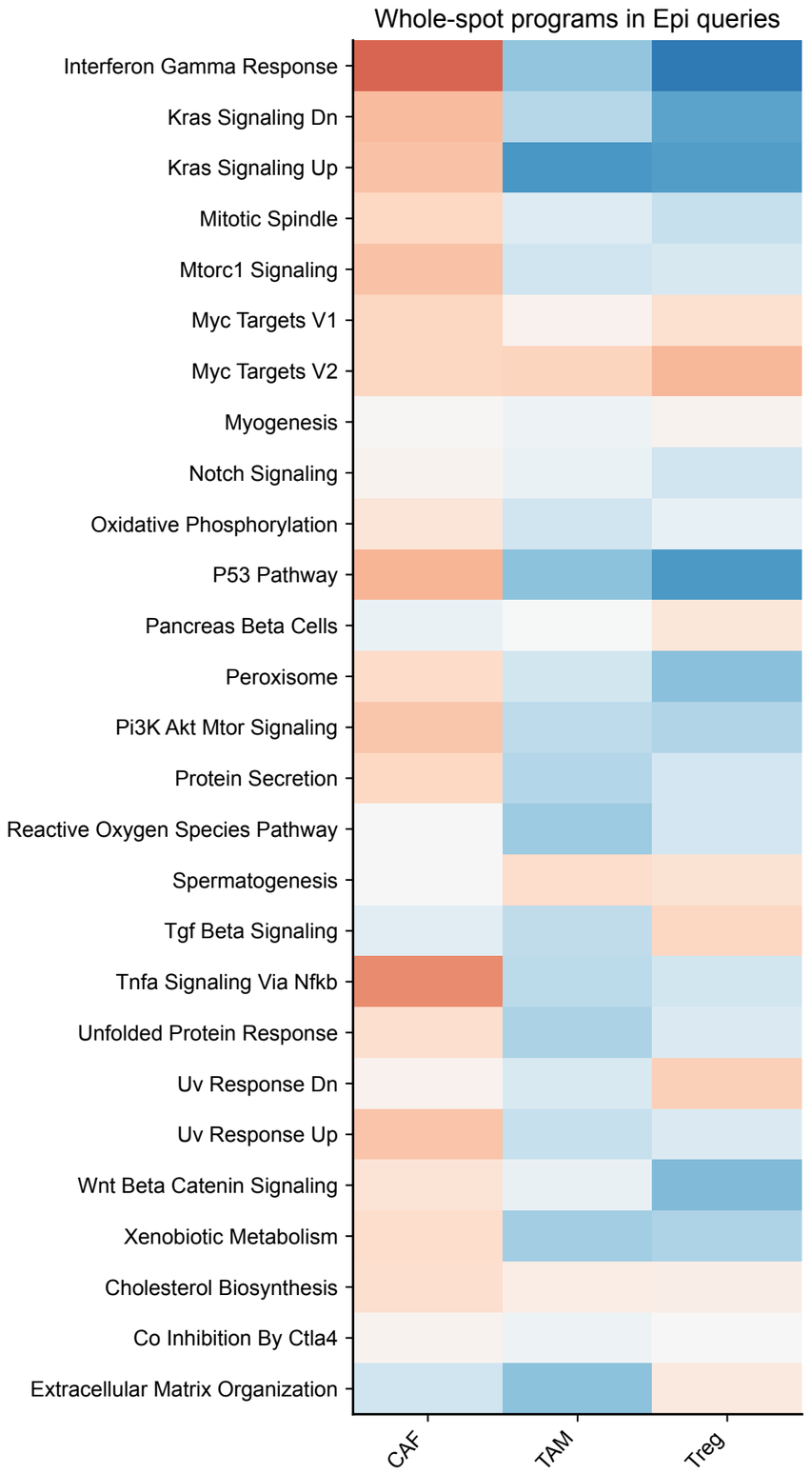




GSE238145_B10 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

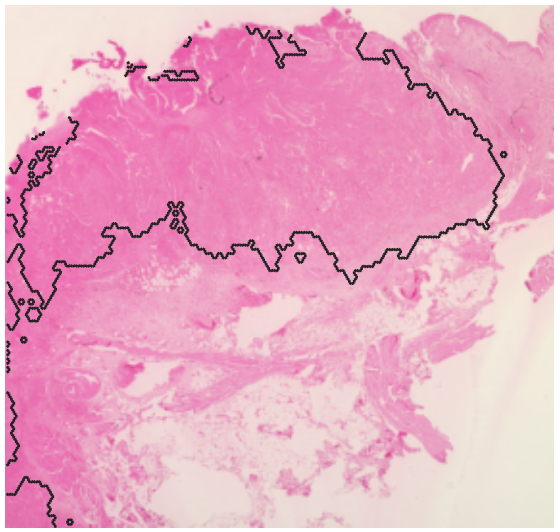




GSE238145_B11 | Tissue context

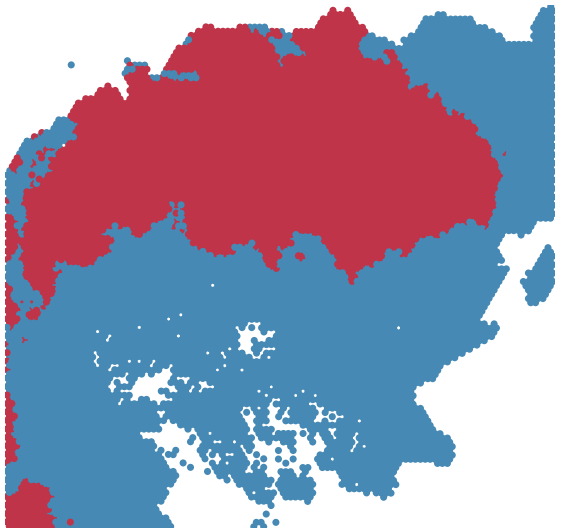
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



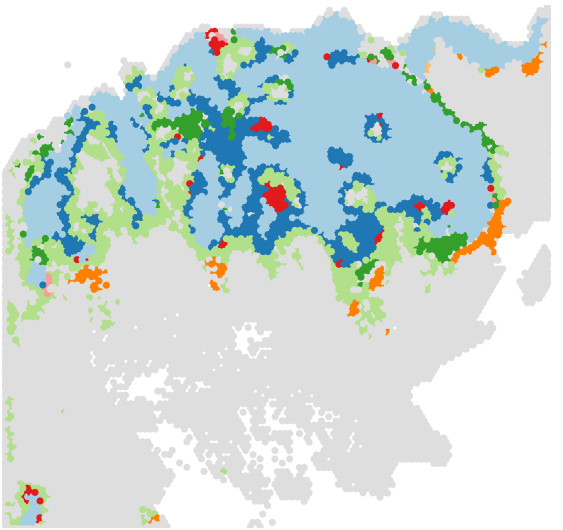
Black line: observed candidate outer edge

Final candidate location



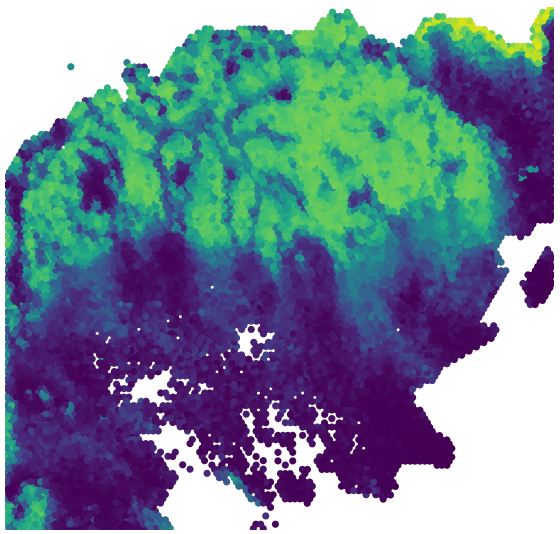
inside outside

Accepted R12 (original 0–8)



0 1 2 3 4 5 6 7

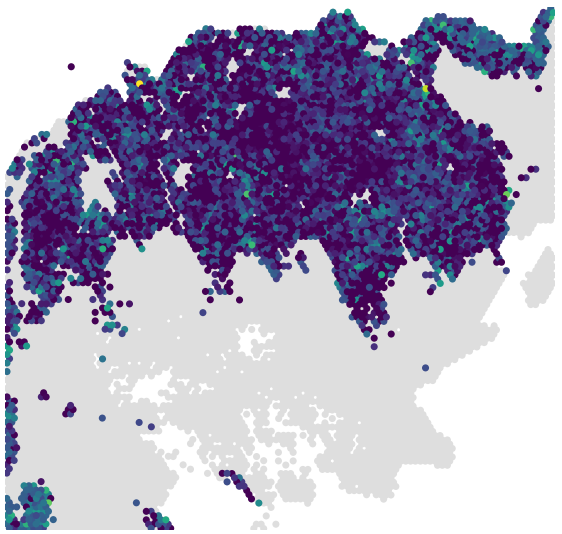
Epithelial support



0 1

Fraction of all 30

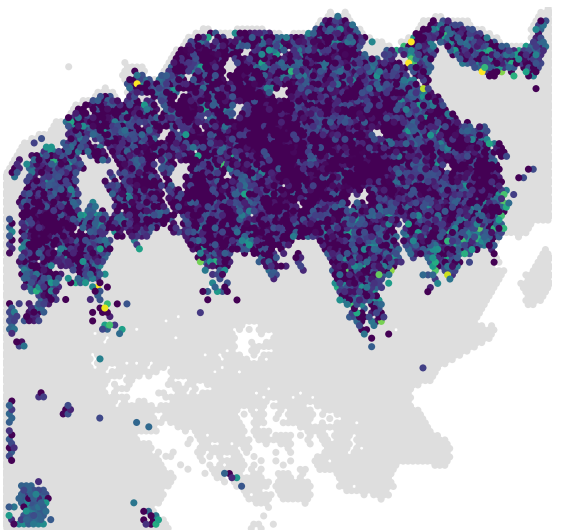
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



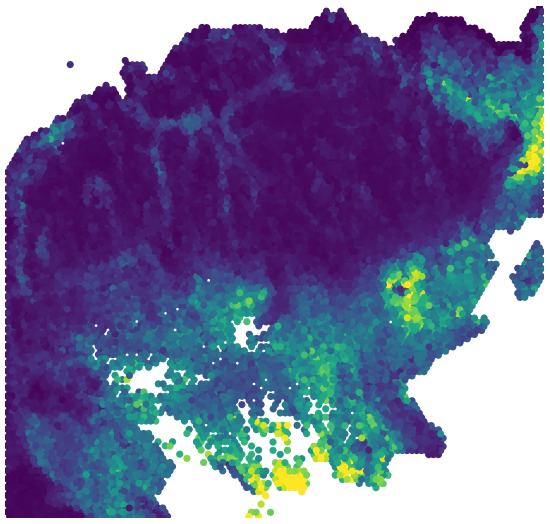
0 0.14

Matched-model AUCell

GSE238145_B11 | Candidate neighborhoods

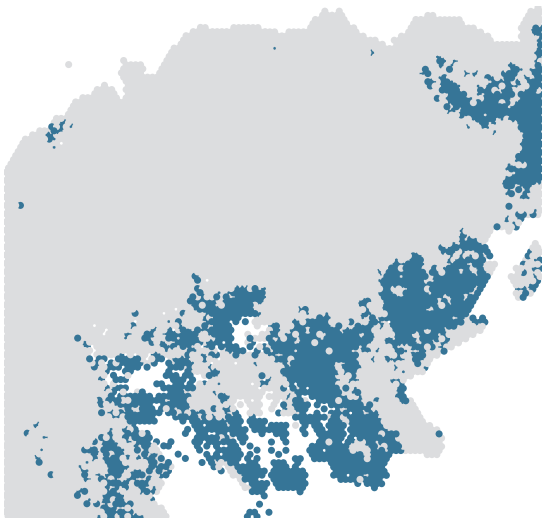
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



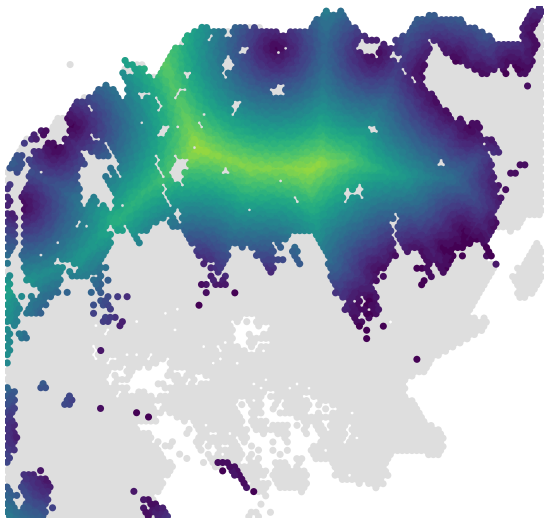
0 0.418
Fraction of all 30

CAF P80 targets



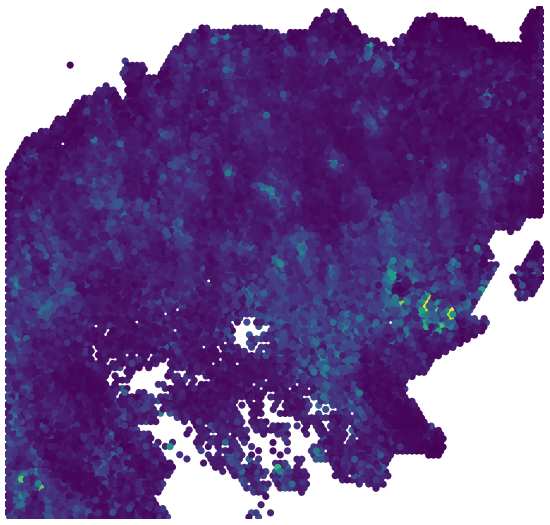
False True

CAF distance



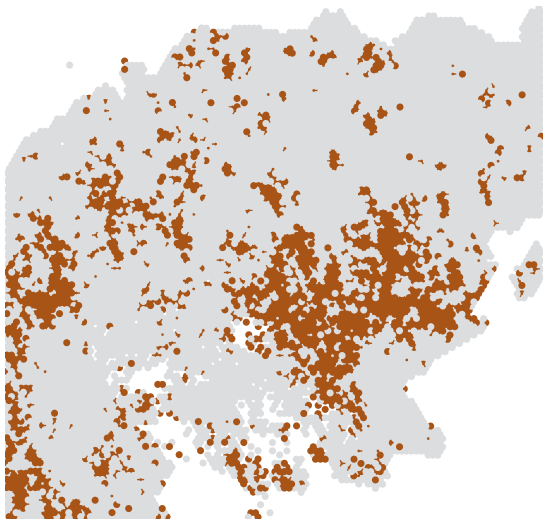
0 3.1e+03
Nearest spot center (µm)

TAM composition



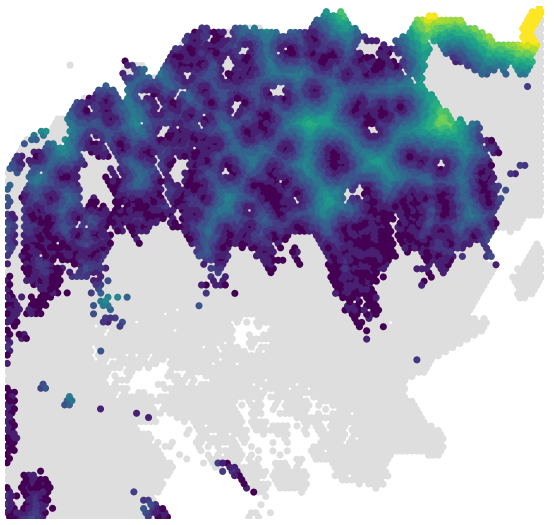
0 0.291
Fraction of all 30

TAM P80 targets



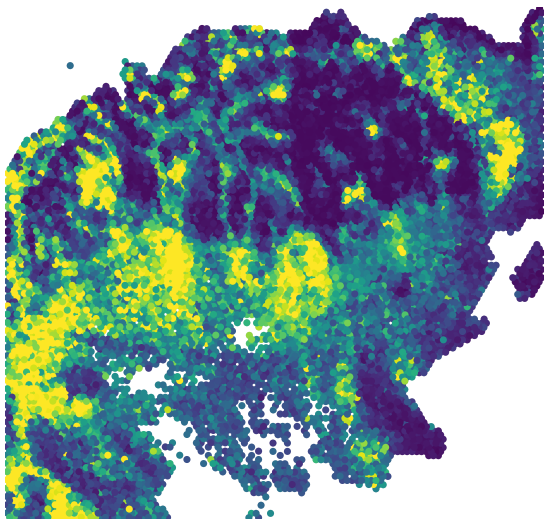
False True

TAM distance



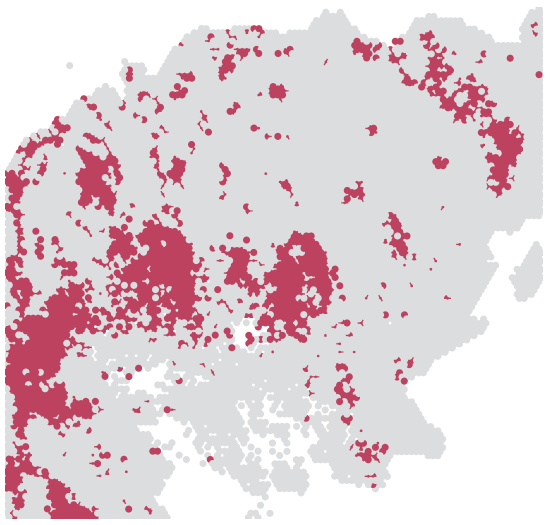
0 1.13e+03
Nearest spot center (µm)

Treg composition



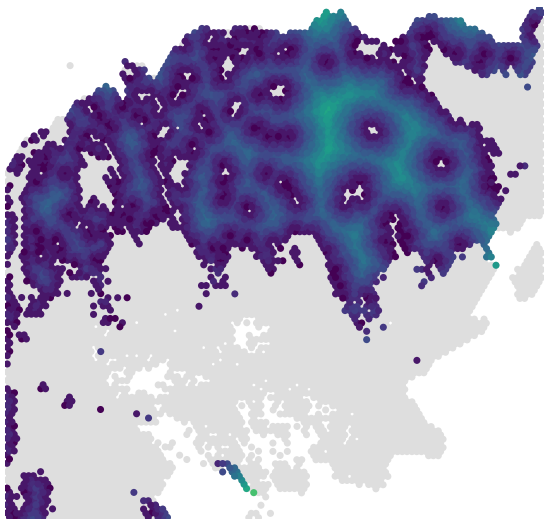
0 0.102
Fraction of all 30

Treg P80 targets



False True

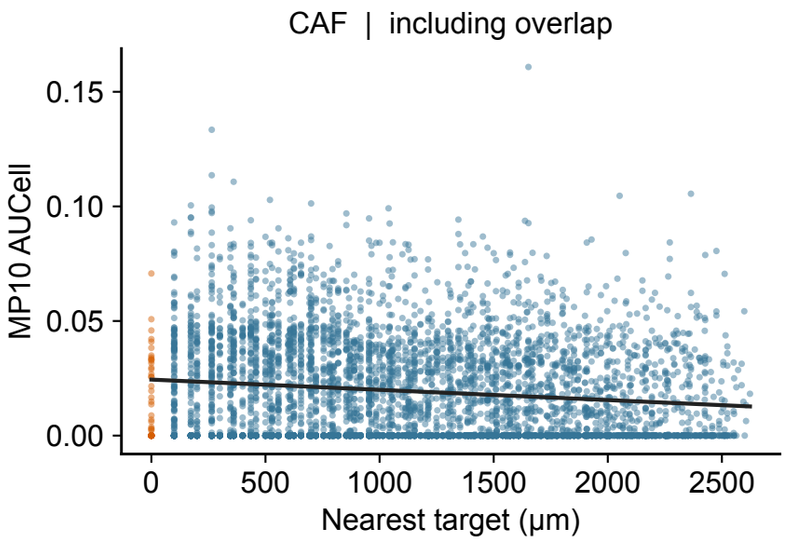
Treg distance



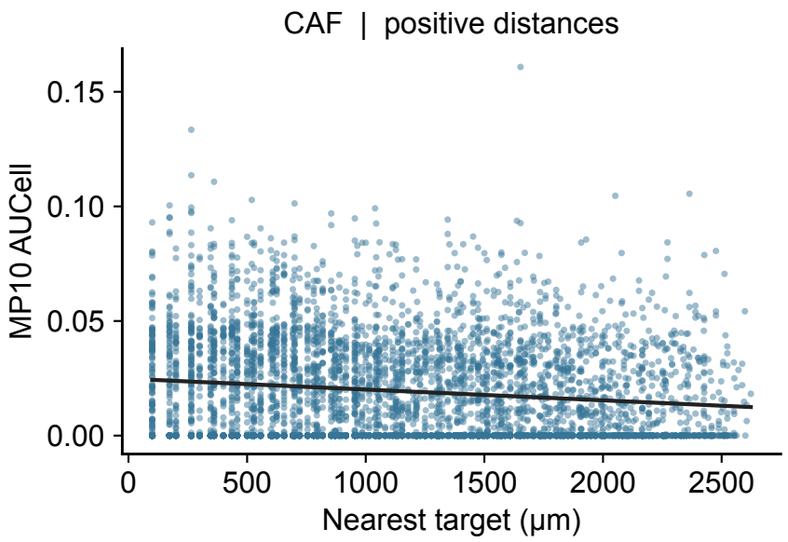
0 1.61e+03
Nearest spot center (µm)

GSE238145_B11 | MP10 versus nearest target

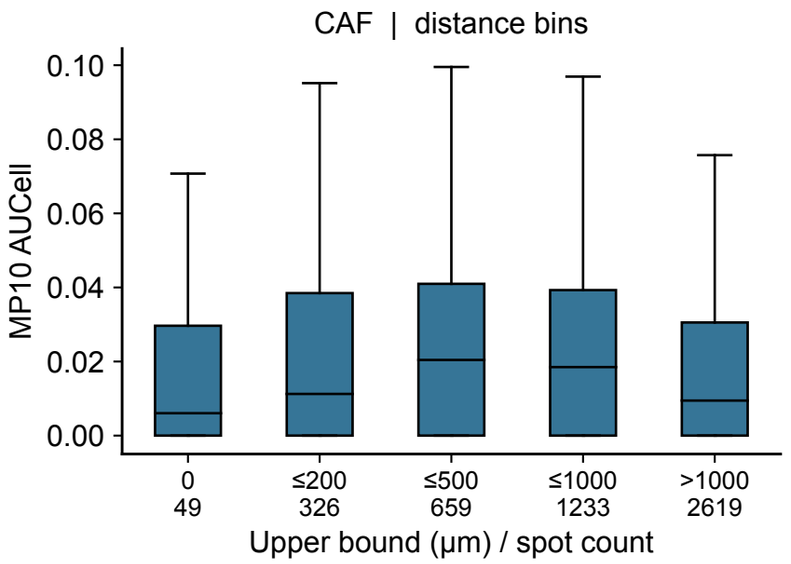
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



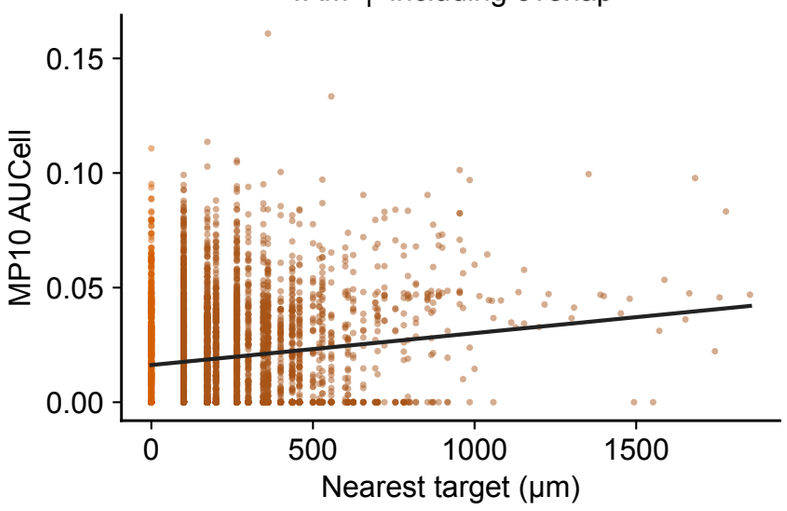
r -0.14 ρ -0.12
n 4886/4887 queries • 2062 targets



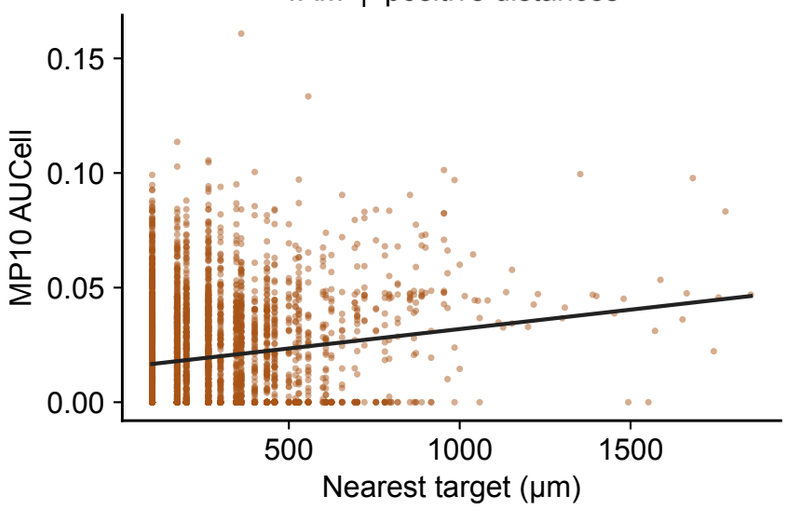
r -0.14 ρ -0.12
n 4837/4887 queries • 2062 targets



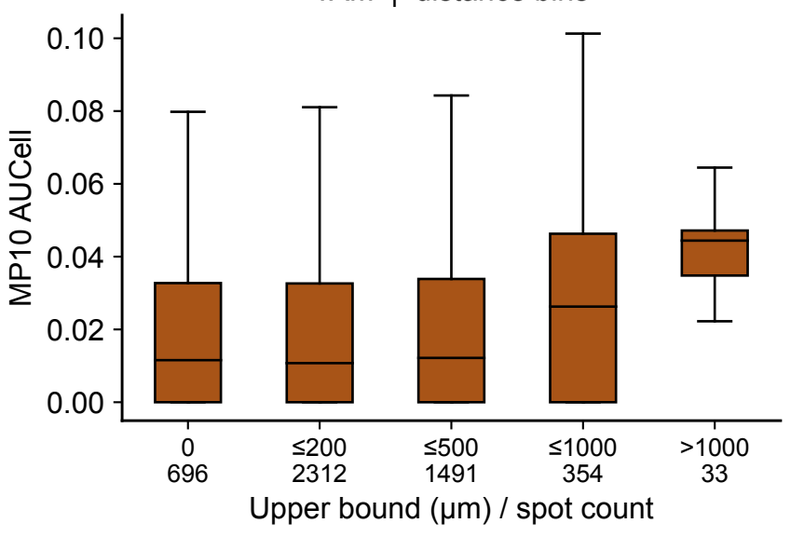
Orange = zero distance
Line = within-section linear fit



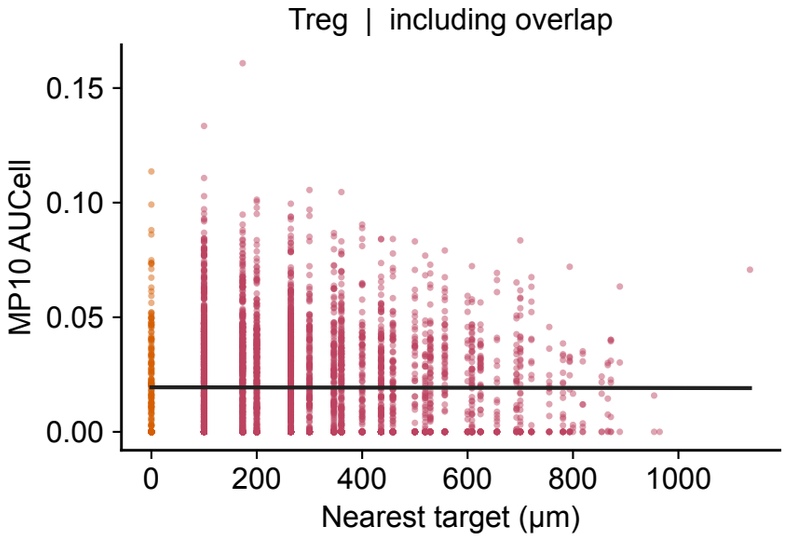
r +0.13 ρ +0.07
n 4886/4887 queries • 2062 targets



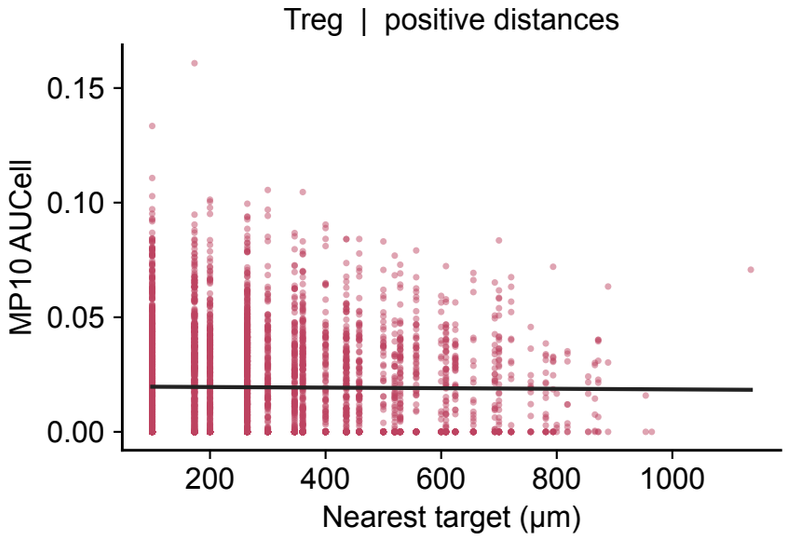
r +0.15 ρ +0.08
n 4190/4887 queries • 2062 targets



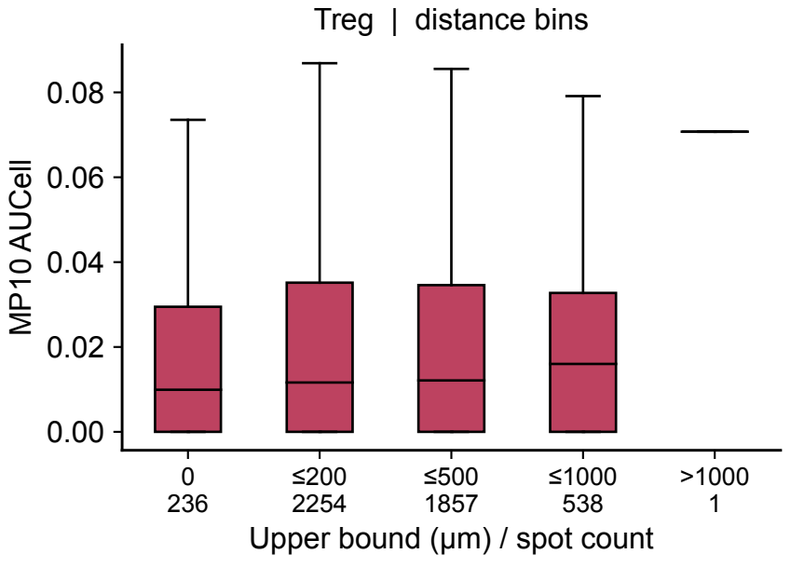
Orange = zero distance
Line = within-section linear fit



r -0.00 ρ +0.02
n 4886/4887 queries • 2062 targets



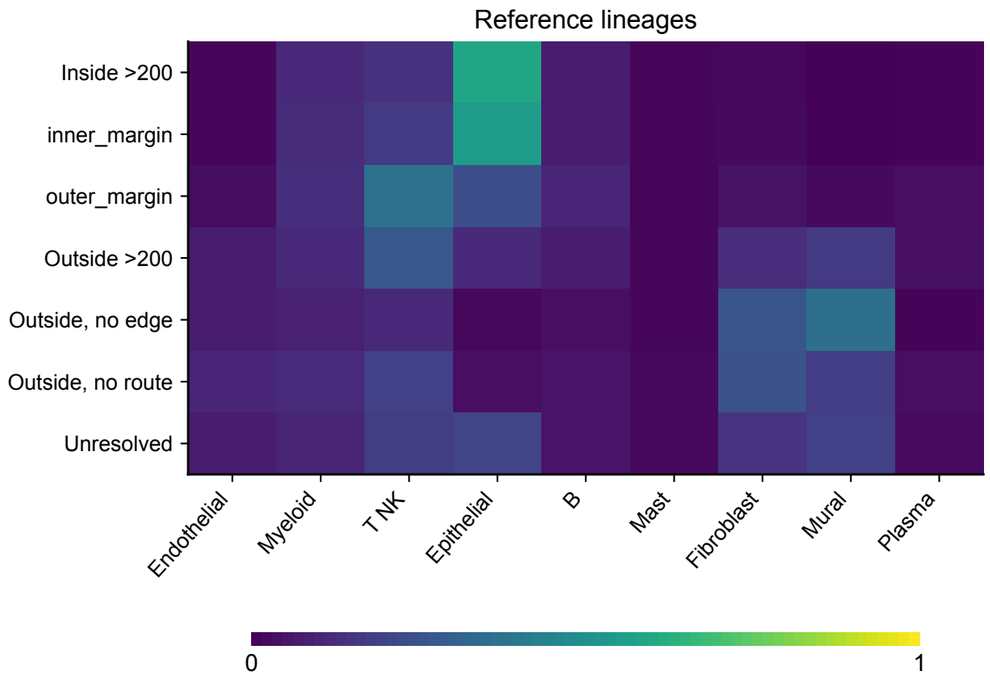
r -0.01 ρ +0.01
n 4650/4887 queries • 2062 targets

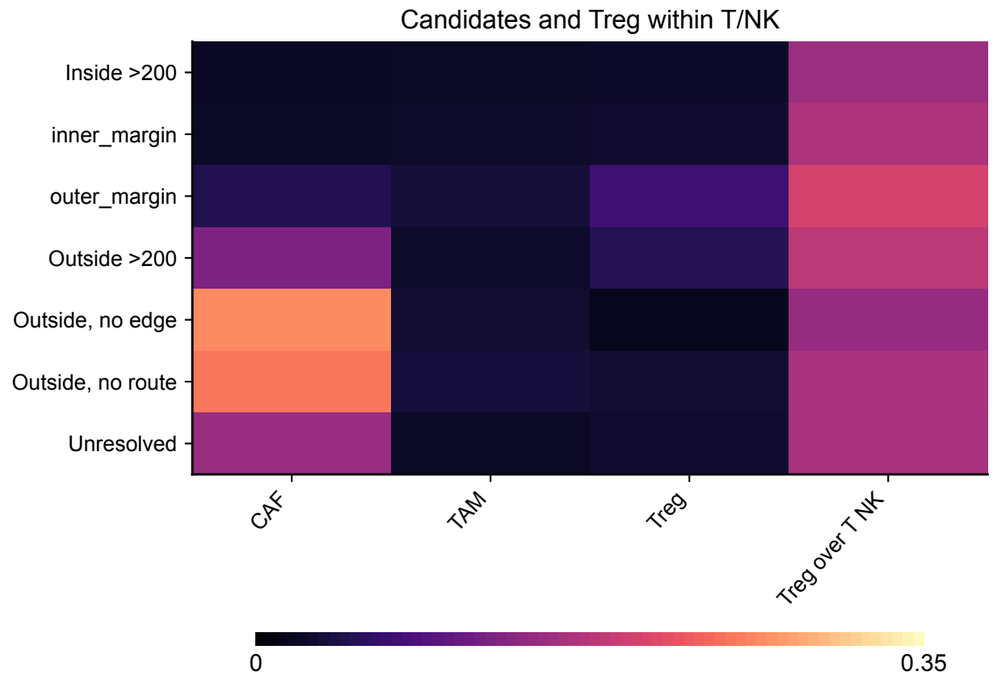


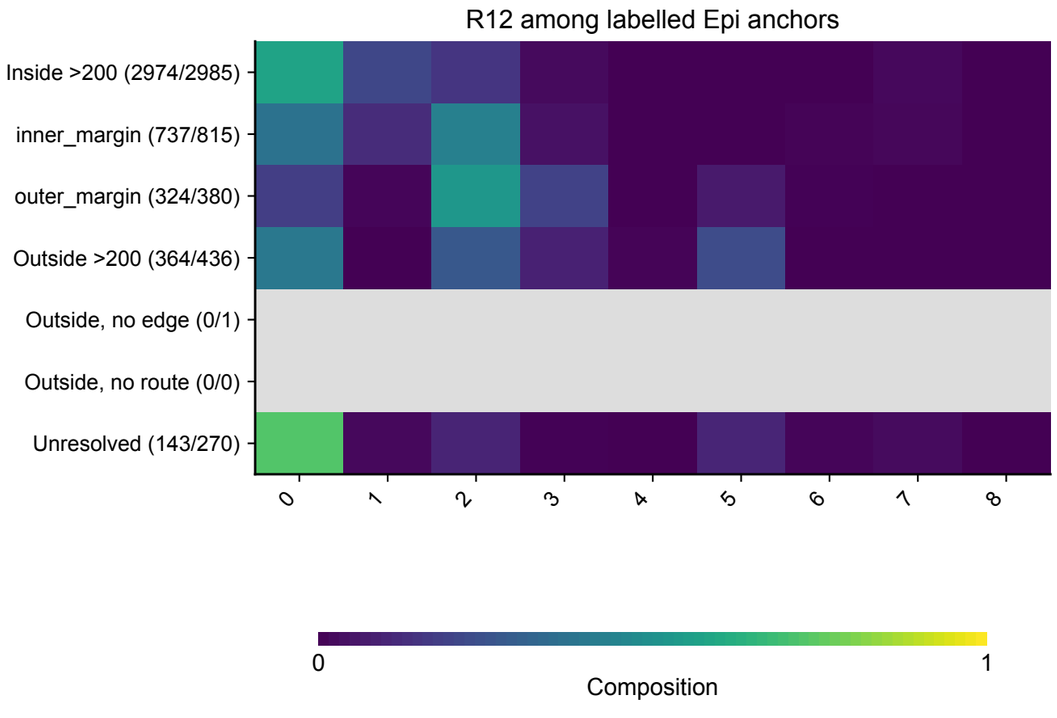
Orange = zero distance
Line = within-section linear fit

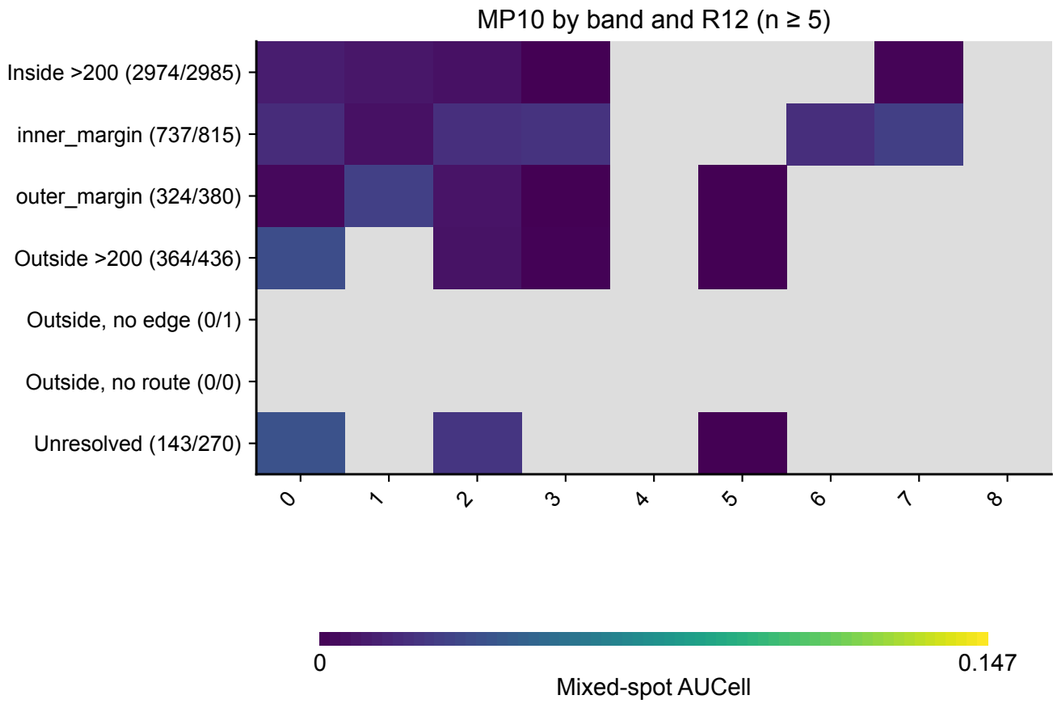
GSE238145_B11 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





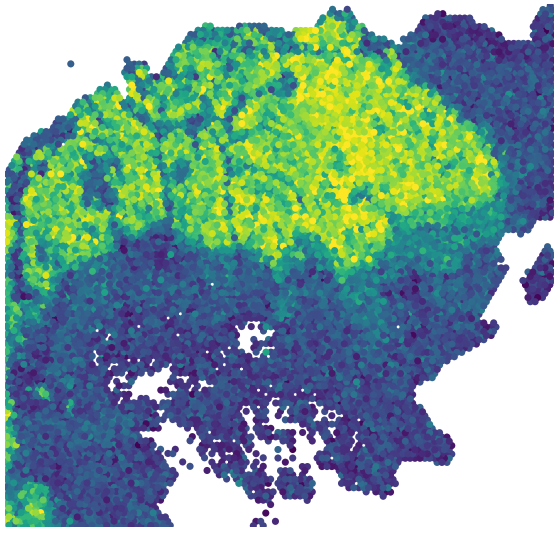




GSE238145_B11 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

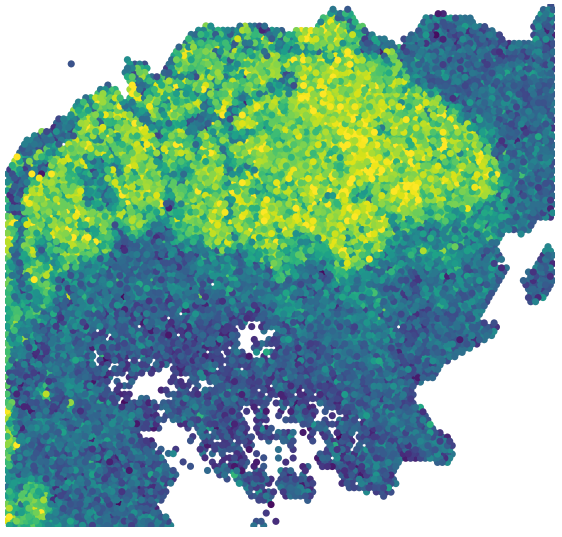
E2F Targets



0 0.152

UCell score

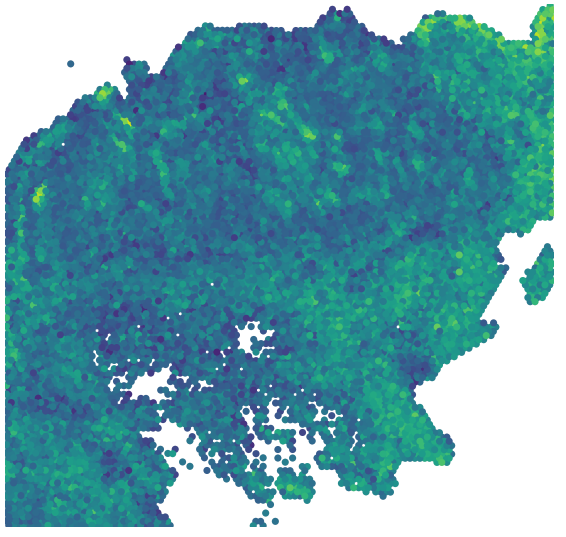
G2M Checkpoint



0 0.134

UCell score

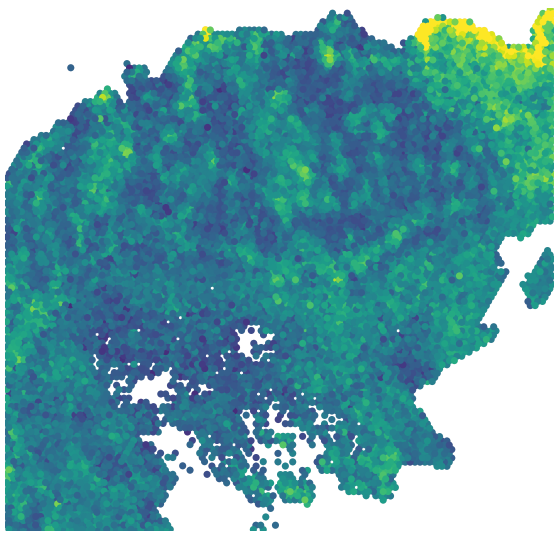
Hypoxia



0 0.211

UCell score

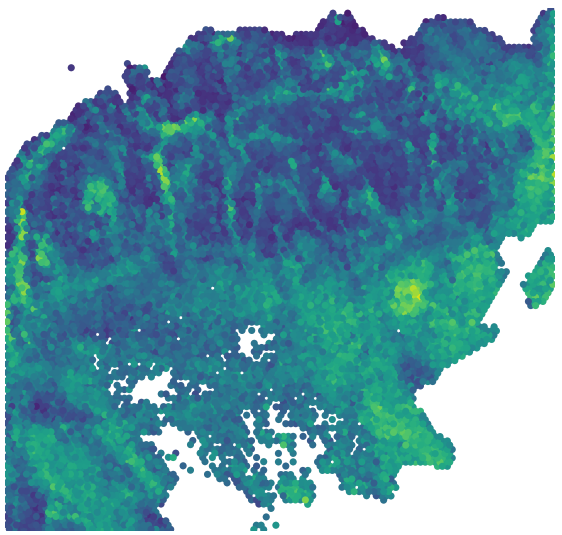
Tnfa Signaling Via Nfkb



0 0.275

UCell score

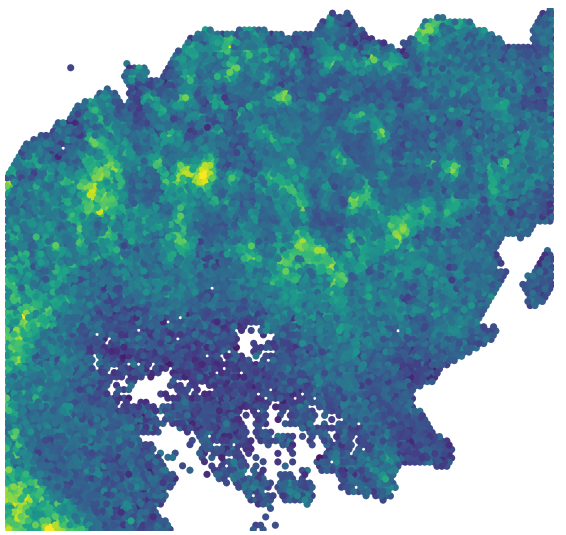
Emt



0 0.386

UCell score

Interferon Gamma Response



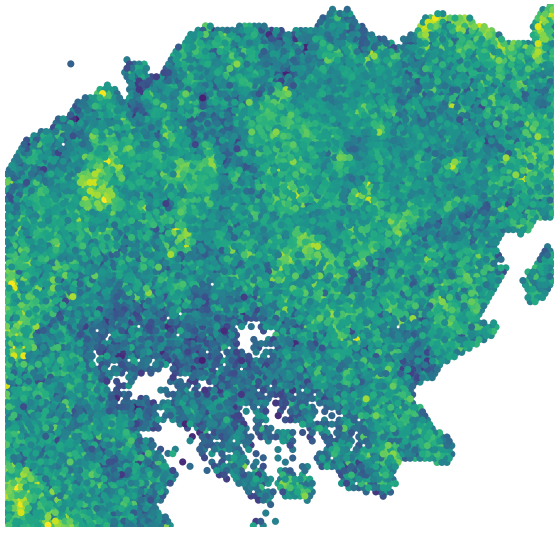
0 0.275

UCell score

GSE238145_B11 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

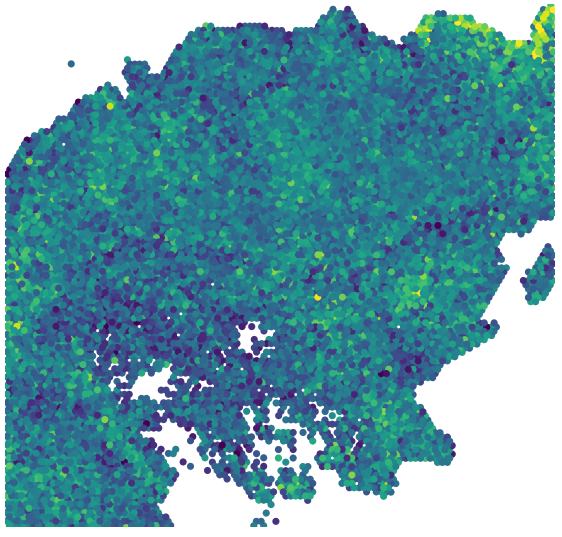
Il2 Stat5 Signaling



0 0.144

UCell score

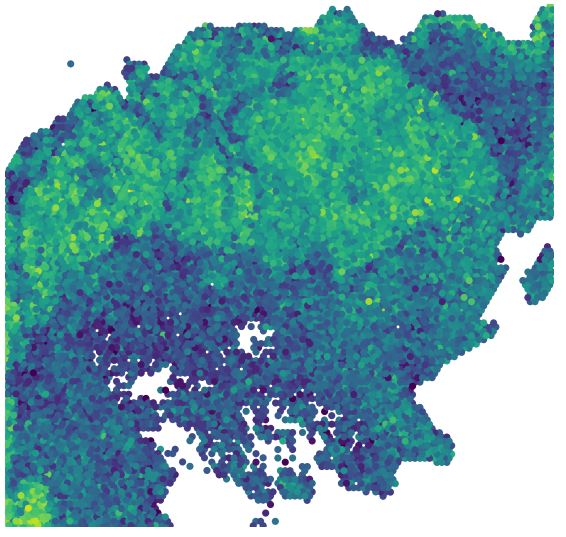
Tgf Beta Signaling



0 0.232

UCell score

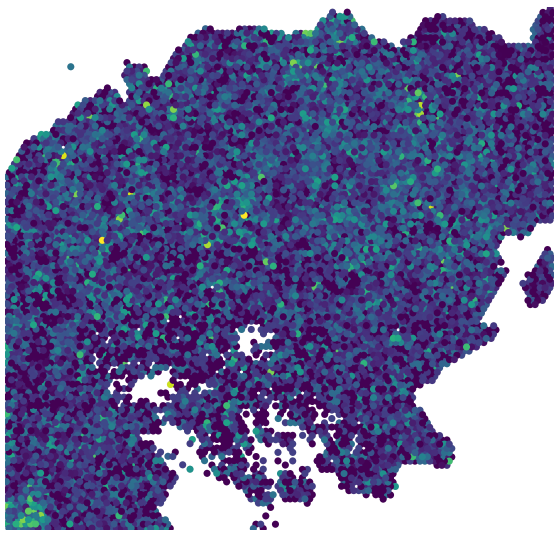
Cholesterol Homeostasis



0 0.21

UCell score

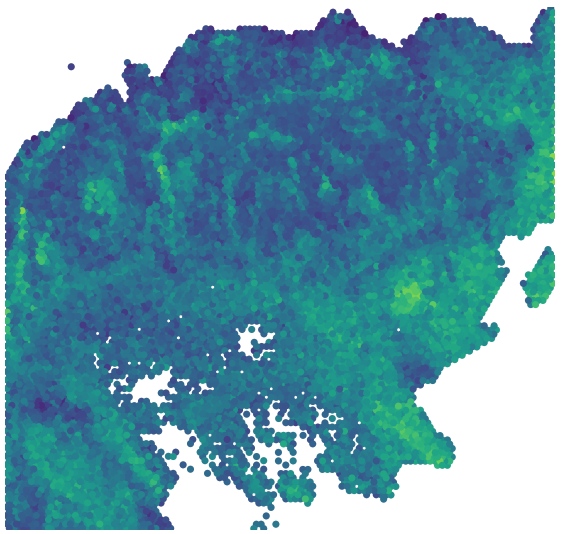
Cholesterol Biosynthesis



0 0.166

UCell score

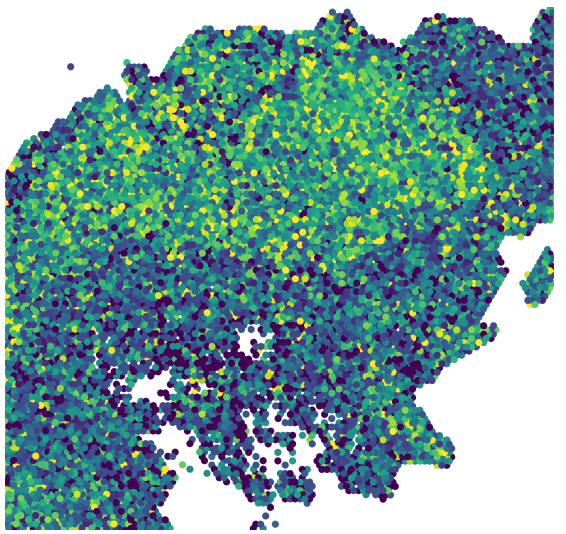
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



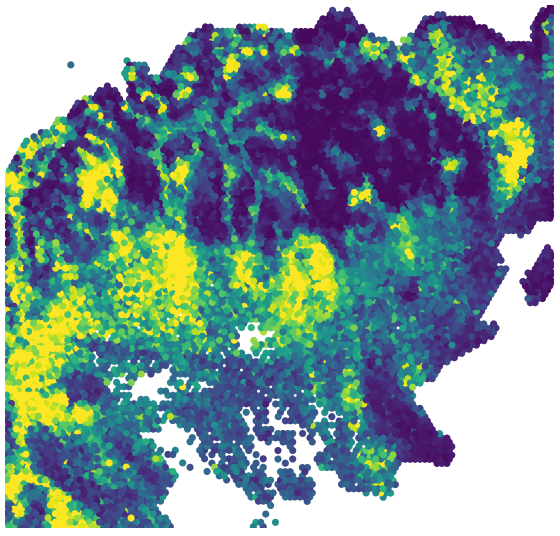
0 0.151

UCell score

GSE238145_B11 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

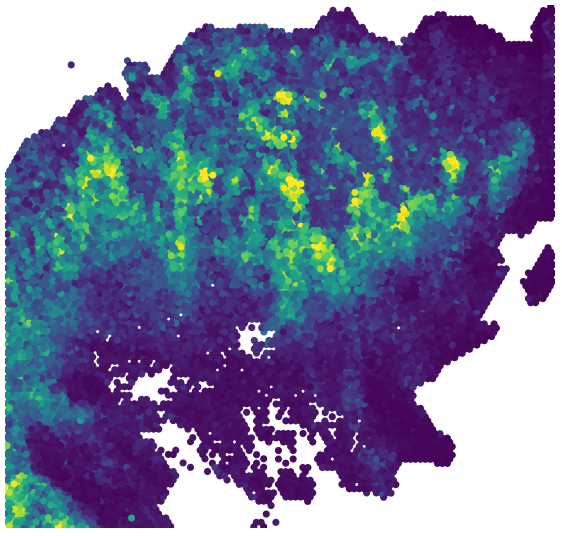
Treg composition



0 0.102

Fraction of all 30

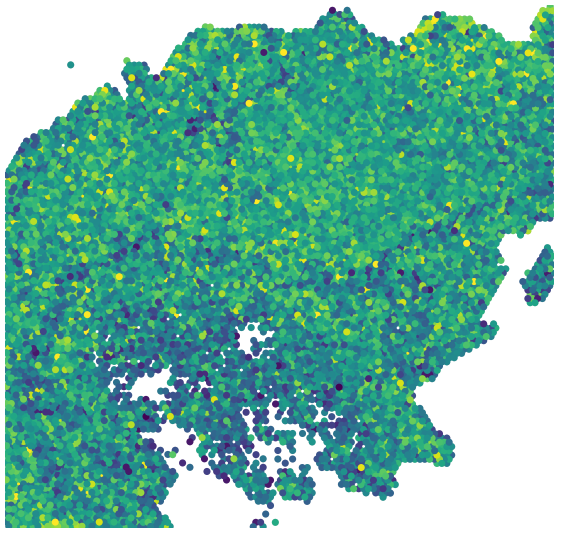
Treg estimated abundance



0 2.85

Posterior mean estimate

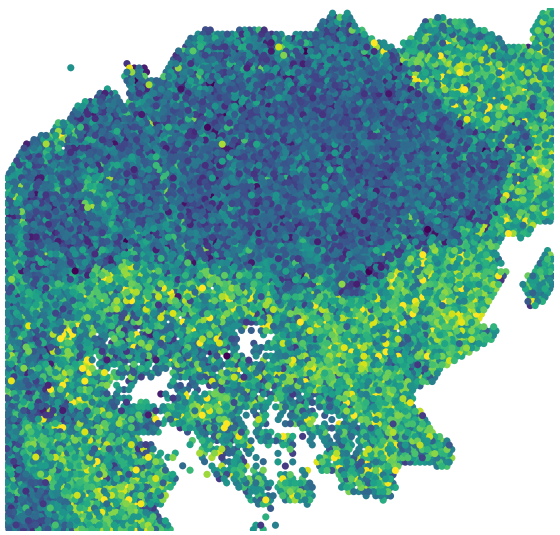
Treg Resting Vs Activated Dn



0 0.0778

UCell score

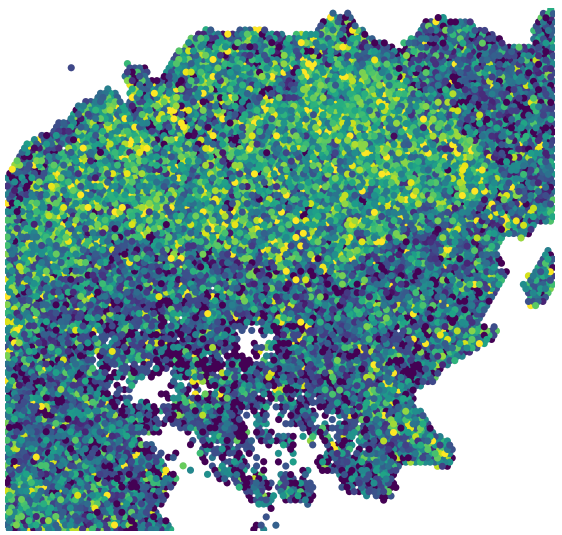
Treg Resting Vs Activated Up



0 0.0609

UCell score

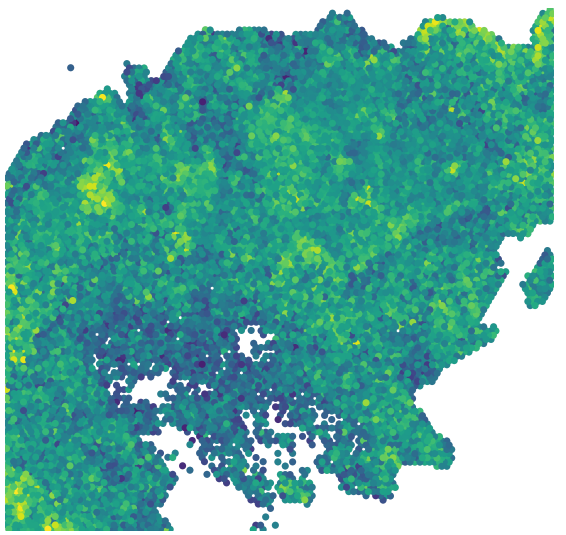
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

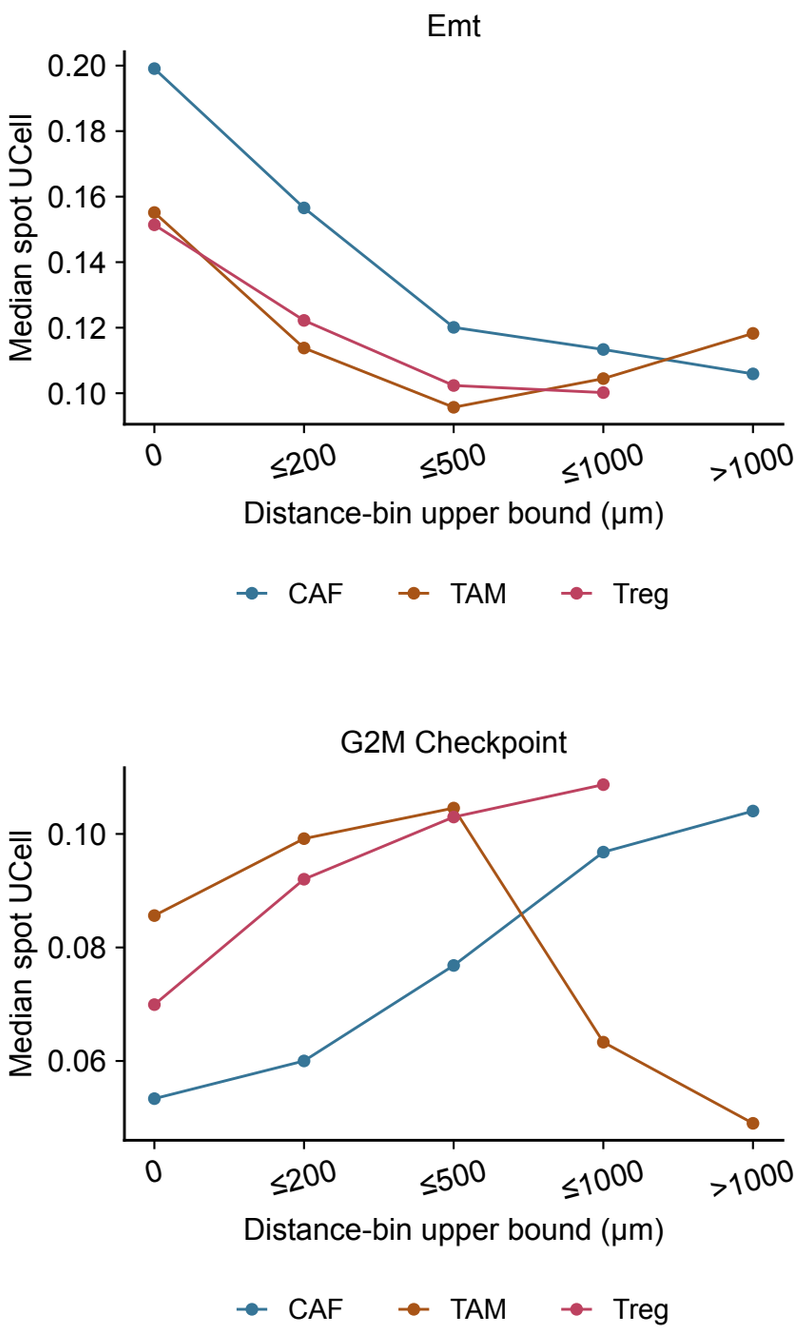


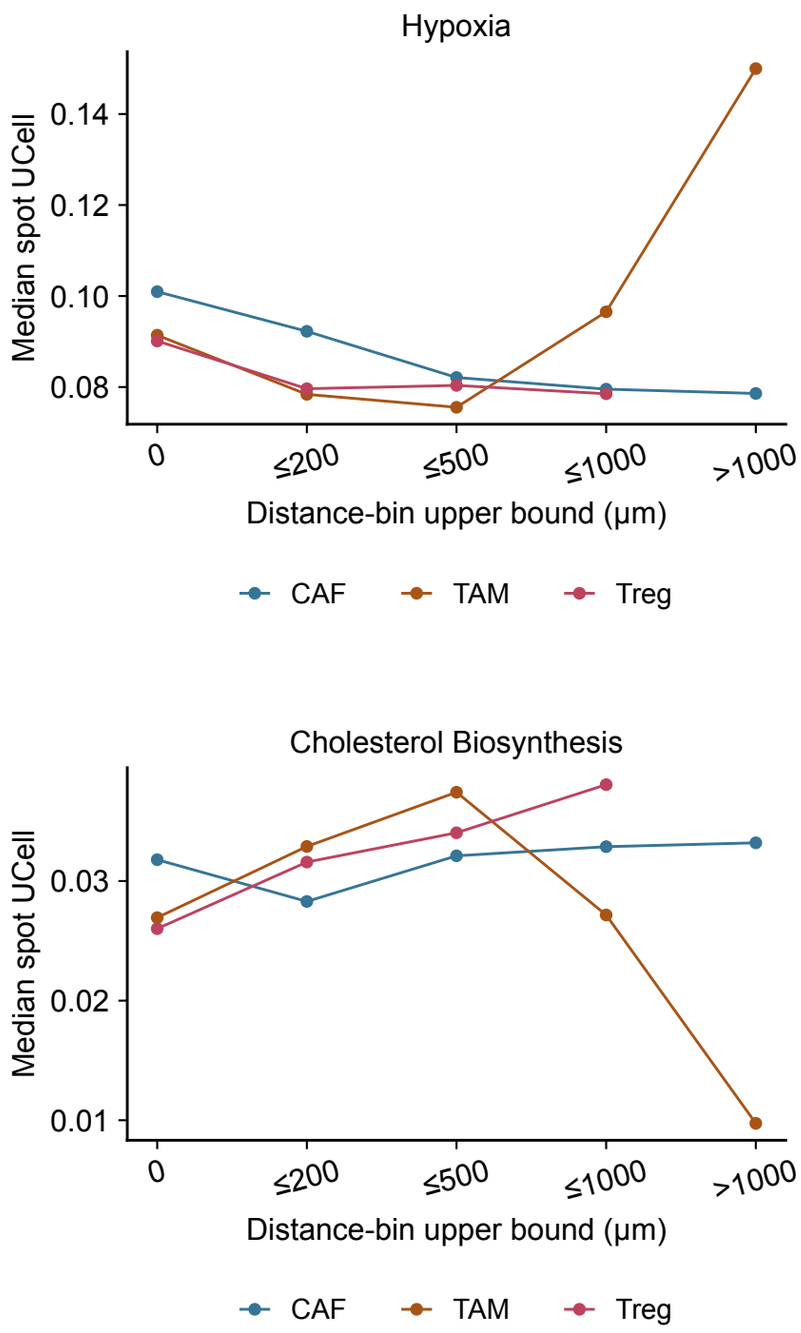
0 0.144

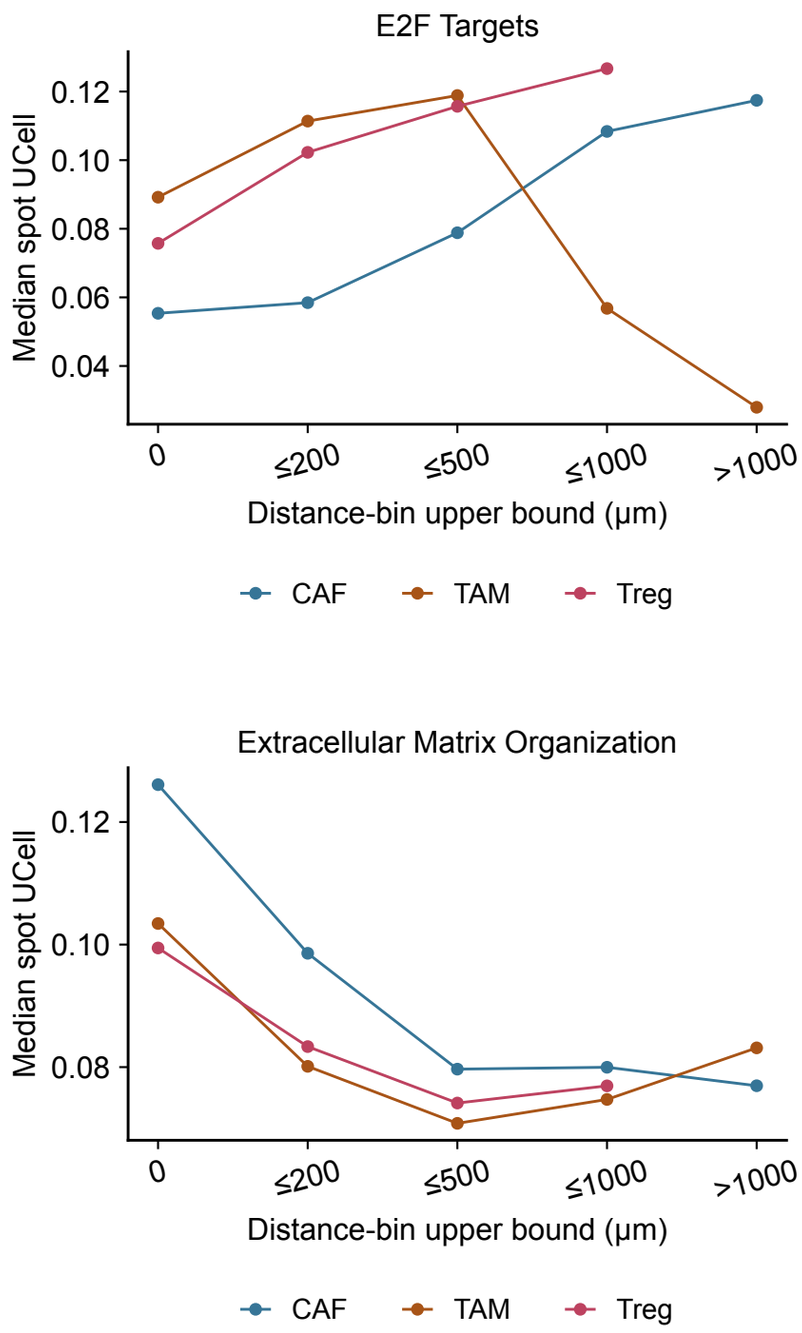
UCell score

GSE238145_B11 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

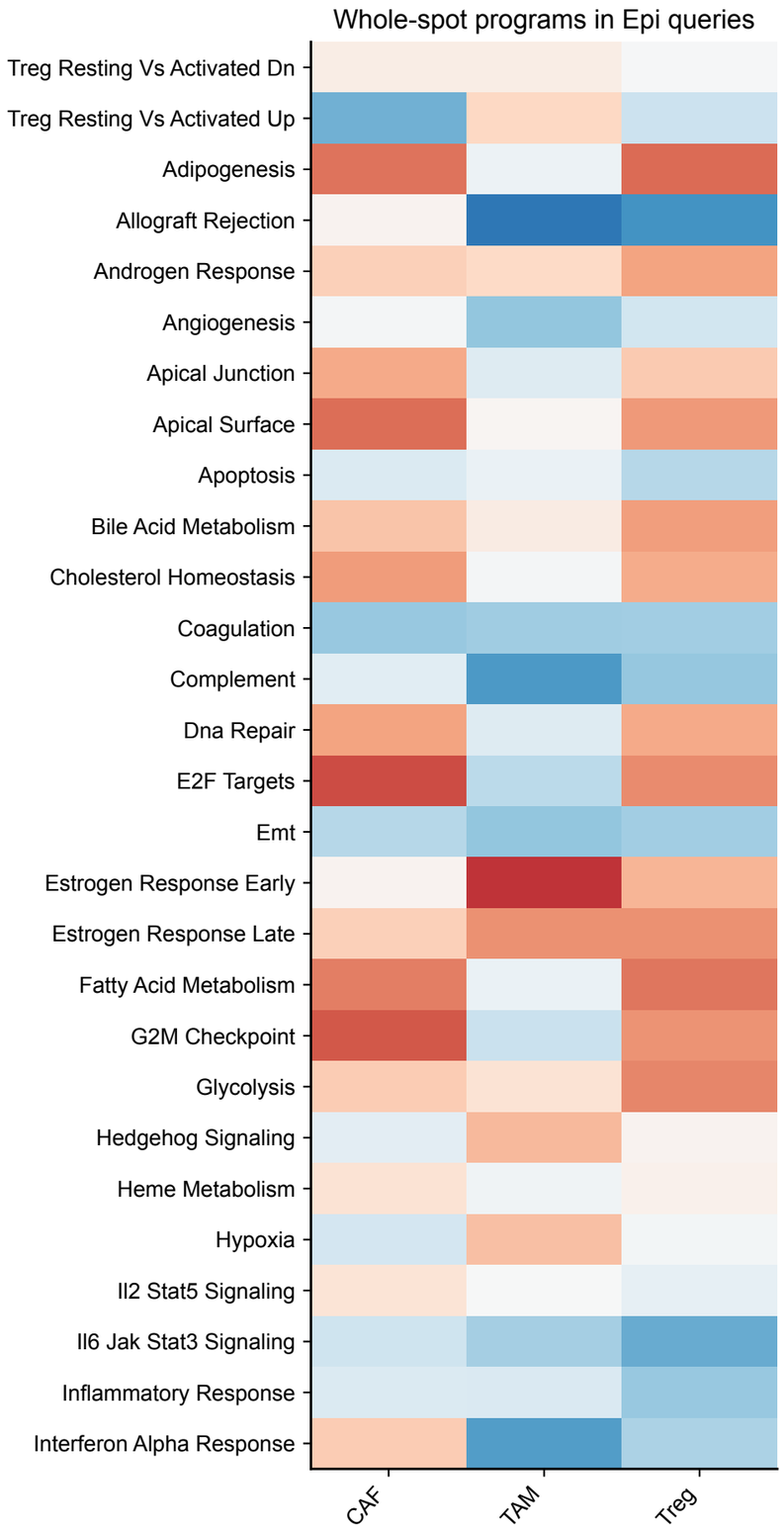


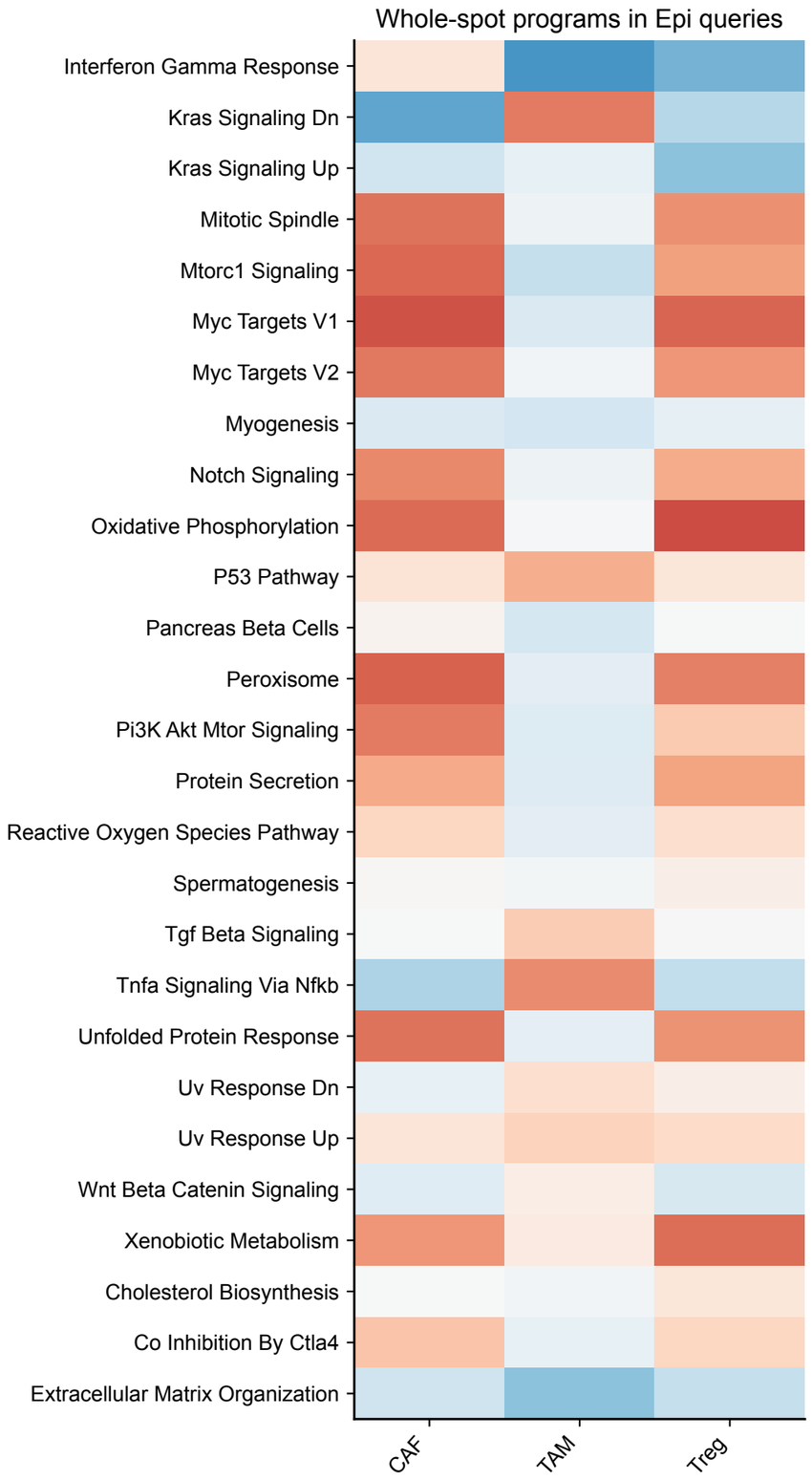




GSE238145_B11 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

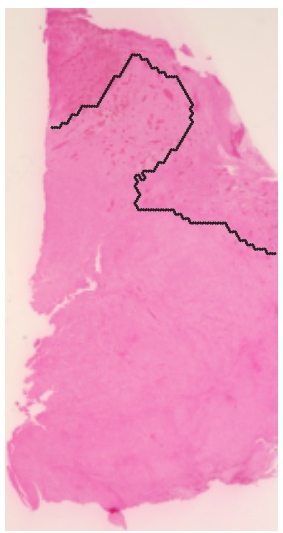




GSE238145_B13 | Tissue context

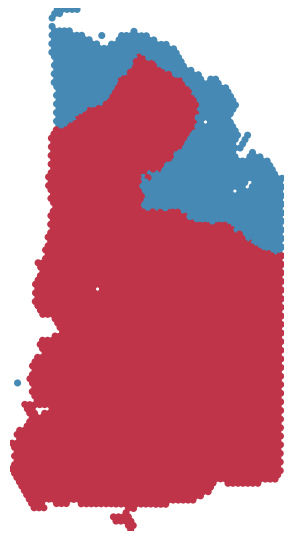
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location



Accepted R12 (original 0–8)

No covered labels

Epithelial support

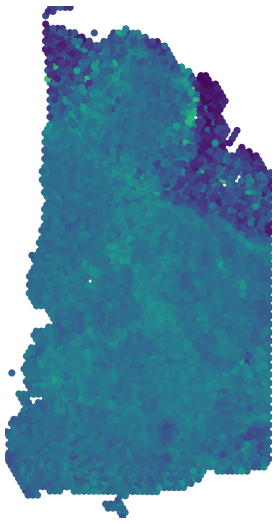
MP10 in Epi queries

Epi-component MP10 (valid only)

GSE238145_B13 | Candidate neighborhoods

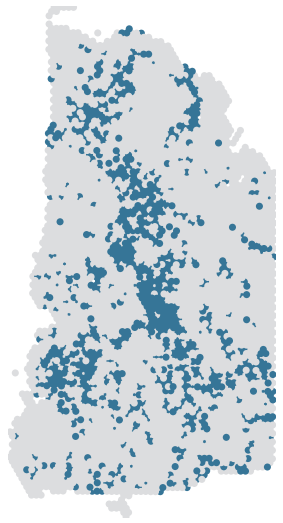
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



0 0.418
Fraction of all 30

CAF P80 targets

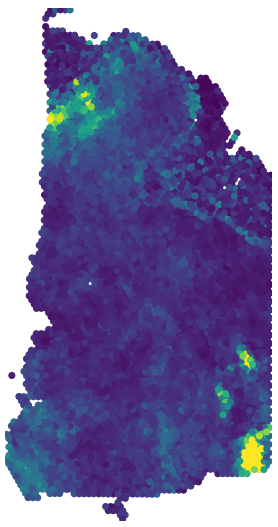


False True

CAF distance

0 3.1e+03
Nearest spot center (µm)

TAM composition



0 0.291
Fraction of all 30

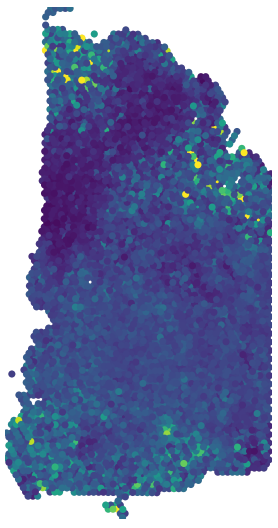
TAM P80 targets

False True

TAM distance

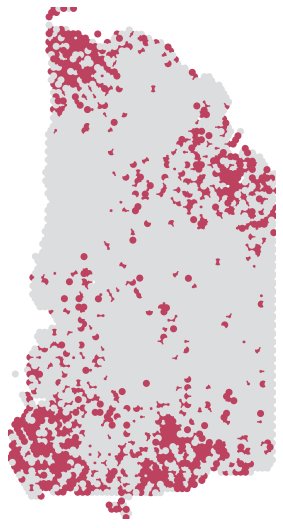
0 1.13e+03
Nearest spot center (µm)

Treg composition



0 0.102
Fraction of all 30

Treg P80 targets



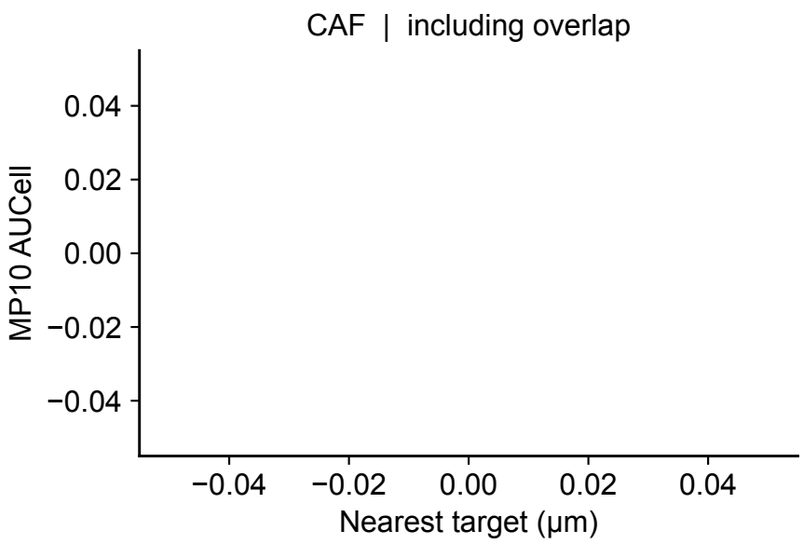
False True

Treg distance

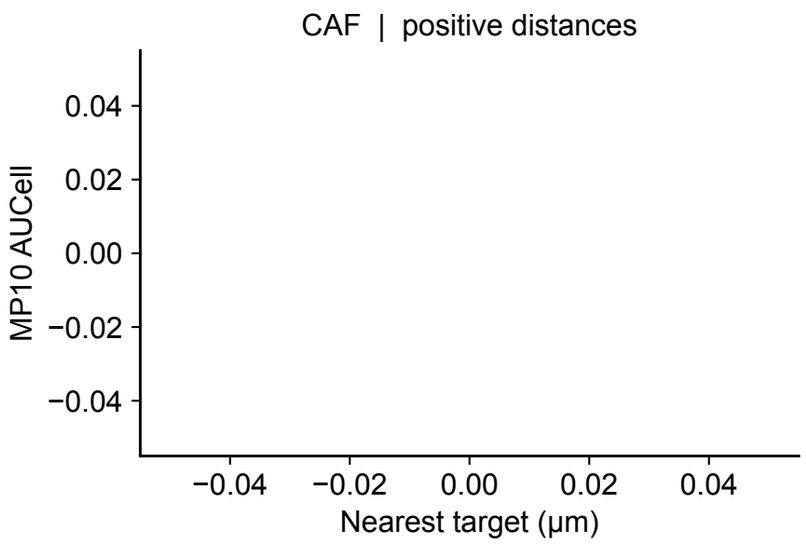
0 1.61e+03
Nearest spot center (µm)

GSE238145_B13 | MP10 versus nearest target

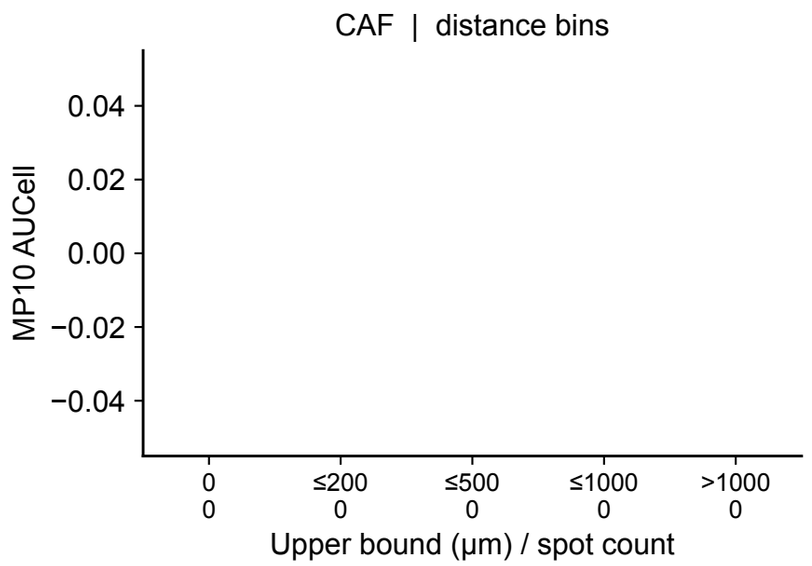
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



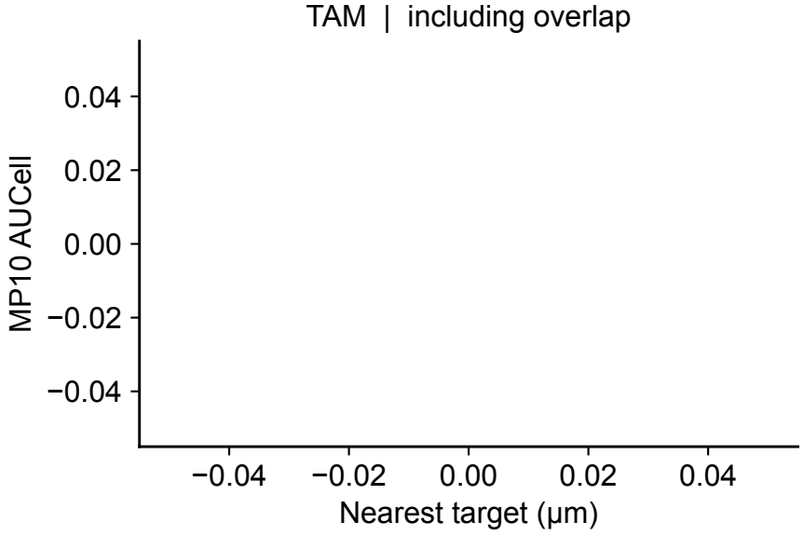
r +nan ρ +nan
n 0/0 queries • 977 targets



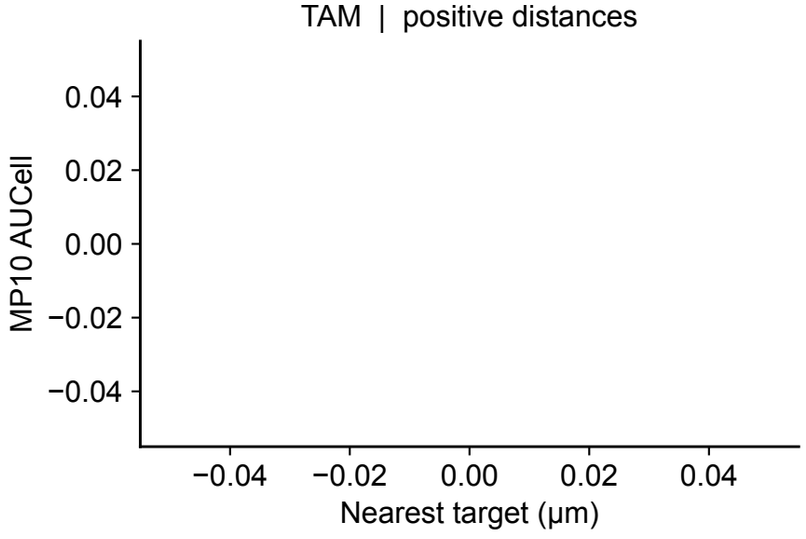
r +nan ρ +nan
n 0/0 queries • 977 targets



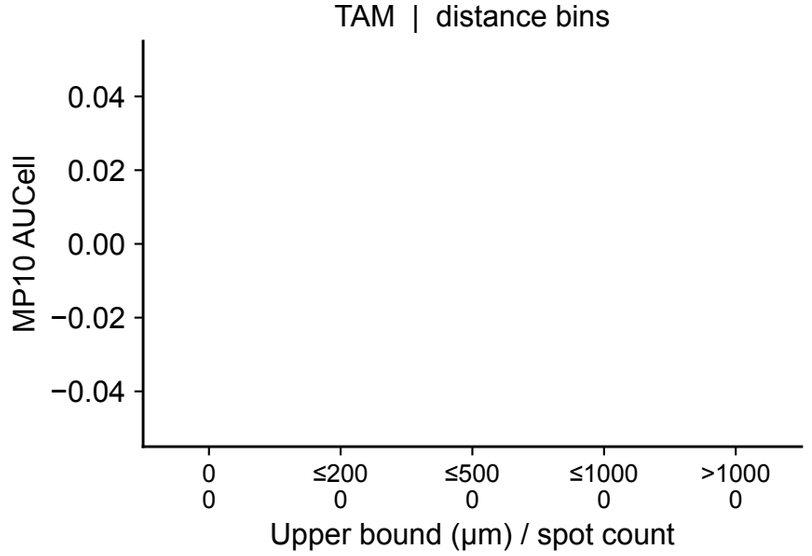
Orange = zero distance
Line = within-section linear fit



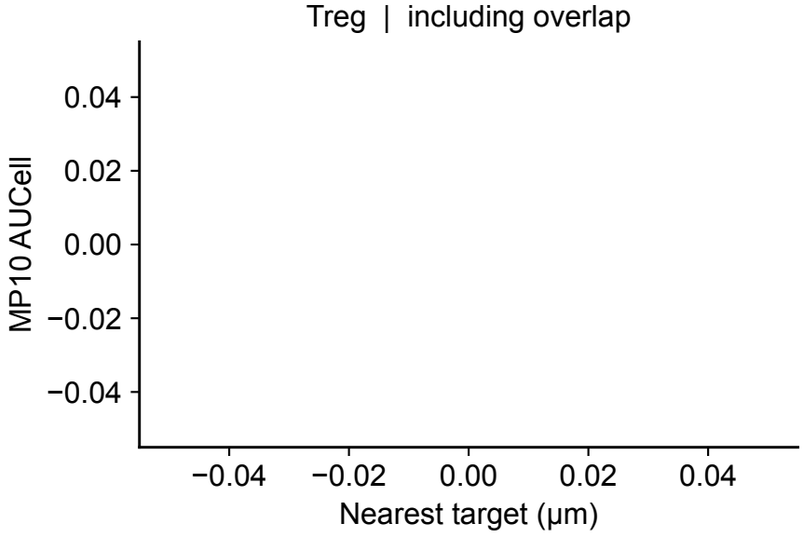
r +nan ρ +nan
n 0/0 queries • 977 targets



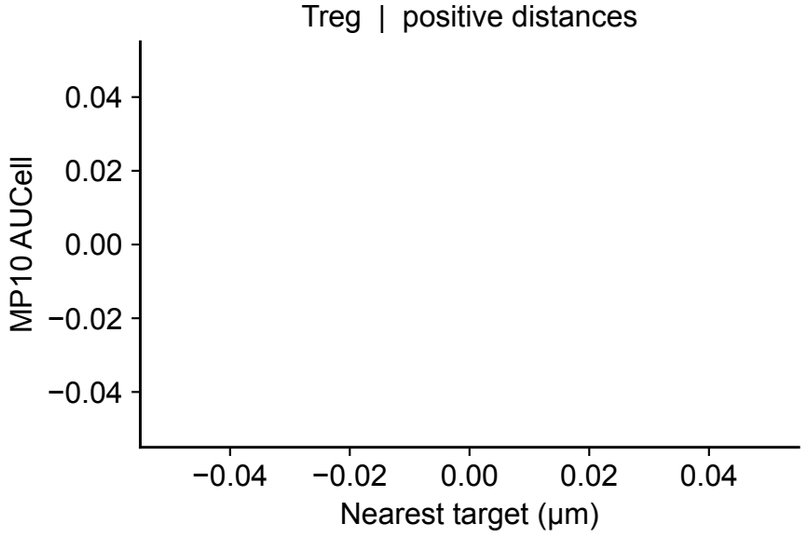
r +nan ρ +nan
n 0/0 queries • 977 targets



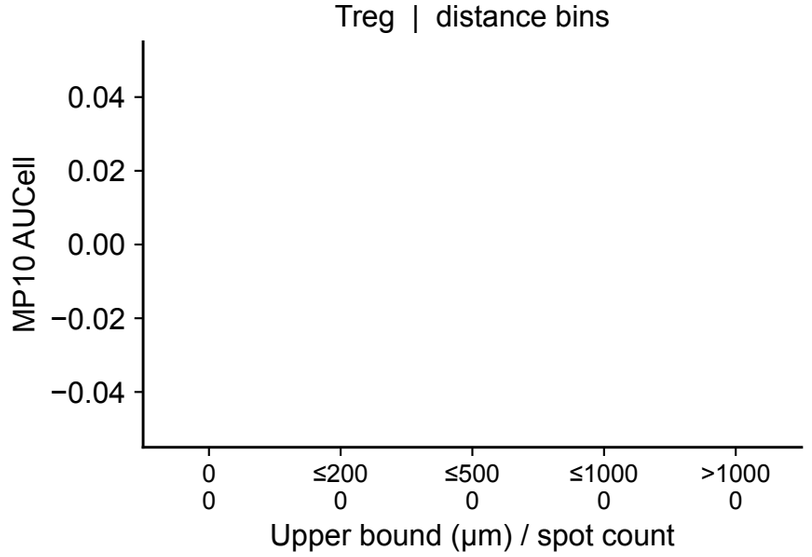
Orange = zero distance
Line = within-section linear fit



r +nan ρ +nan
n 0/0 queries • 977 targets



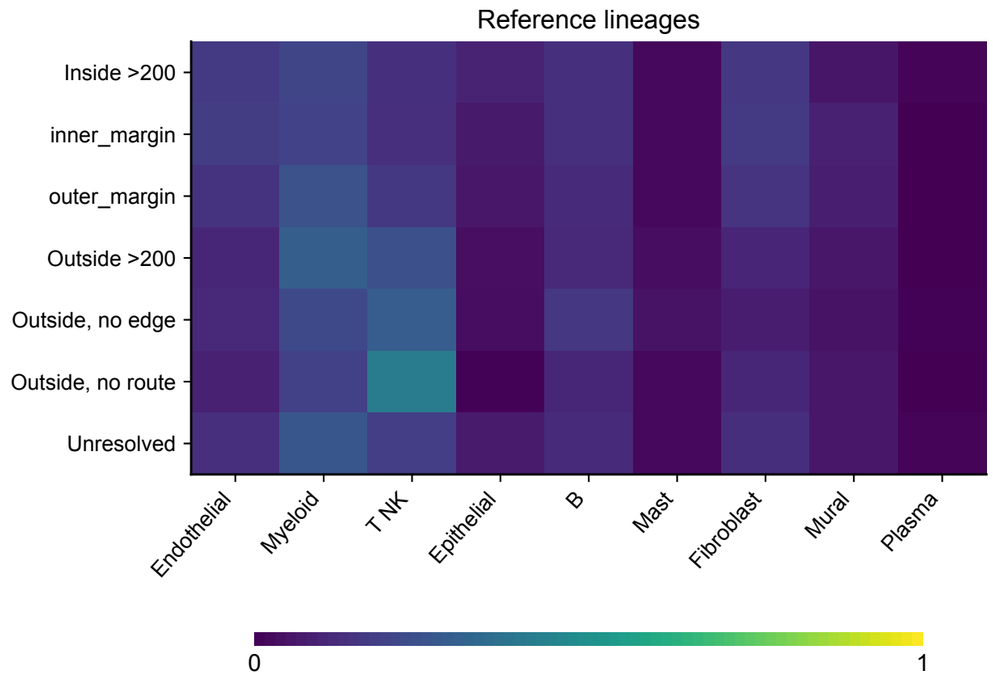
r +nan ρ +nan
n 0/0 queries • 977 targets

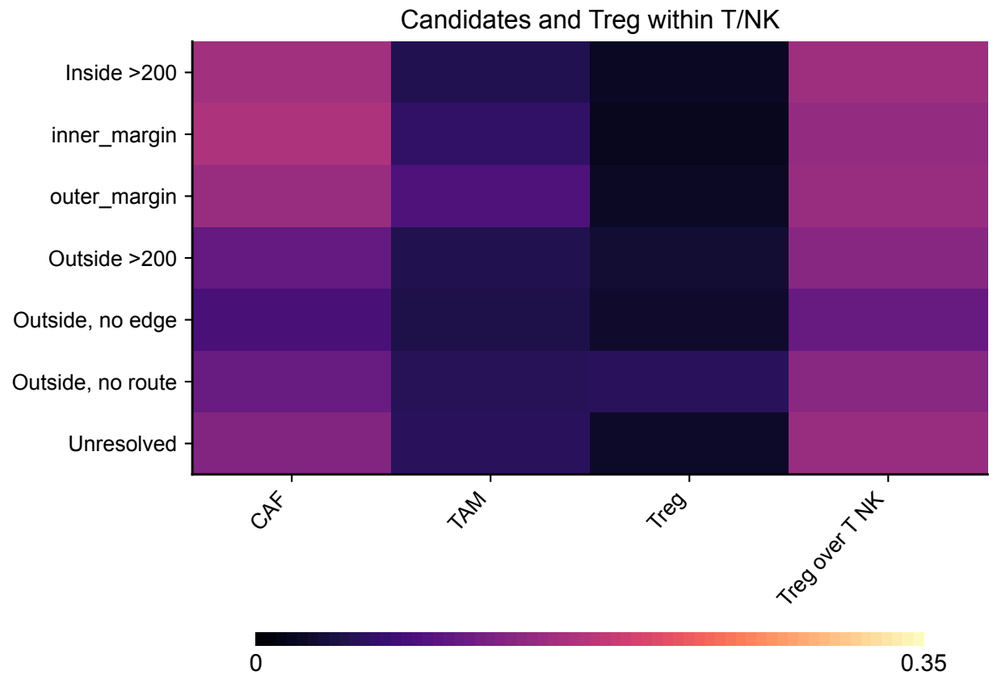


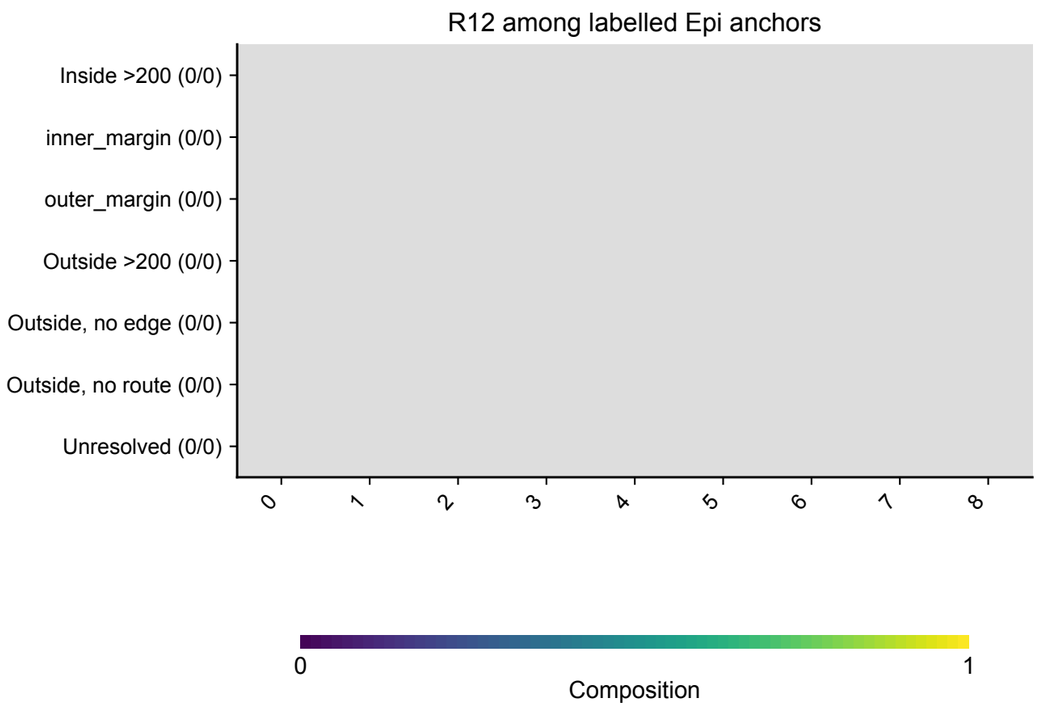
Orange = zero distance
Line = within-section linear fit

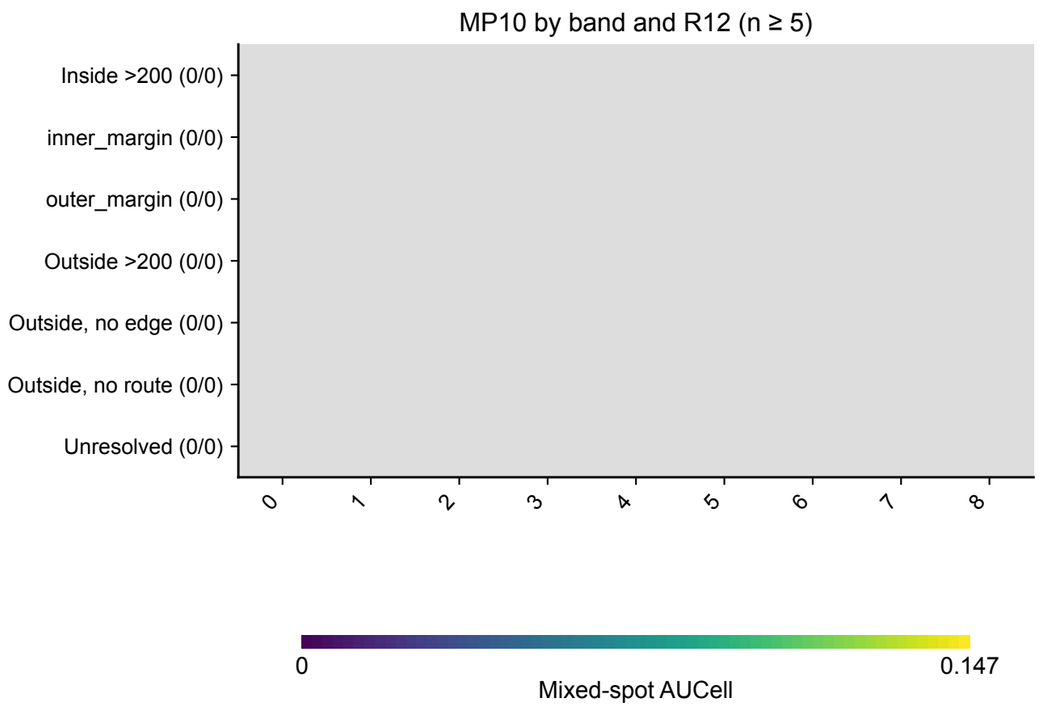
GSE238145_B13 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





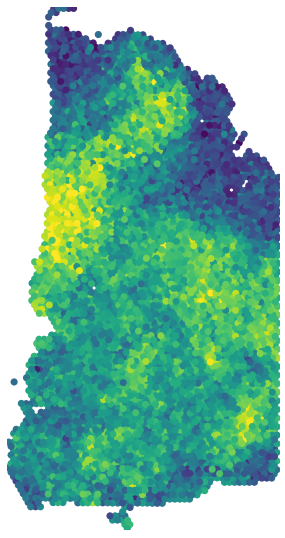




GSE238145_B13 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

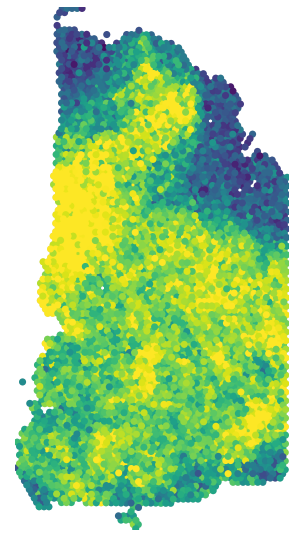
E2F Targets



0 0.152

UCell score

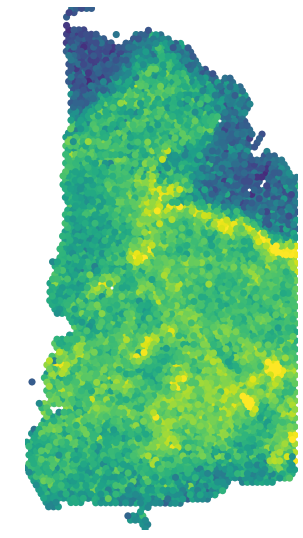
G2M Checkpoint



0 0.134

UCell score

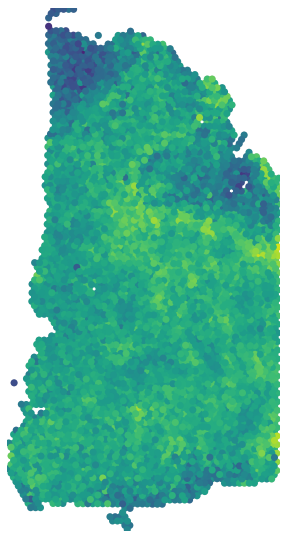
Hypoxia



0 0.211

UCell score

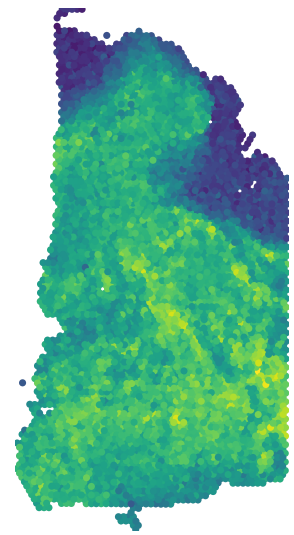
Tnfa Signaling Via Nfkb



0 0.275

UCell score

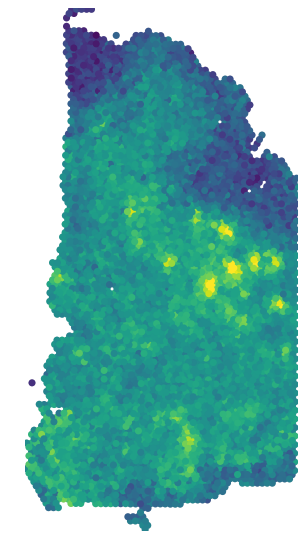
Emt



0 0.386

UCell score

Interferon Gamma Response



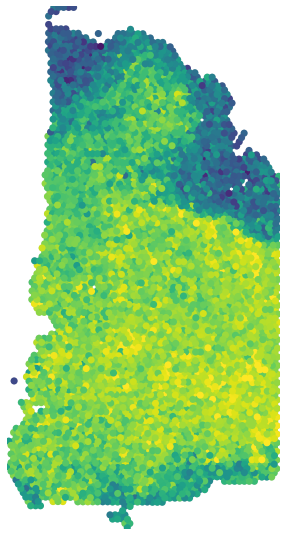
0 0.275

UCell score

GSE238145_B13 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

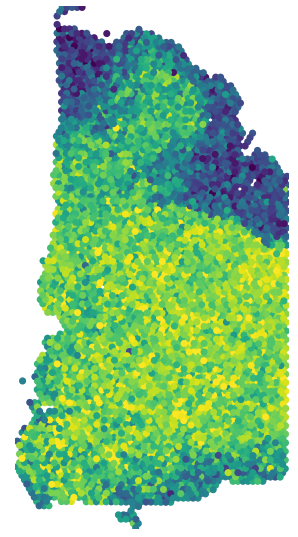
Il2 Stat5 Signaling



0 0.144

UCell score

Tgf Beta Signaling



0 0.232

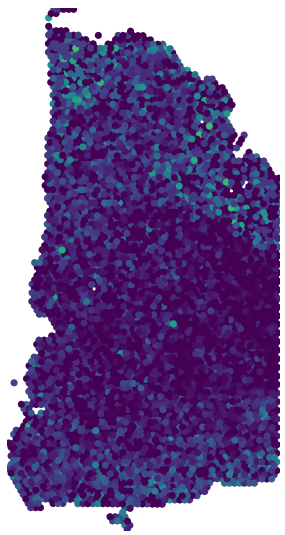
UCell score

Cholesterol Homeostasis

0 0.21

UCell score

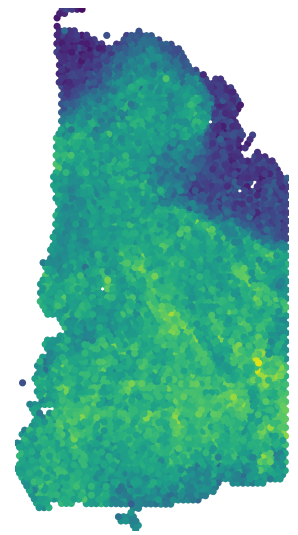
Cholesterol Biosynthesis



0 0.166

UCell score

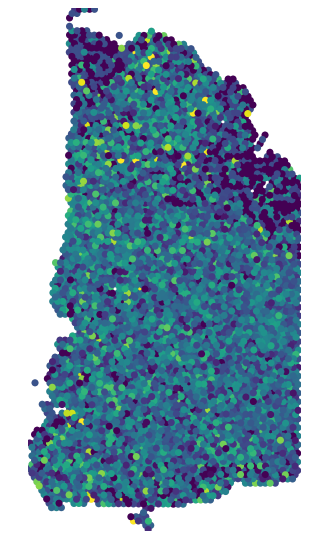
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



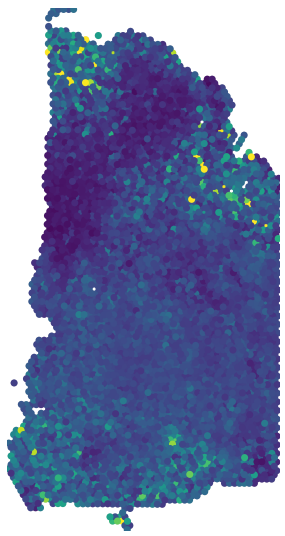
0 0.151

UCell score

GSE238145_B13 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

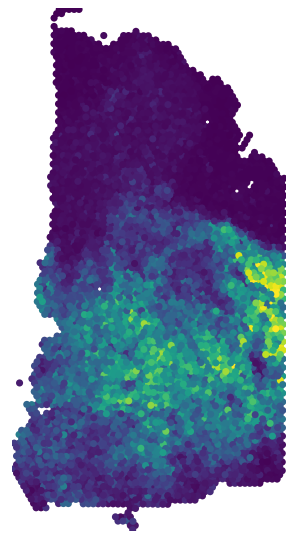
Treg composition



0 0.102

Fraction of all 30

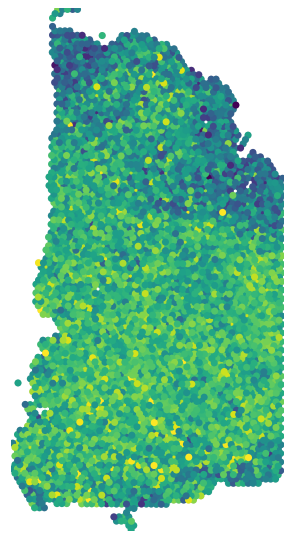
Treg estimated abundance



0 2.85

Posterior mean estimate

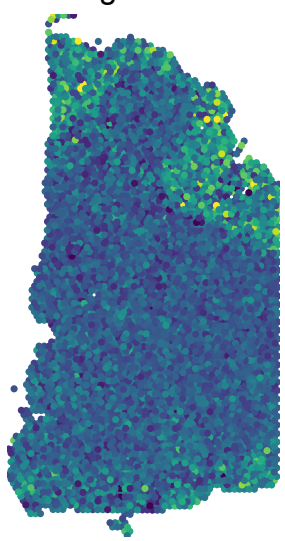
Treg Resting Vs Activated Dn



0 0.0778

UCell score

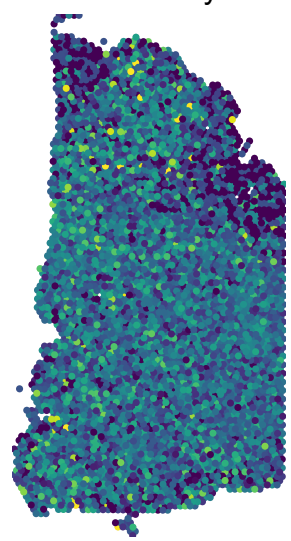
Treg Resting Vs Activated Up



0 0.0609

UCell score

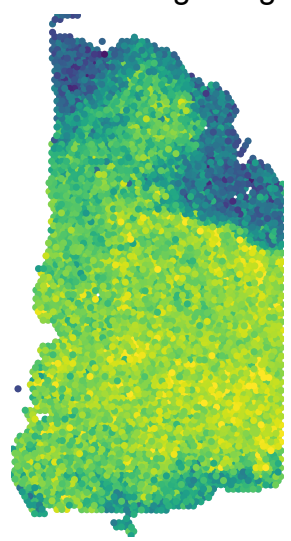
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

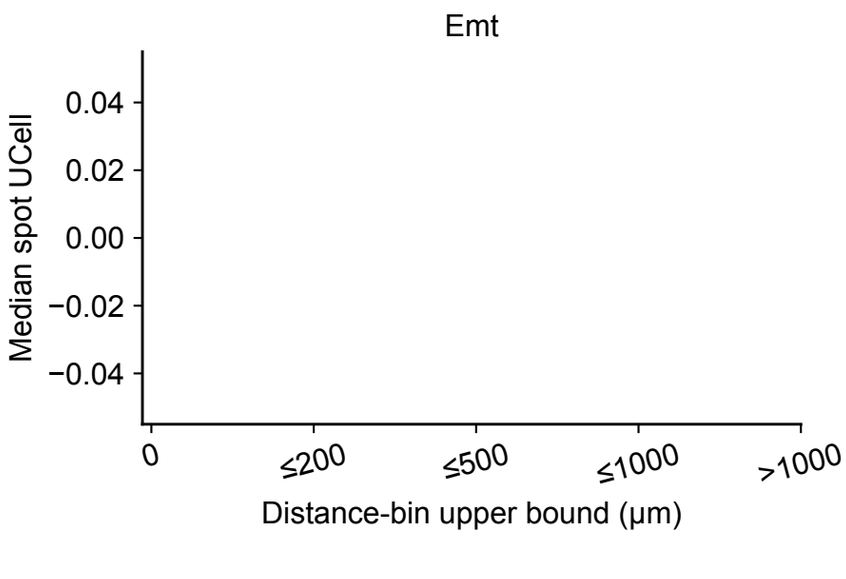


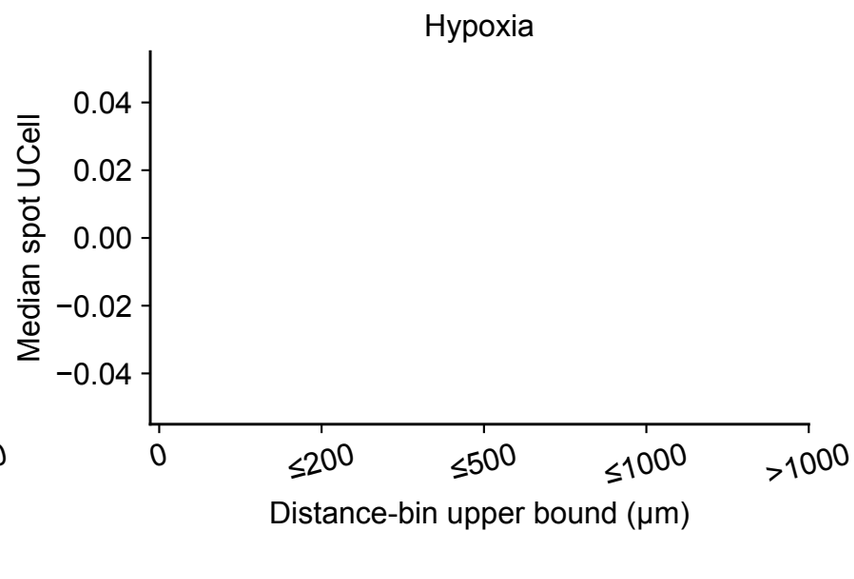
0 0.144

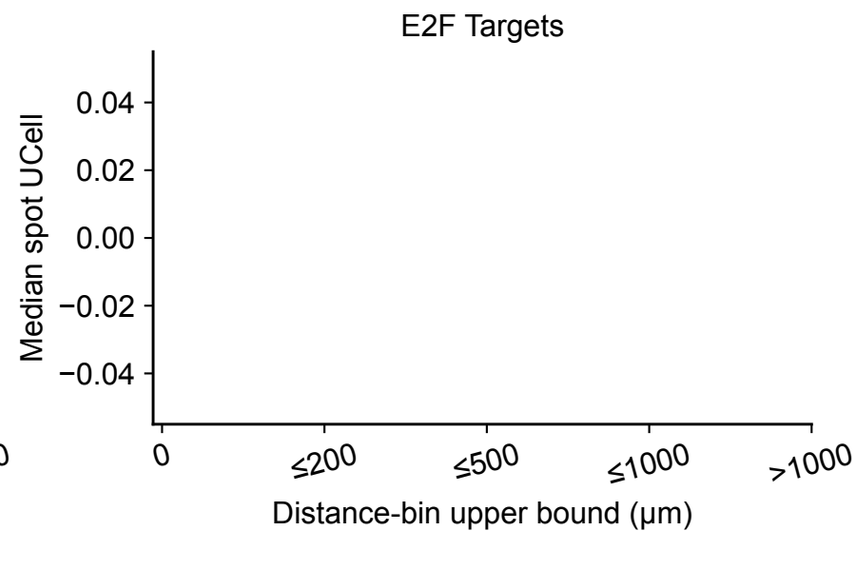
UCell score

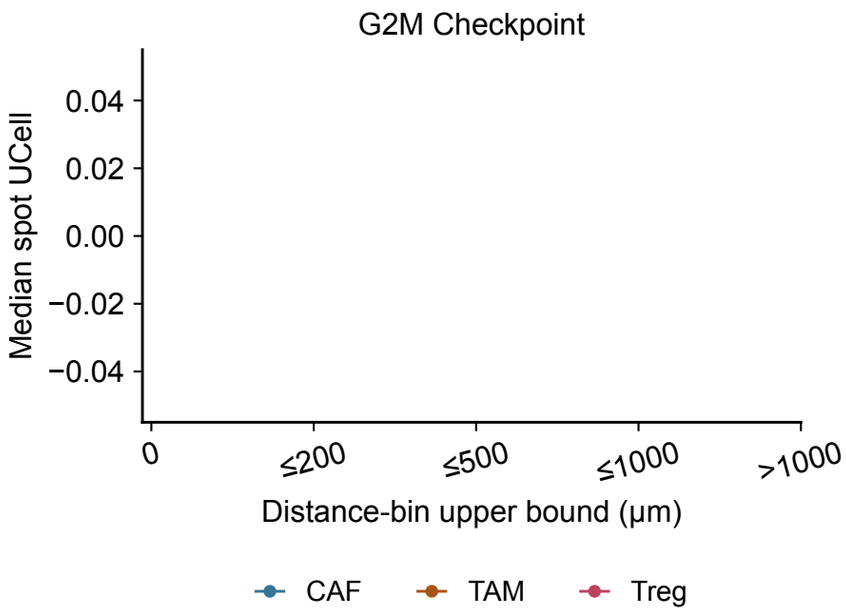
GSE238145_B13 | Programs by target distance

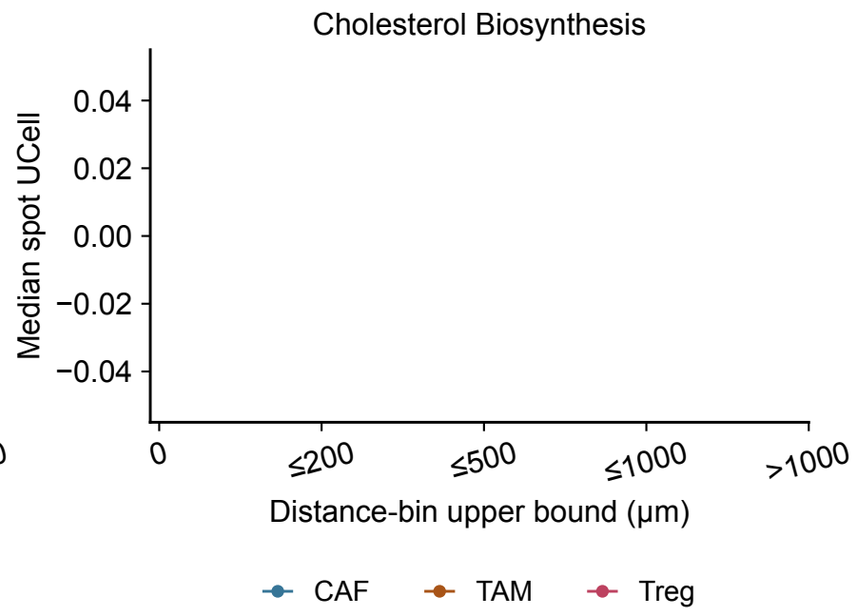
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

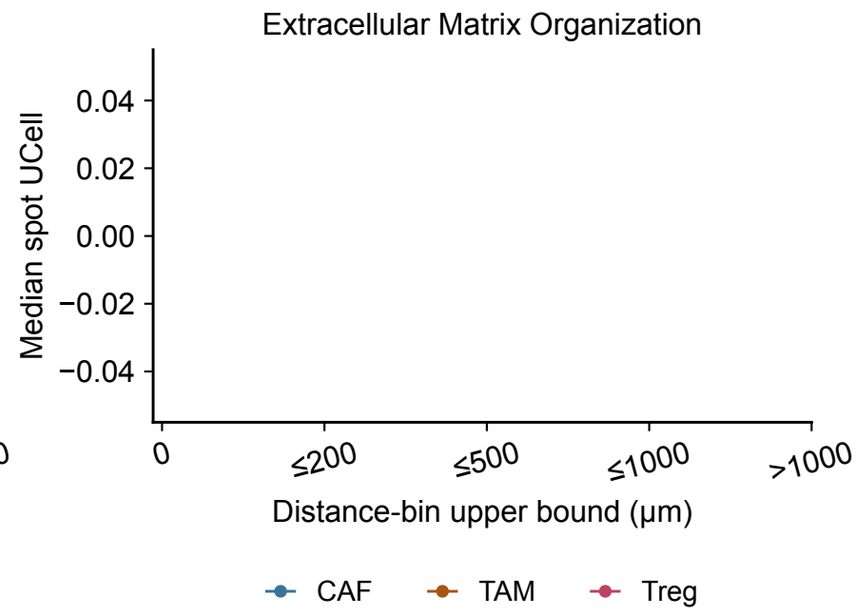






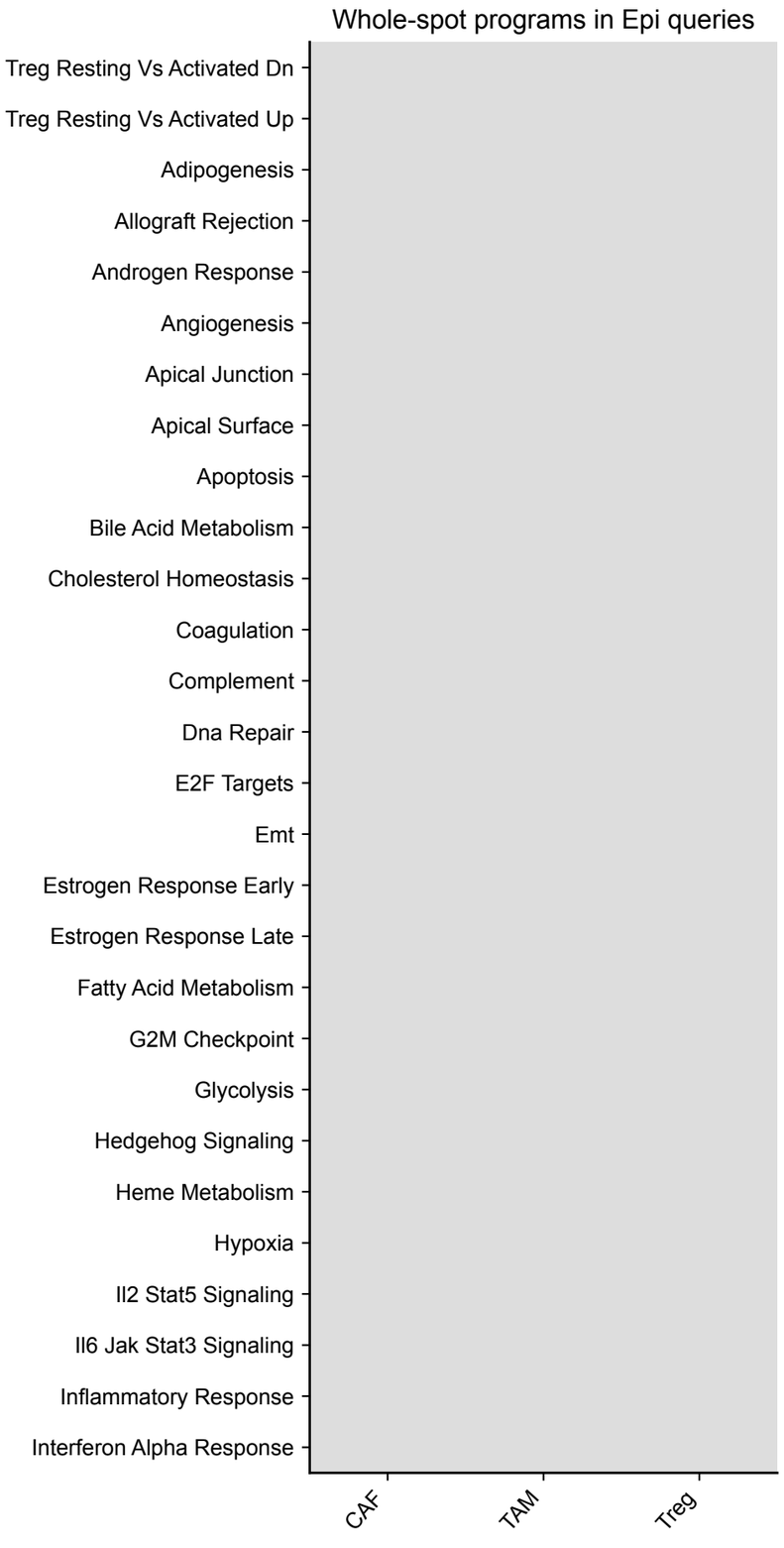


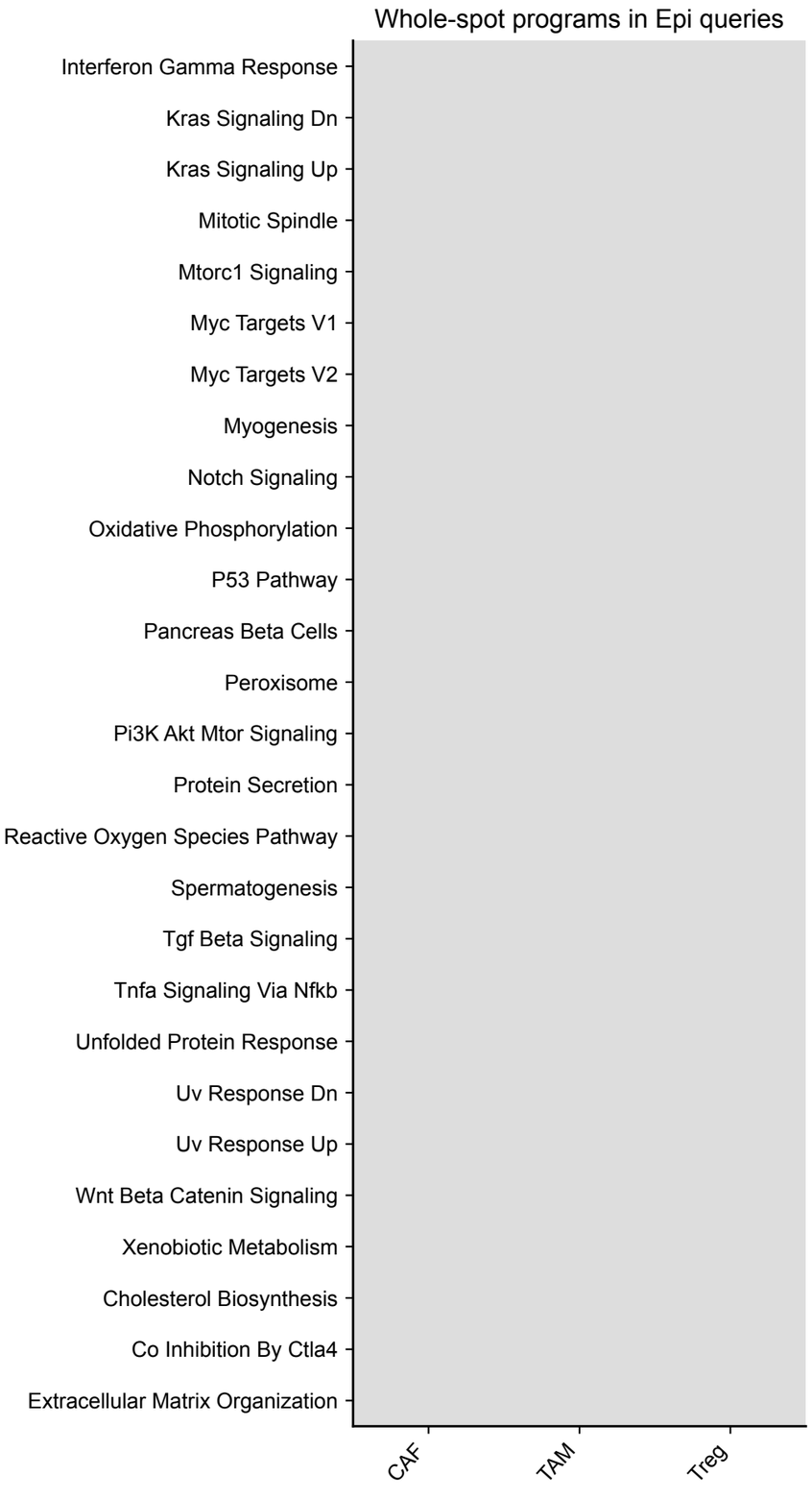




GSE238145_B13 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

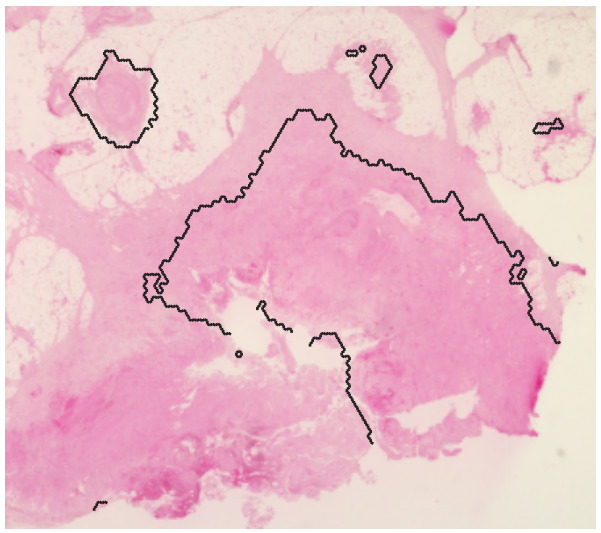




GSE238145_B14 | Tissue context

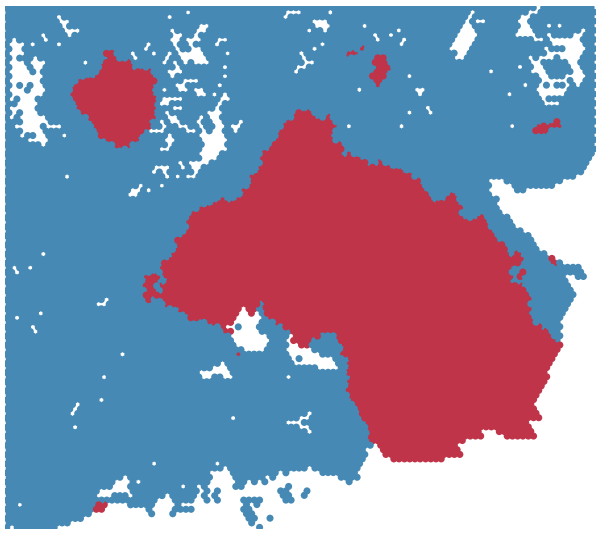
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



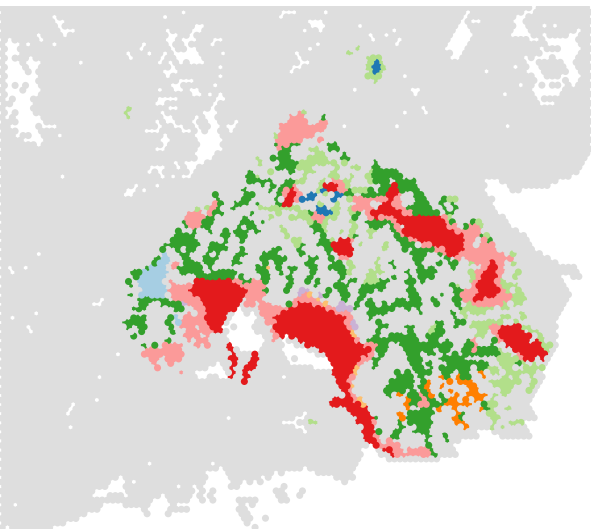
Black line: observed candidate outer edge

Final candidate location



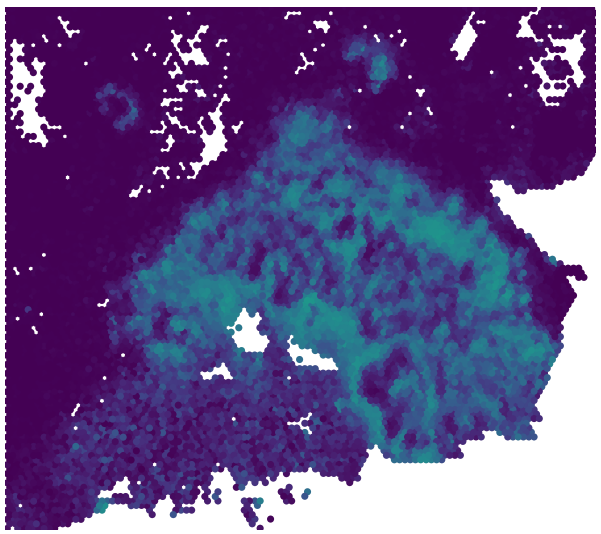
inside outside

Accepted R12 (original 0–8)



0 2 4 6 8
1 3 5 7

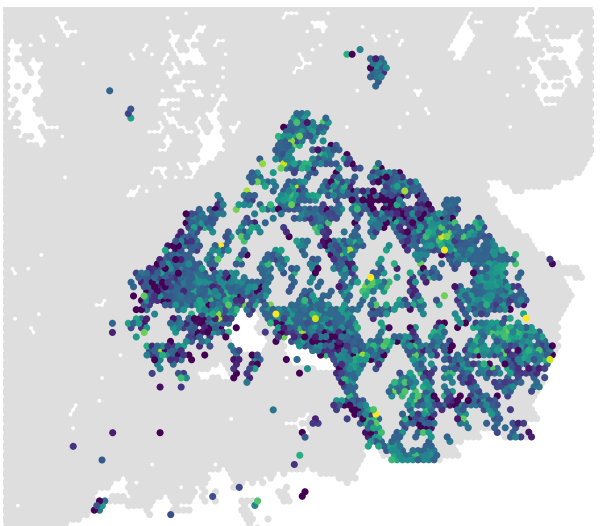
Epithelial support



0 1

Fraction of all 30

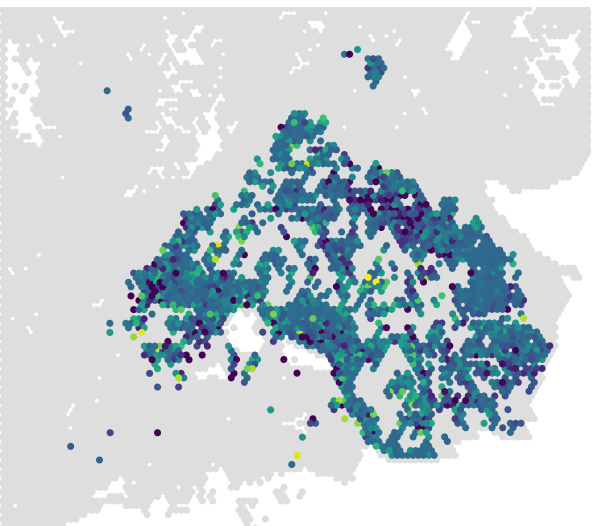
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



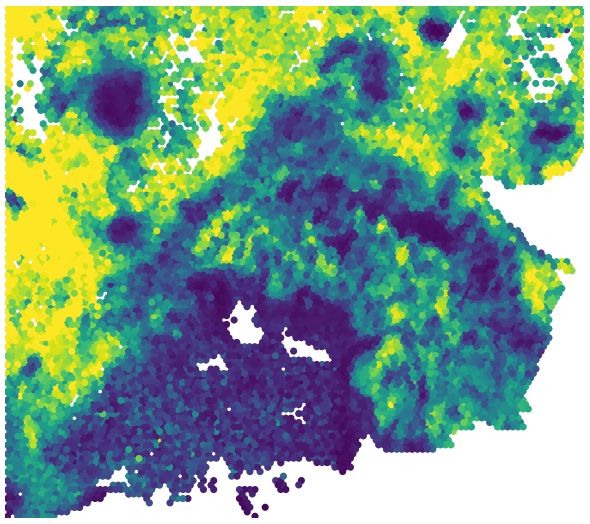
0 0.14

Matched-model AUCell

GSE238145_B14 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

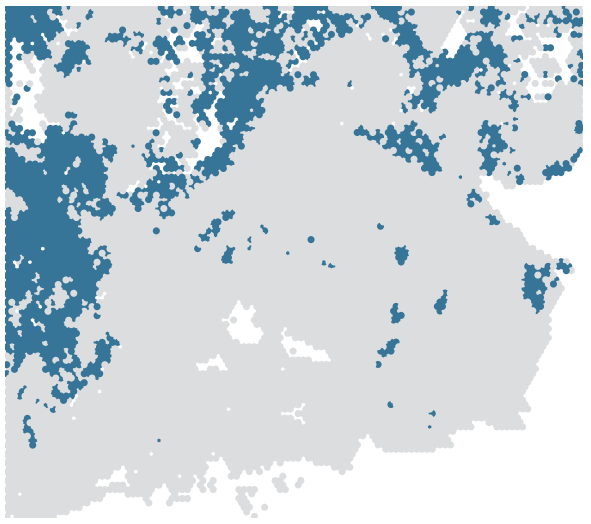
CAF composition



0 0.418

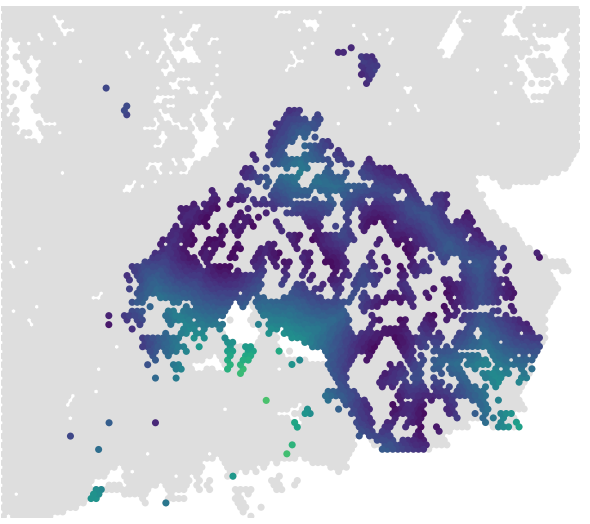
Fraction of all 30

CAF P80 targets



False True

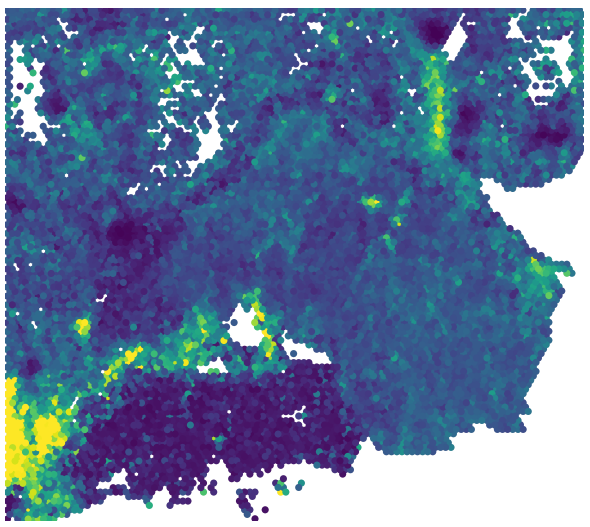
CAF distance



0 3.1e+03

Nearest spot center (µm)

TAM composition



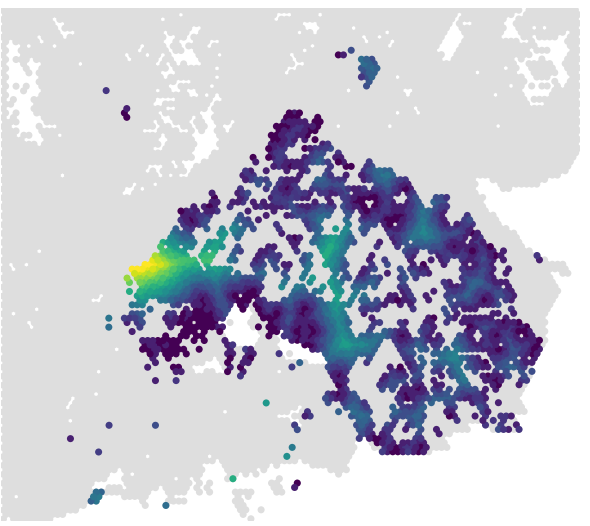
0 0.291

Fraction of all 30

TAM P80 targets

False True

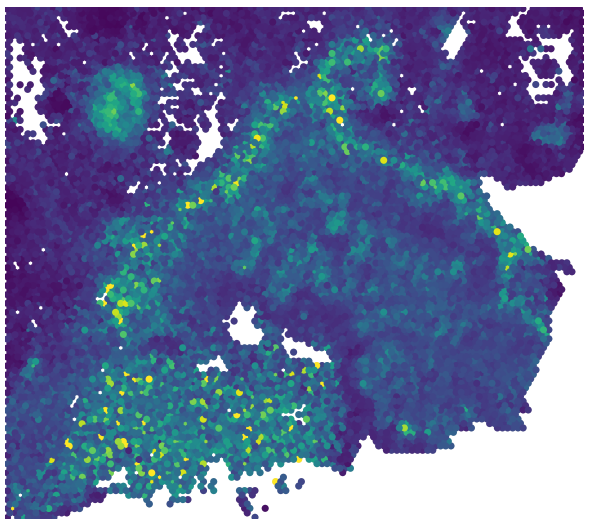
TAM distance



0 1.13e+03

Nearest spot center (µm)

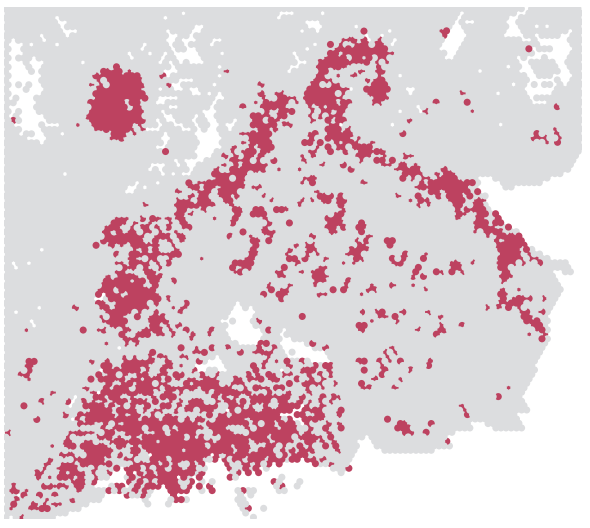
Treg composition



0 0.102

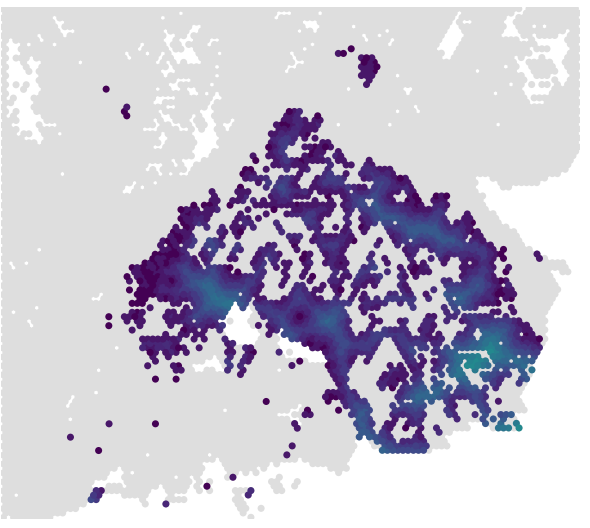
Fraction of all 30

Treg P80 targets



False True

Treg distance

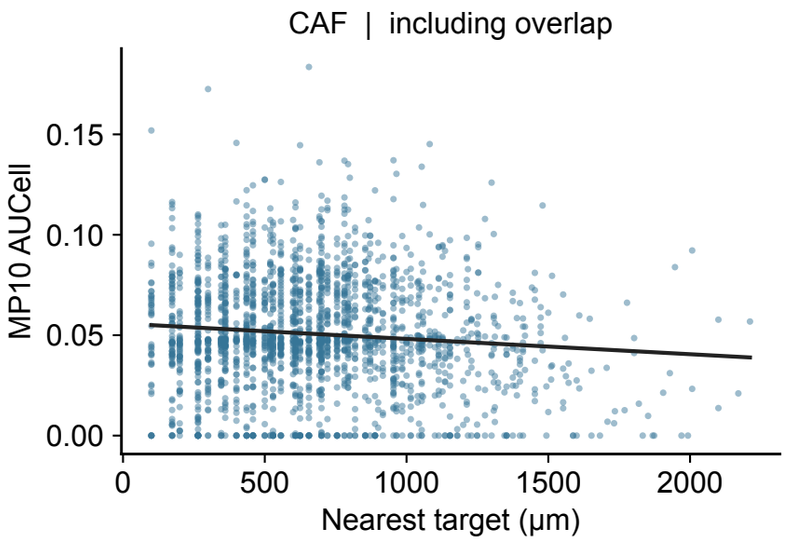


0 1.61e+03

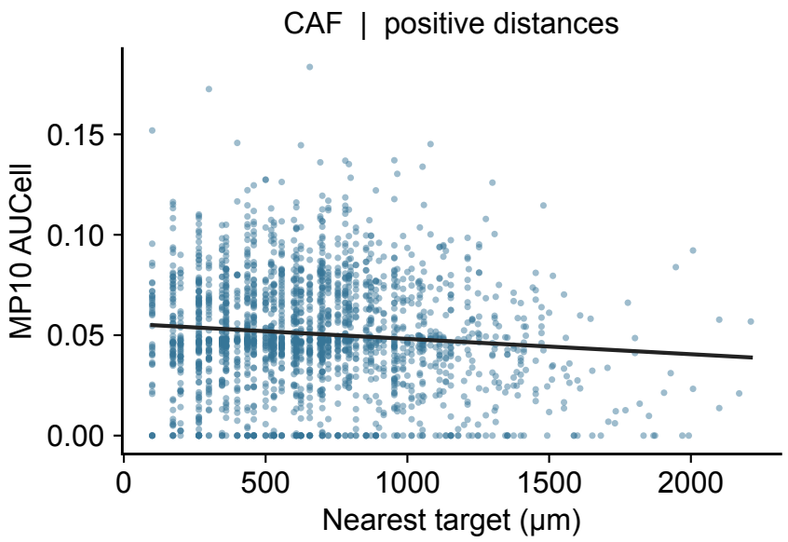
Nearest spot center (µm)

GSE238145_B14 | MP10 versus nearest target

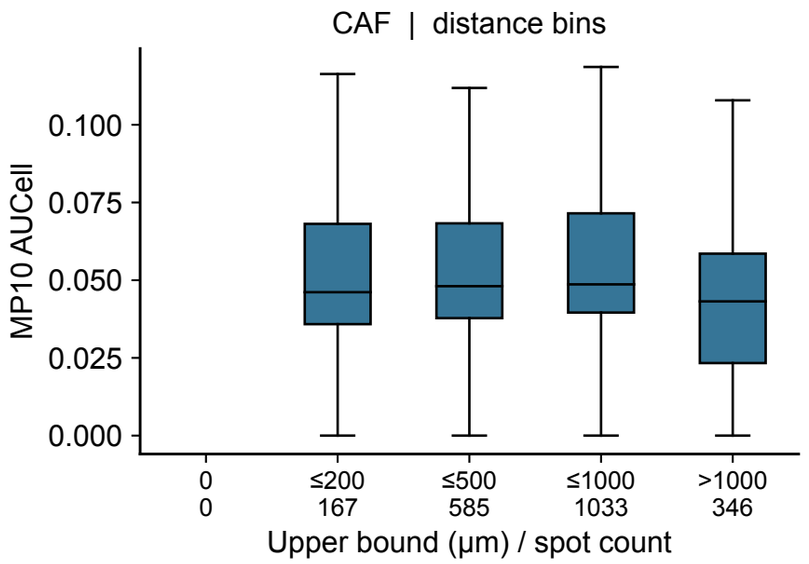
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



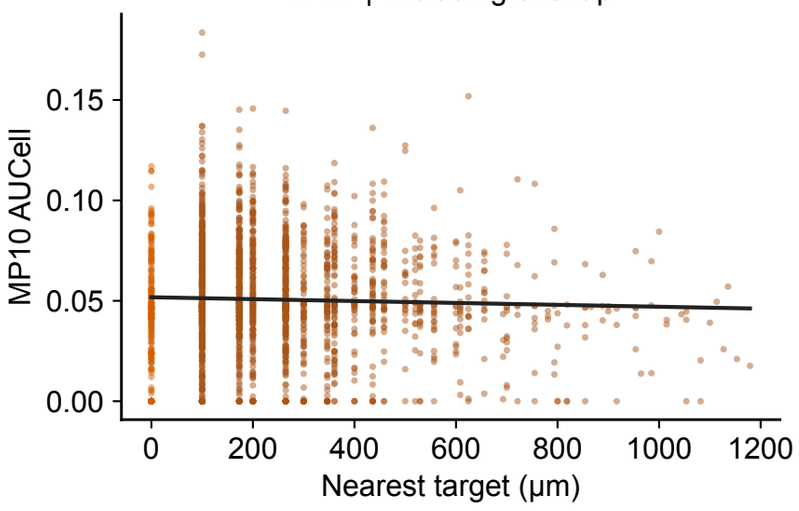
r -0.10 ρ -0.06
n 2131/2138 queries • 2087 targets



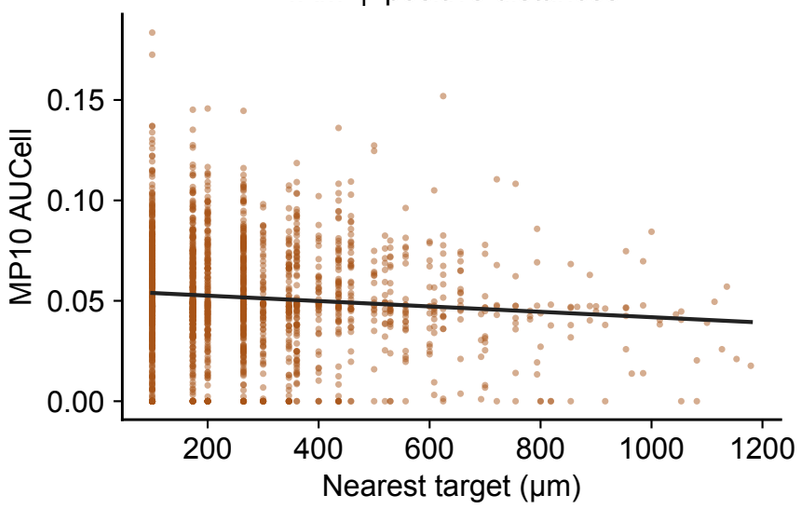
r -0.10 ρ -0.06
n 2131/2138 queries • 2087 targets



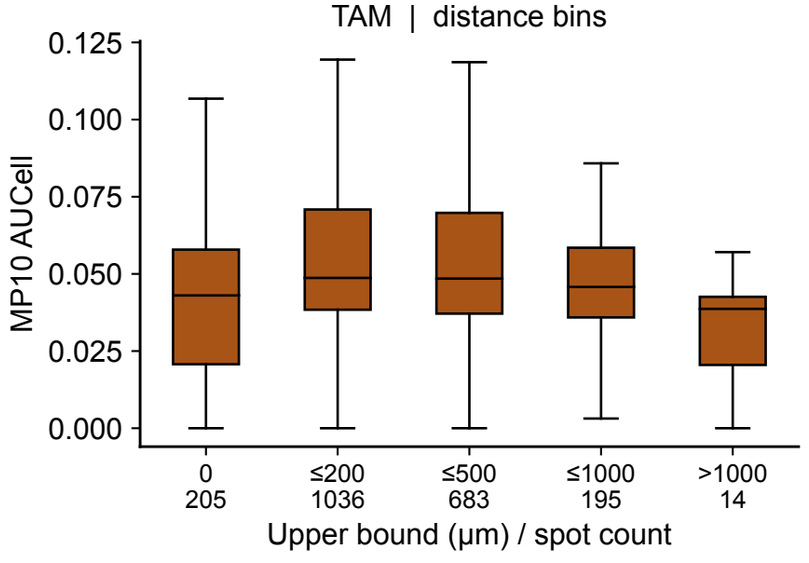
Orange = zero distance
Line = within-section linear fit



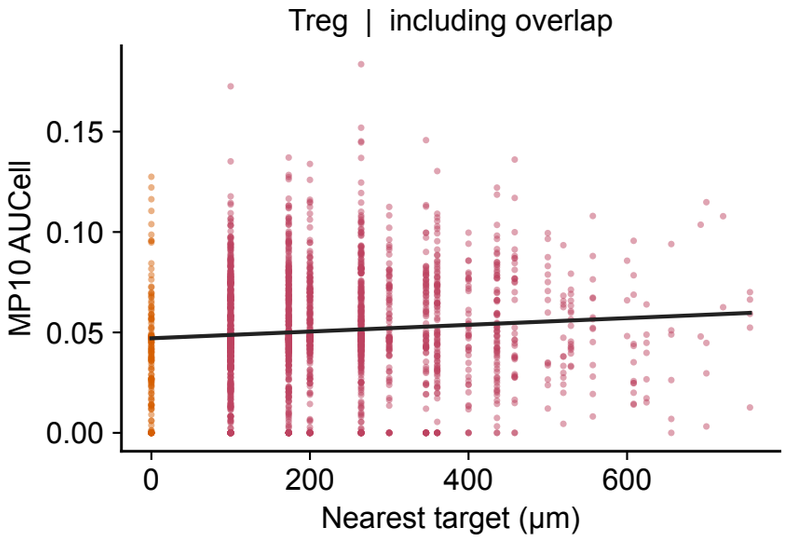
r -0.03 ρ +0.01
n 2133/2138 queries • 2087 targets



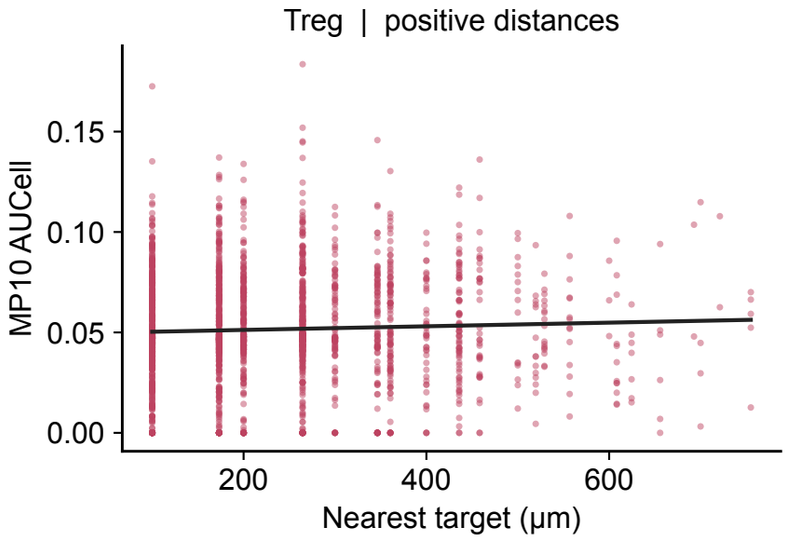
r -0.09 ρ -0.06
n 1928/2138 queries • 2087 targets



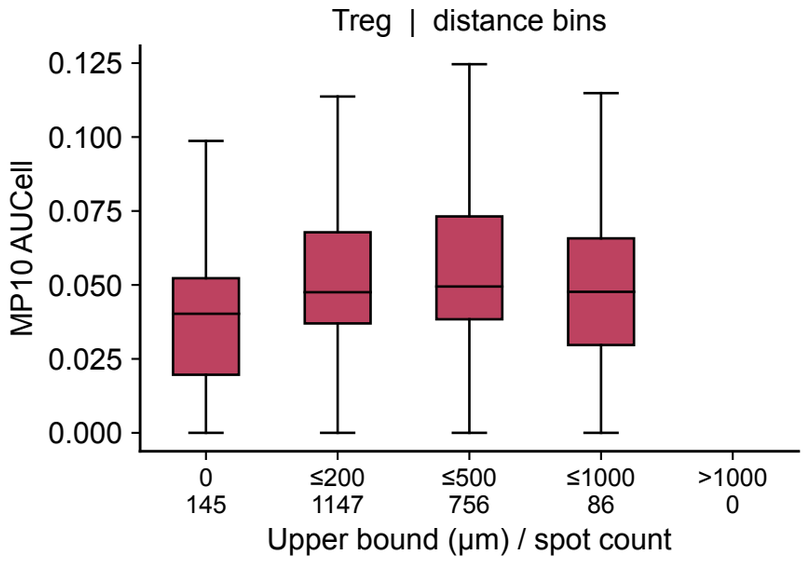
Orange = zero distance
Line = within-section linear fit



r +0.08 ρ +0.10
n 2134/2138 queries • 2087 targets



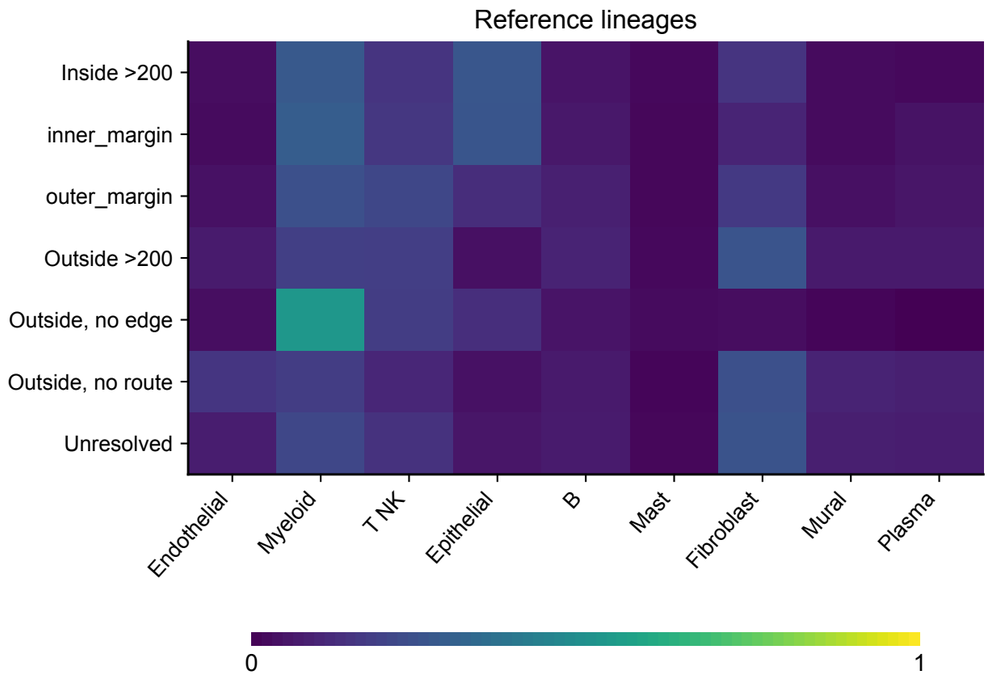
r +0.04 ρ +0.06
n 1989/2138 queries • 2087 targets

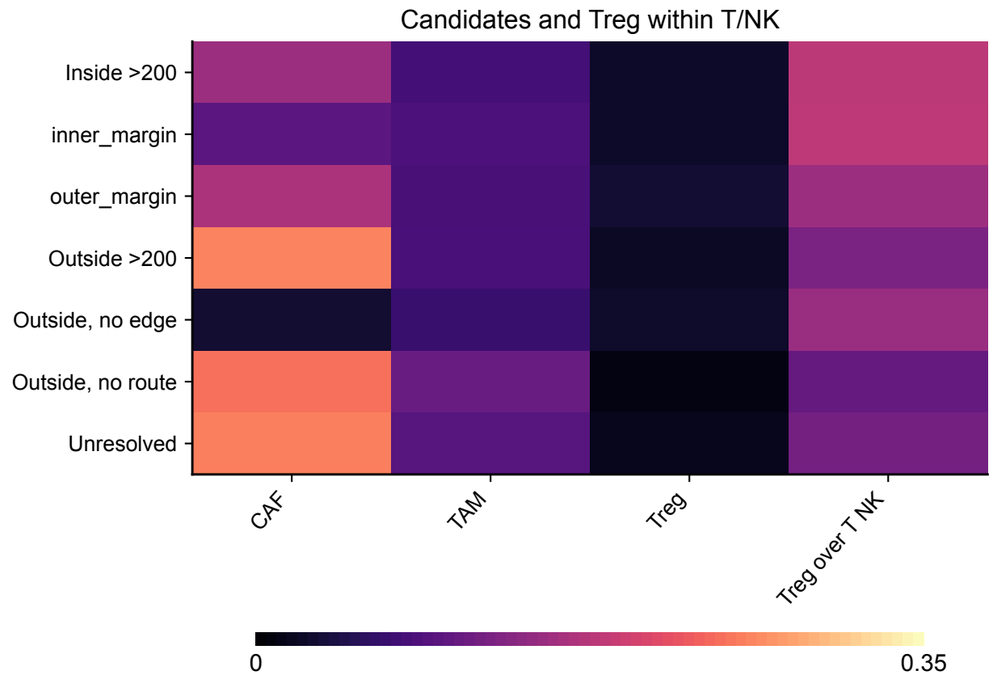


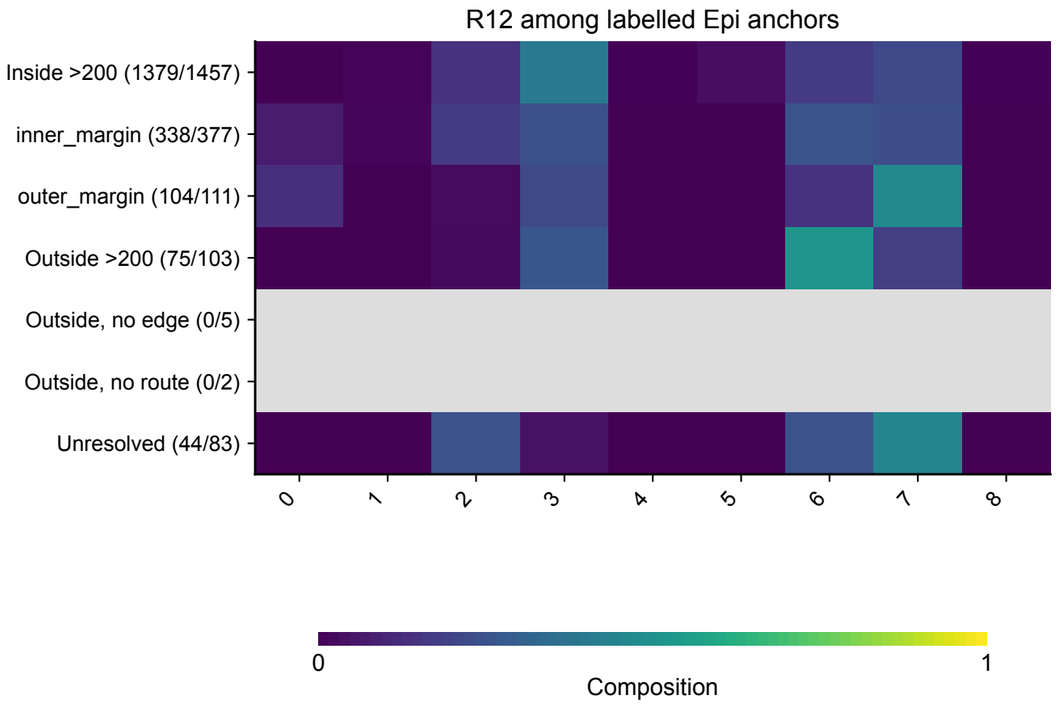
Orange = zero distance
Line = within-section linear fit

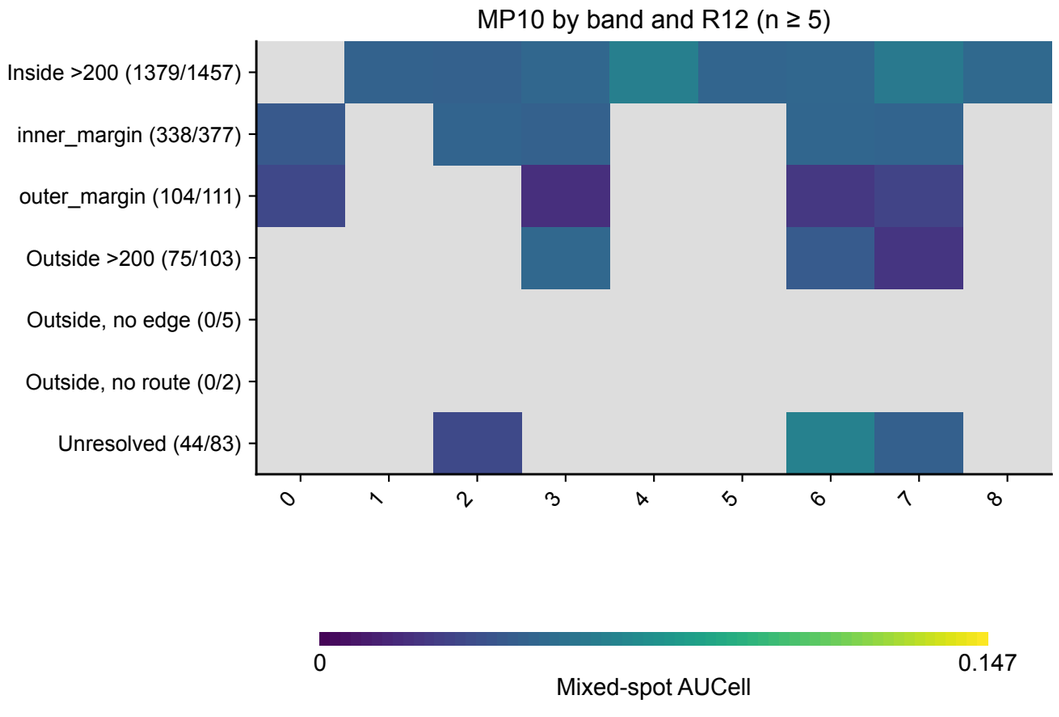
GSE238145_B14 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





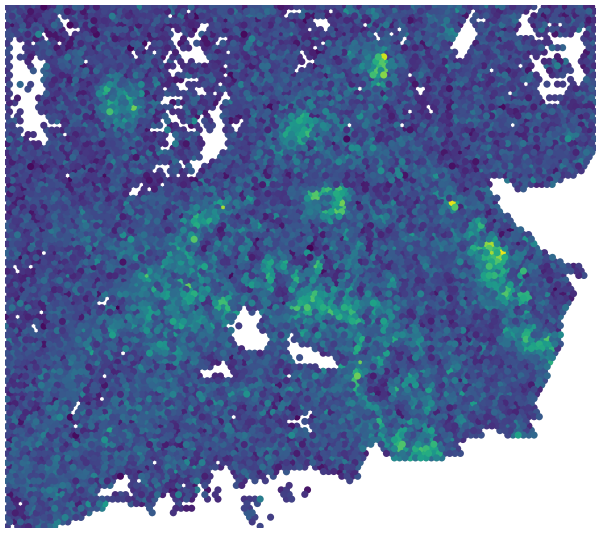




GSE238145_B14 | Growth, stress and inflammation

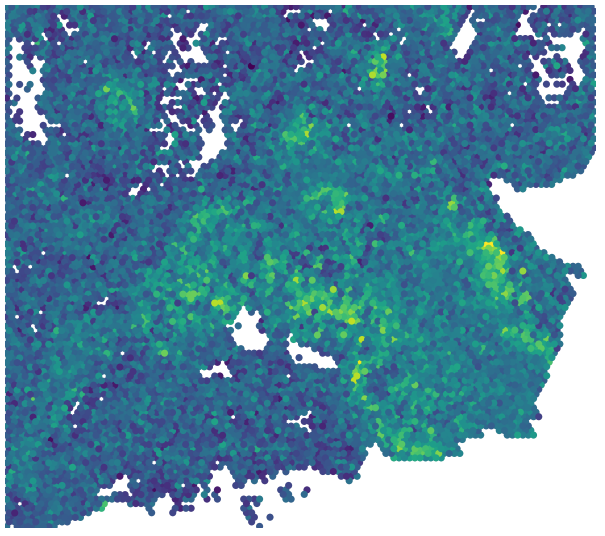
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets



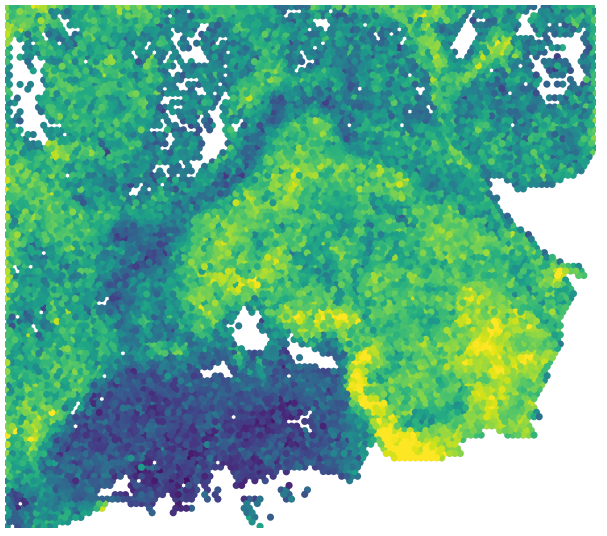
0 0.152 UCell score

G2M Checkpoint



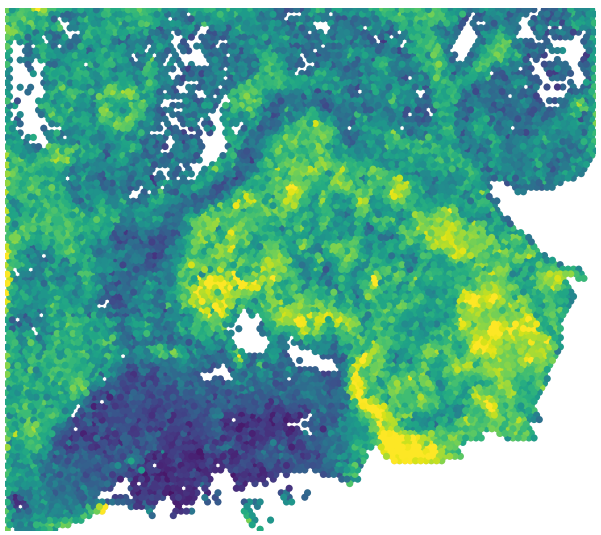
0 0.134 UCell score

Hypoxia



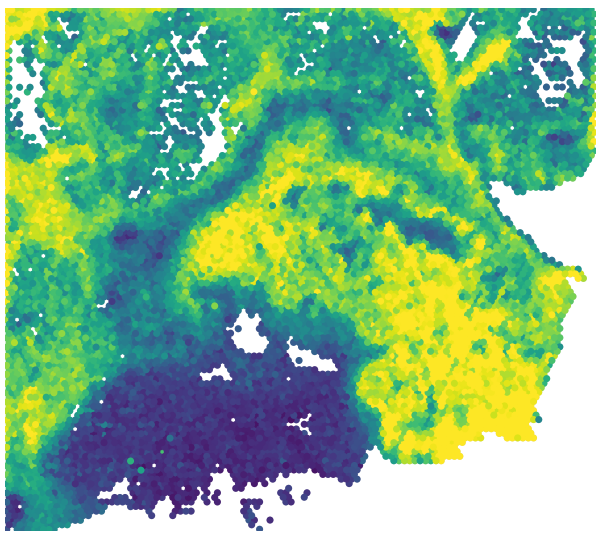
0 0.211 UCell score

Tnfa Signaling Via Nfkb



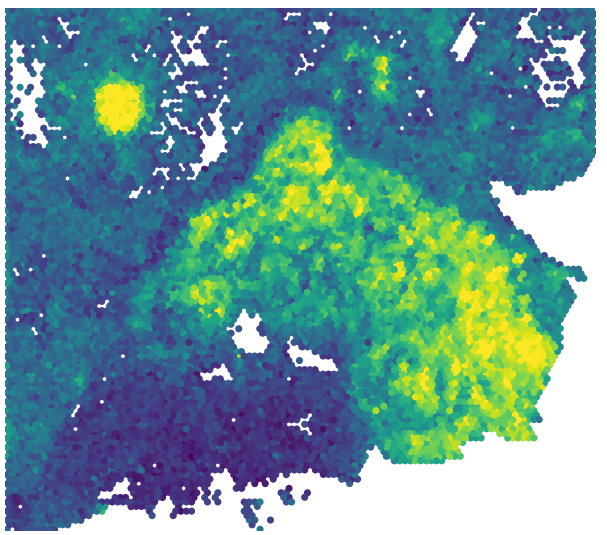
0 0.275 UCell score

Emt



0 0.386 UCell score

Interferon Gamma Response

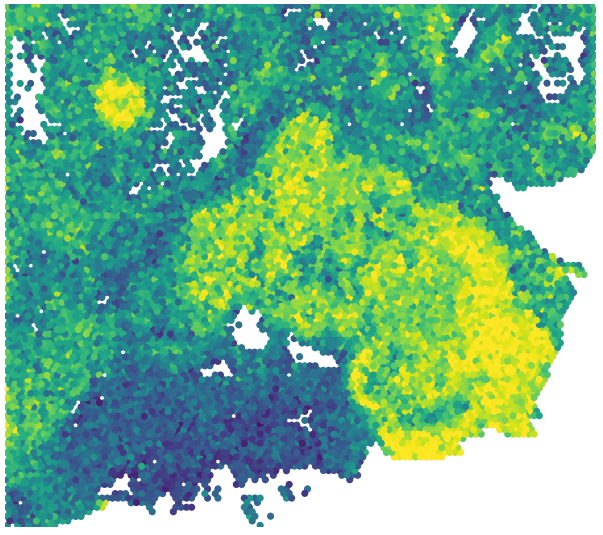


0 0.275 UCell score

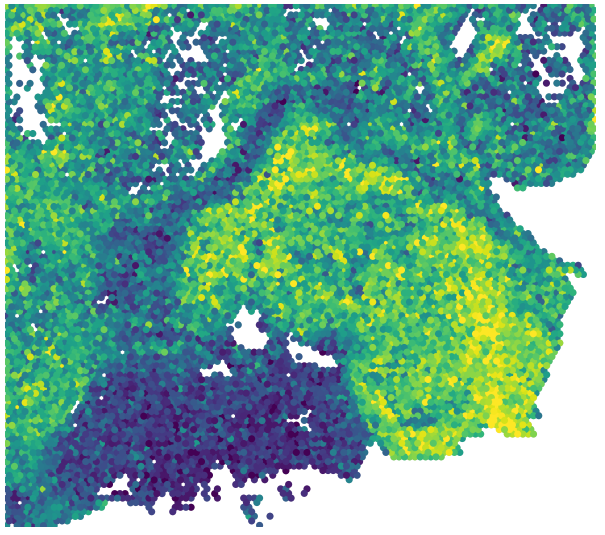
GSE238145_B14 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

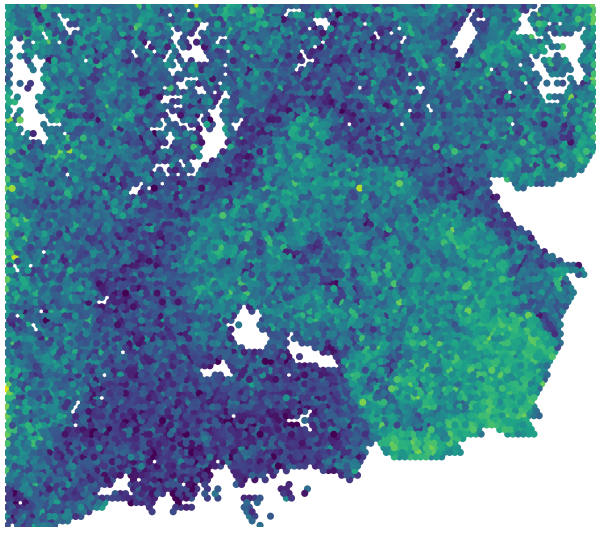
Il2 Stat5 Signaling



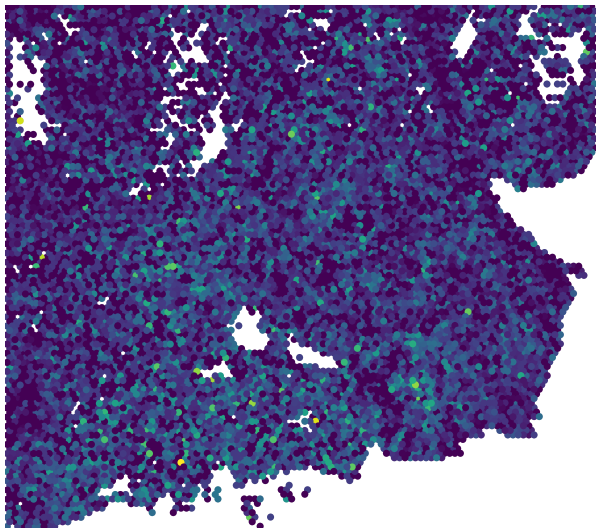
Tgf Beta Signaling

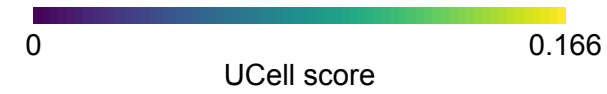


Cholesterol Homeostasis

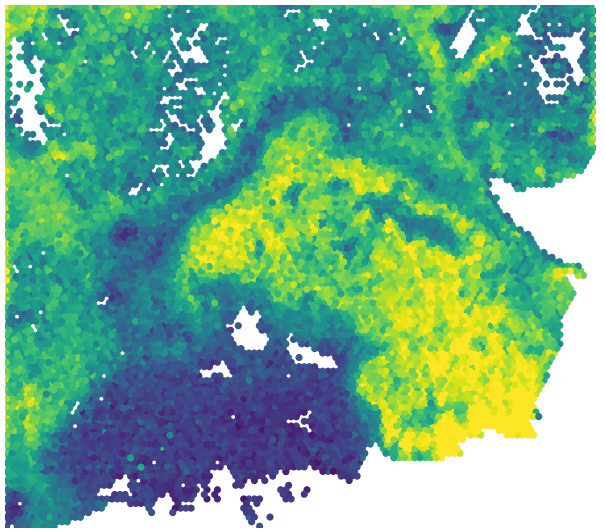


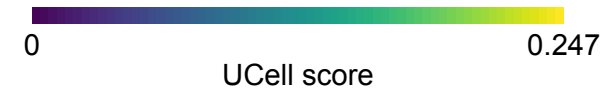
Cholesterol Biosynthesis



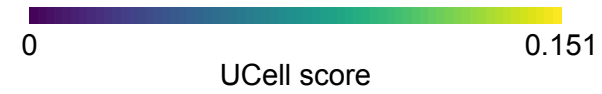


Extracellular Matrix Organization





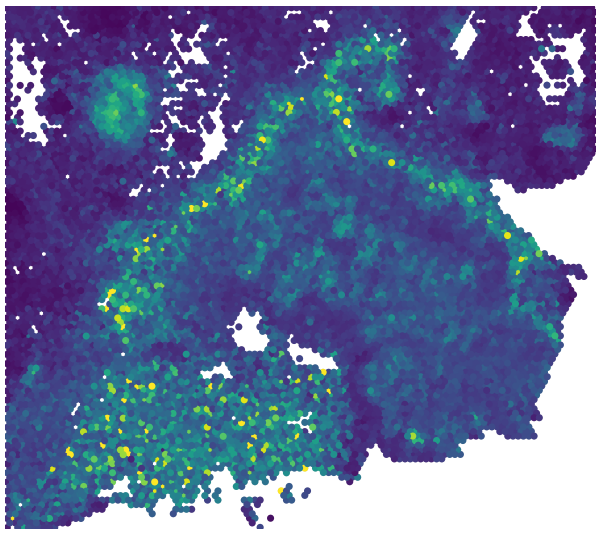
Co Inhibition By Ctla4



GSE238145_B14 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

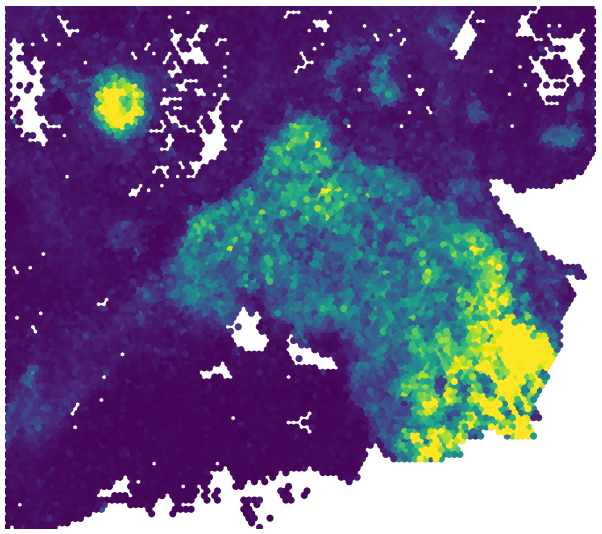
Treg composition



0 0.102

Fraction of all 30

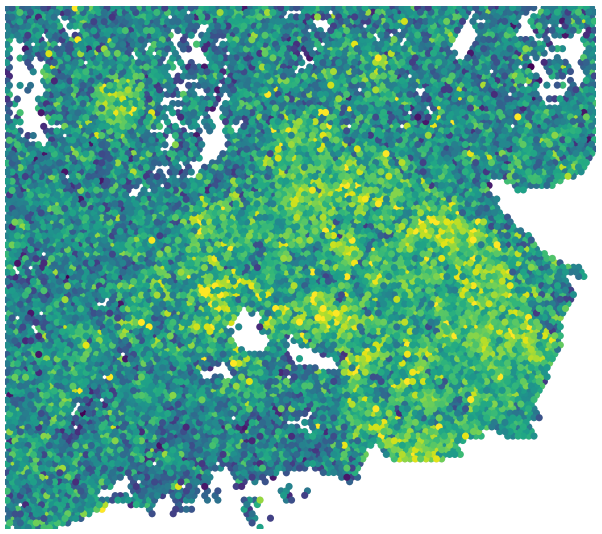
Treg estimated abundance



0 2.85

Posterior mean estimate

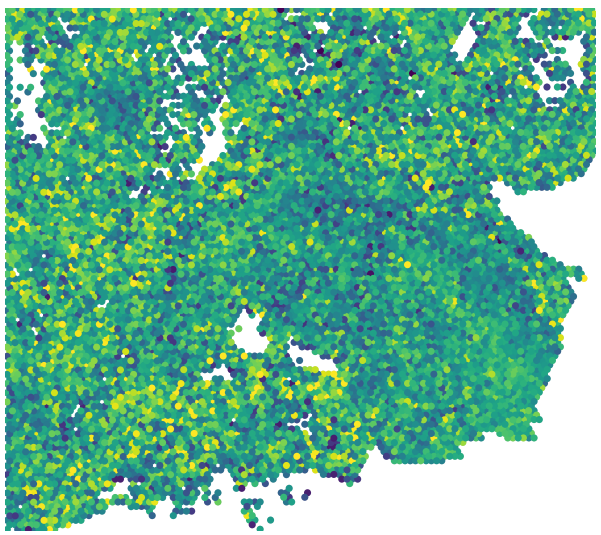
Treg Resting Vs Activated Dn



0 0.0778

UCell score

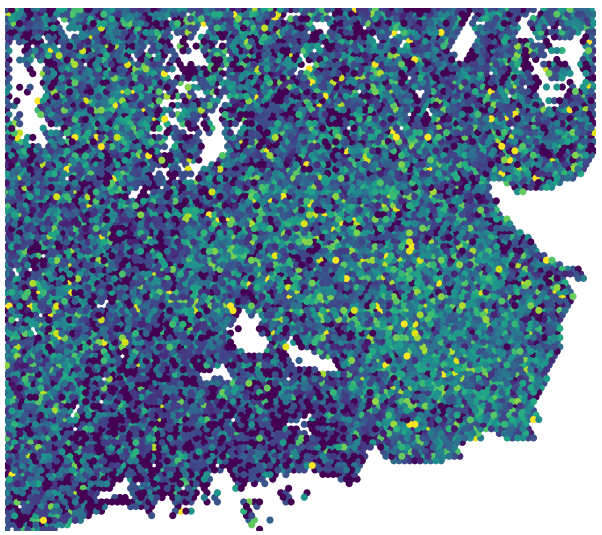
Treg Resting Vs Activated Up



0 0.0609

UCell score

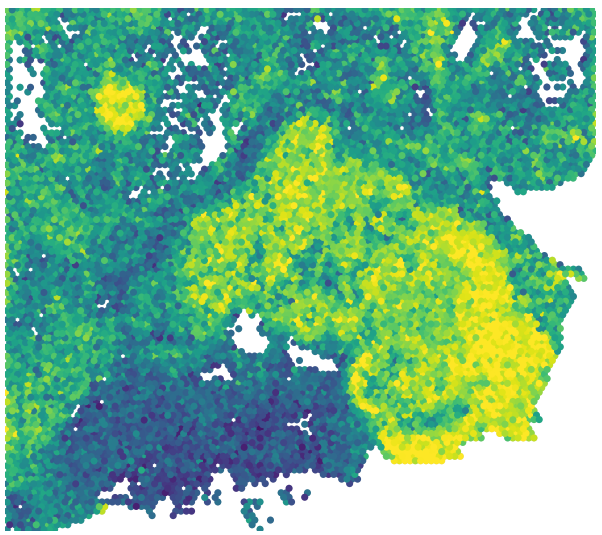
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

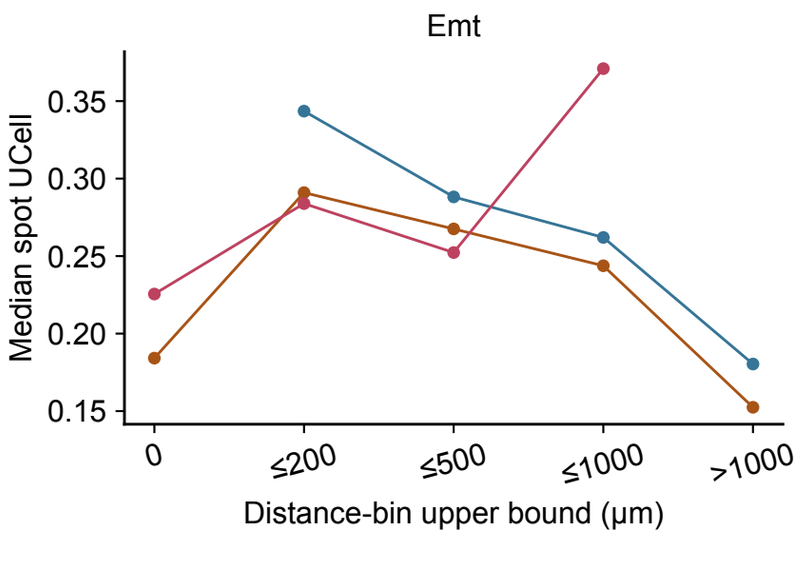


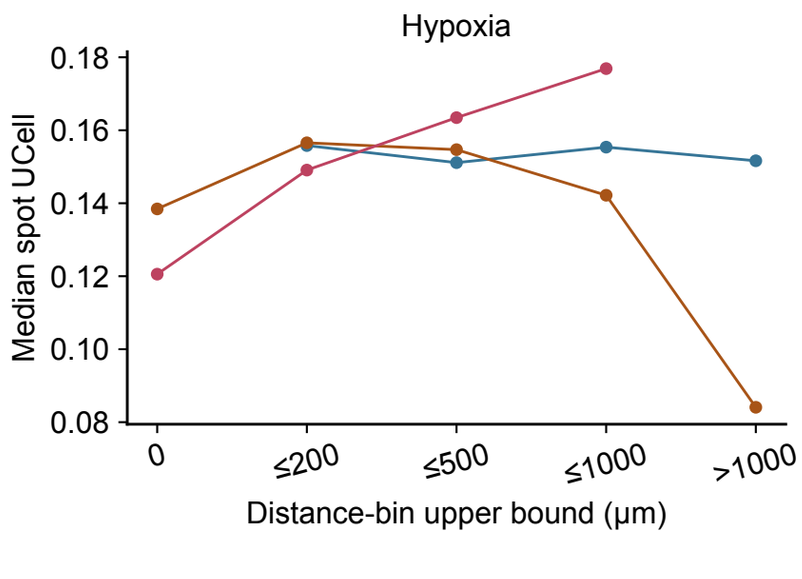
0 0.144

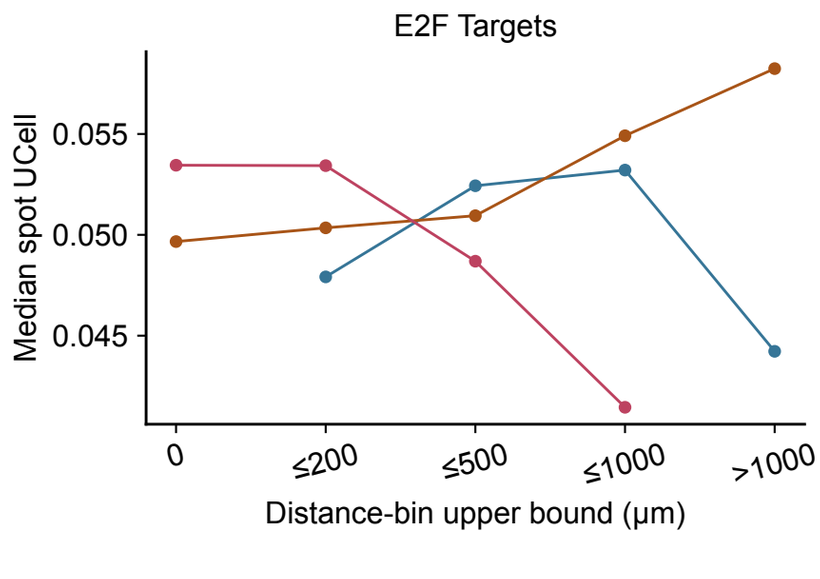
UCell score

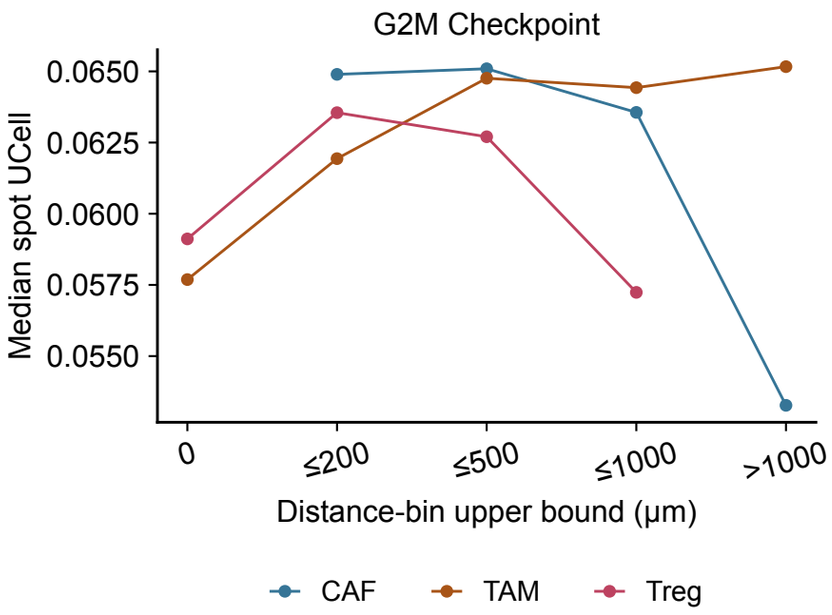
GSE238145_B14 | Programs by target distance

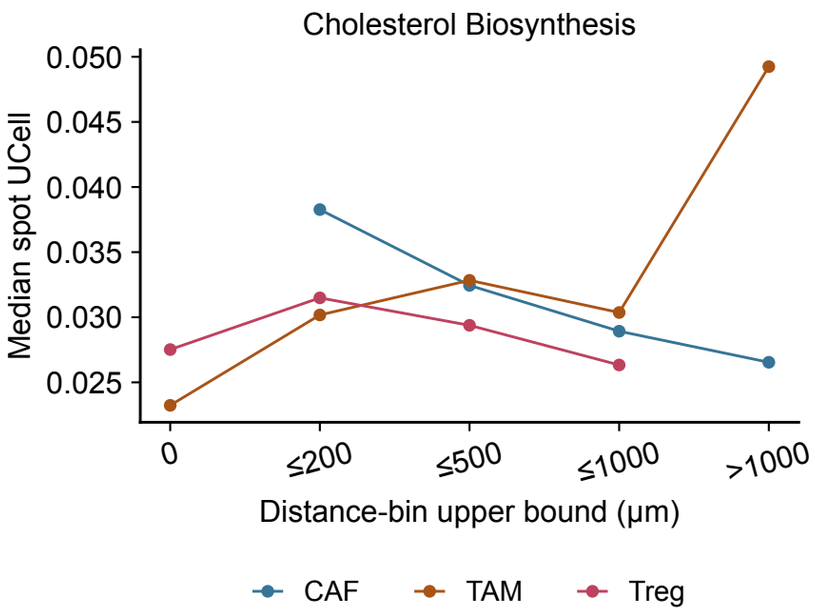
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

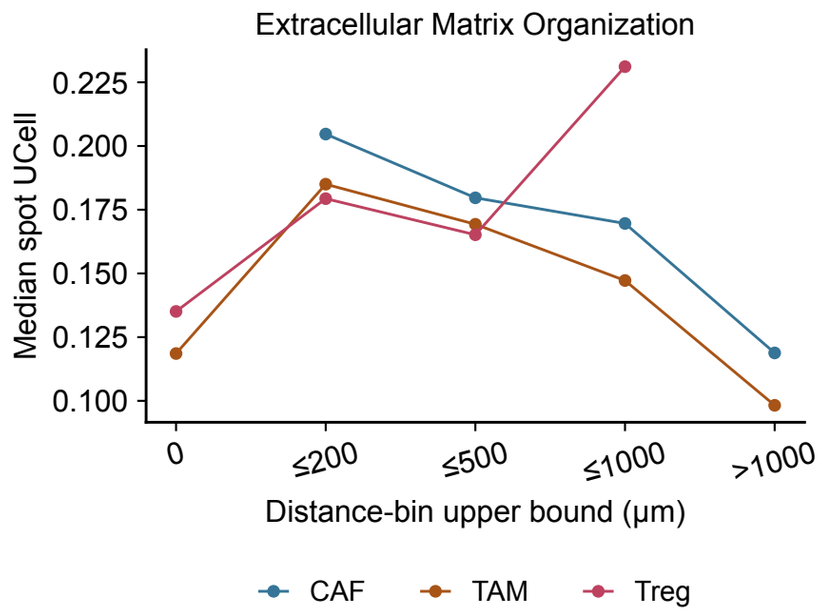






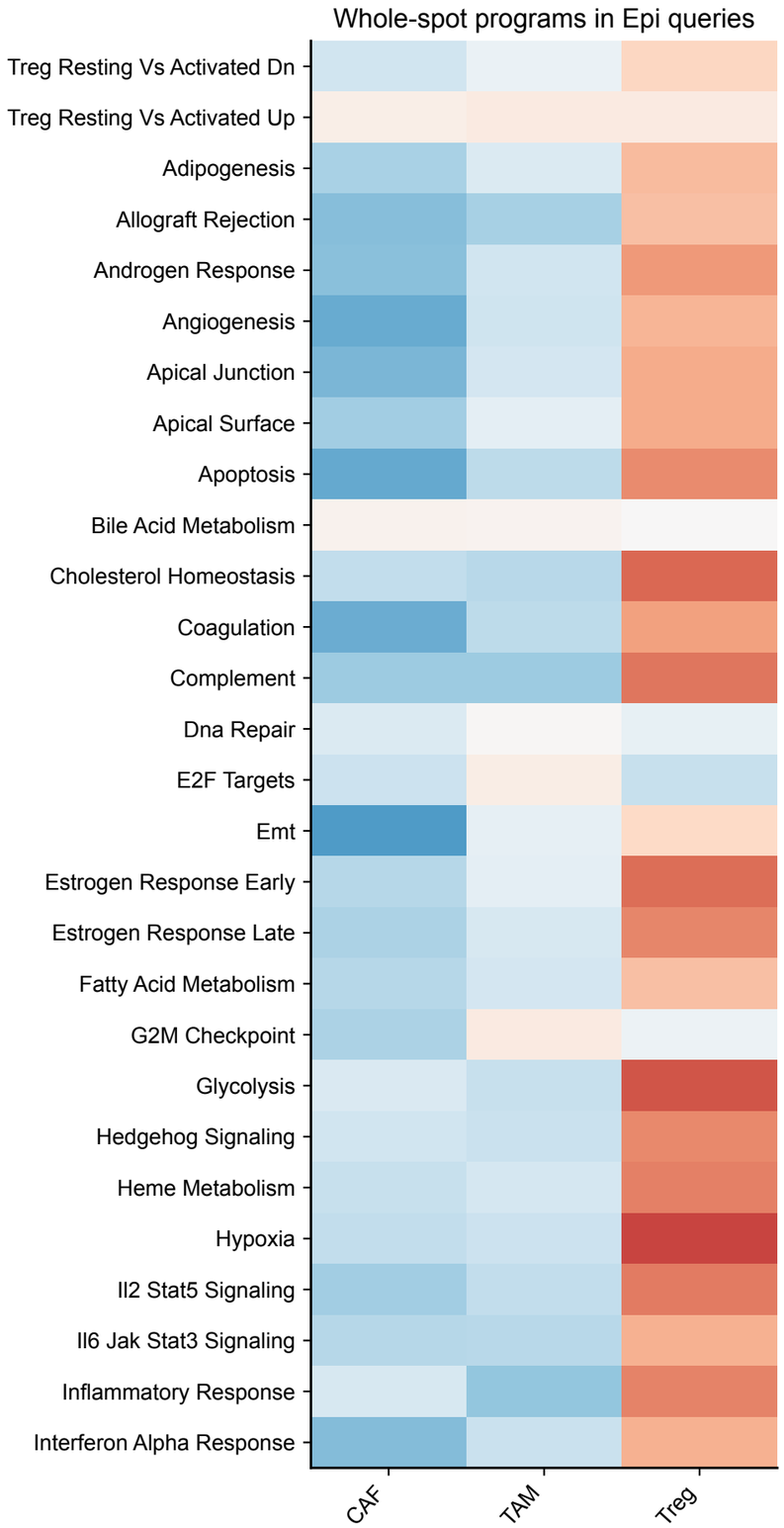


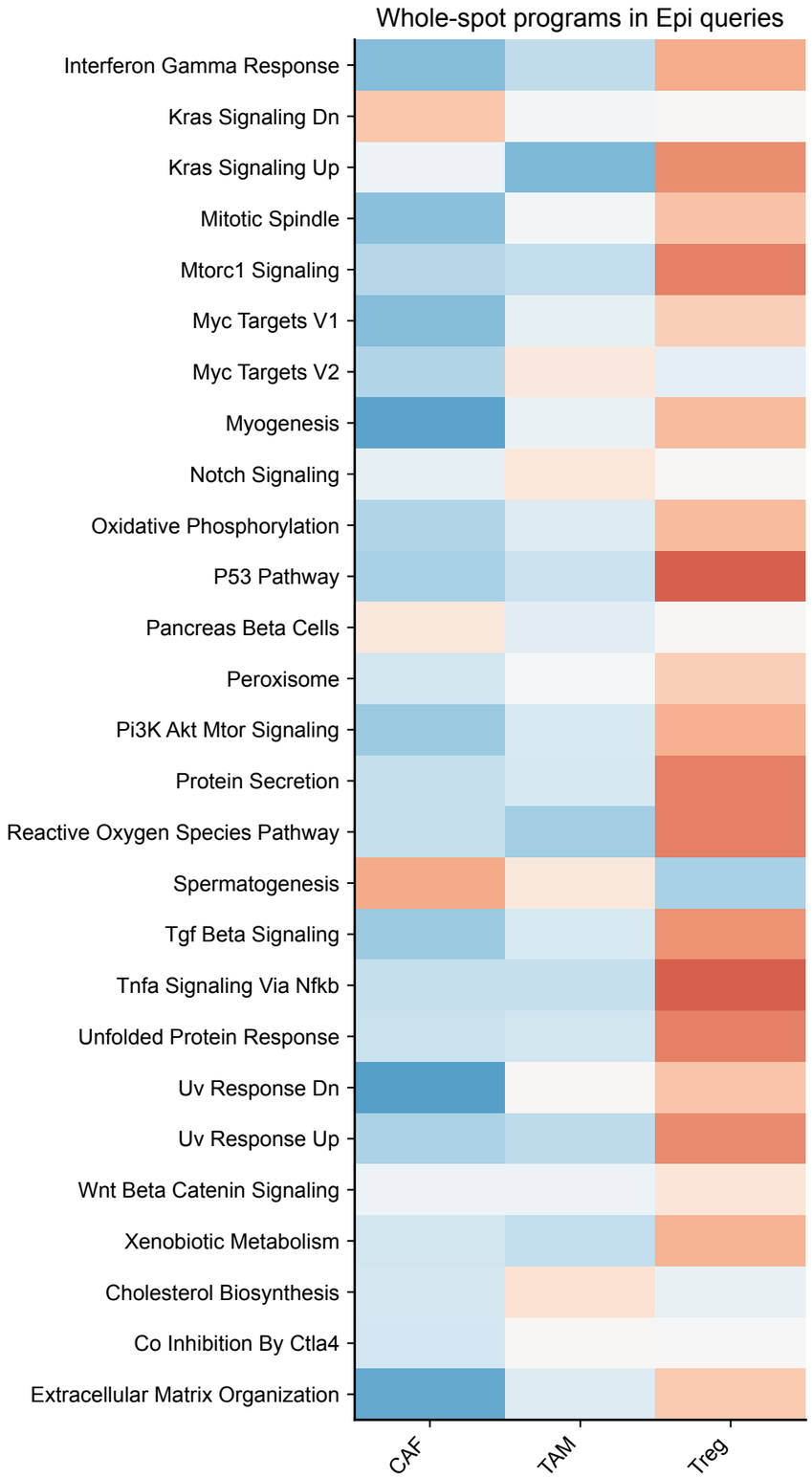




GSE238145_B14 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

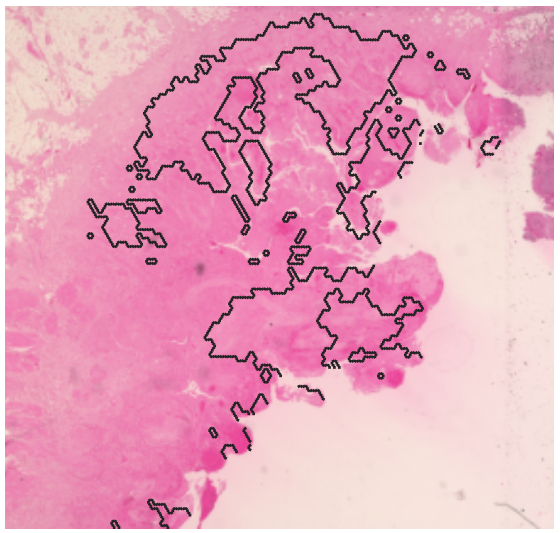




GSE238145_B15 | Tissue context

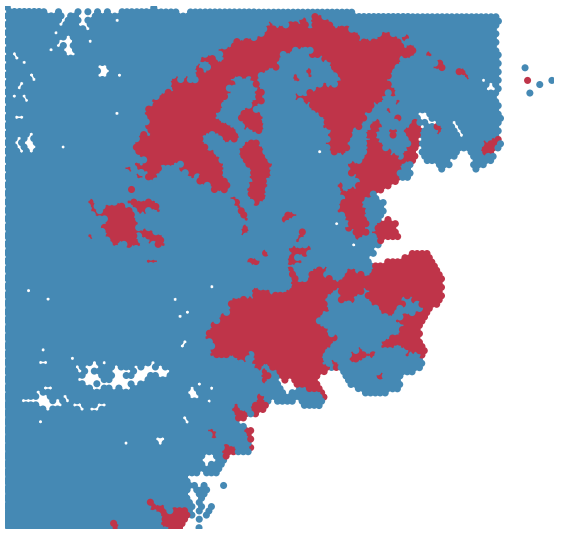
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



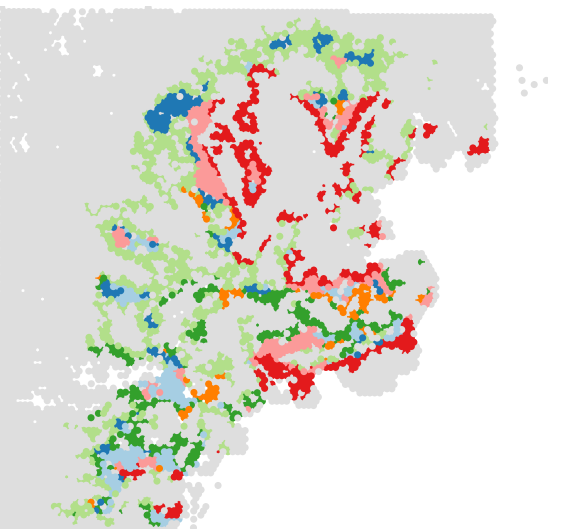
Black line: observed candidate outer edge

Final candidate location



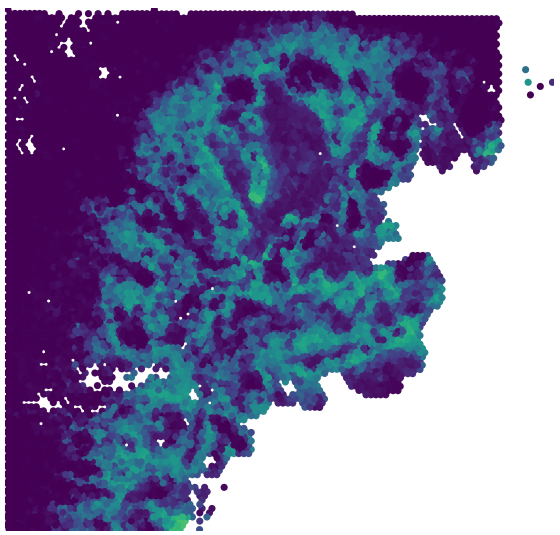
inside outside

Accepted R12 (original 0–8)



0 2 5 6 7
1 3

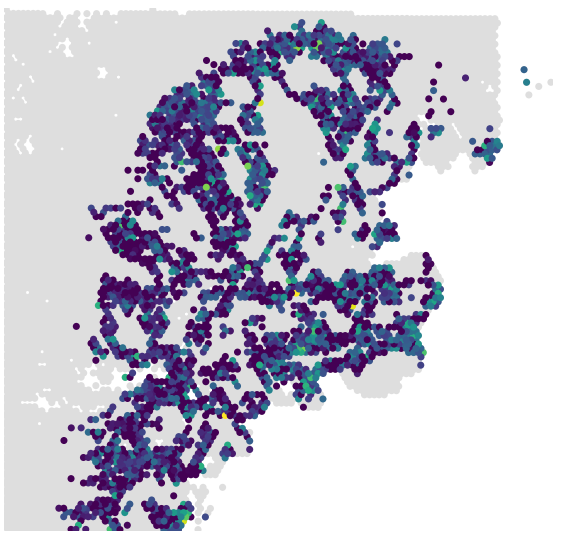
Epithelial support



0 1

Fraction of all 30

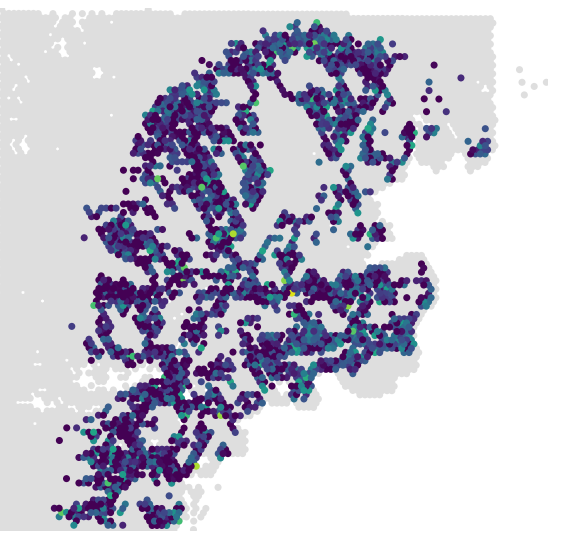
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



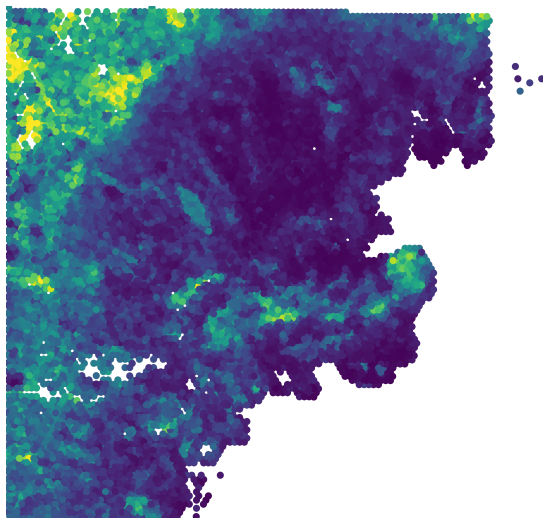
0 0.14

Matched-model AUCell

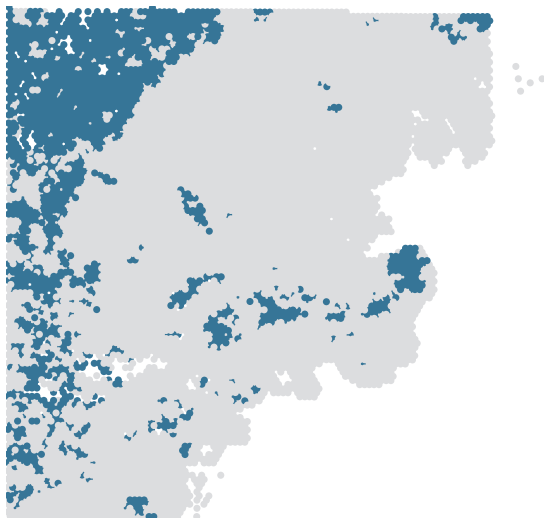
GSE238145_B15 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

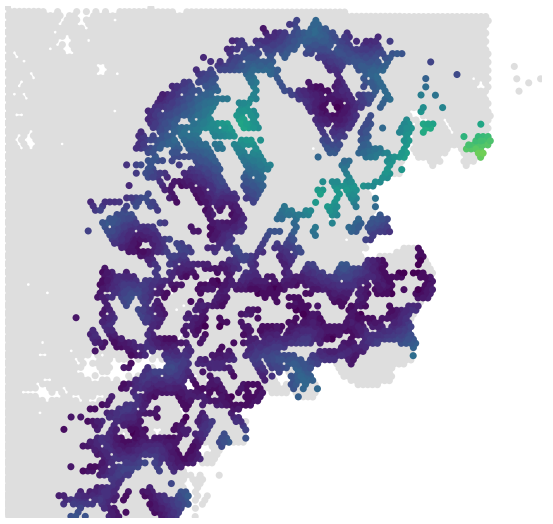
CAF composition



CAF P80 targets



CAF distance

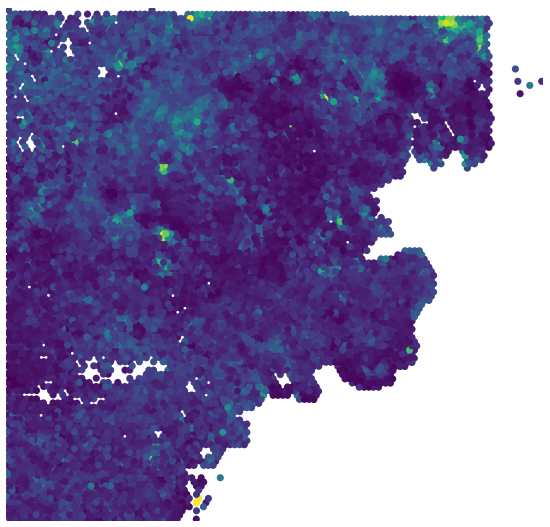


0 0.418
Fraction of all 30

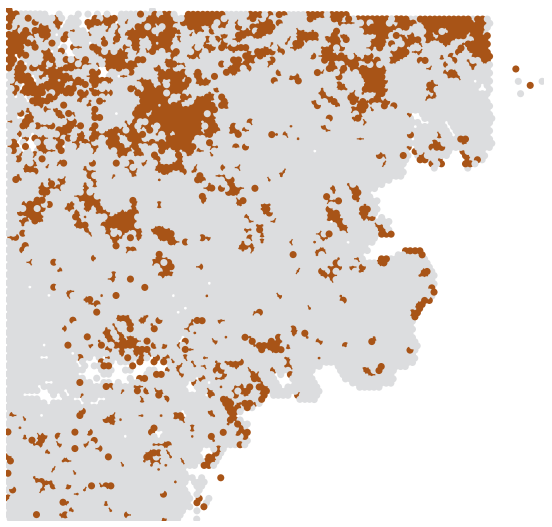
False True

0 3.1e+03
Nearest spot center (µm)

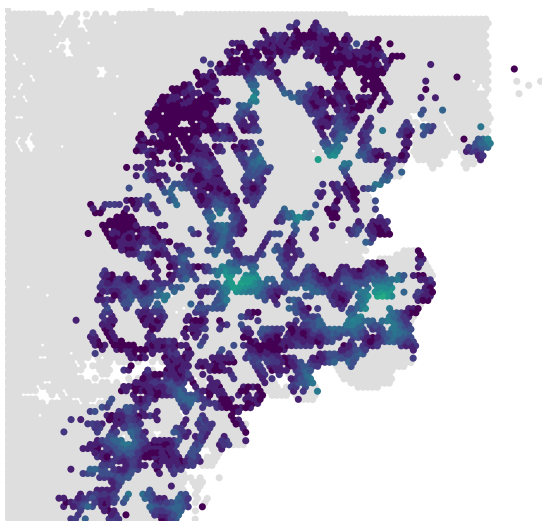
TAM composition



TAM P80 targets



TAM distance

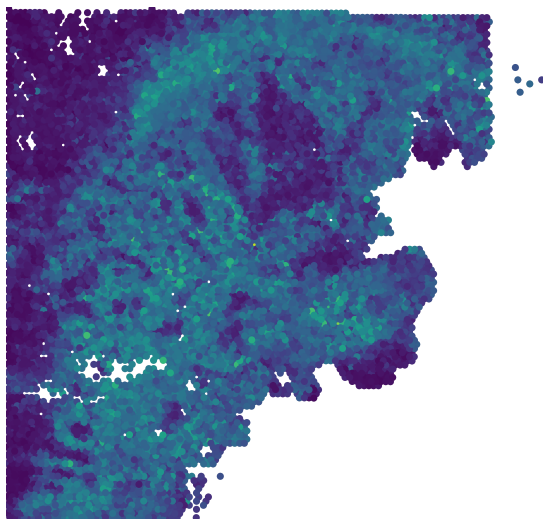


0 0.291
Fraction of all 30

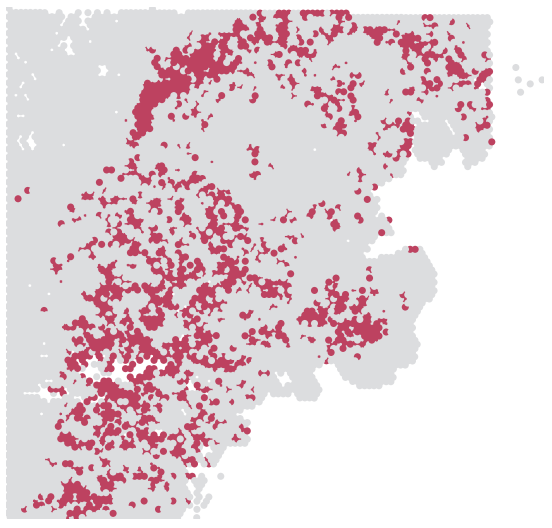
False True

0 1.13e+03
Nearest spot center (µm)

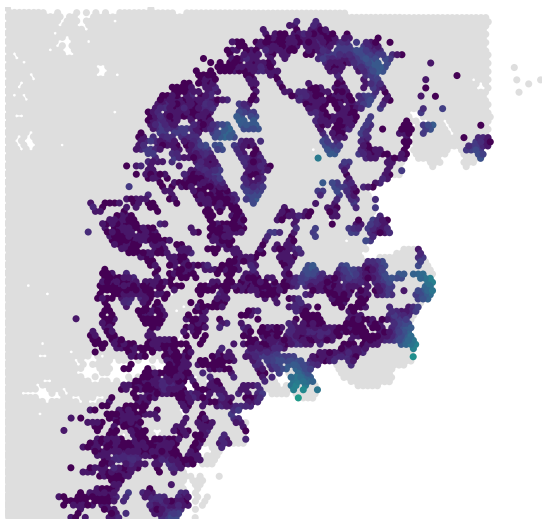
Treg composition



Treg P80 targets



Treg distance



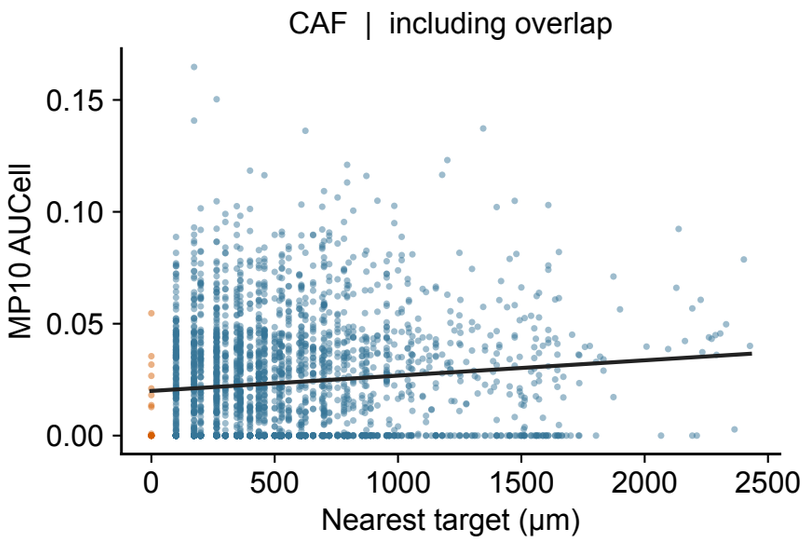
0 0.102
Fraction of all 30

False True

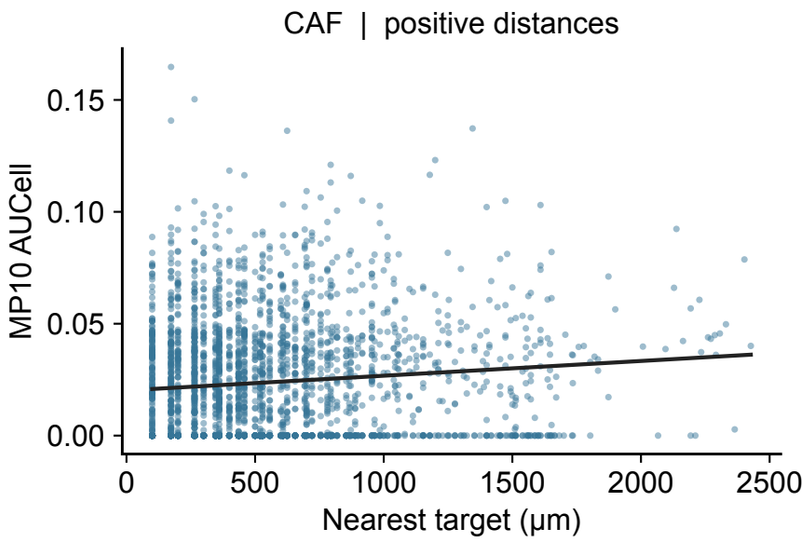
0 1.61e+03
Nearest spot center (µm)

GSE238145_B15 | MP10 versus nearest target

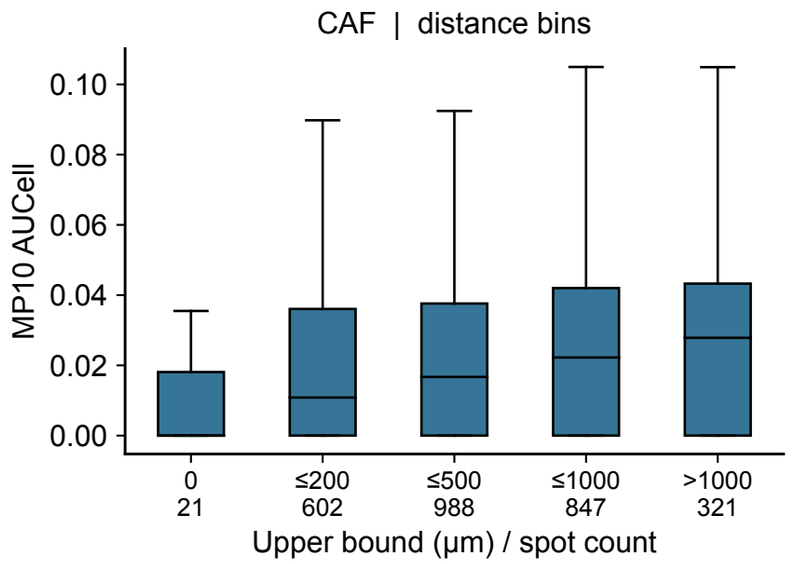
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



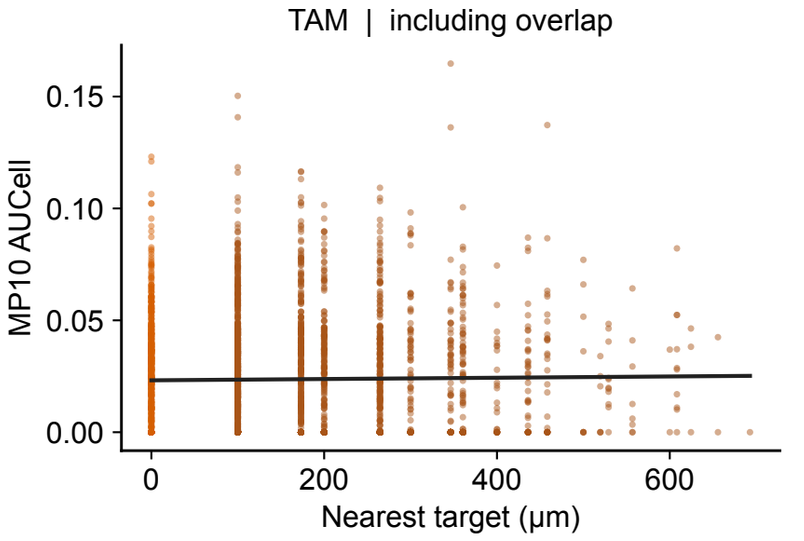
r +0.11 ρ +0.11
n 2779/2782 queries • 1878 targets



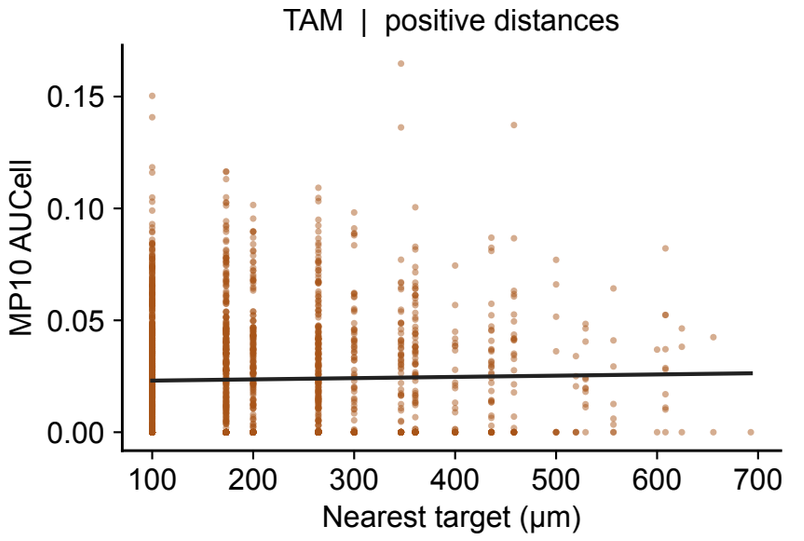
r +0.10 ρ +0.11
n 2758/2782 queries • 1878 targets



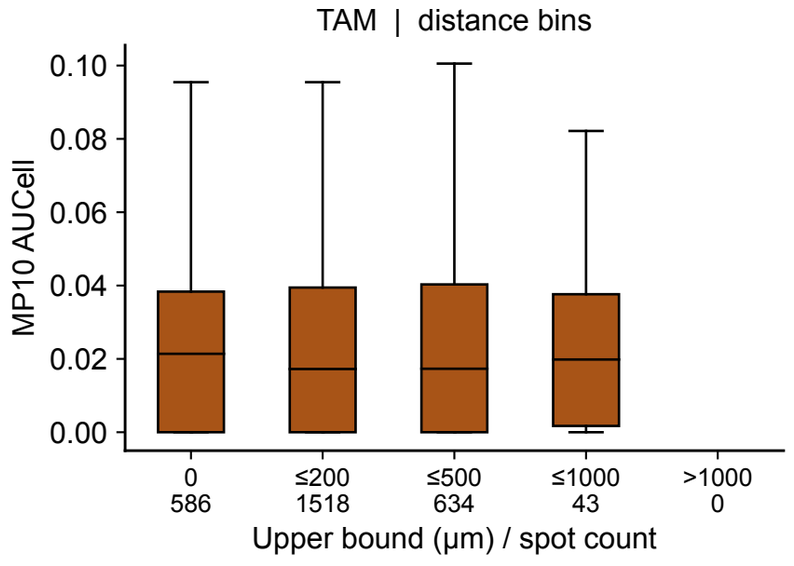
Orange = zero distance
Line = within-section linear fit



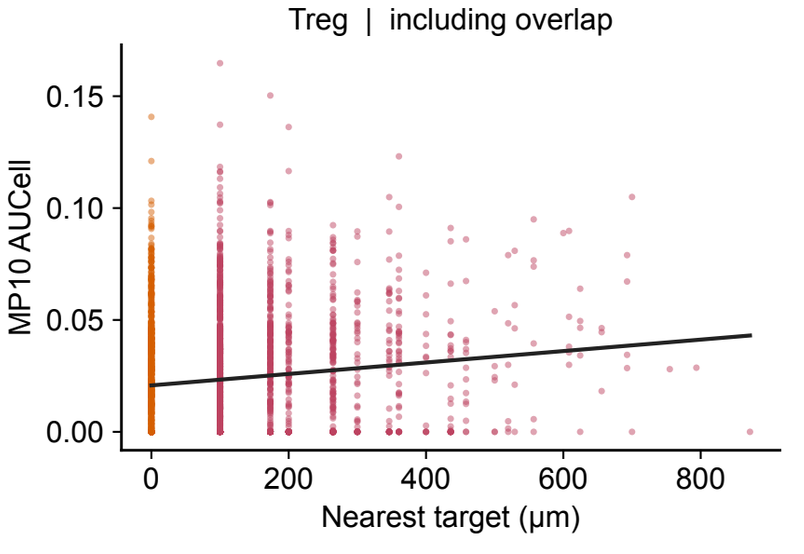
r +0.01 ρ -0.01
n 2781/2782 queries • 1878 targets



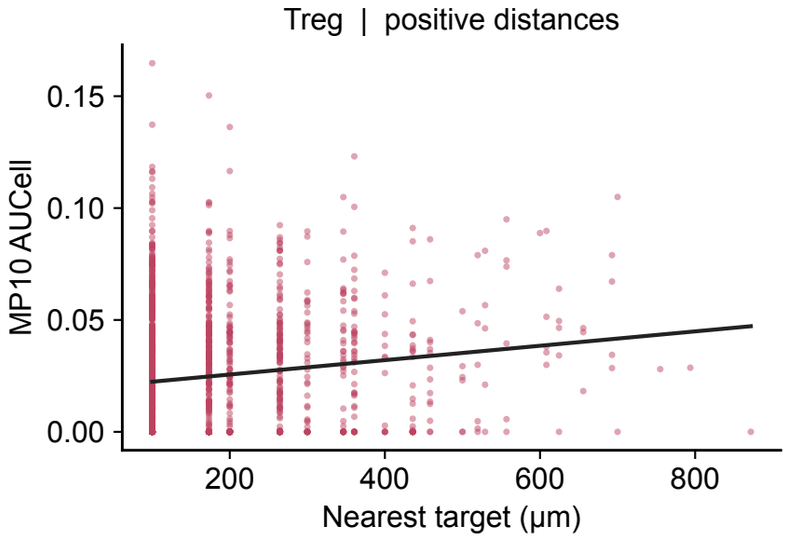
r +0.02 ρ +0.01
n 2195/2782 queries • 1878 targets



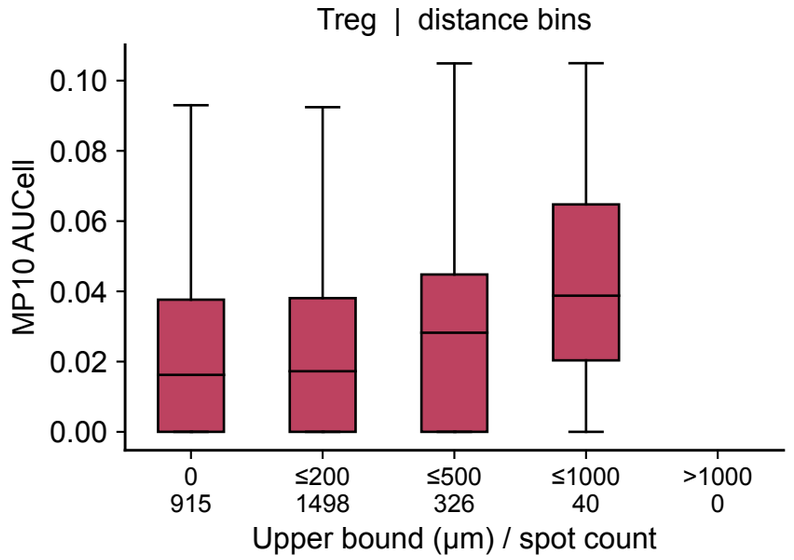
Orange = zero distance
Line = within-section linear fit



r +0.12 ρ +0.10
n 2779/2782 queries • 1878 targets



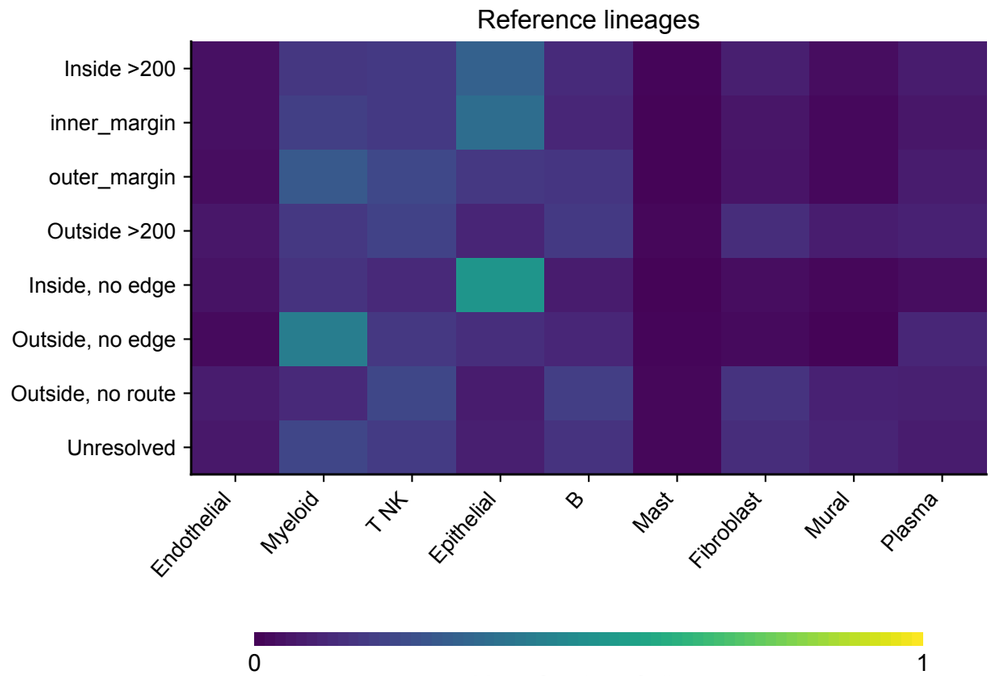
r +0.14 ρ +0.14
n 1864/2782 queries • 1878 targets

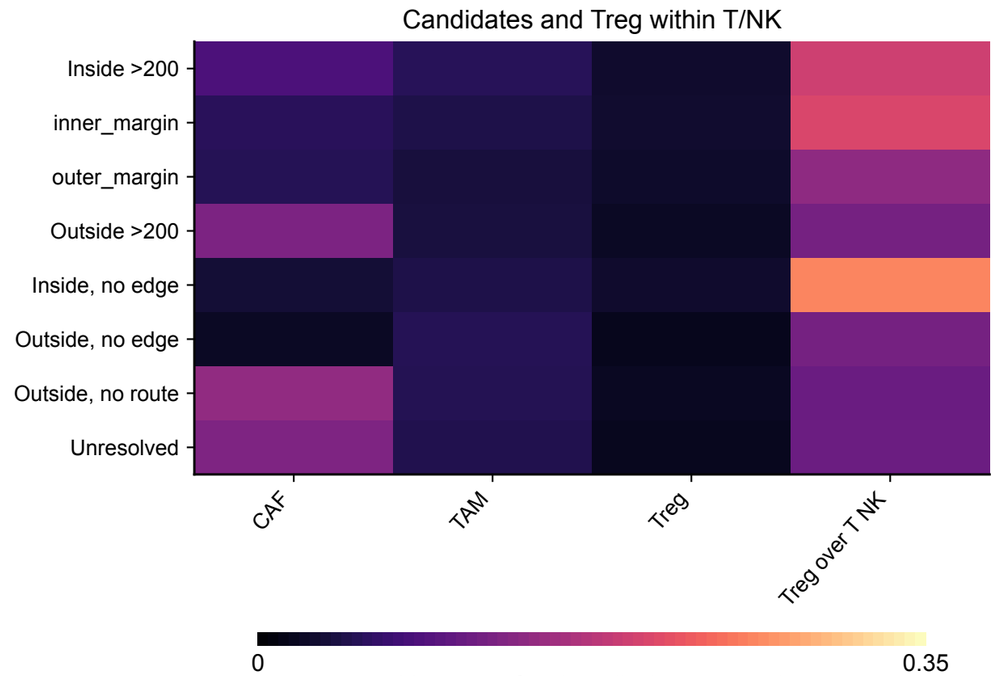


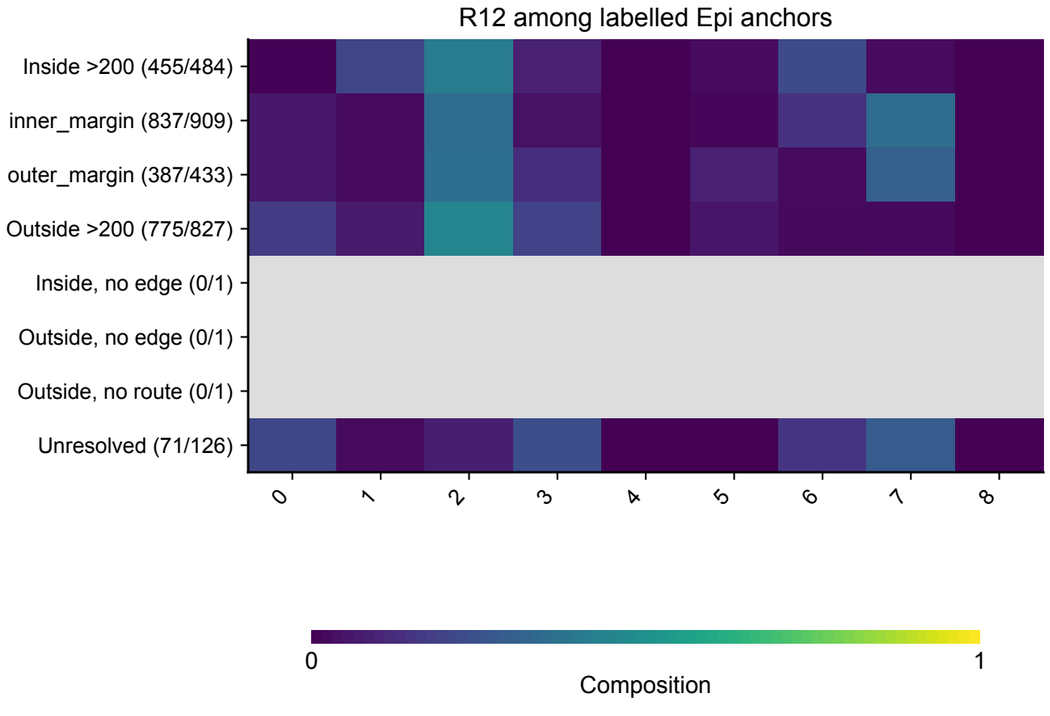
Orange = zero distance
Line = within-section linear fit

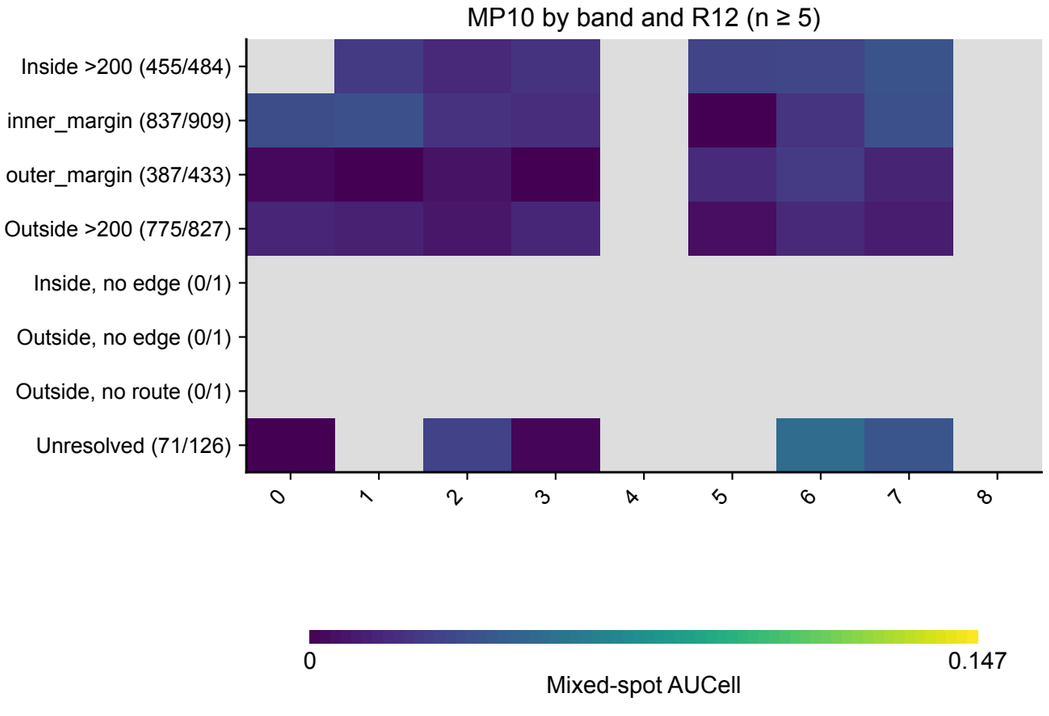
GSE238145_B15 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





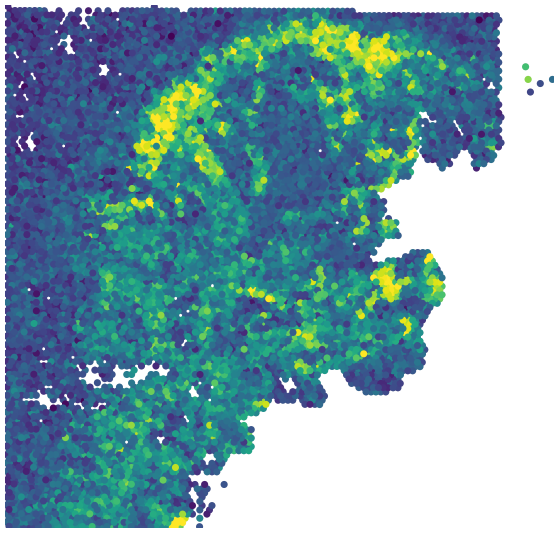




GSE238145_B15 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

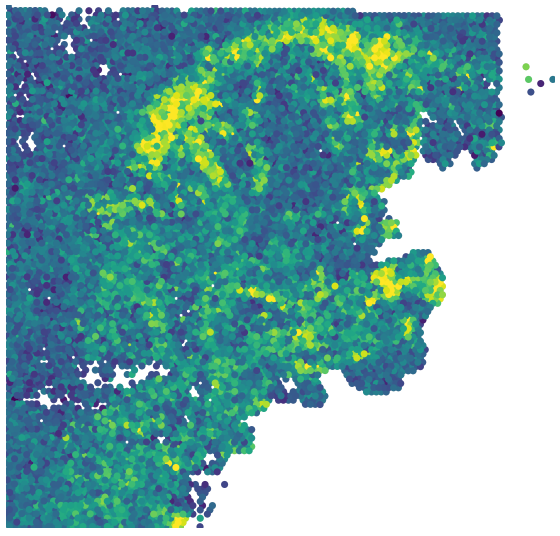
E2F Targets



0 0.152

UCell score

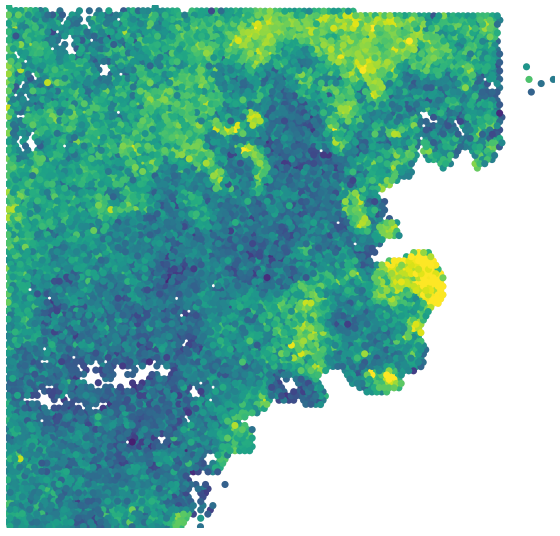
G2M Checkpoint



0 0.134

UCell score

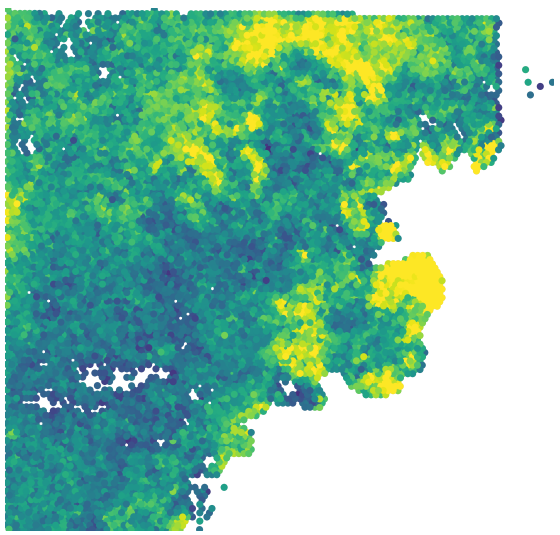
Hypoxia



0 0.211

UCell score

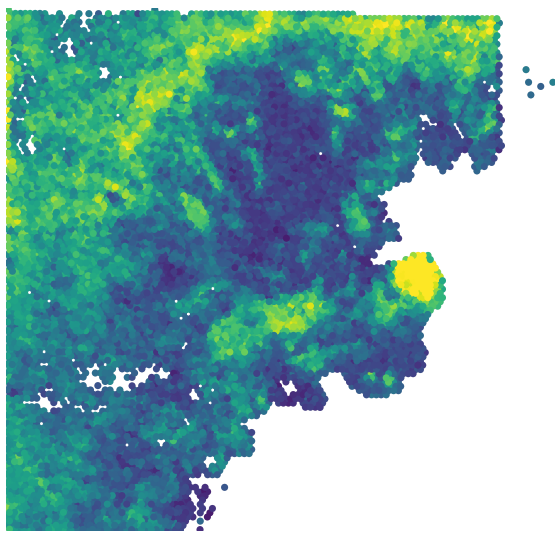
Tnfa Signaling Via Nfkb



0 0.275

UCell score

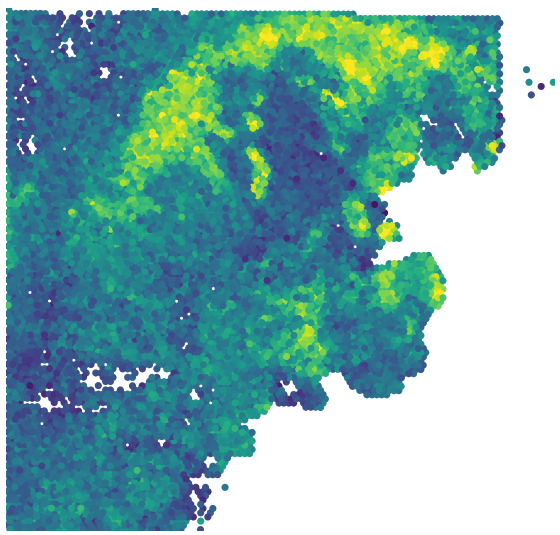
Emt



0 0.386

UCell score

Interferon Gamma Response



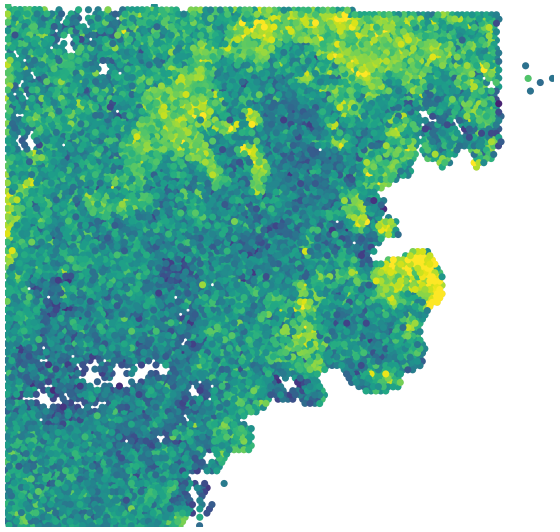
0 0.275

UCell score

GSE238145_B15 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

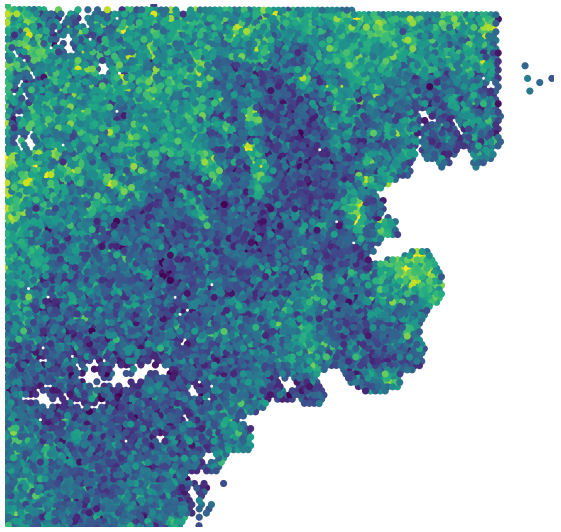
Il2 Stat5 Signaling



0 0.144

UCell score

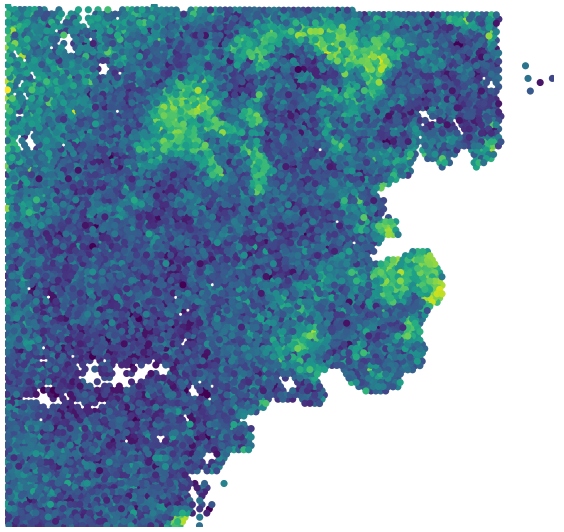
Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

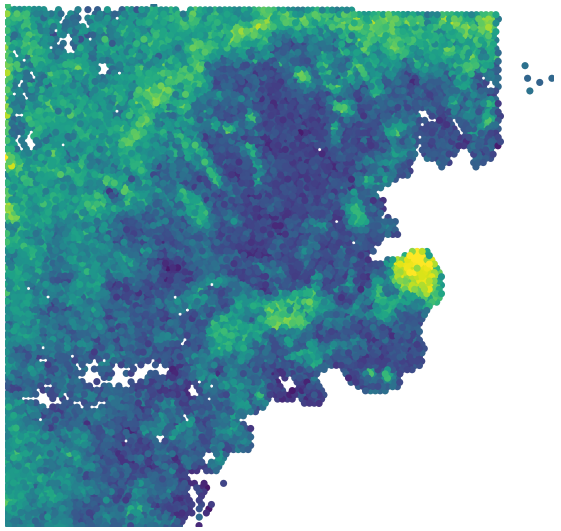
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4

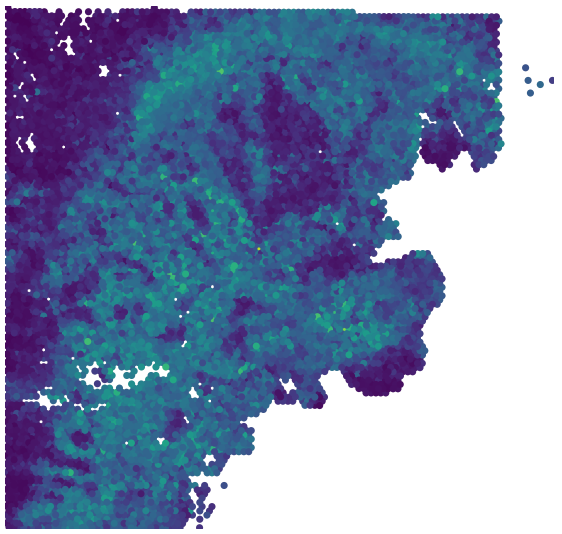
0 0.151

UCell score

GSE238145_B15 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

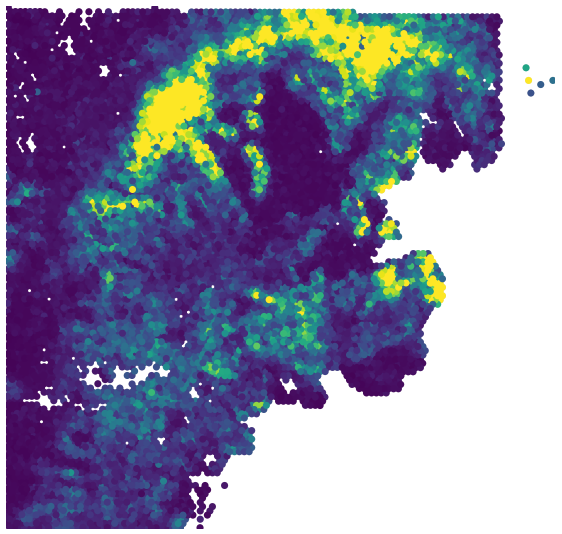
Treg composition



0 0.102

Fraction of all 30

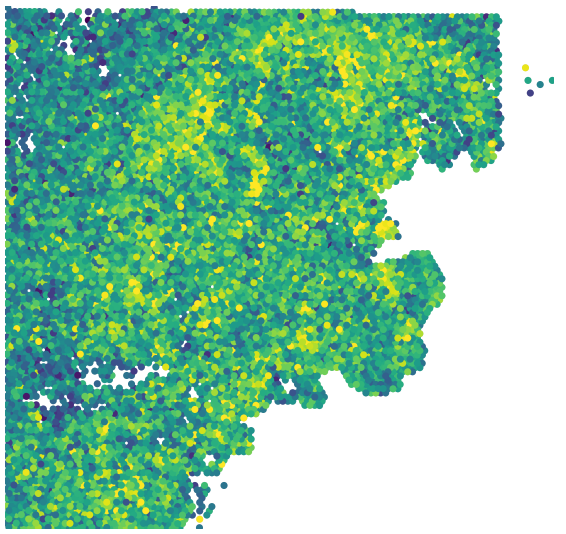
Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn



0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

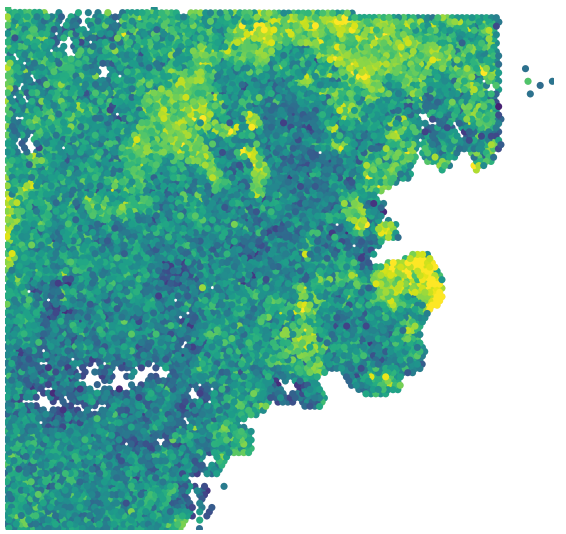
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

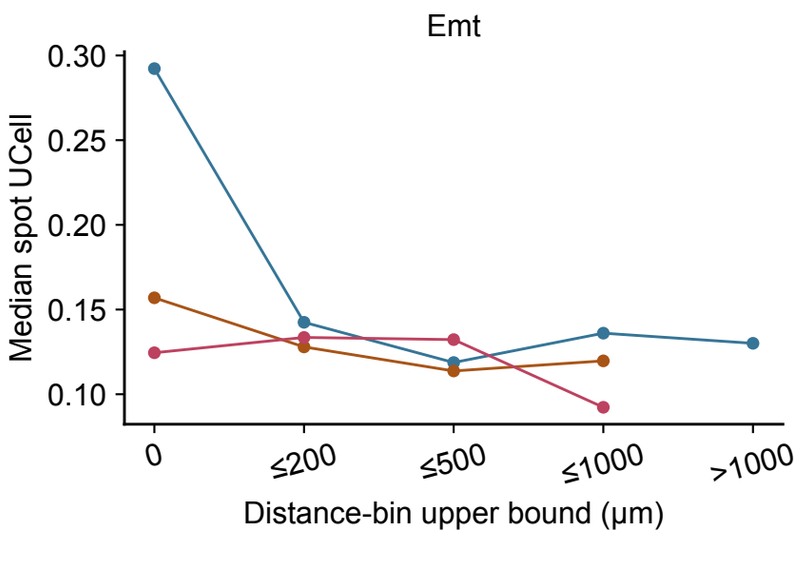


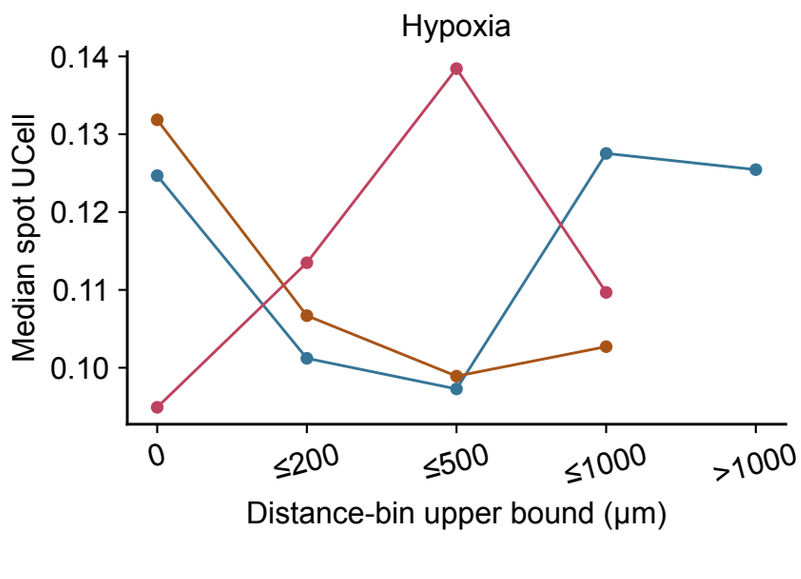
0 0.144

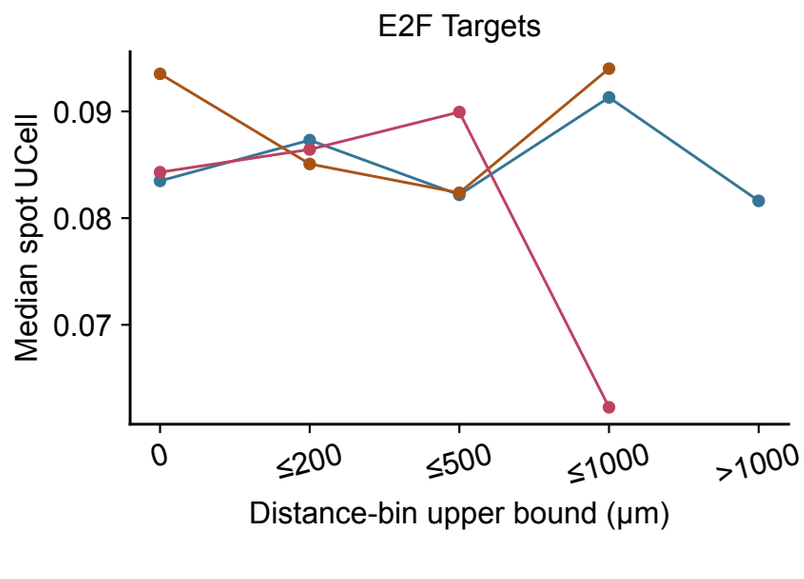
UCell score

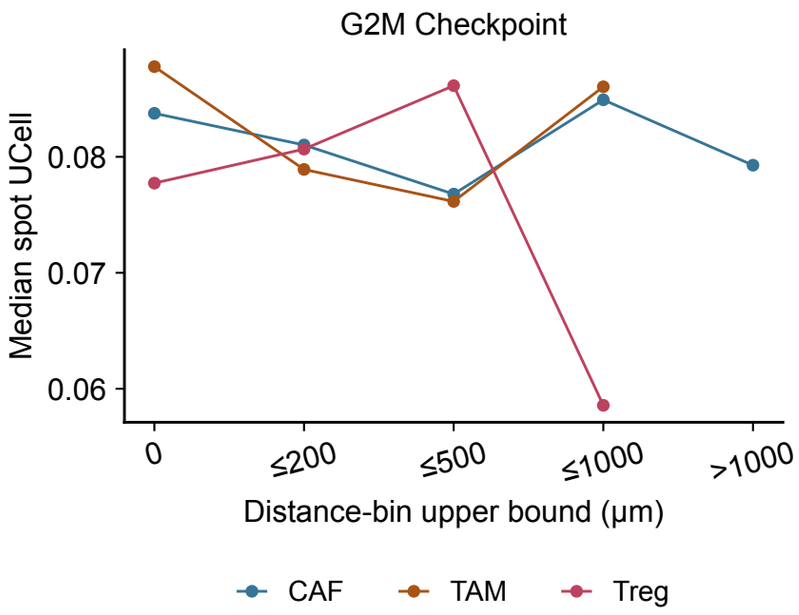
GSE238145_B15 | Programs by target distance

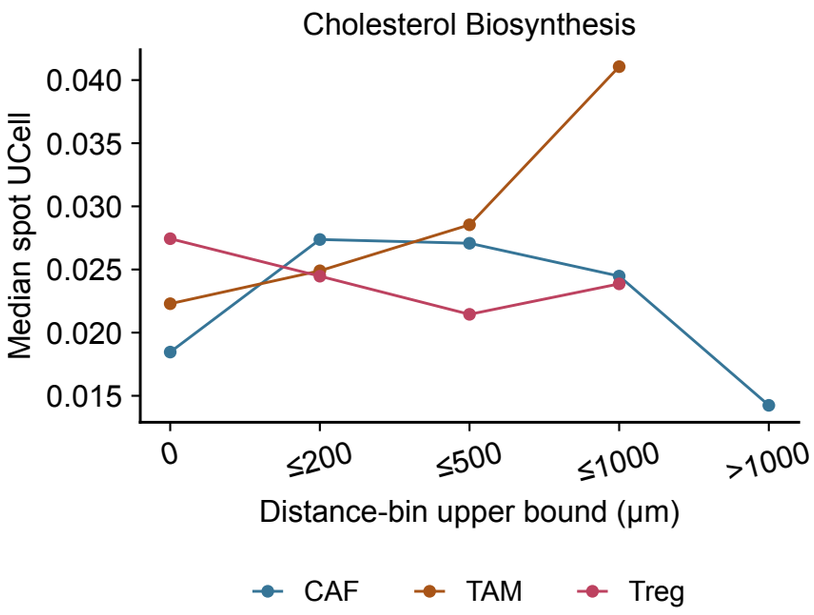
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

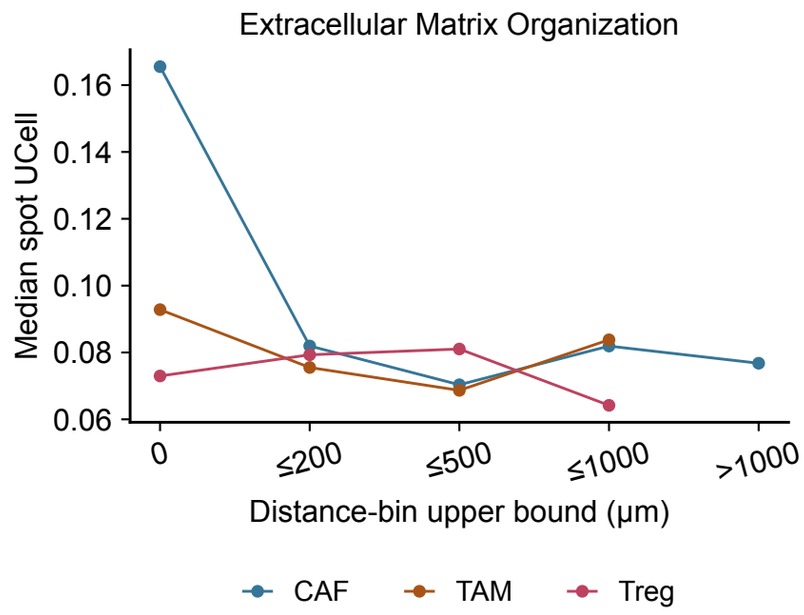






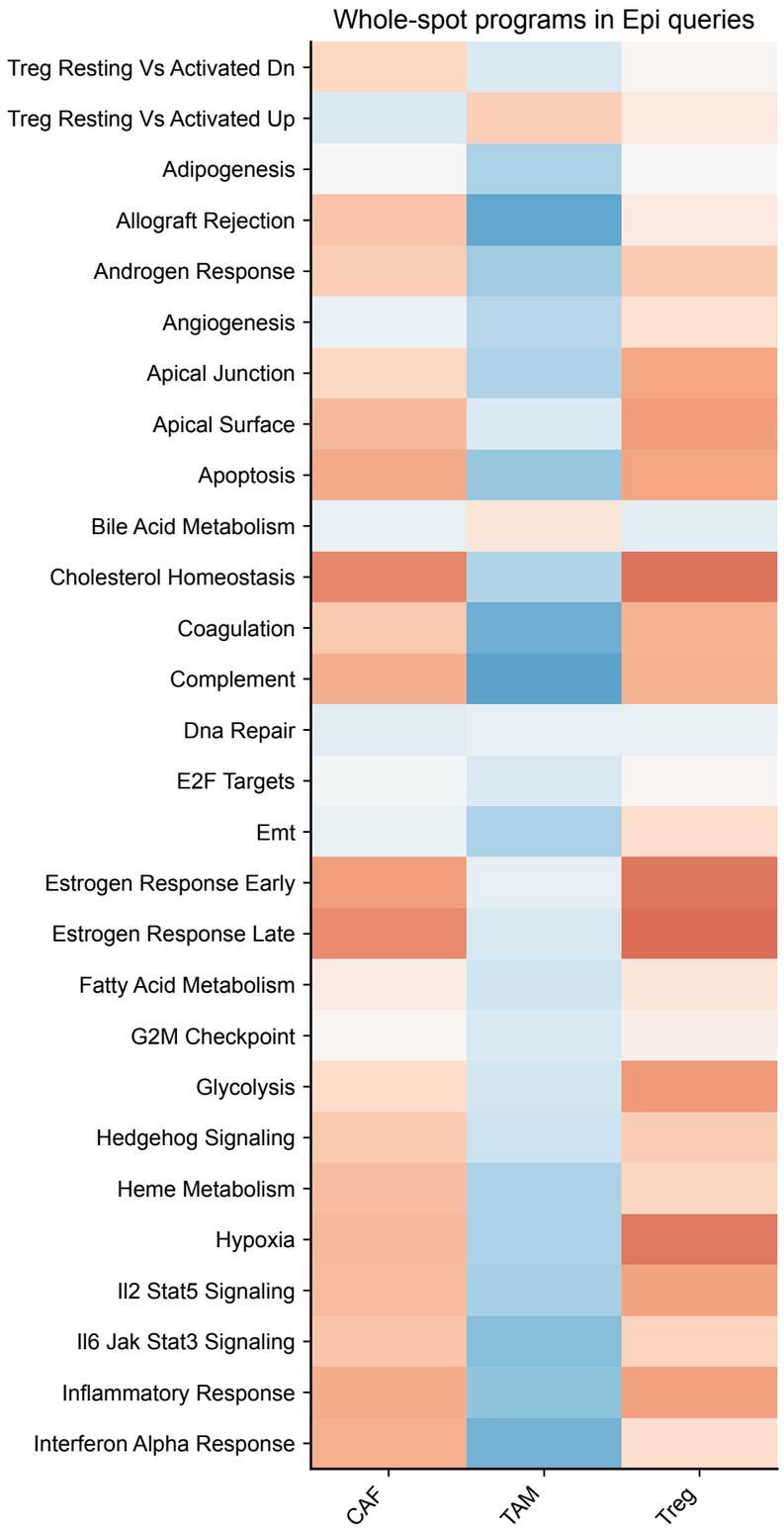


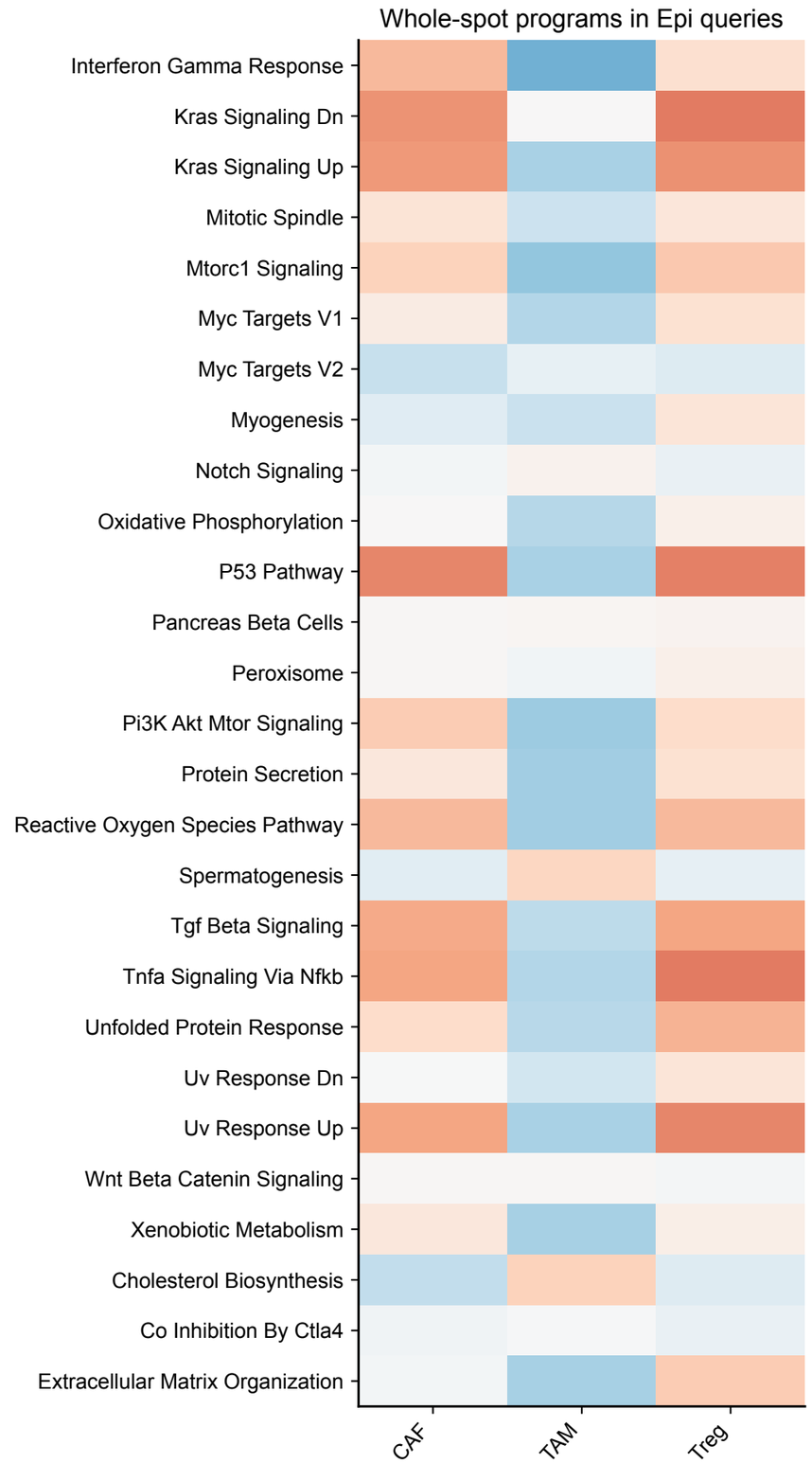




GSE238145_B15 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

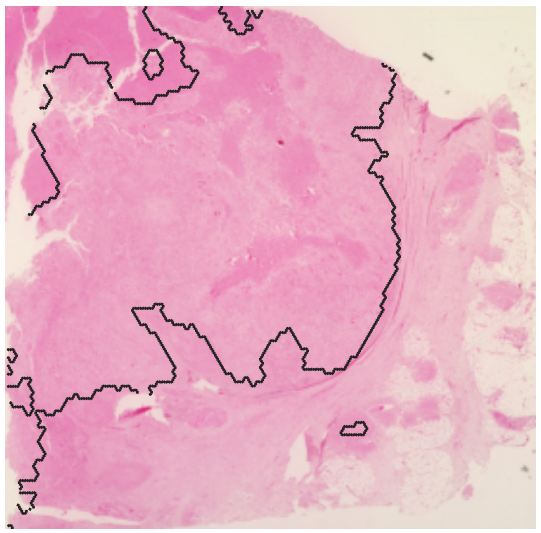




GSE238145_B16 | Tissue context

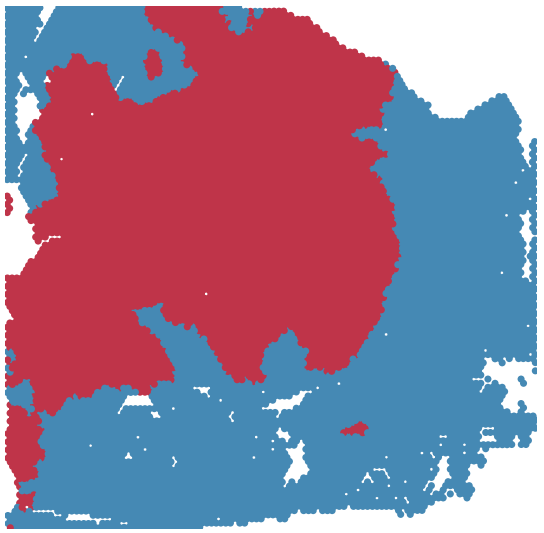
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



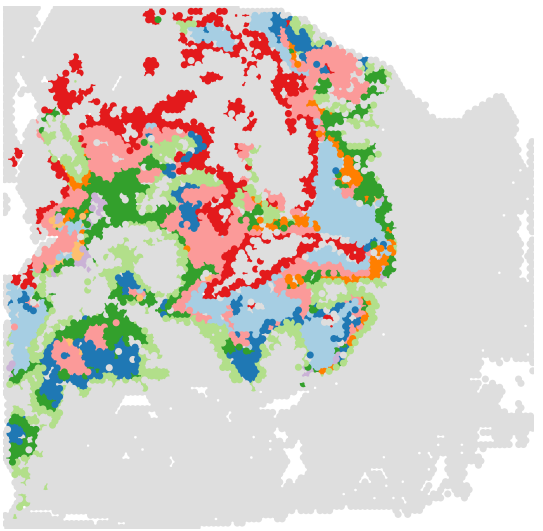
Black line: observed candidate outer edge

Final candidate location



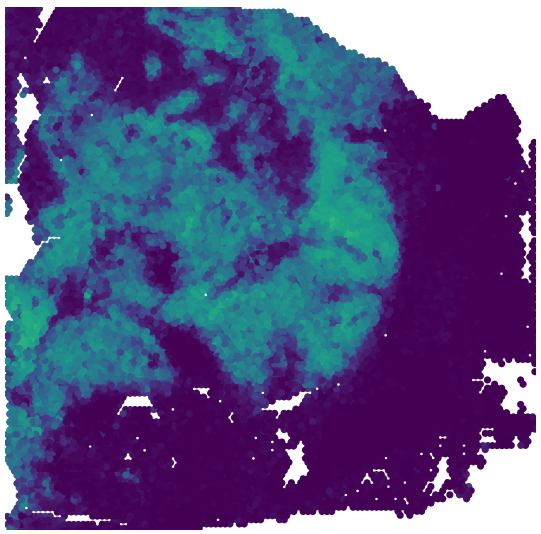
inside outside

Accepted R12 (original 0–8)



0 1 2 3 4 5 6 7 8

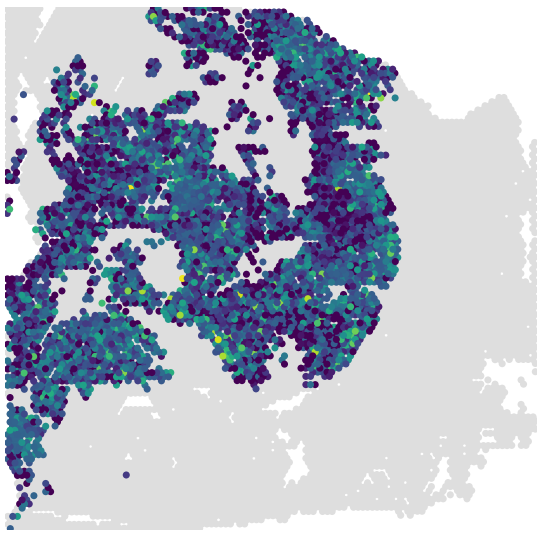
Epithelial support



0 1

Fraction of all 30

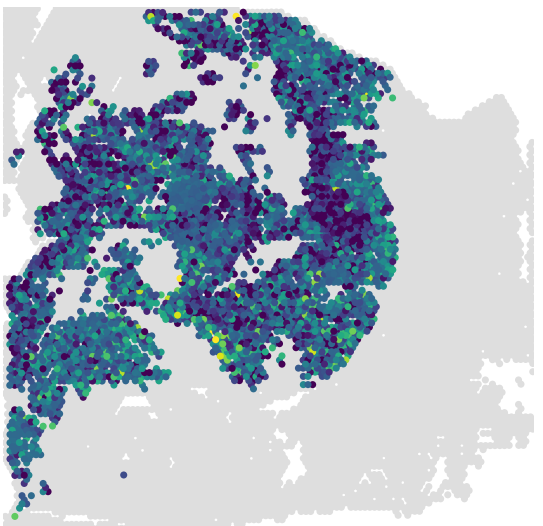
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



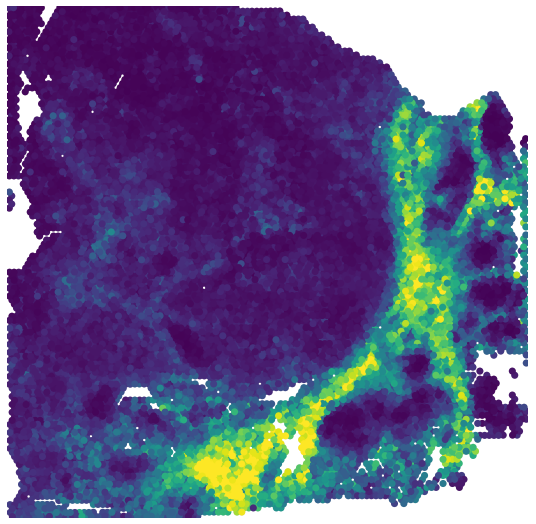
0 0.14

Matched-model AUCell

GSE238145_B16 | Candidate neighborhoods

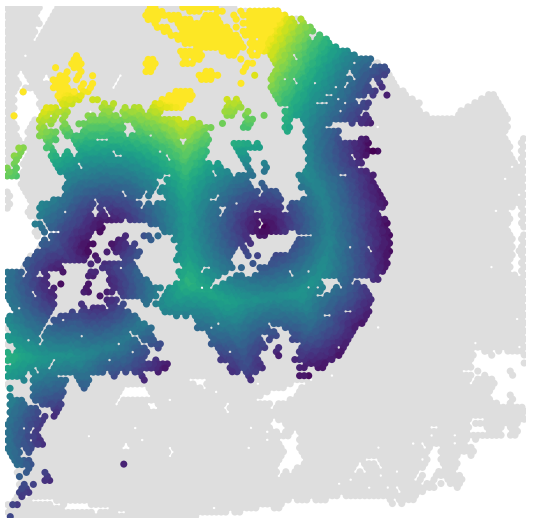
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



CAF P80 targets

CAF distance

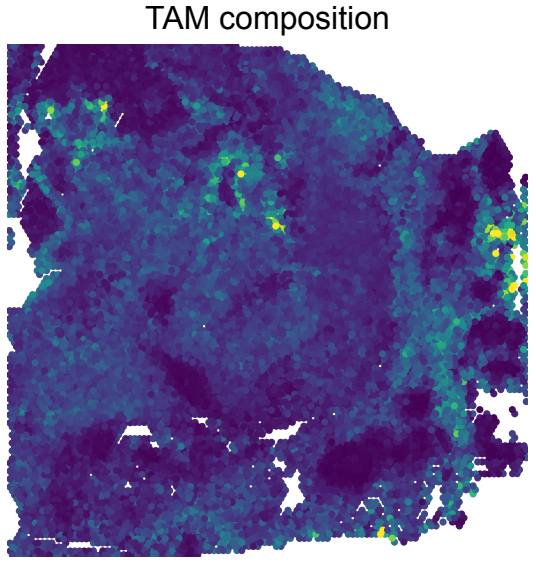


0 0.418
Fraction of all 30

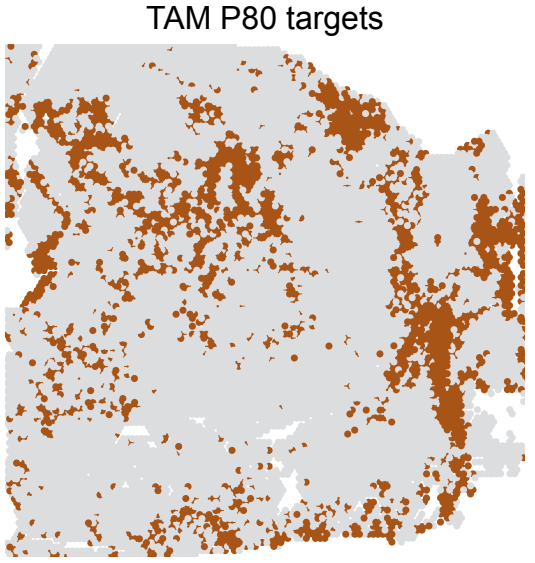
False True

0 3.1e+03
Nearest spot center (µm)

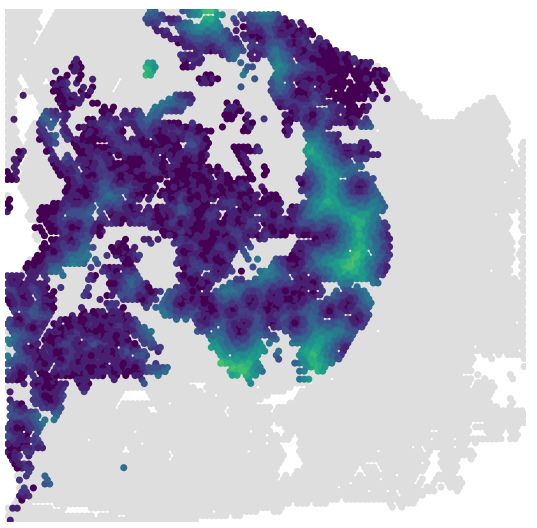
TAM composition



TAM P80 targets



TAM distance

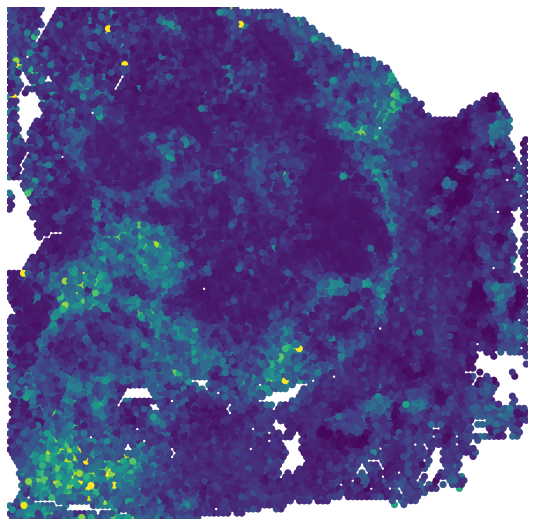


0 0.291
Fraction of all 30

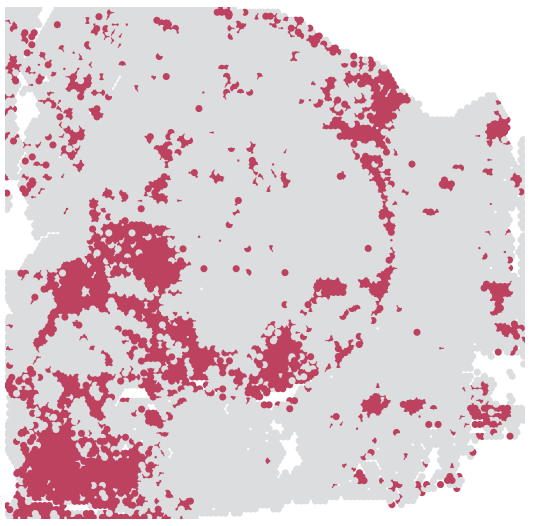
False True

0 1.13e+03
Nearest spot center (µm)

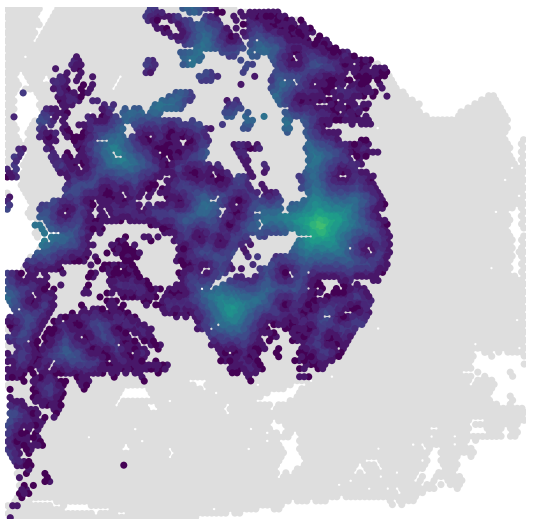
Treg composition



Treg P80 targets



Treg distance



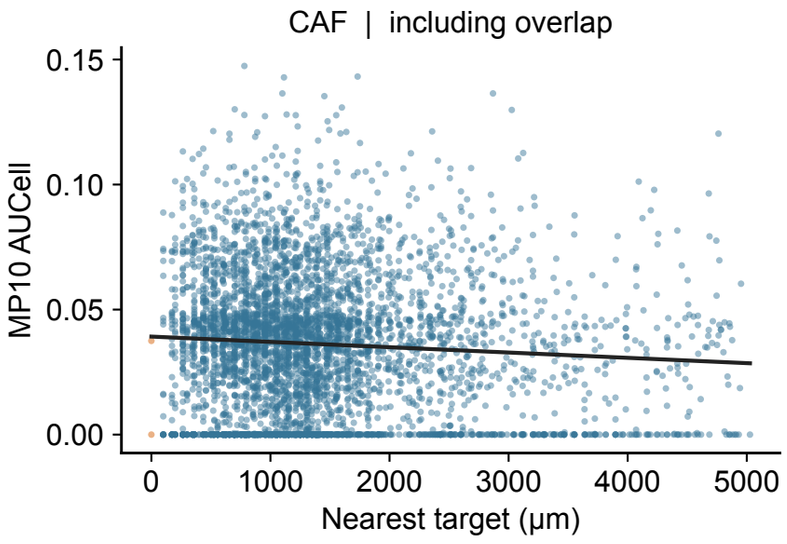
0 0.102
Fraction of all 30

False True

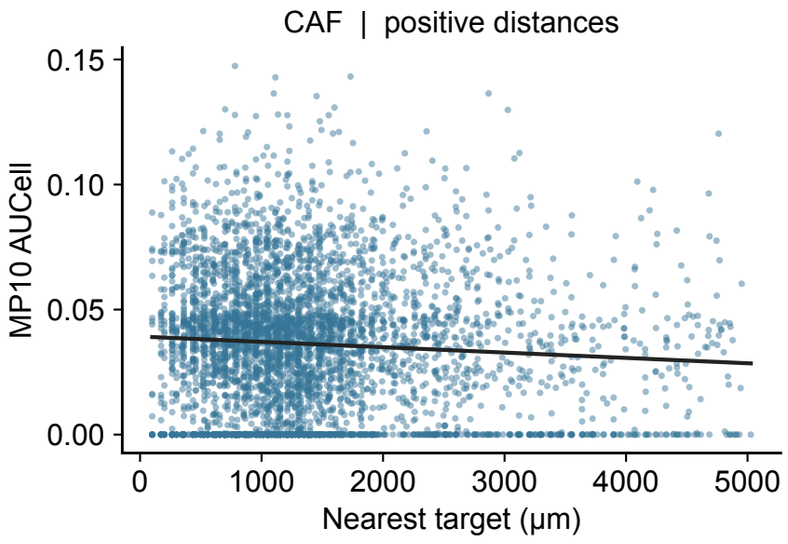
0 1.61e+03
Nearest spot center (µm)

GSE238145_B16 | MP10 versus nearest target

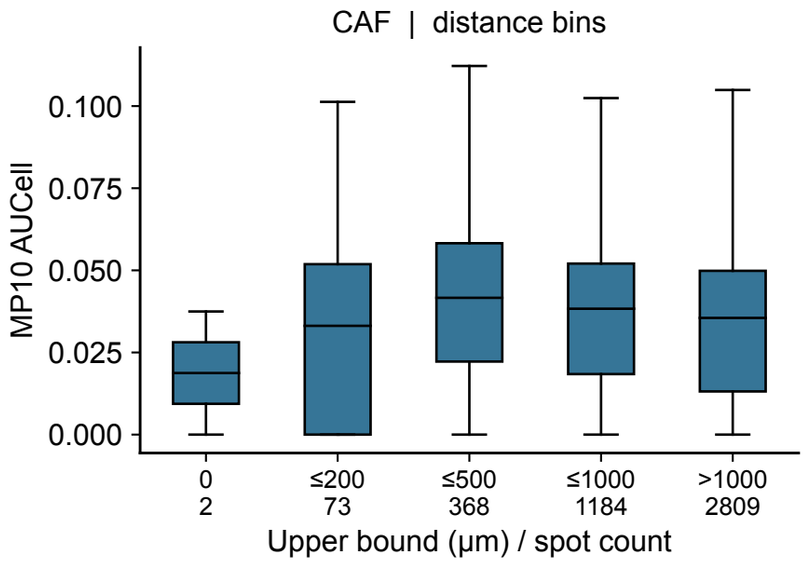
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



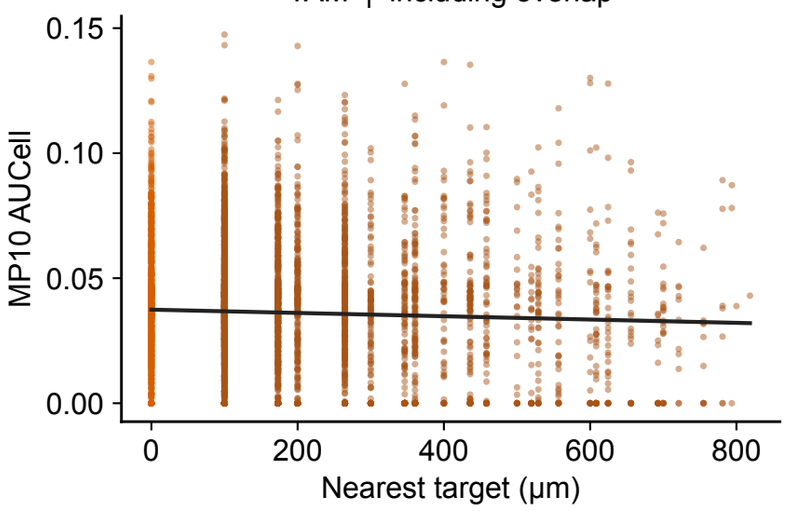
r -0.07 ρ -0.06
n 4436/4440 queries • 2446 targets



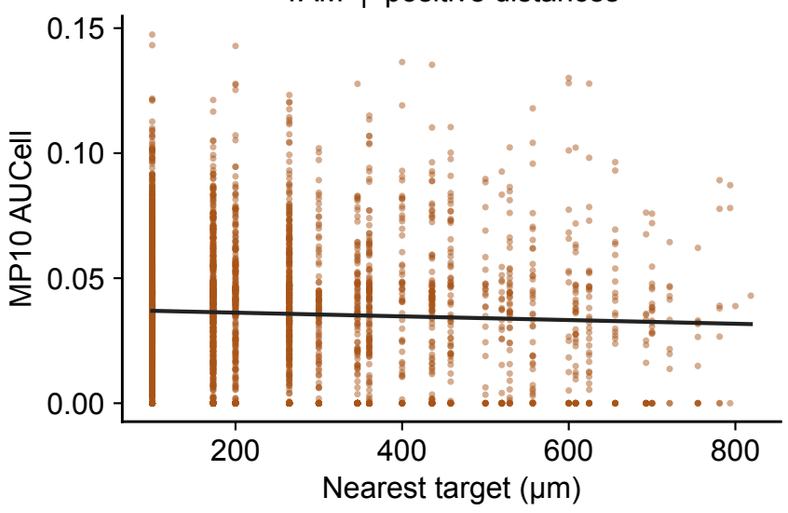
r -0.07 ρ -0.06
n 4434/4440 queries • 2446 targets



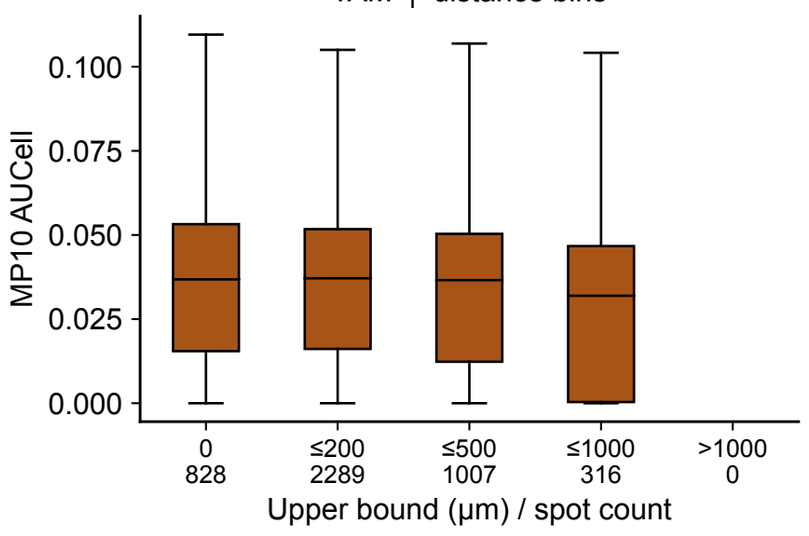
Orange = zero distance
Line = within-section linear fit



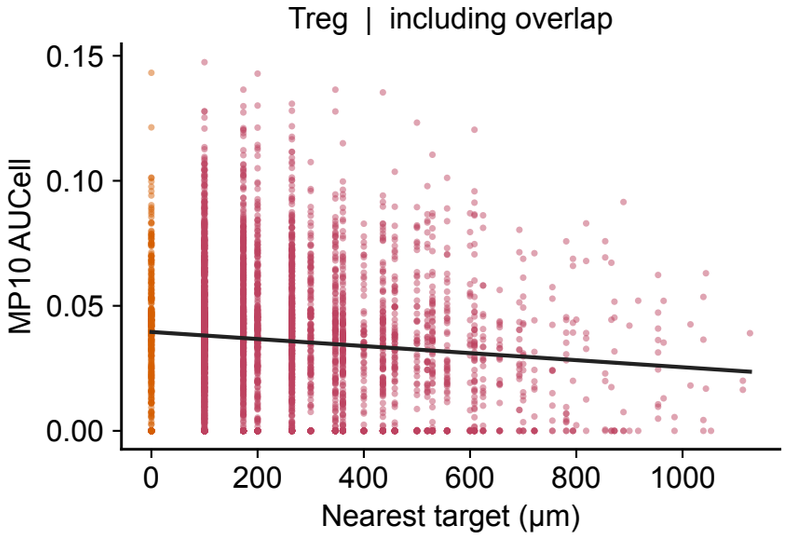
r -0.04 ρ -0.04
n 4440/4440 queries • 2446 targets



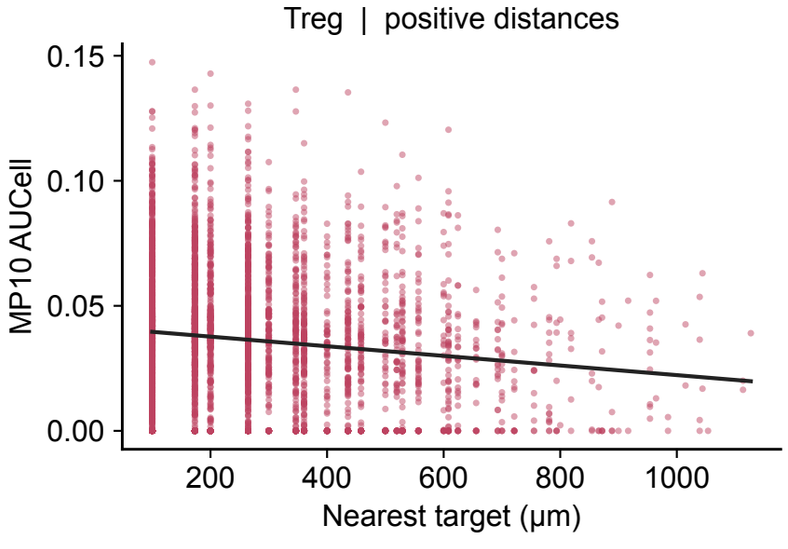
r -0.04 ρ -0.05
n 3612/4440 queries • 2446 targets



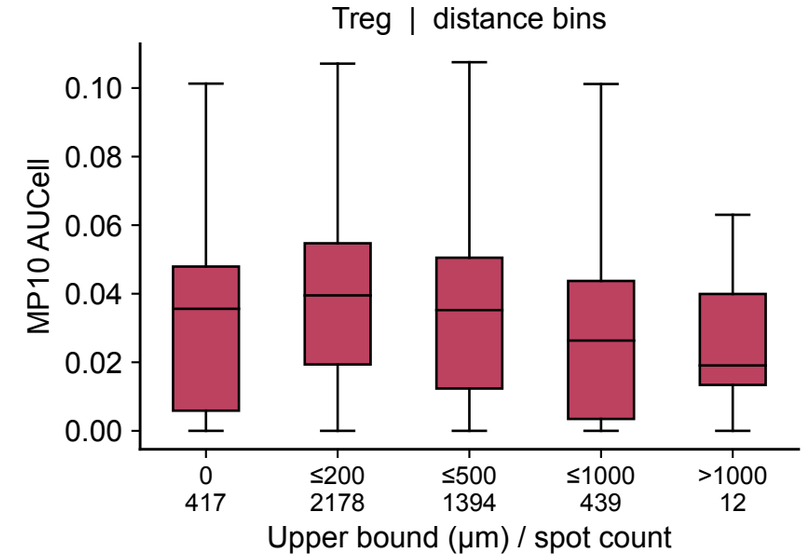
Orange = zero distance
Line = within-section linear fit



r -0.10 ρ -0.08
n 4440/4440 queries • 2446 targets



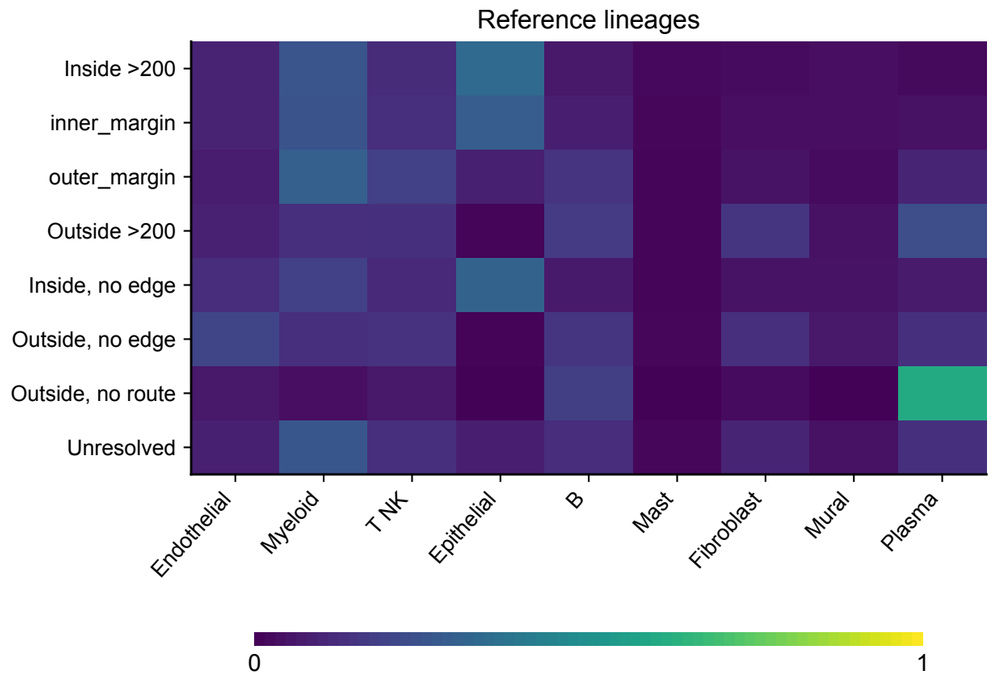
r -0.13 ρ -0.12
n 4023/4440 queries • 2446 targets

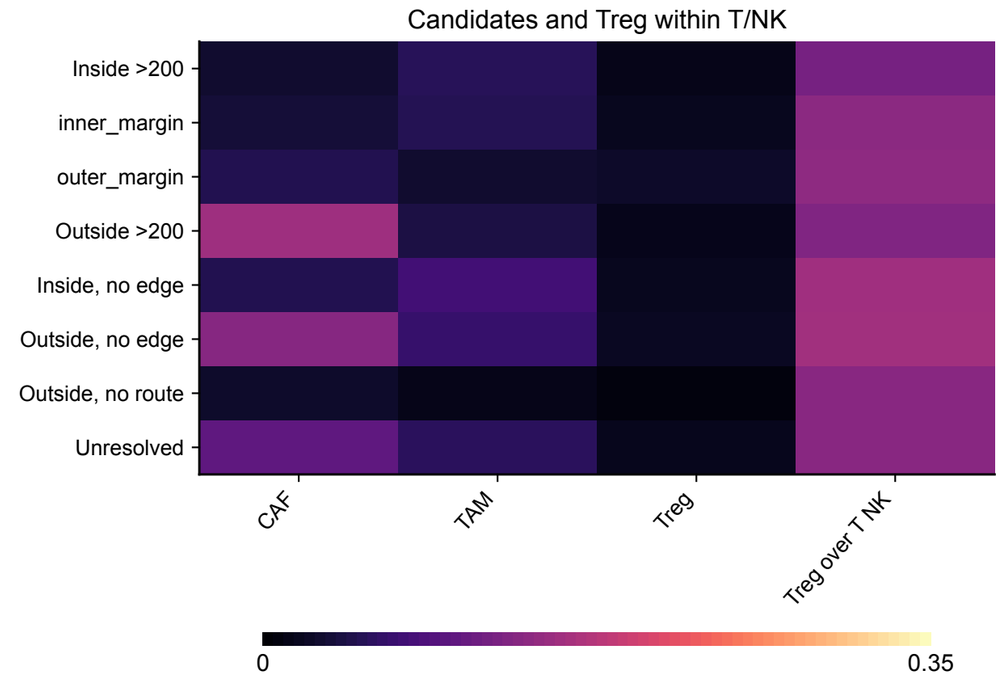


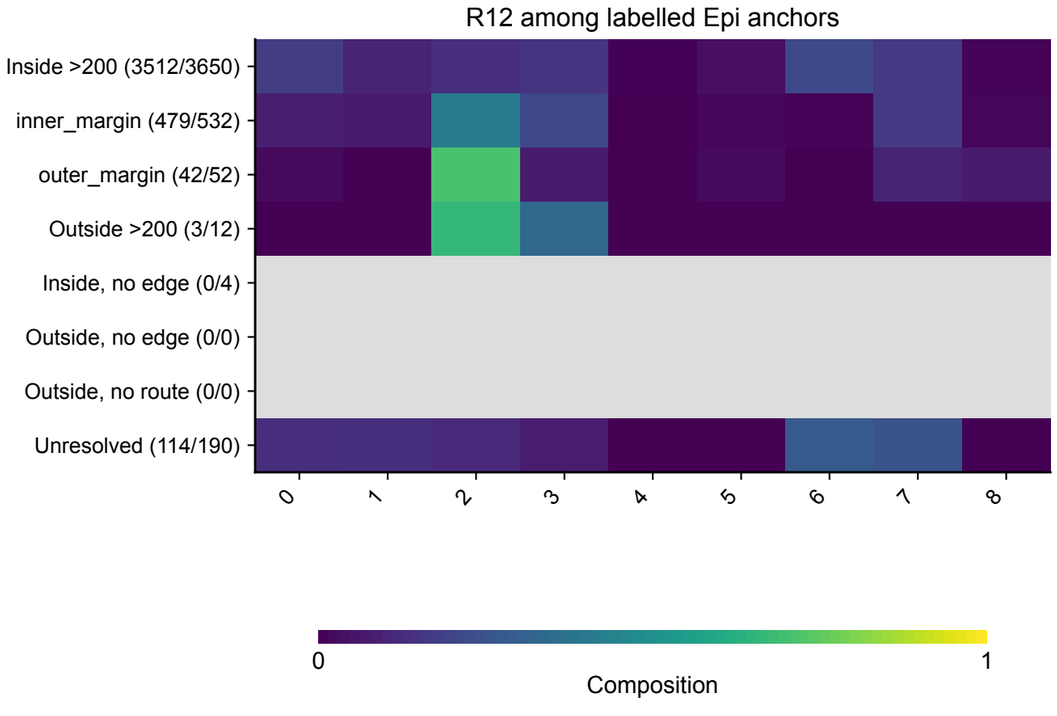
Orange = zero distance
Line = within-section linear fit

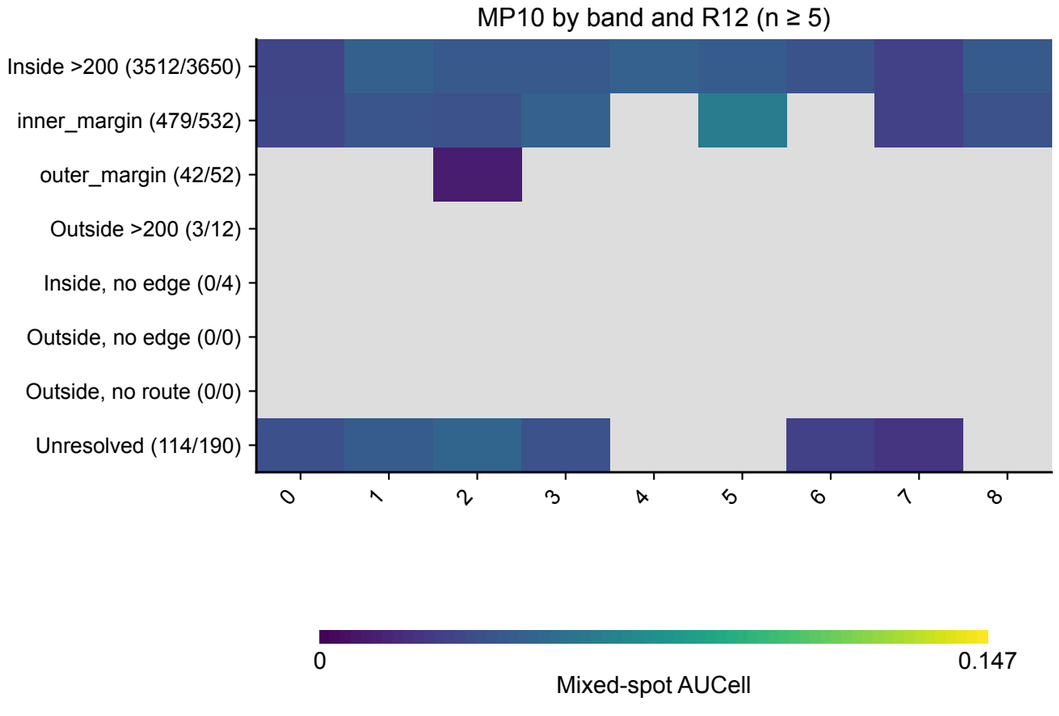
GSE238145_B16 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





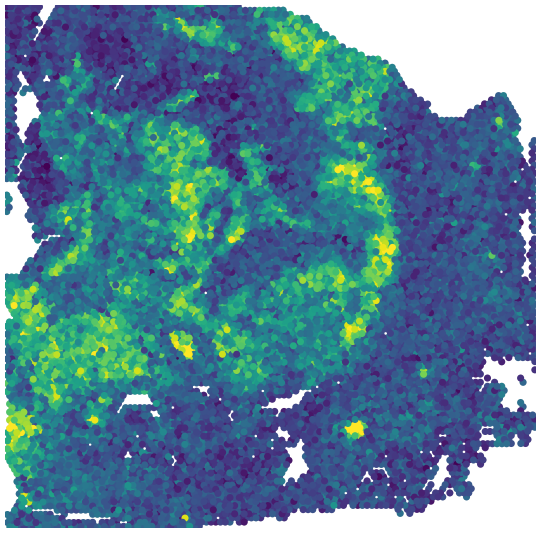




GSE238145_B16 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

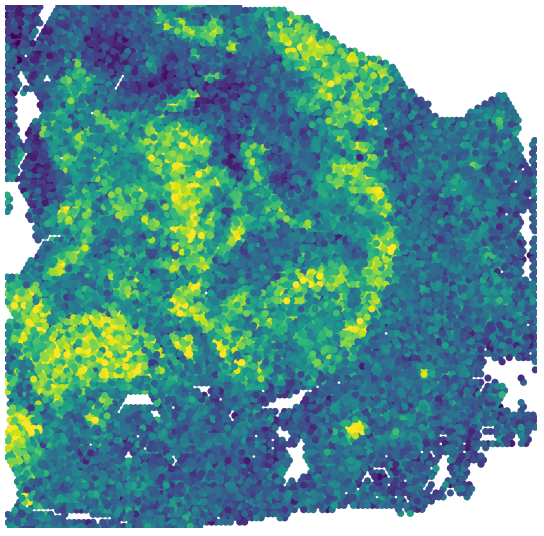
E2F Targets



0 0.152

UCell score

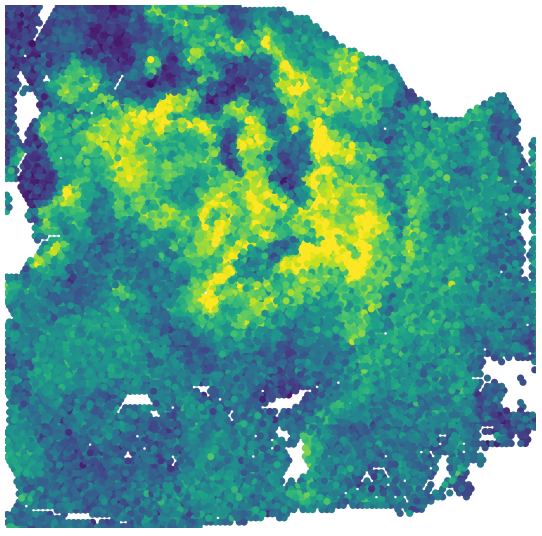
G2M Checkpoint



0 0.134

UCell score

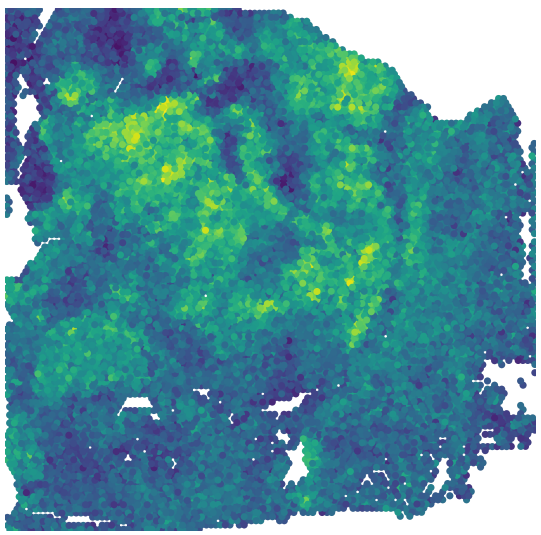
Hypoxia



0 0.211

UCell score

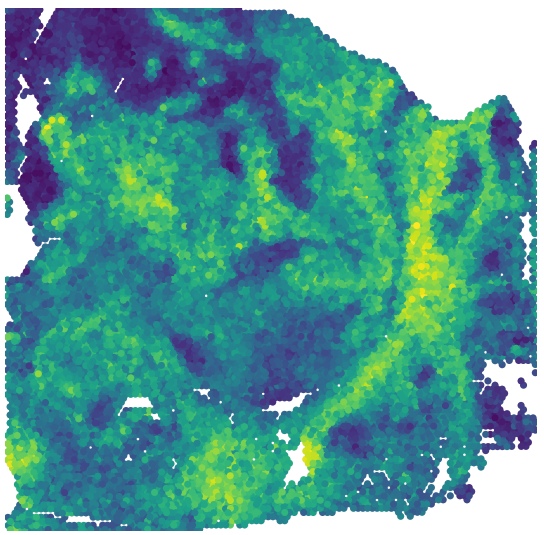
Tnfa Signaling Via Nfkb



0 0.275

UCell score

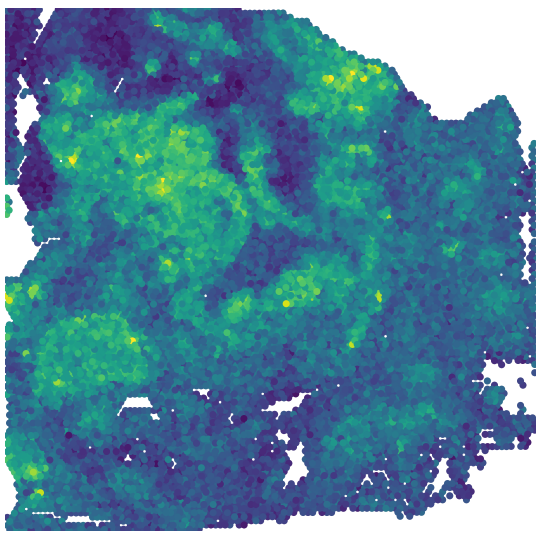
Emt



0 0.386

UCell score

Interferon Gamma Response



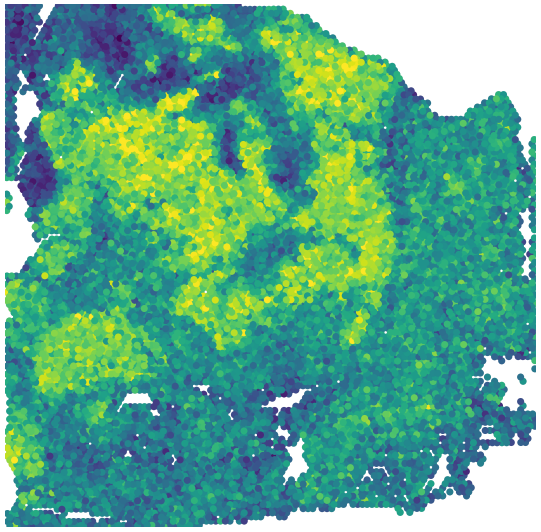
0 0.275

UCell score

GSE238145_B16 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

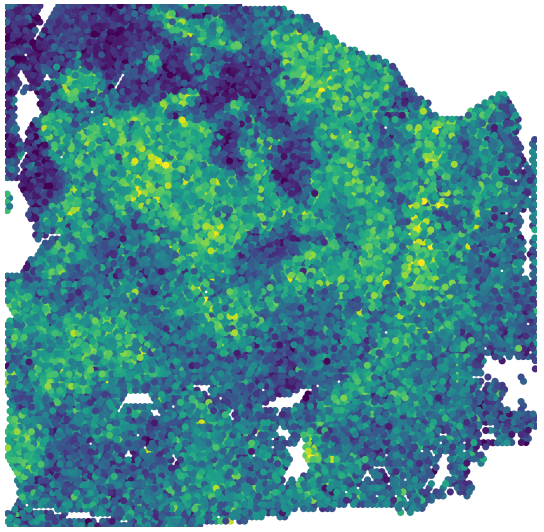
Il2 Stat5 Signaling



0 0.144

UCell score

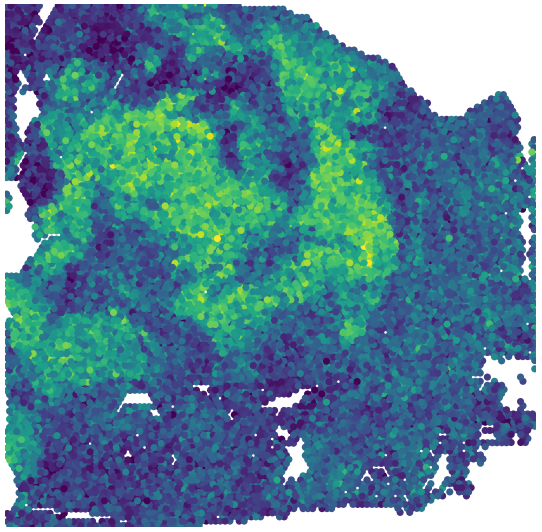
Tgf Beta Signaling



0 0.232

UCell score

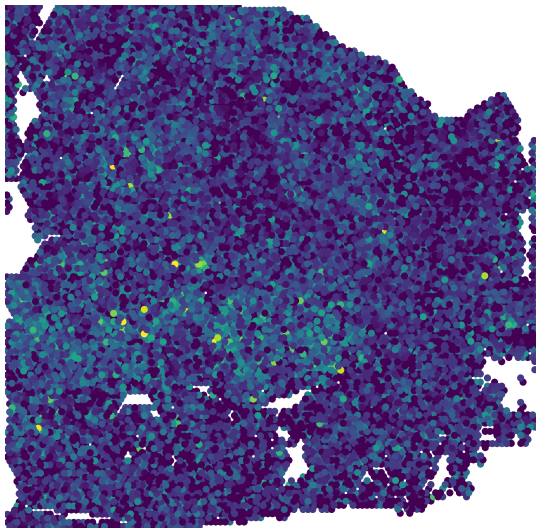
Cholesterol Homeostasis



0 0.21

UCell score

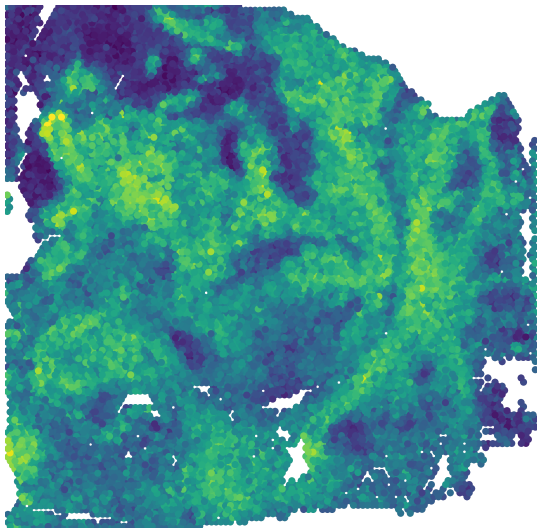
Cholesterol Biosynthesis



0 0.166

UCell score

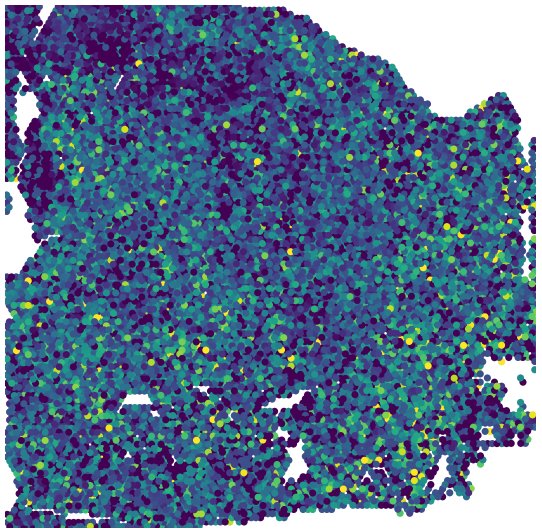
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



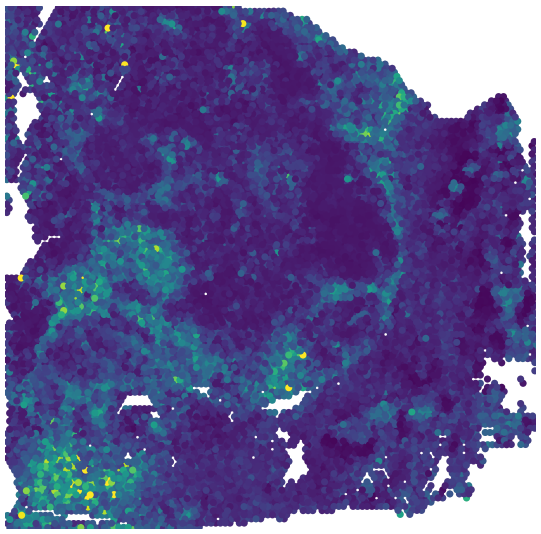
0 0.151

UCell score

GSE238145_B16 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

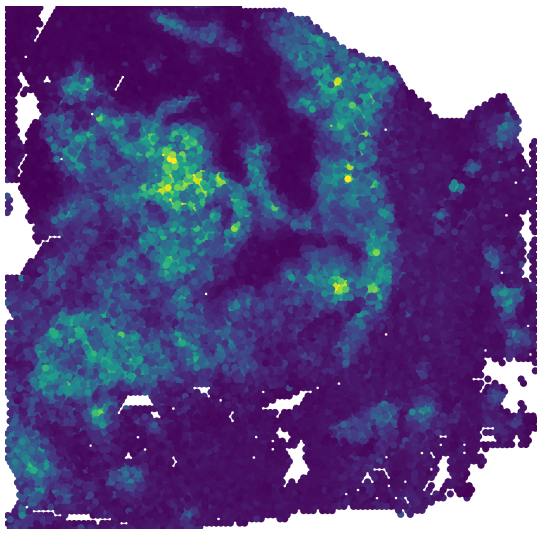
Treg composition



0 0.102

Fraction of all 30

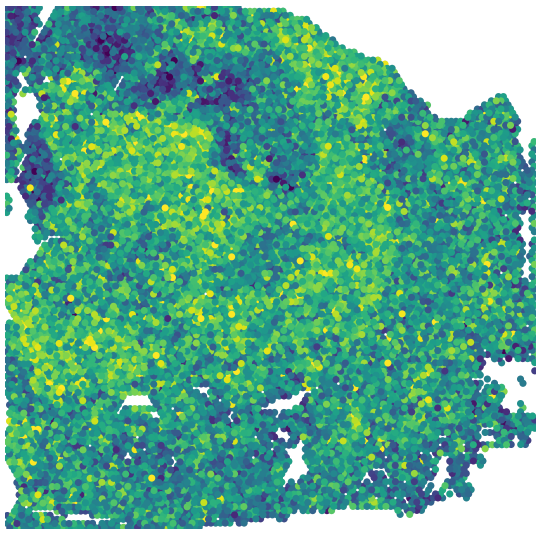
Treg estimated abundance



0 2.85

Posterior mean estimate

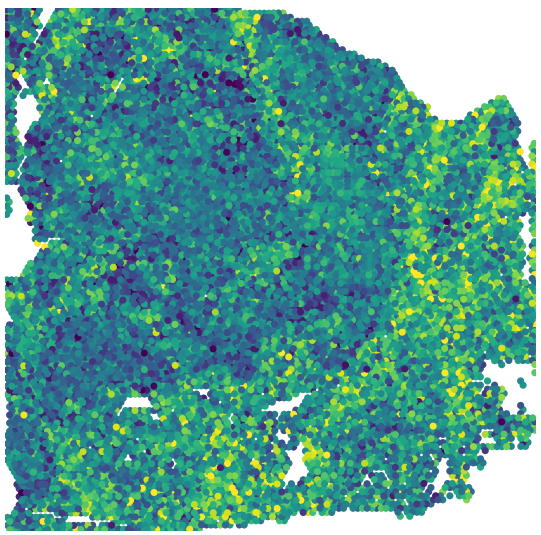
Treg Resting Vs Activated Dn



0 0.0778

UCell score

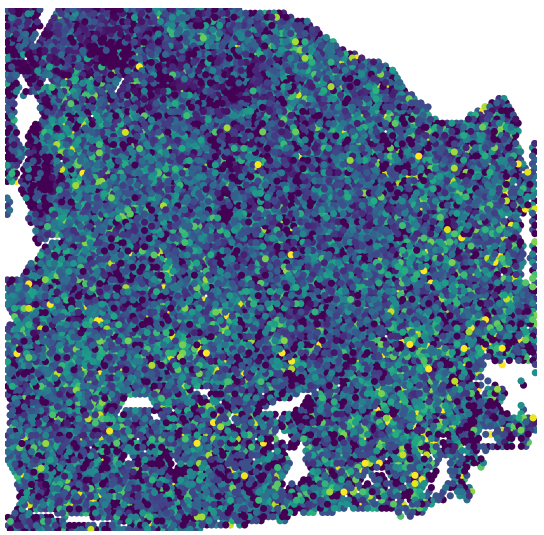
Treg Resting Vs Activated Up



0 0.0609

UCell score

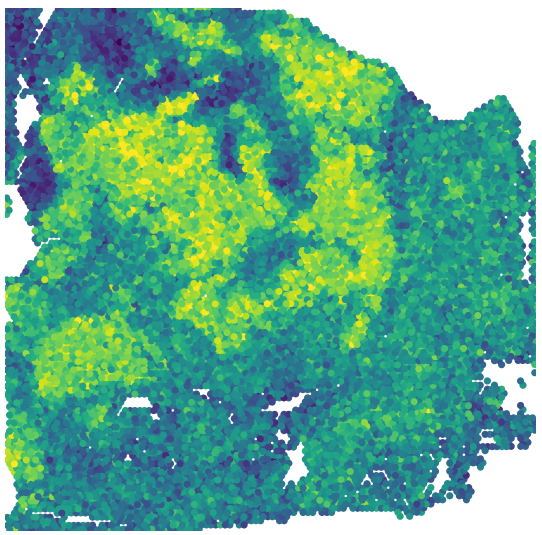
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

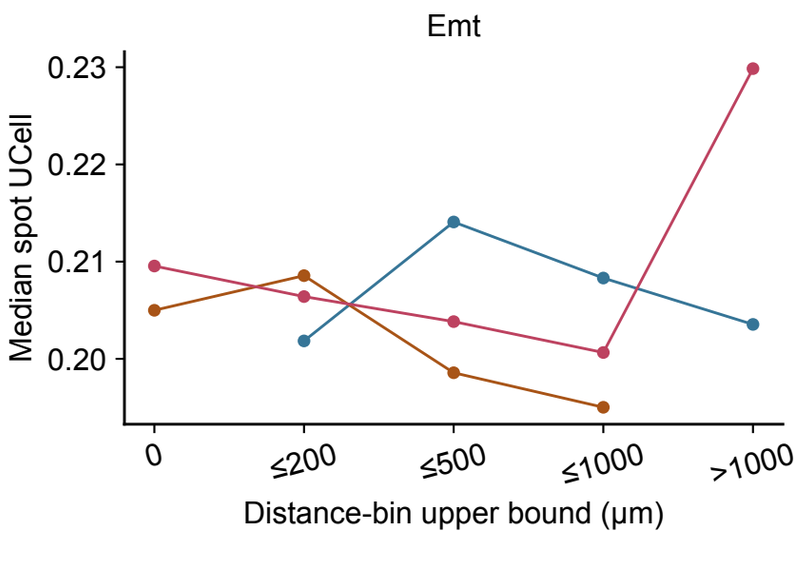


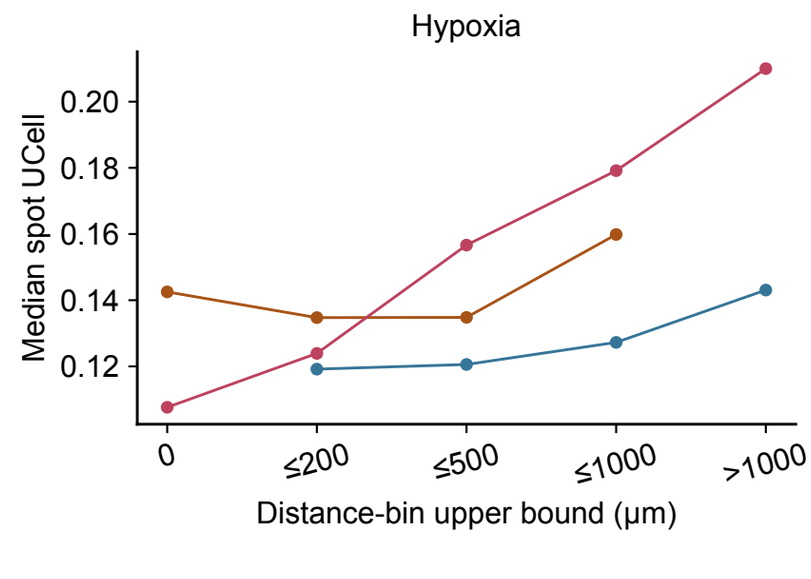
0 0.144

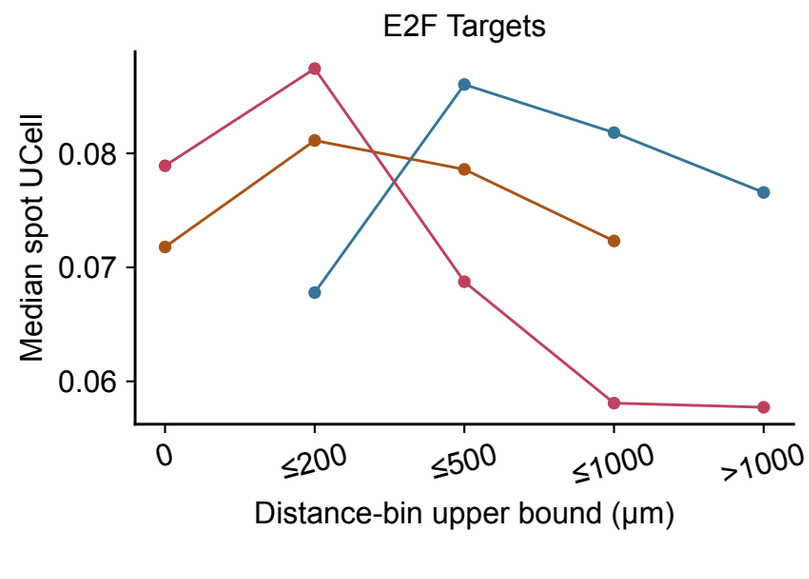
UCell score

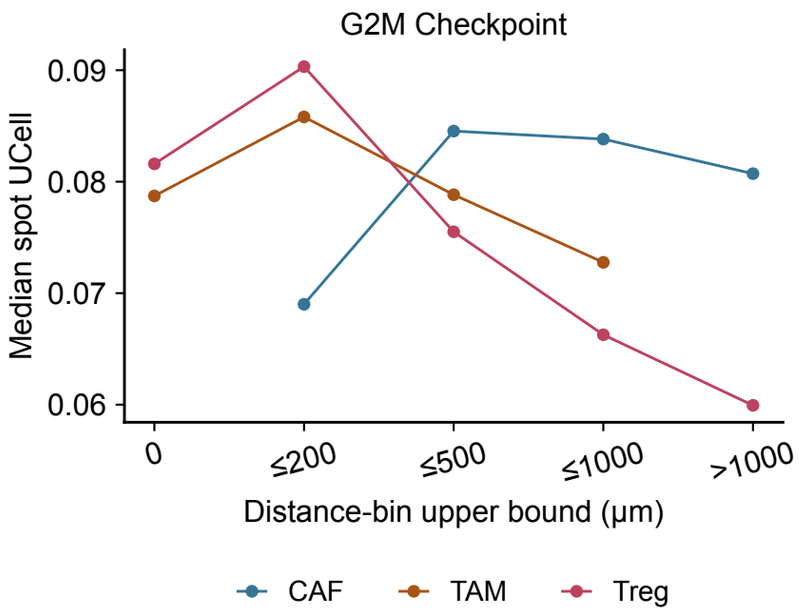
GSE238145_B16 | Programs by target distance

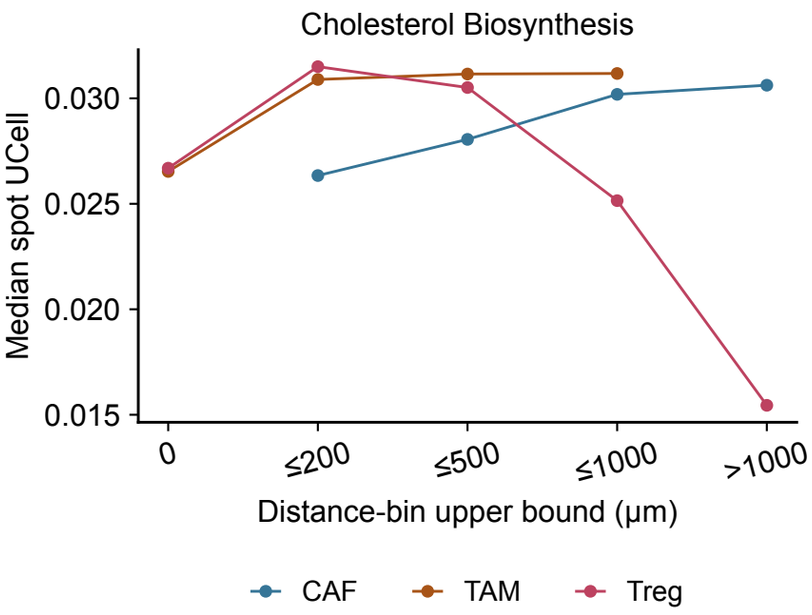
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

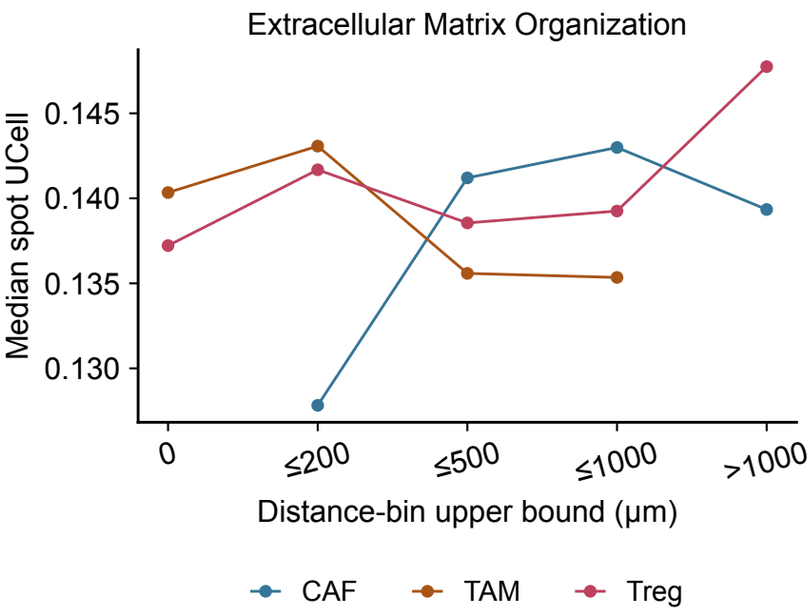






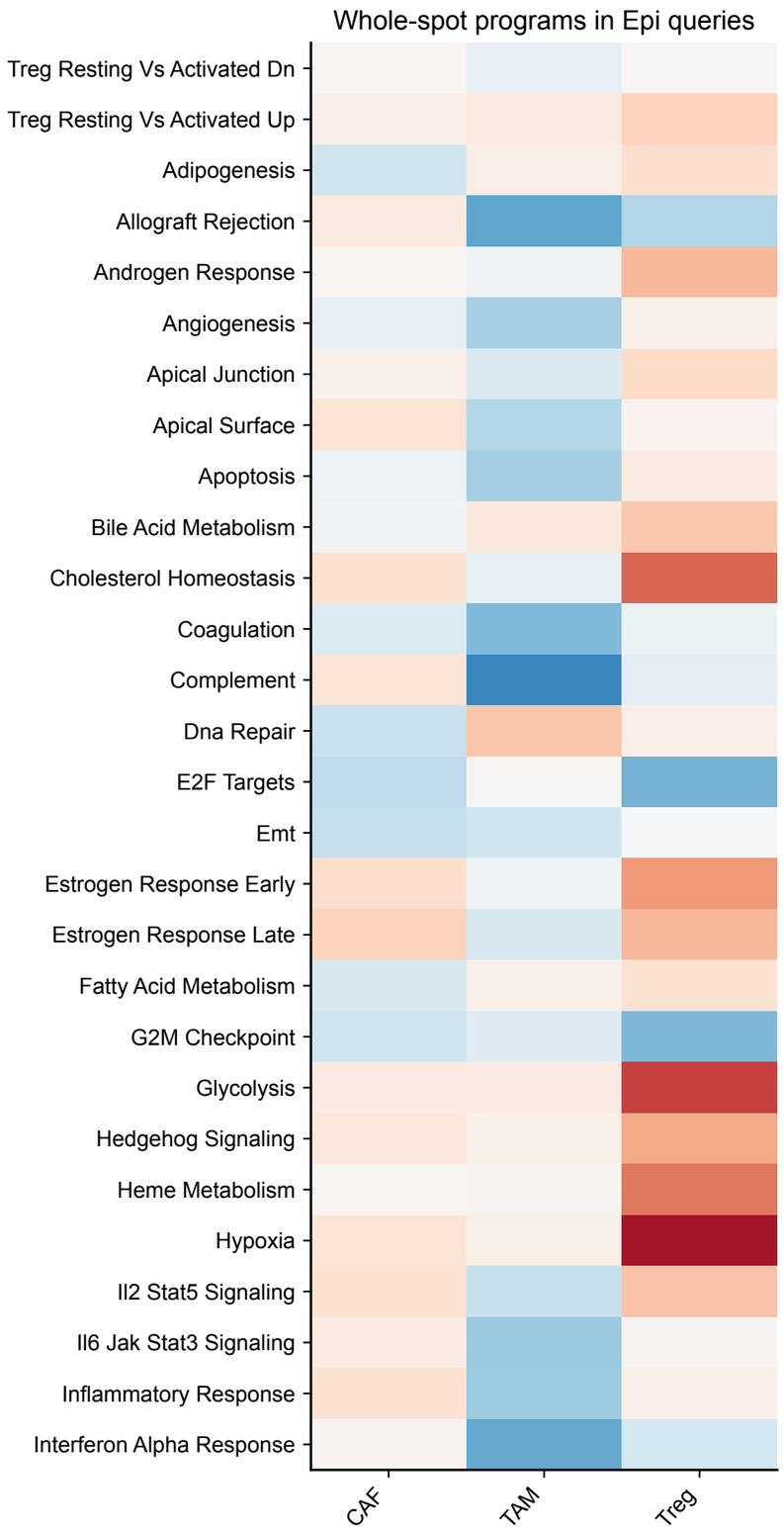


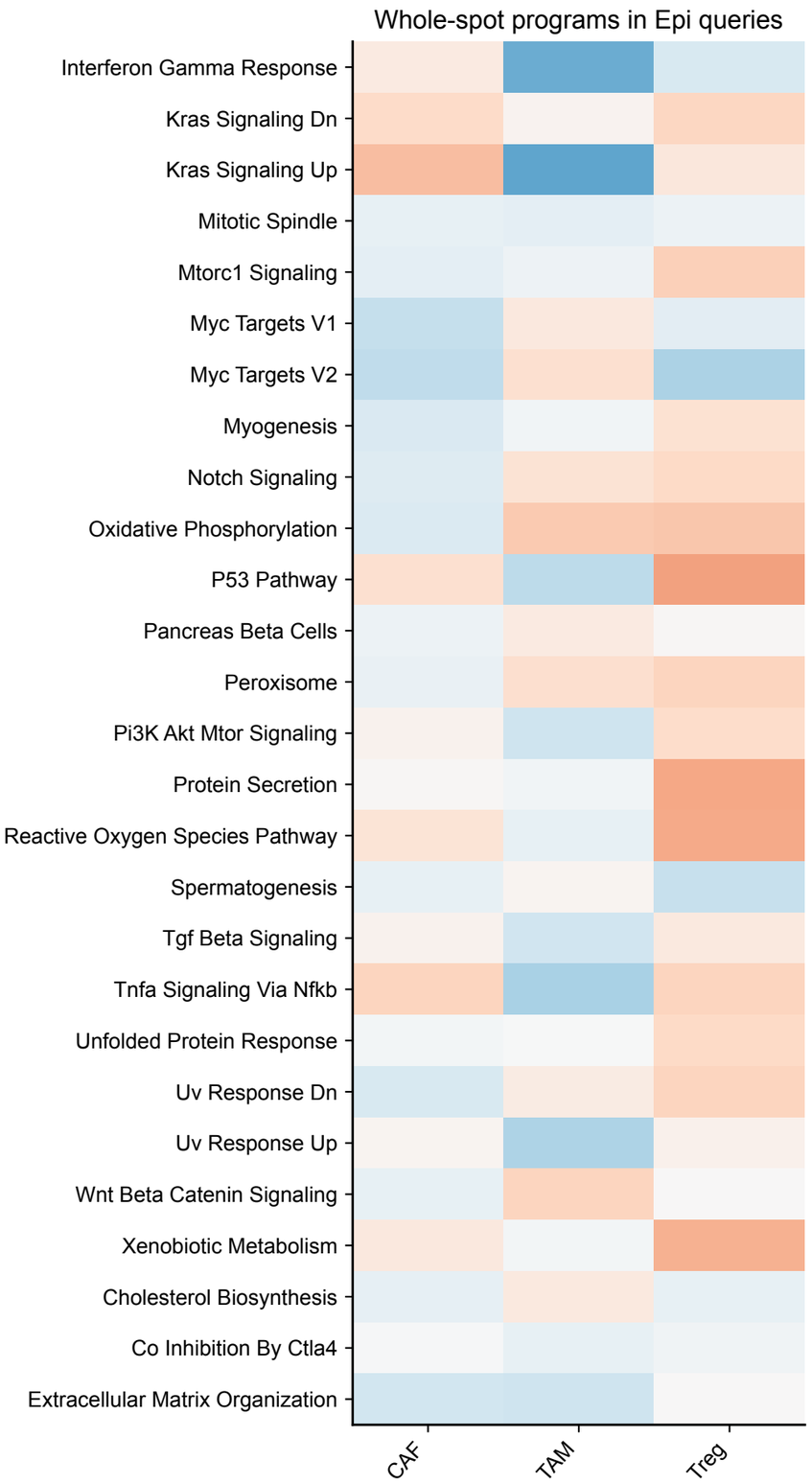




GSE238145_B16 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

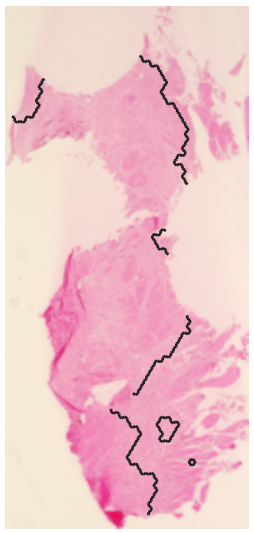




GSE238145_B17 | Tissue context

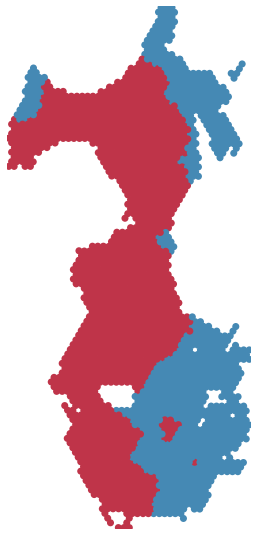
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



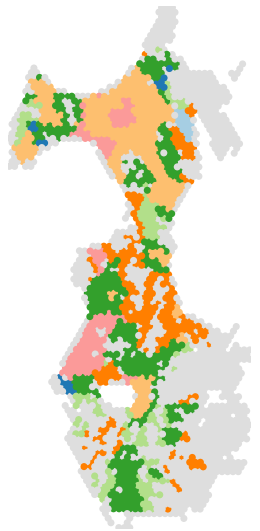
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



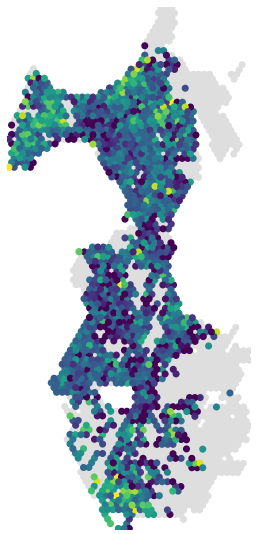
0 1 2 3 4 5 6

Epithelial support

0 1

Fraction of all 30

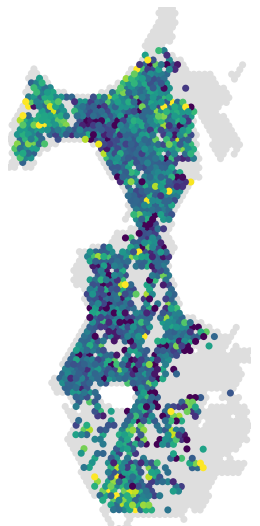
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



0 0.14

Matched-model AUCell

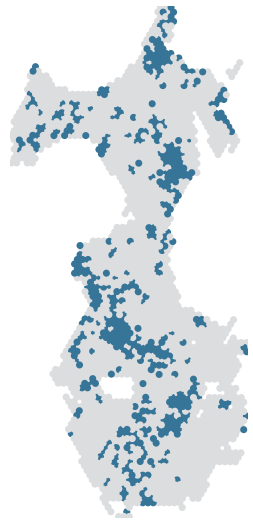
GSE238145_B17 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

0 0.418
Fraction of all 30

CAF P80 targets



False True

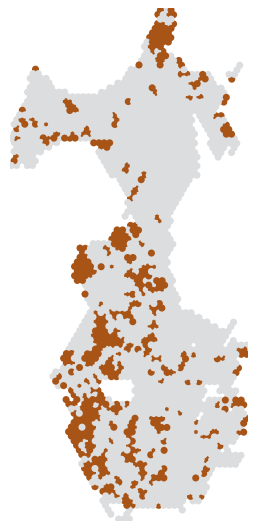
CAF distance

0 3.1e+03
Nearest spot center (μm)

TAM composition

0 0.291
Fraction of all 30

TAM P80 targets



False True

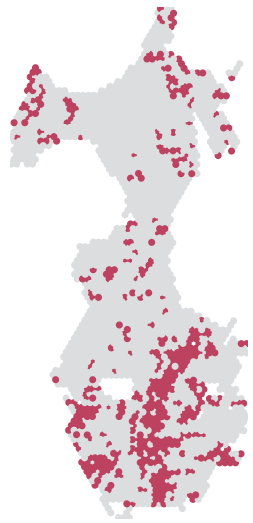
TAM distance

0 1.13e+03
Nearest spot center (μm)

Treg composition

0 0.102
Fraction of all 30

Treg P80 targets



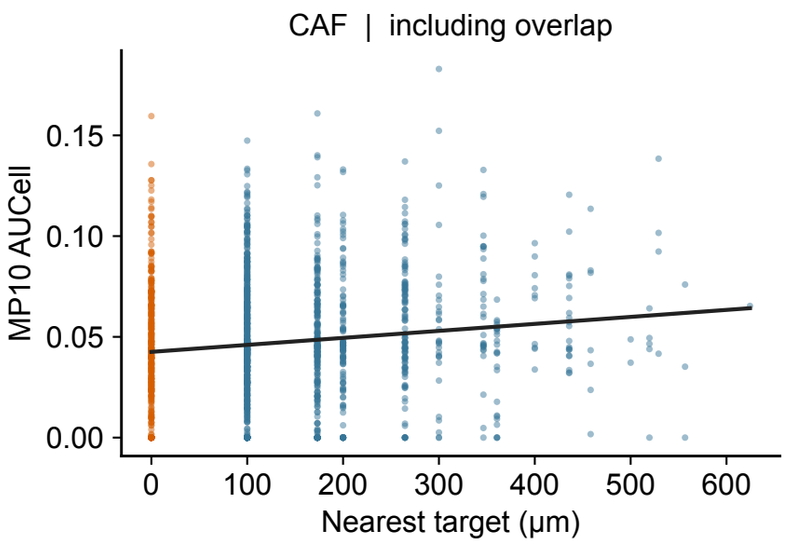
False True

Treg distance

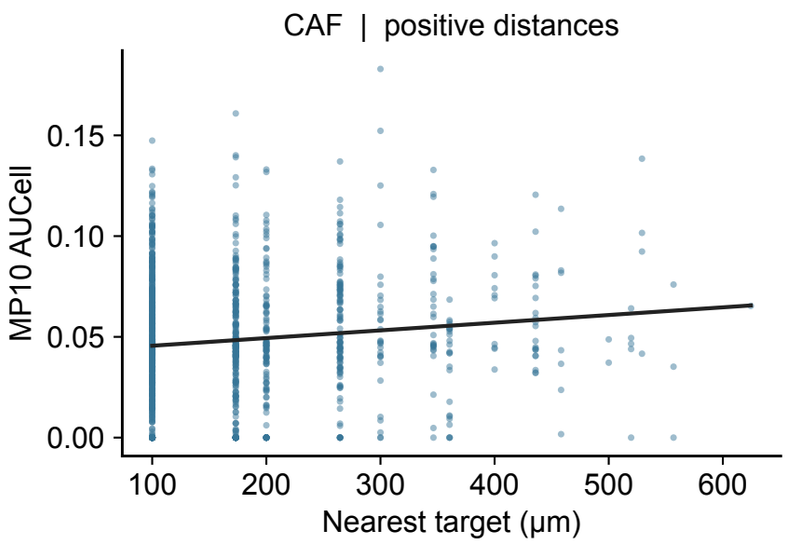
0 1.61e+03
Nearest spot center (μm)

GSE238145_B17 | MP10 versus nearest target

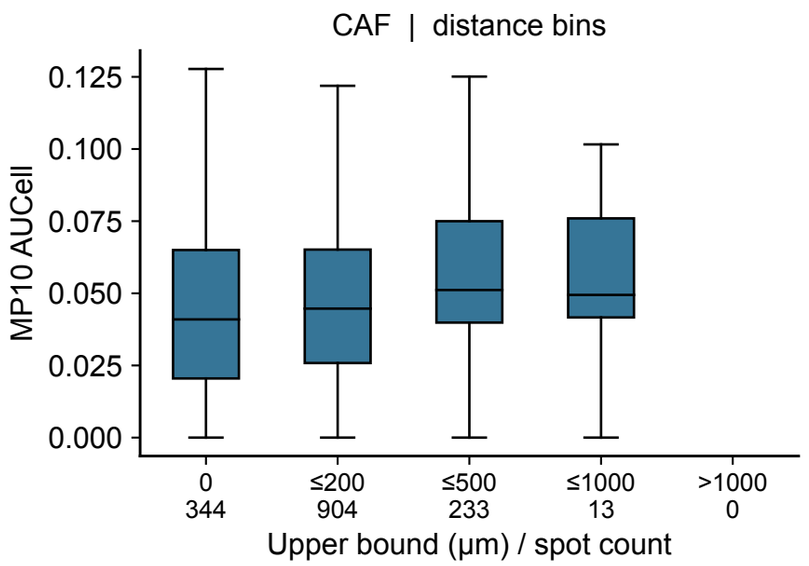
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



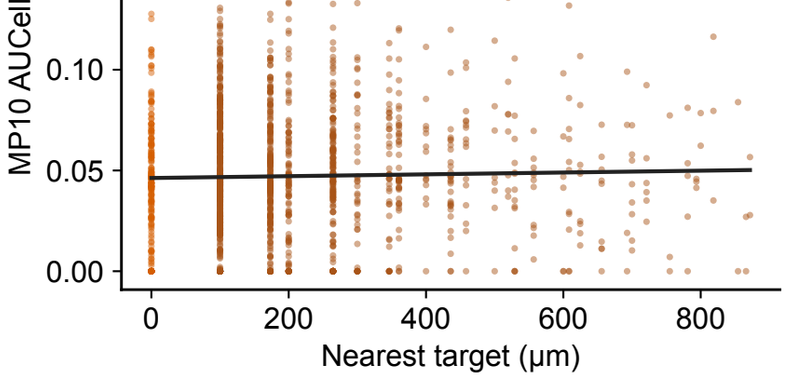
r +0.12 ρ +0.12
n 1494/1494 queries • 480 targets



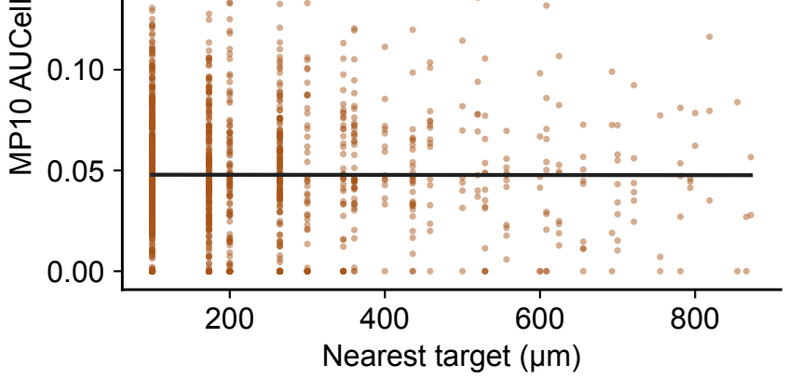
r +0.12 ρ +0.11
n 1150/1494 queries • 480 targets



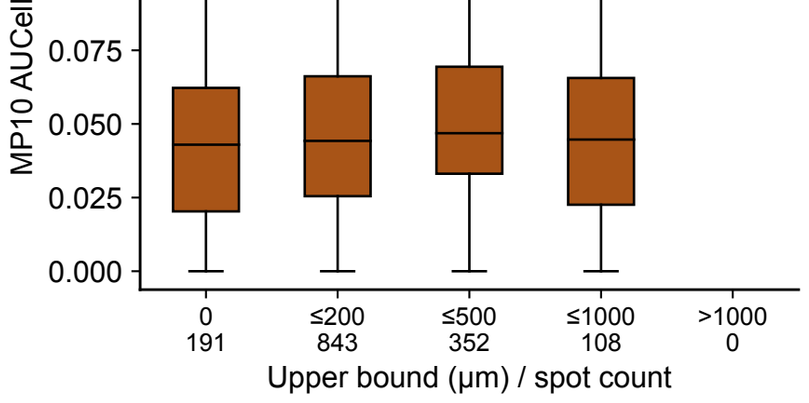
Orange = zero distance
Line = within-section linear fit



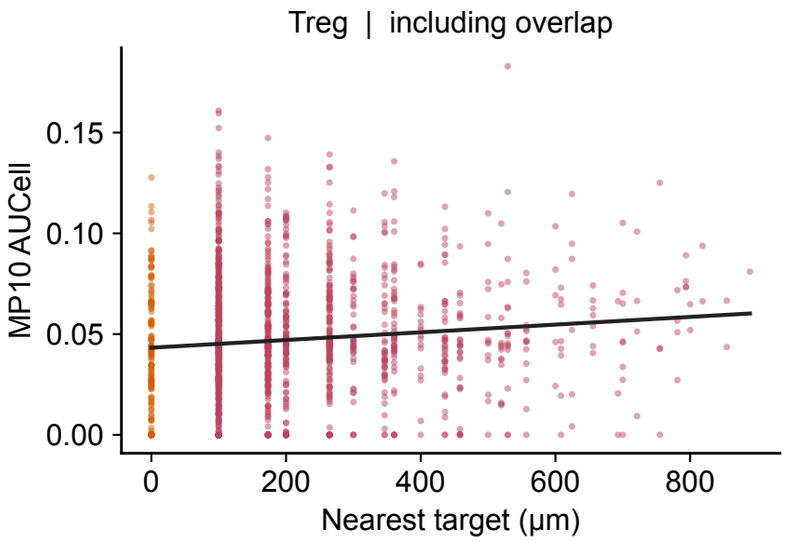
r +0.02 ρ +0.03
n 1494/1494 queries • 480 targets



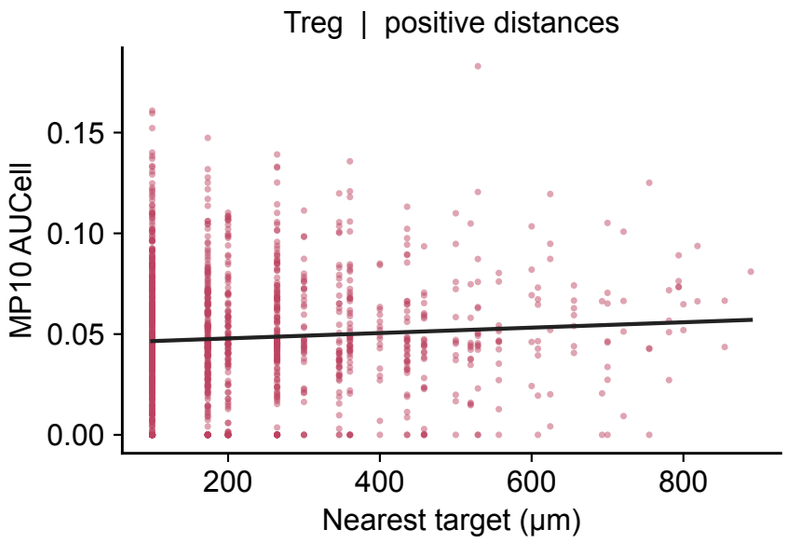
r -0.00 ρ +0.00
n 1303/1494 queries • 480 targets



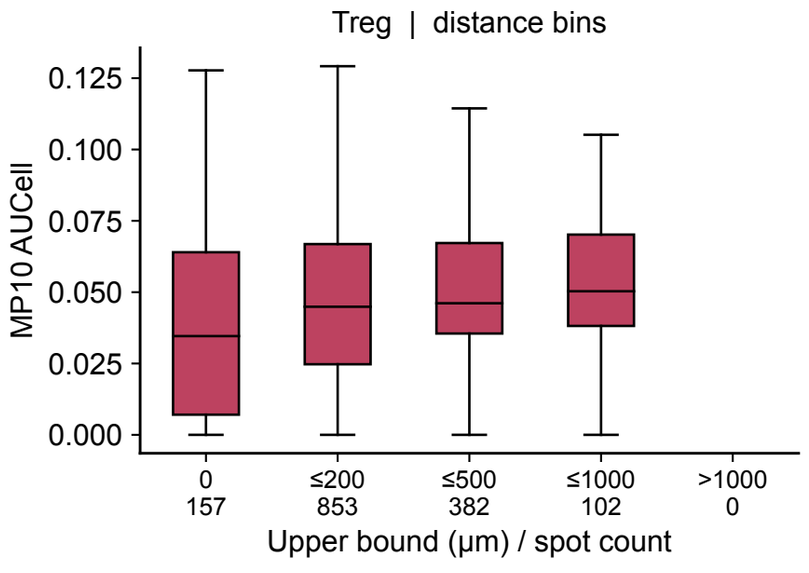
Orange = zero distance
Line = within-section linear fit



r +0.10 ρ +0.11
n 1494/1494 queries • 480 targets



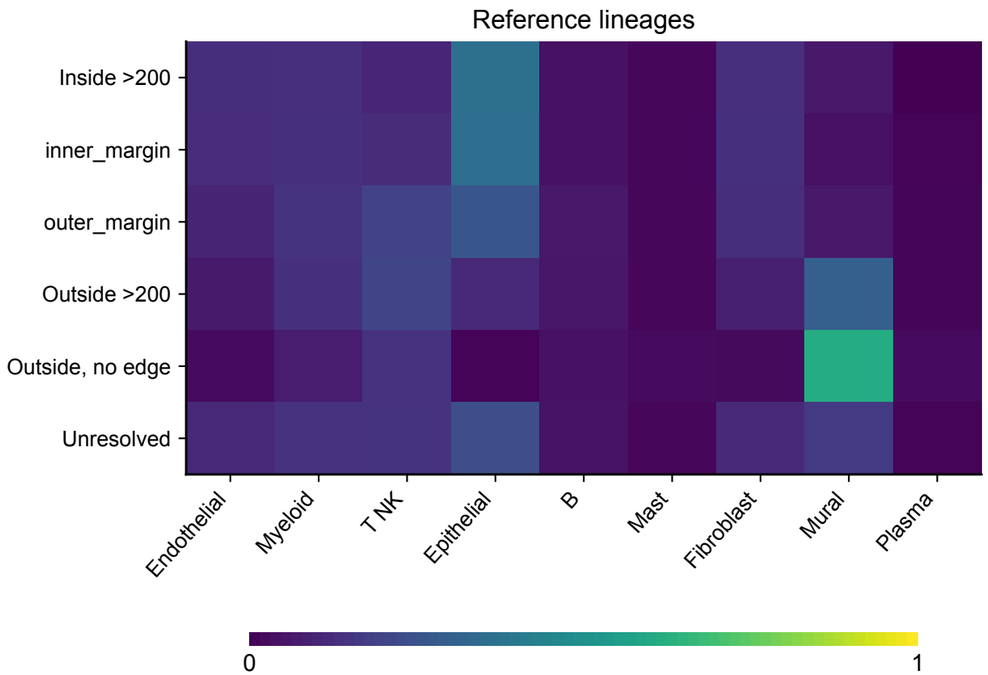
r +0.07 ρ +0.07
n 1337/1494 queries • 480 targets

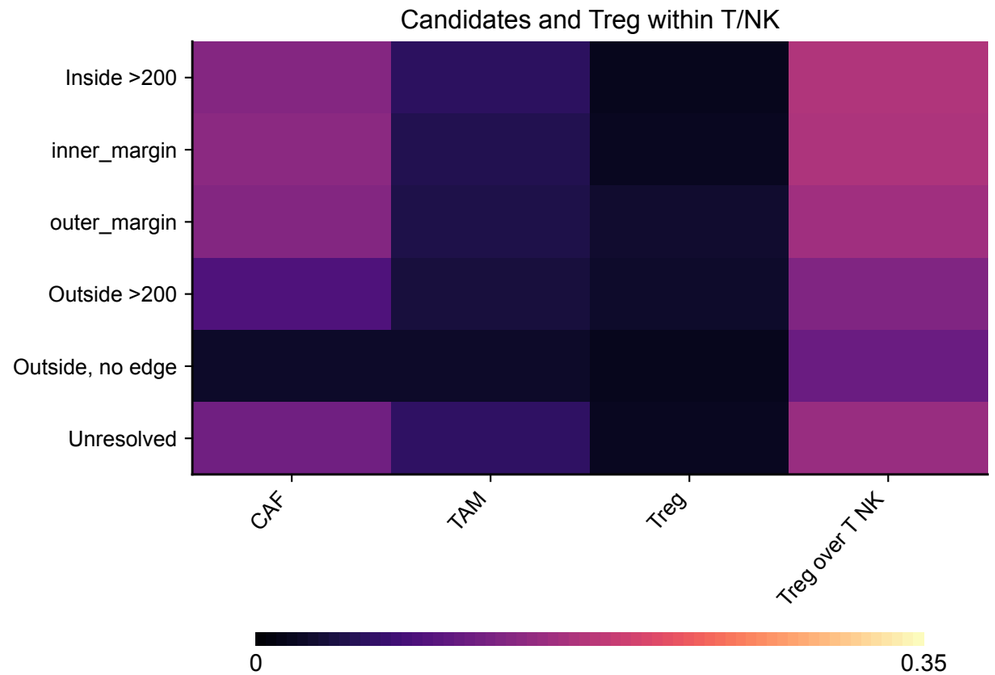


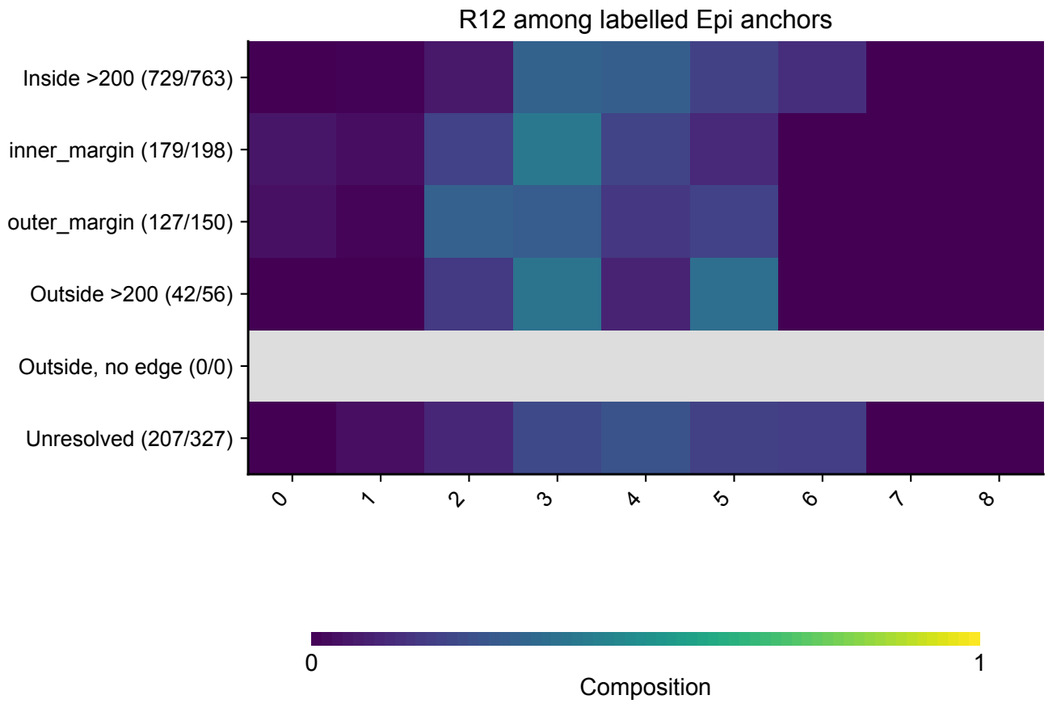
Orange = zero distance
Line = within-section linear fit

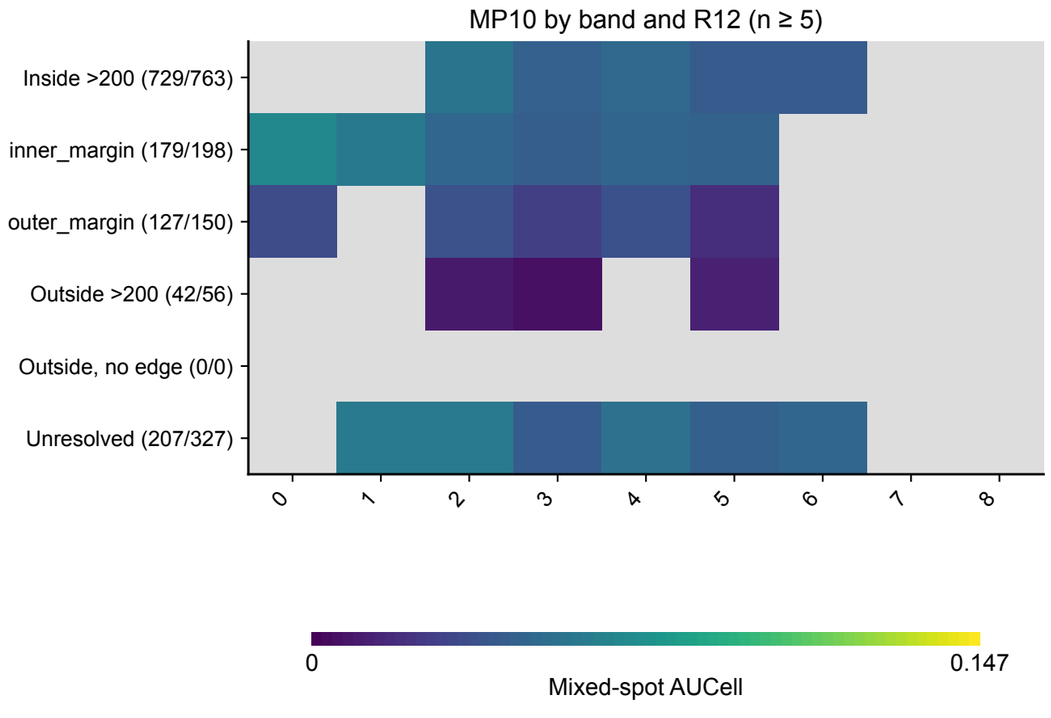
GSE238145_B17 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable









GSE238145_B17 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets

0 0.152

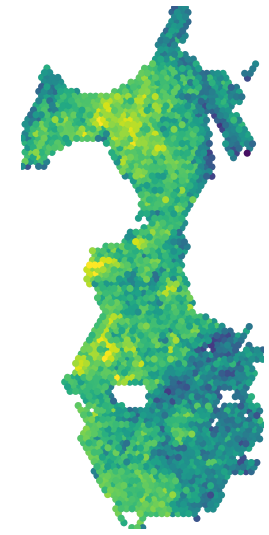
UCell score

G2M Checkpoint

0 0.134

UCell score

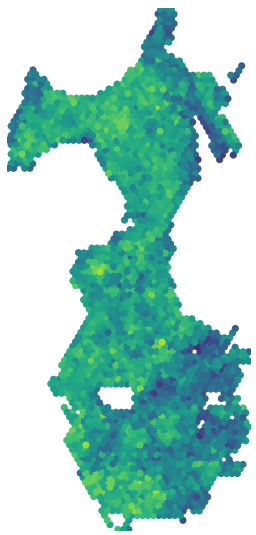
Hypoxia



0 0.211

UCell score

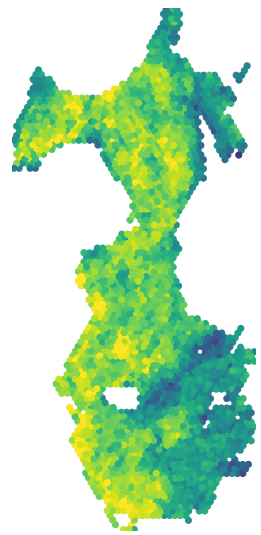
Tnfa Signaling Via Nfkb



0 0.275

UCell score

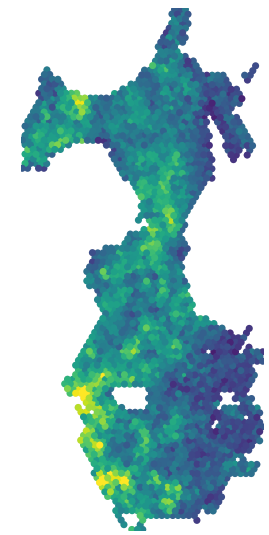
Emt



0 0.386

UCell score

Interferon Gamma Response



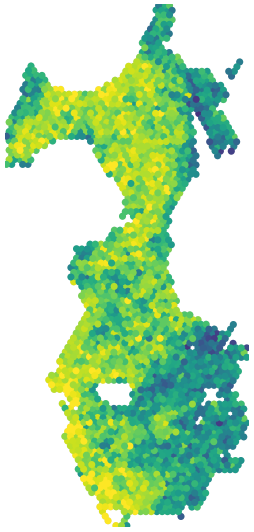
0 0.275

UCell score

GSE238145_B17 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

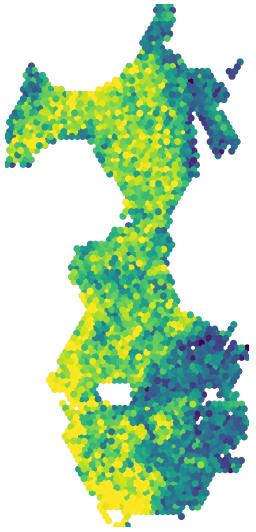
Il2 Stat5 Signaling



0 0.144

UCell score

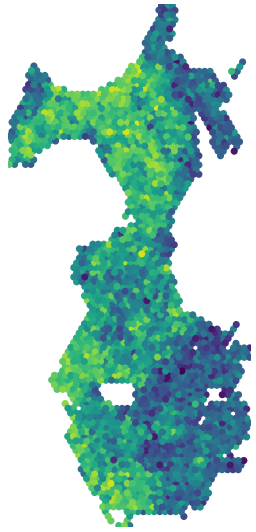
Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

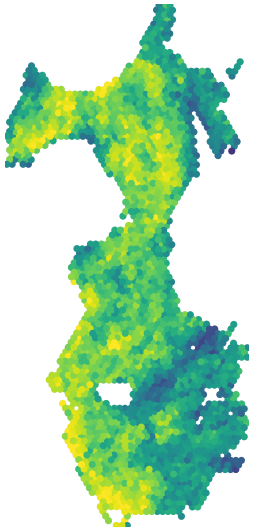
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

GSE238145_B17 | Treg context across the full tissue

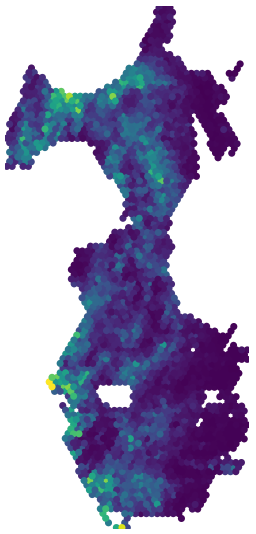
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

Fraction of all 30

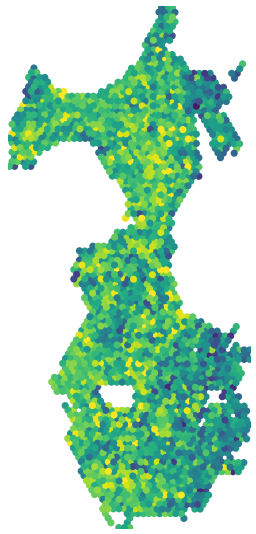
Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn



0 0.0778

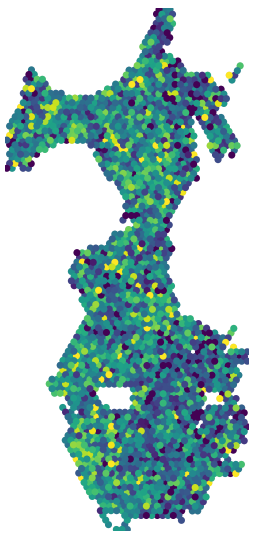
UCell score

Treg Resting Vs Activated Up

0 0.0609

UCell score

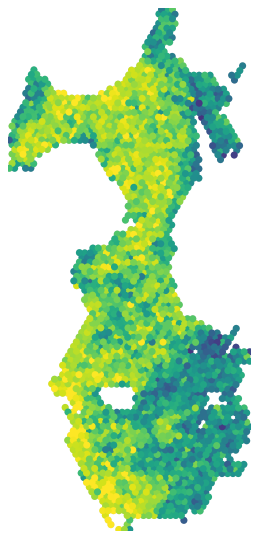
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

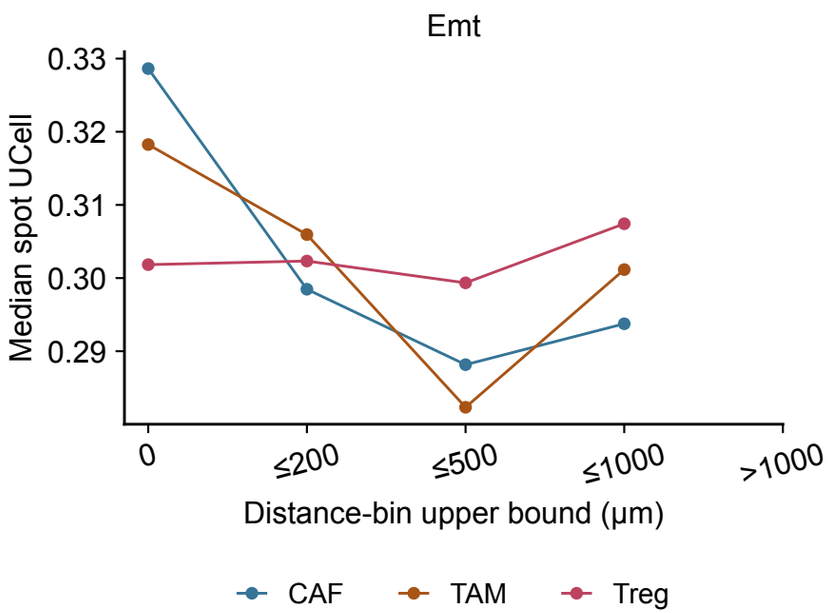


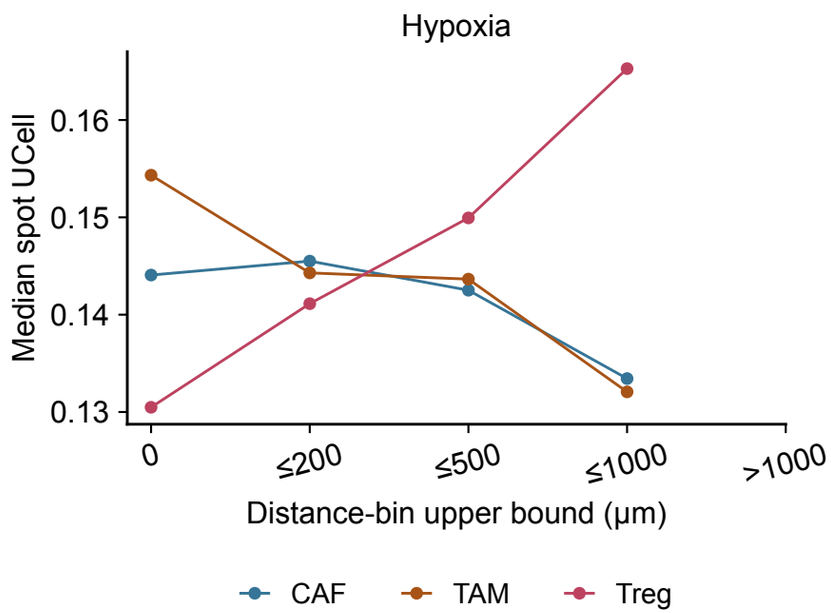
0 0.144

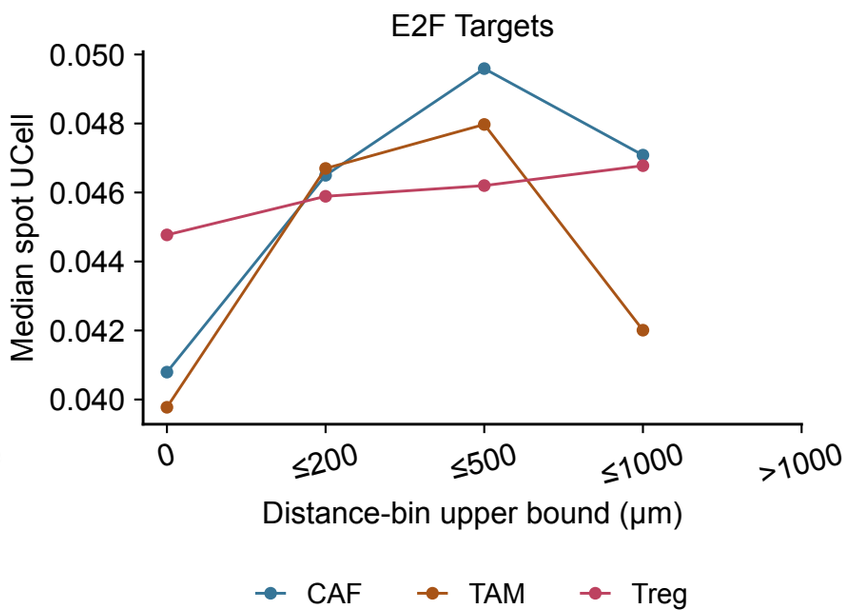
UCell score

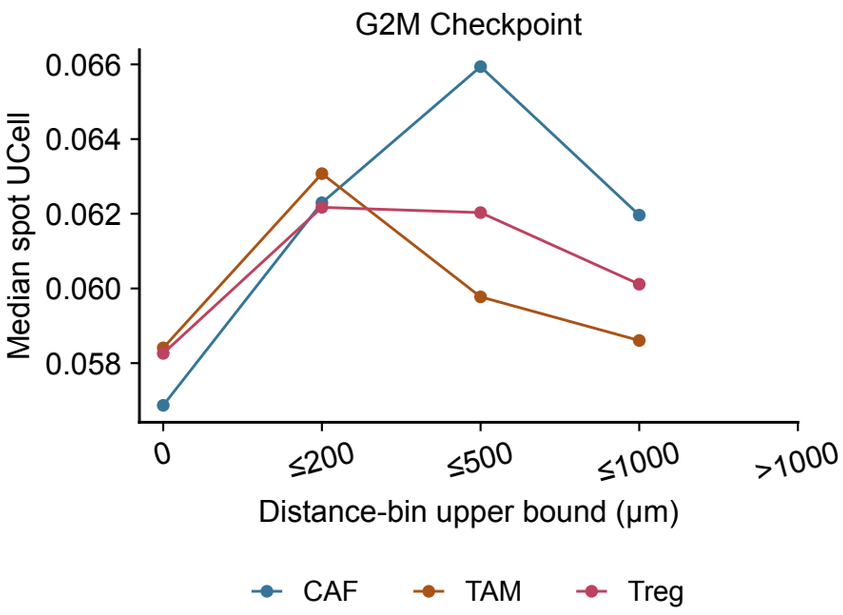
GSE238145_B17 | Programs by target distance

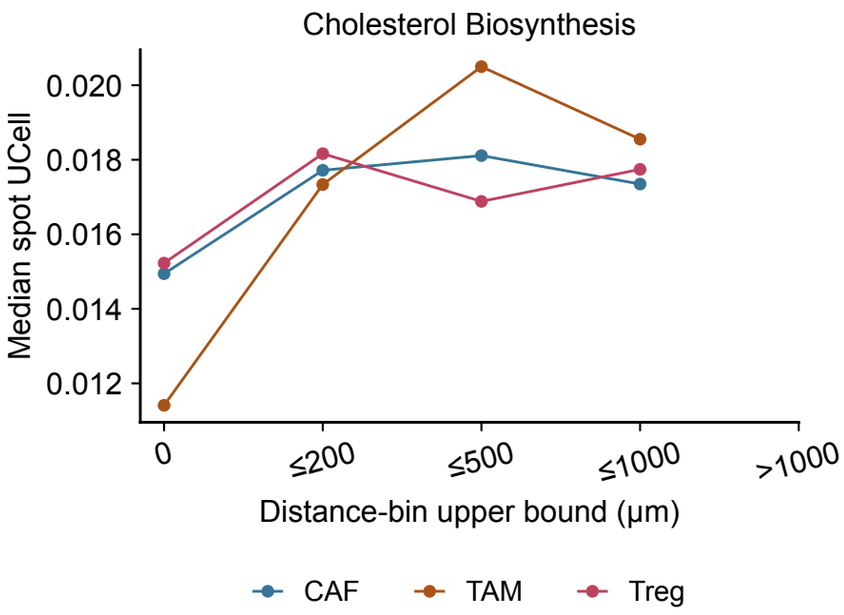
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

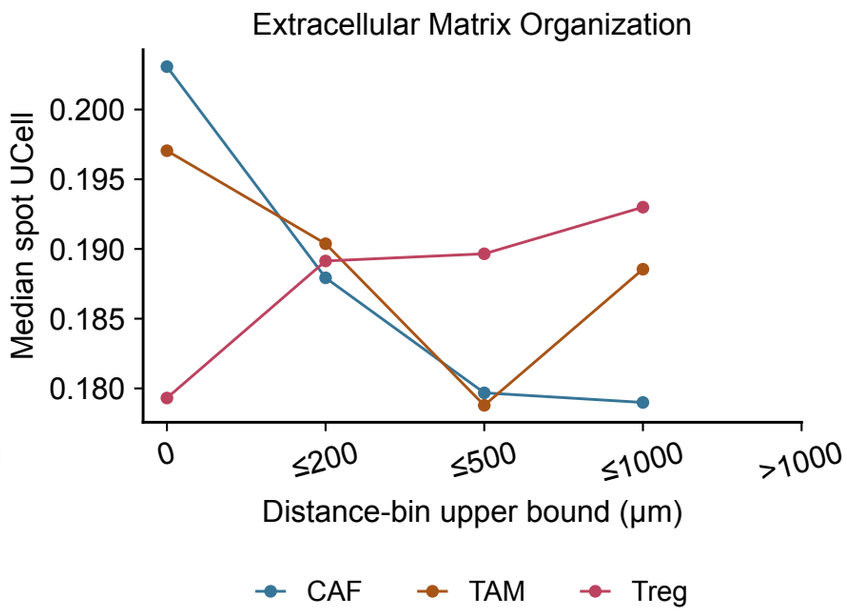






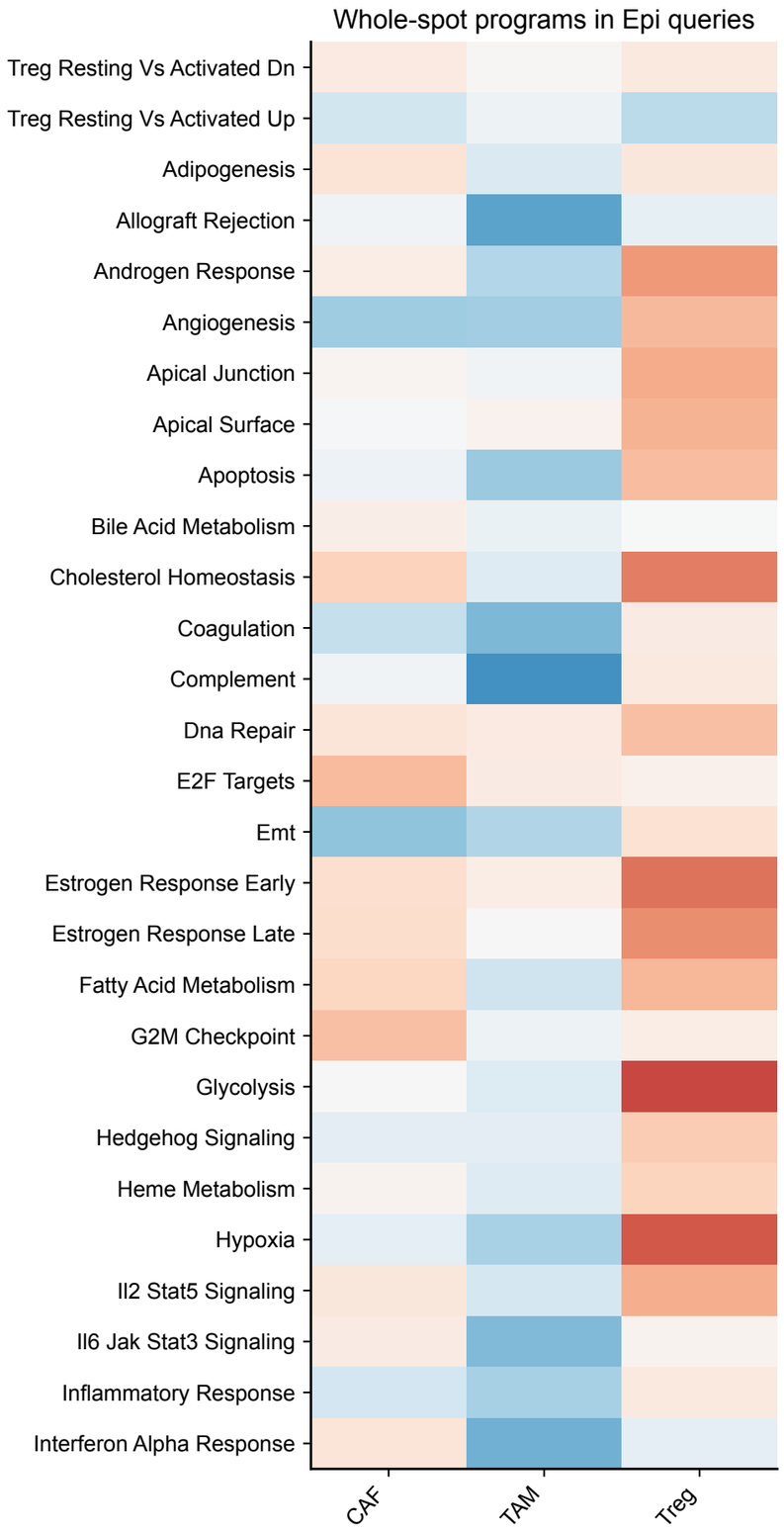


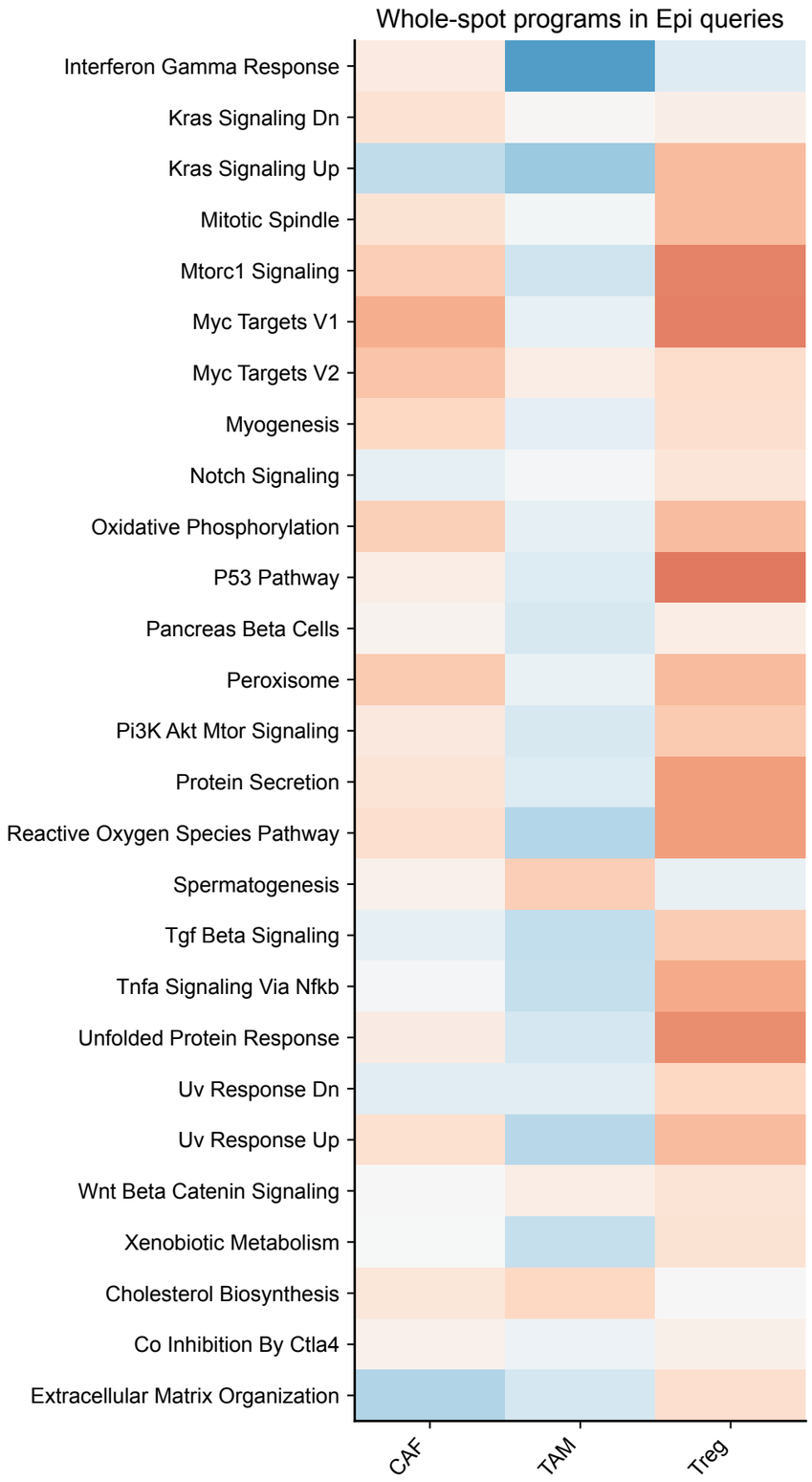




GSE238145_B17 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

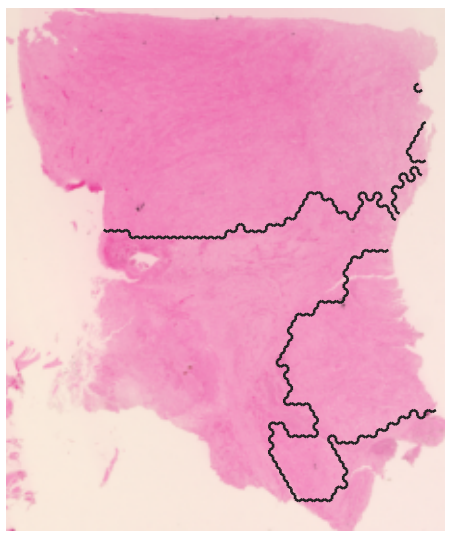




GSE238145_B18 | Tissue context

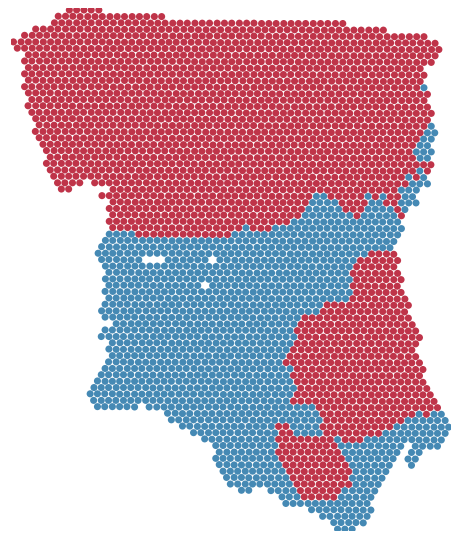
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location

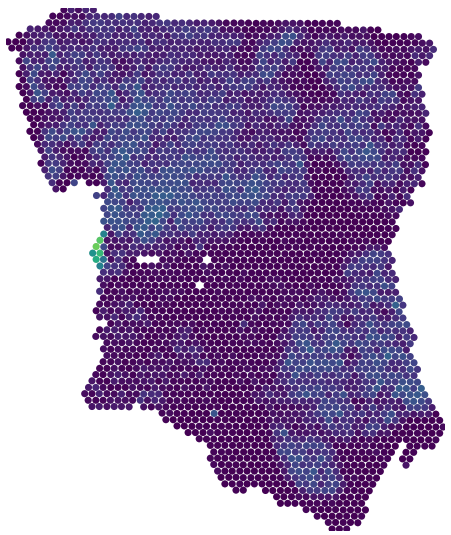


inside outside

Accepted R12 (original 0–8)

2 7

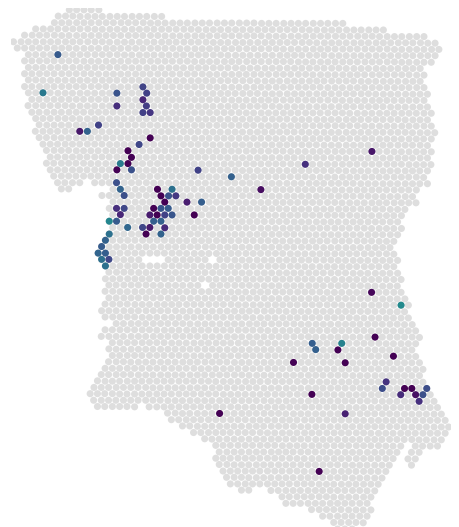
Epithelial support



0 1

Fraction of all 30

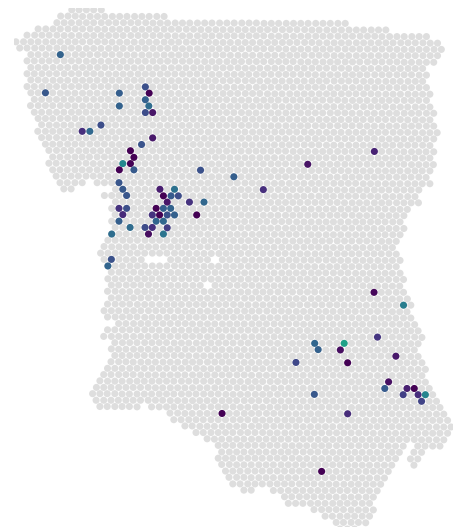
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



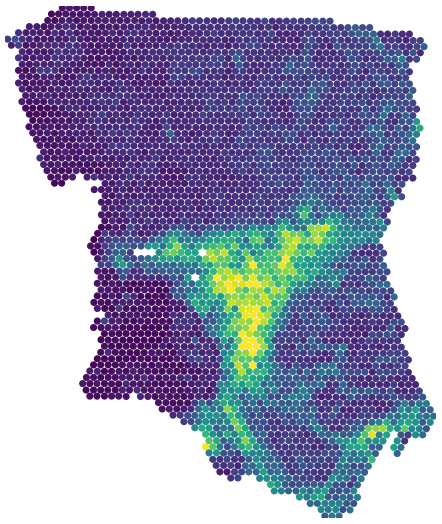
0 0.14

Matched-model AUCell

GSE238145_B18 | Candidate neighborhoods

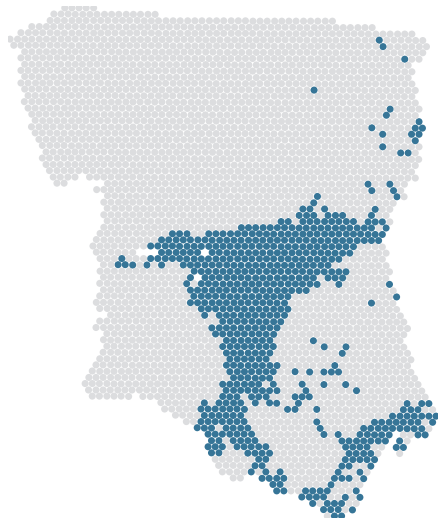
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



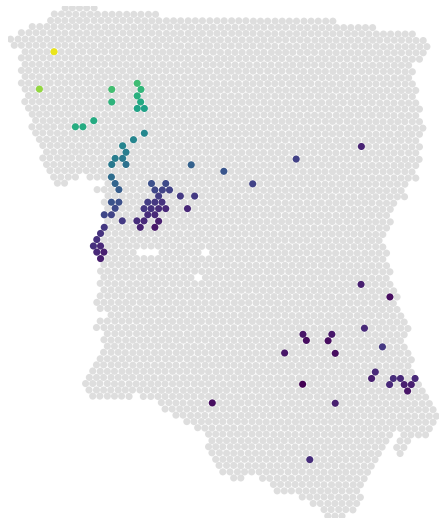
0 0.418
Fraction of all 30

CAF P80 targets



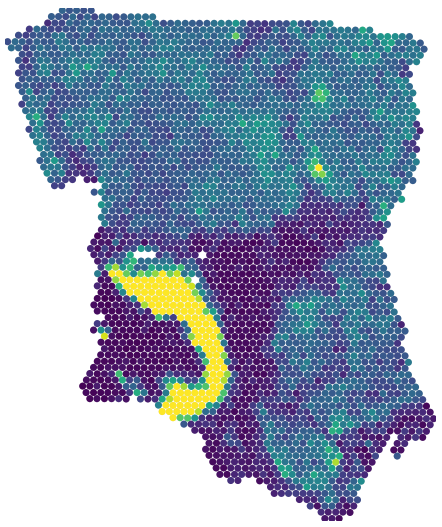
False True

CAF distance



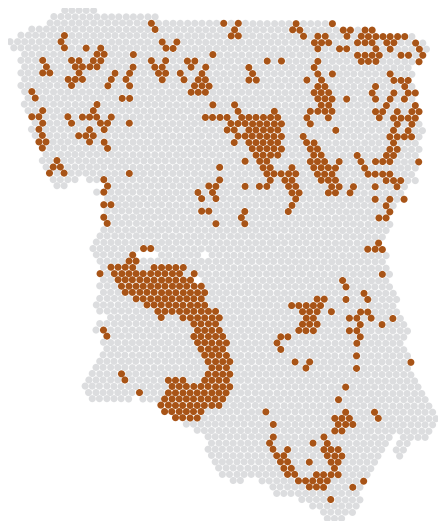
0 3.1e+03
Nearest spot center (μm)

TAM composition



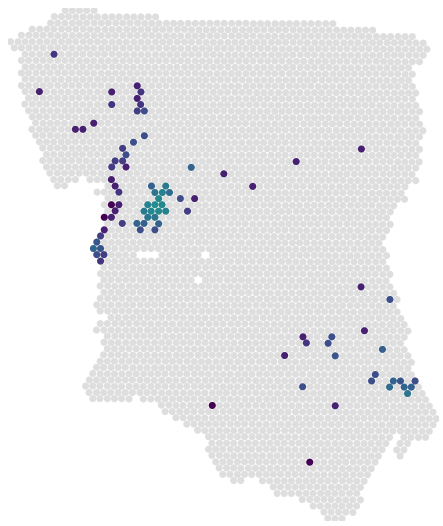
0 0.291
Fraction of all 30

TAM P80 targets



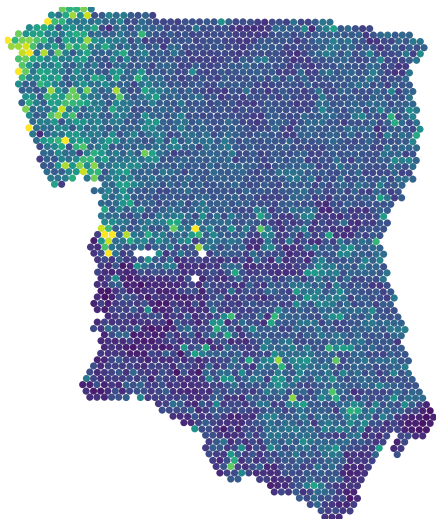
False True

TAM distance



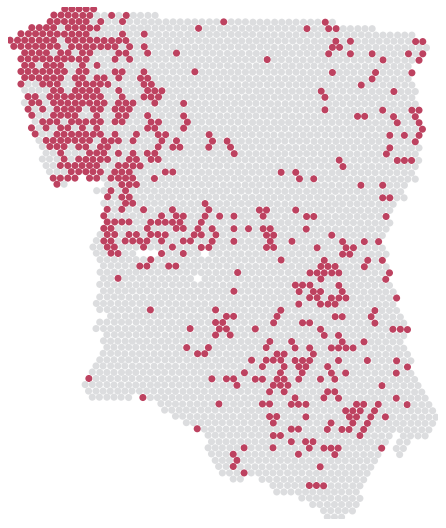
0 1.13e+03
Nearest spot center (μm)

Treg composition



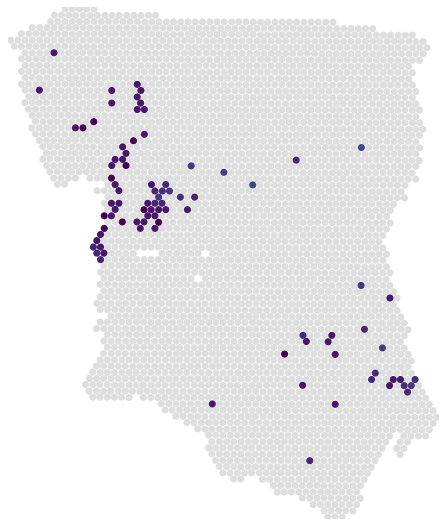
0 0.102
Fraction of all 30

Treg P80 targets



False True

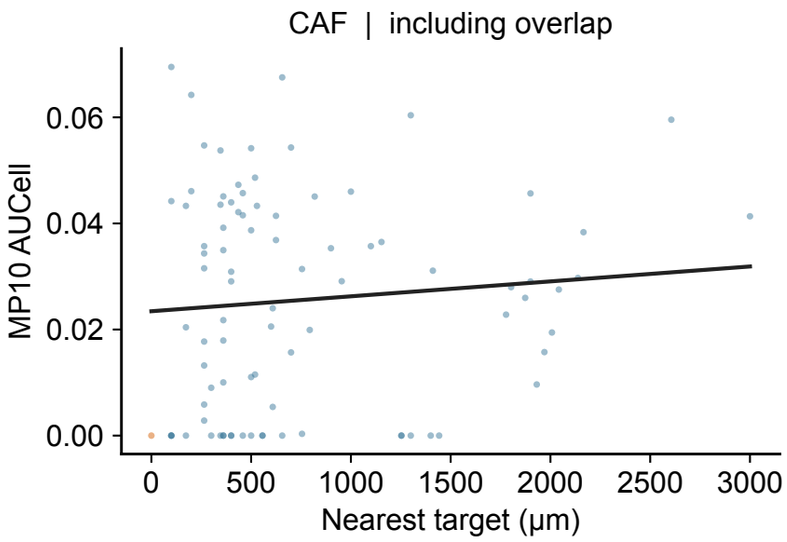
Treg distance



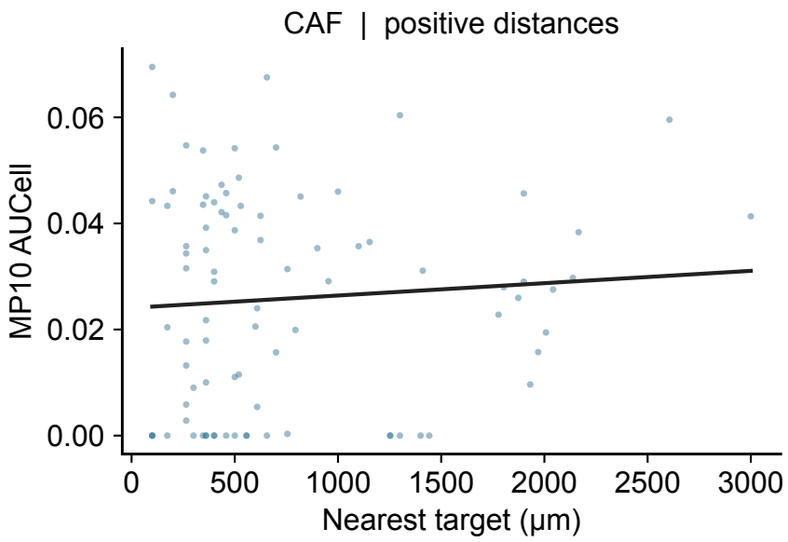
0 1.61e+03
Nearest spot center (μm)

GSE238145_B18 | MP10 versus nearest target

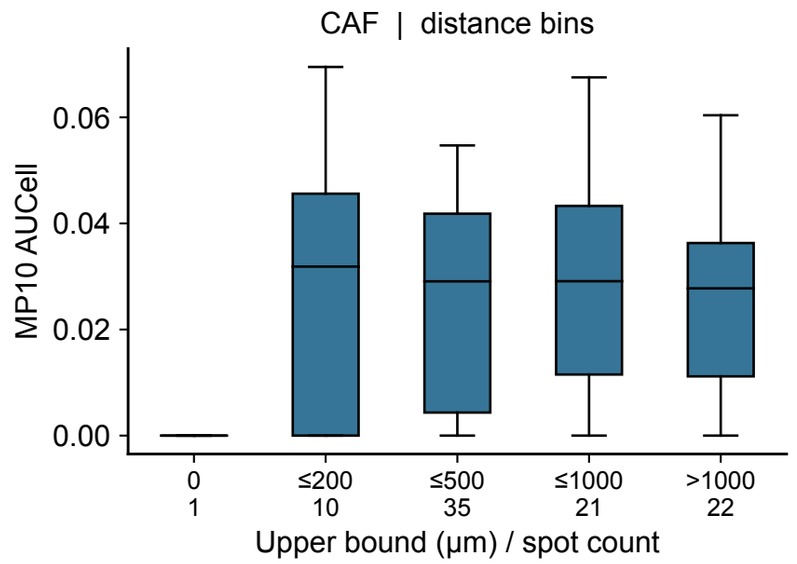
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



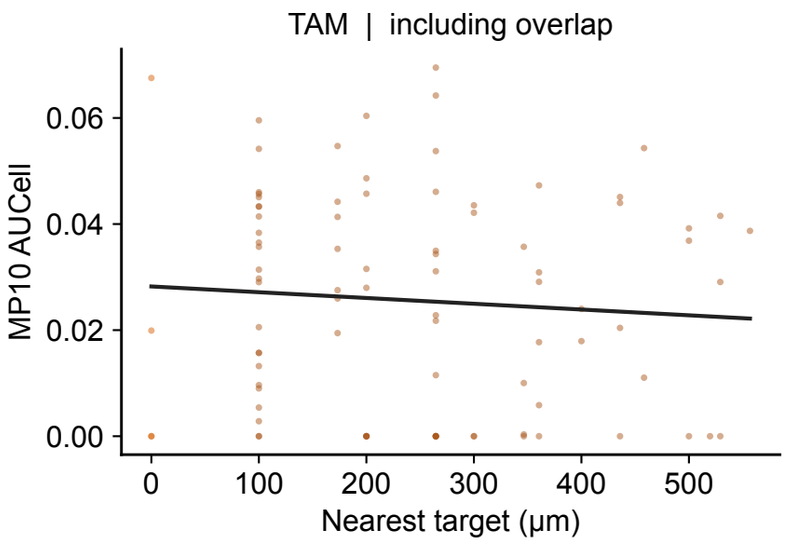
r +0.09 ρ +0.06
n 89/89 queries • 661 targets



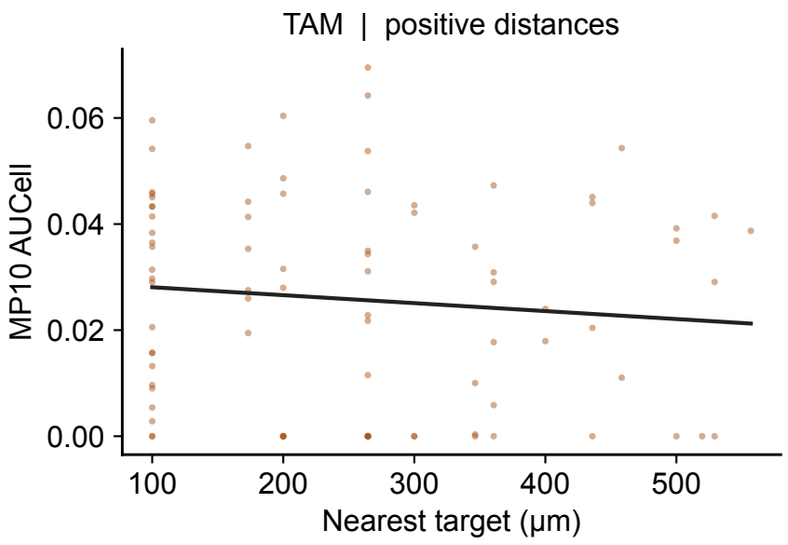
r +0.08 ρ +0.04
n 88/89 queries • 661 targets



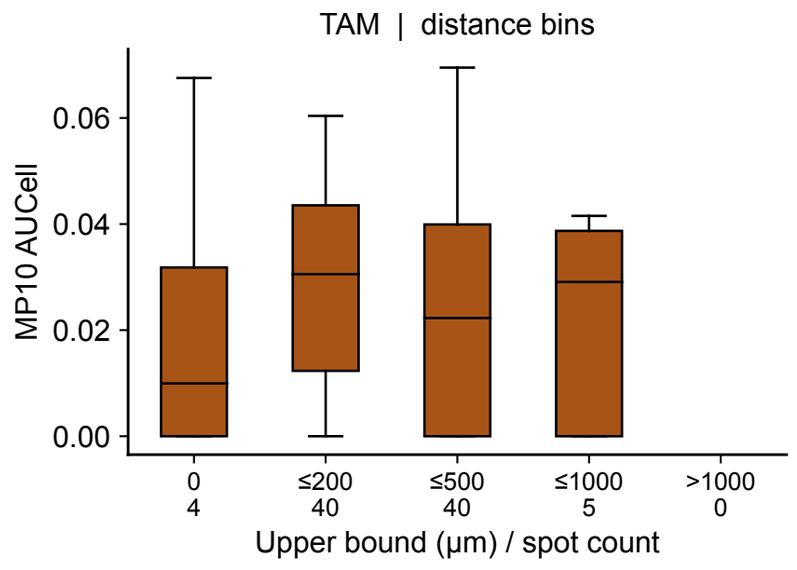
Orange = zero distance
Line = within-section linear fit



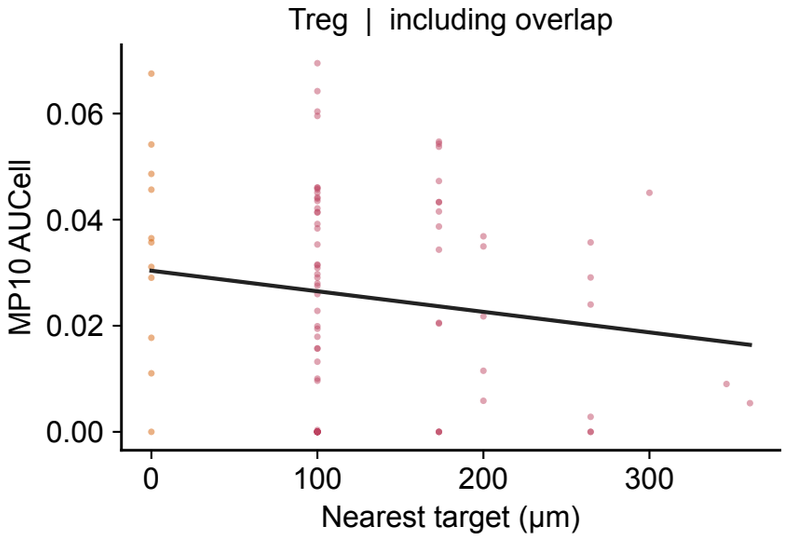
r -0.08 ρ -0.06
n 89/89 queries • 661 targets



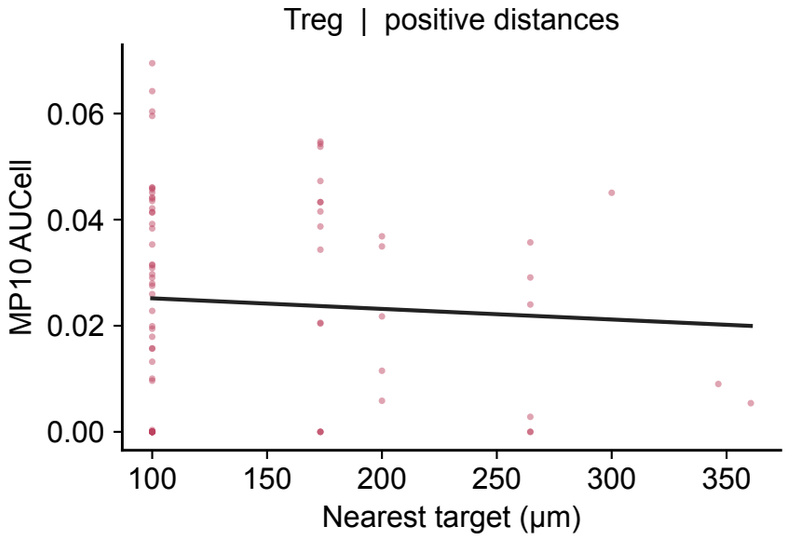
r -0.10 ρ -0.09
n 85/89 queries • 661 targets



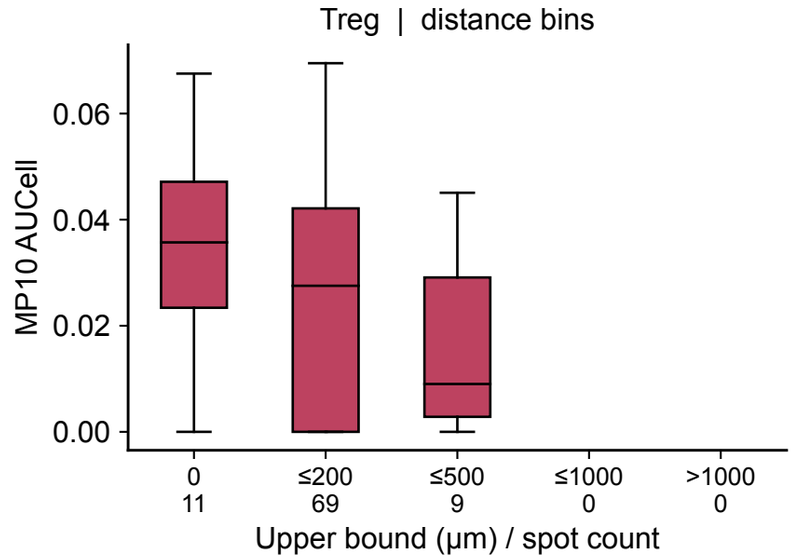
Orange = zero distance
Line = within-section linear fit



r -0.15 ρ -0.04
n 89/89 queries • 661 targets



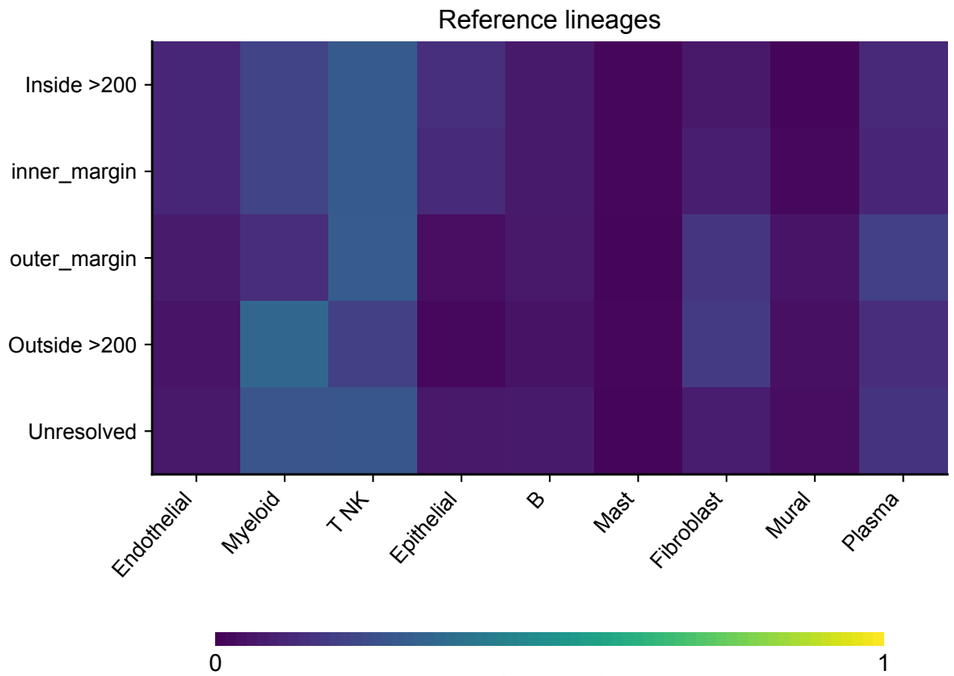
r -0.06 ρ +0.06
n 78/89 queries • 661 targets

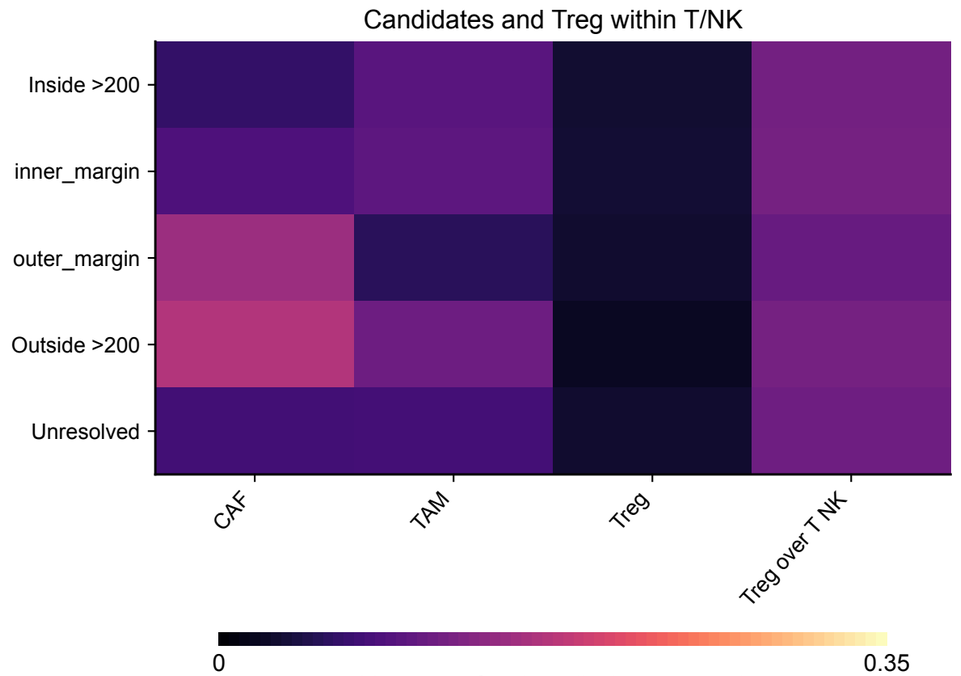


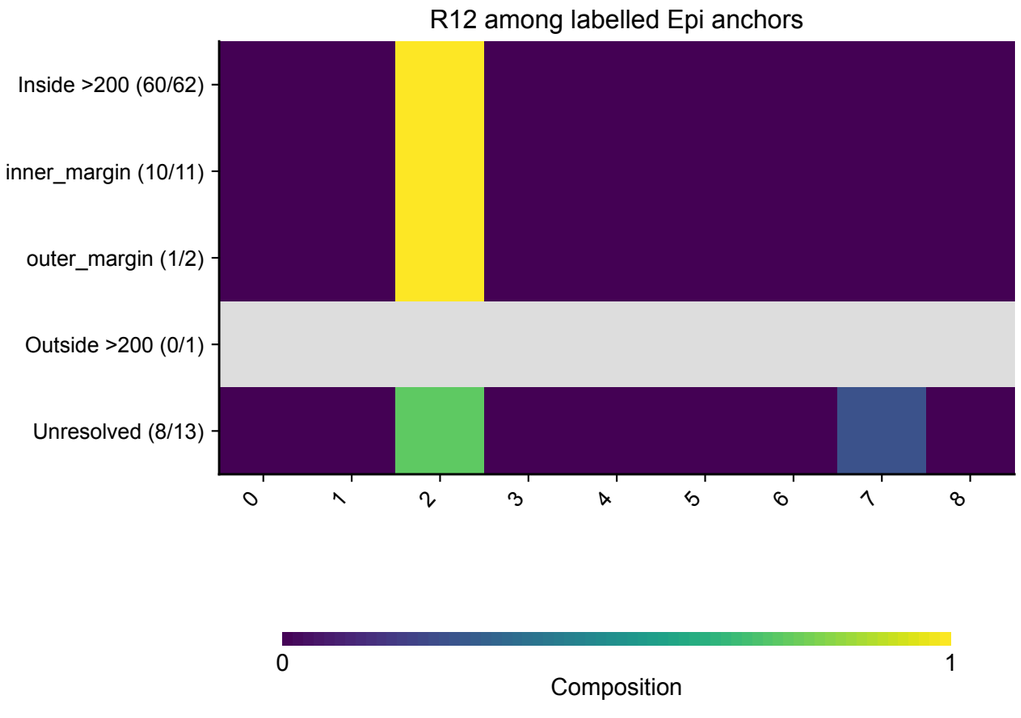
Orange = zero distance
Line = within-section linear fit

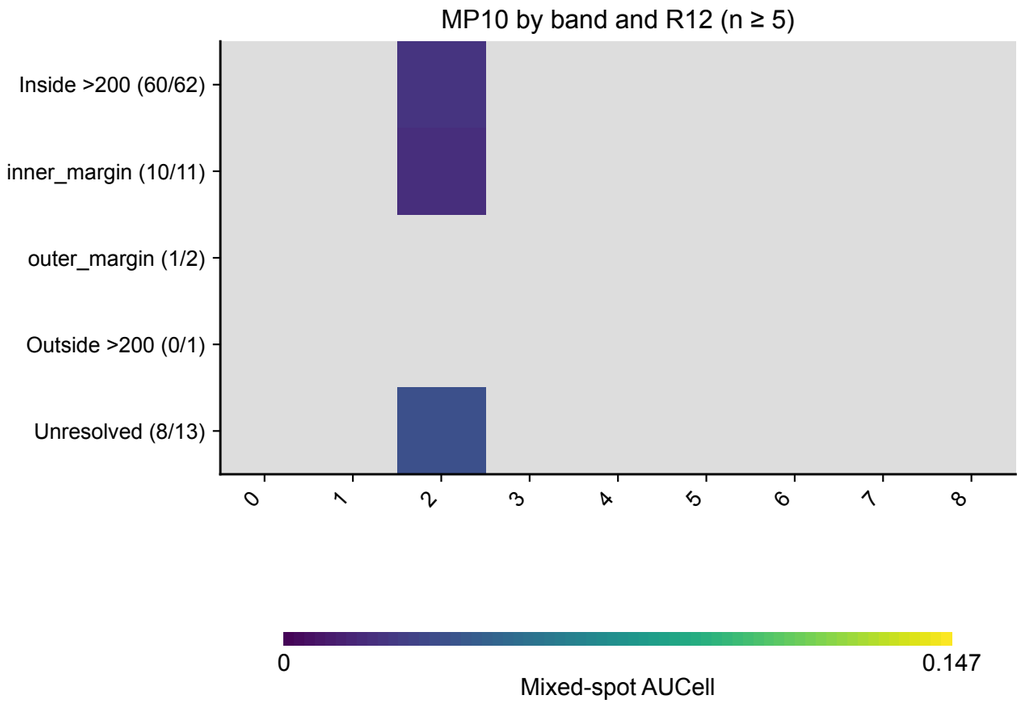
GSE238145_B18 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





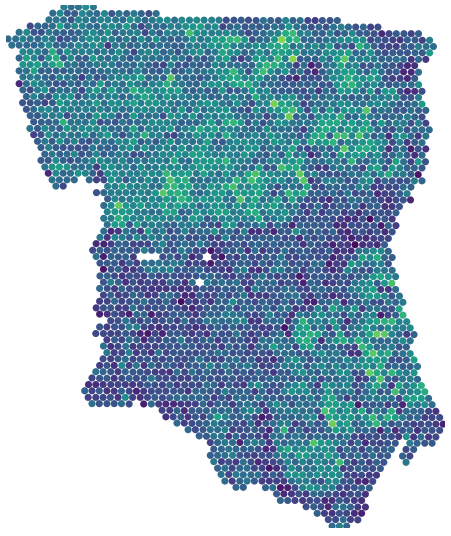




GSE238145_B18 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

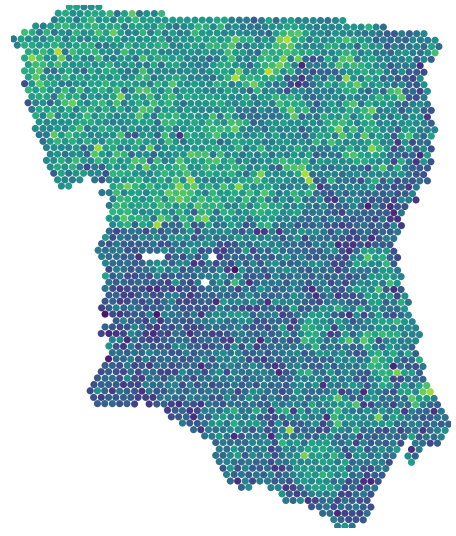
E2F Targets



0 0.152

UCell score

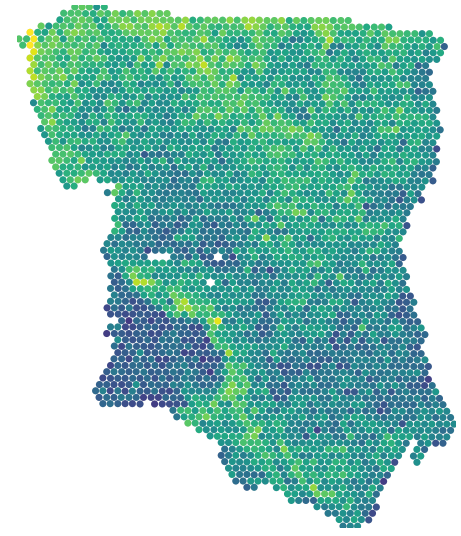
G2M Checkpoint



0 0.134

UCell score

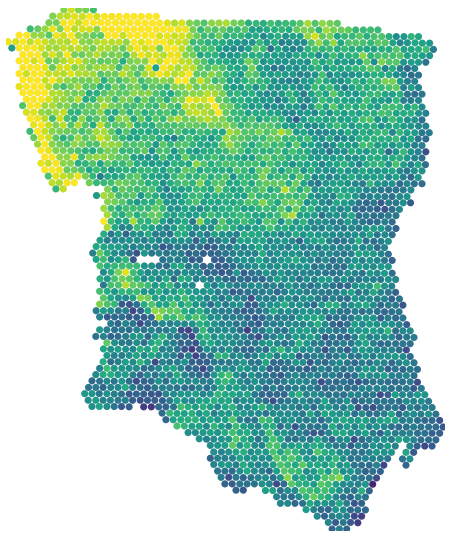
Hypoxia



0 0.211

UCell score

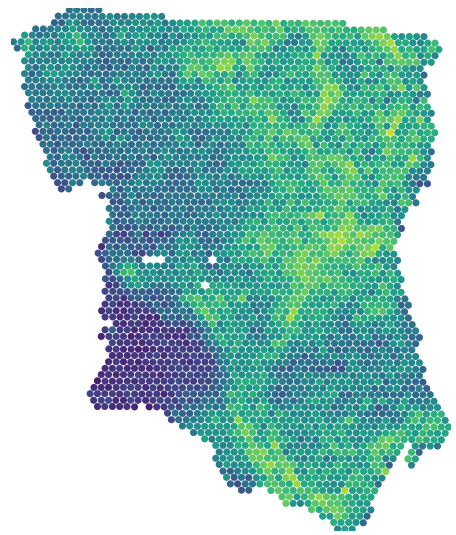
Tnfa Signaling Via Nfkb



0 0.275

UCell score

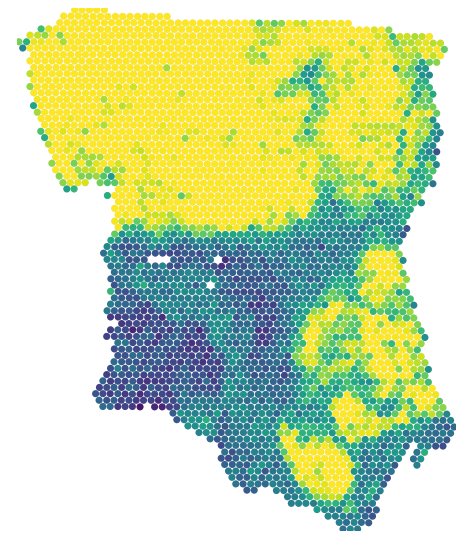
Emt



0 0.386

UCell score

Interferon Gamma Response



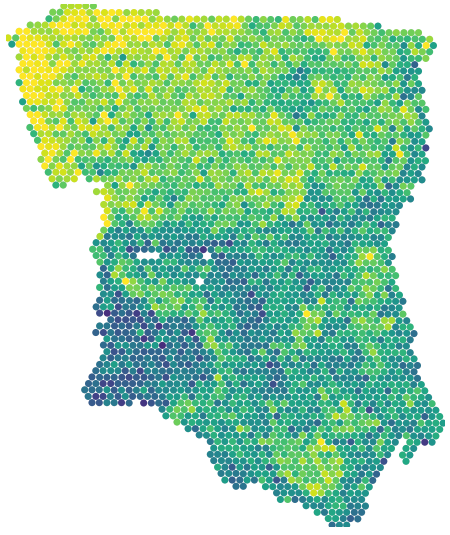
0 0.275

UCell score

GSE238145_B18 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

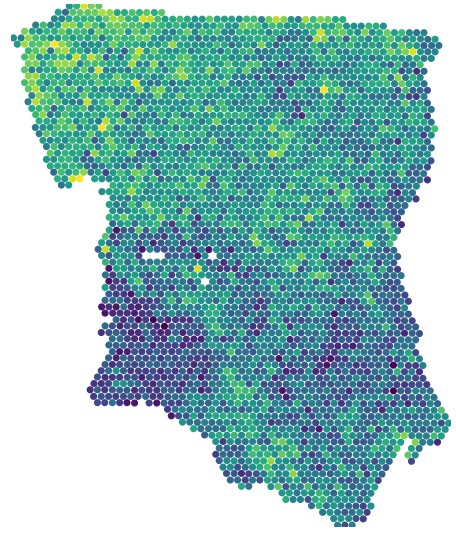
Il2 Stat5 Signaling



0 0.144

UCell score

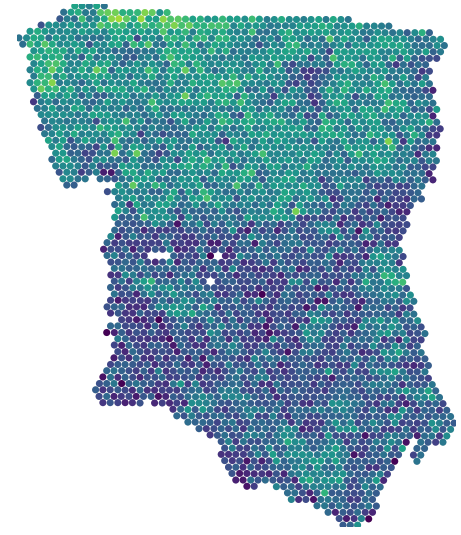
Tgf Beta Signaling



0 0.232

UCell score

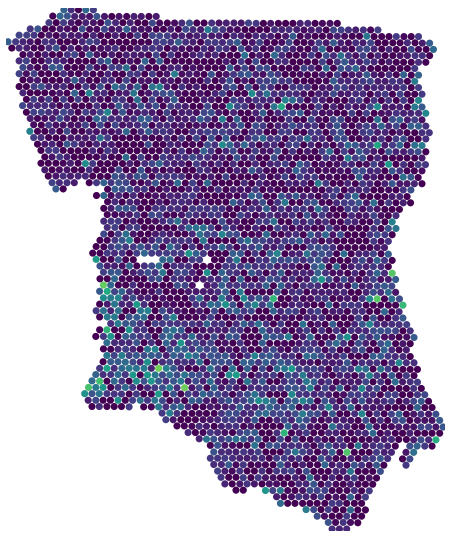
Cholesterol Homeostasis



0 0.21

UCell score

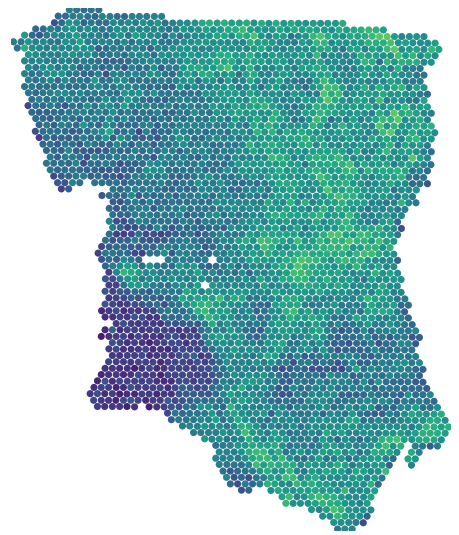
Cholesterol Biosynthesis



0 0.166

UCell score

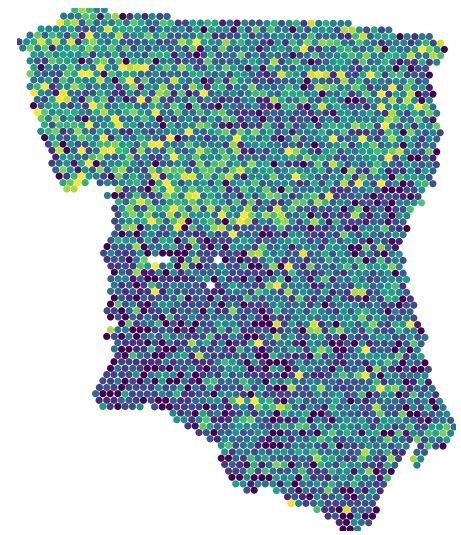
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



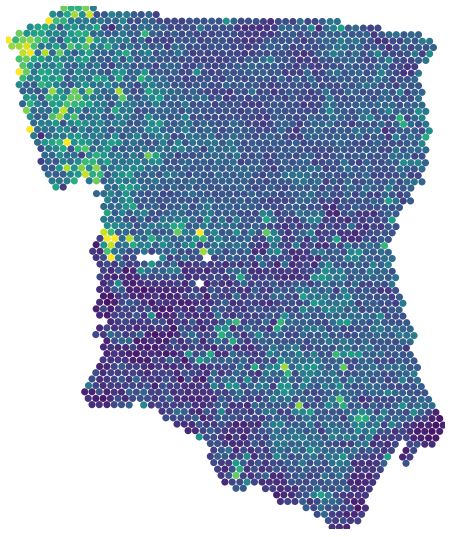
0 0.151

UCell score

GSE238145_B18 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

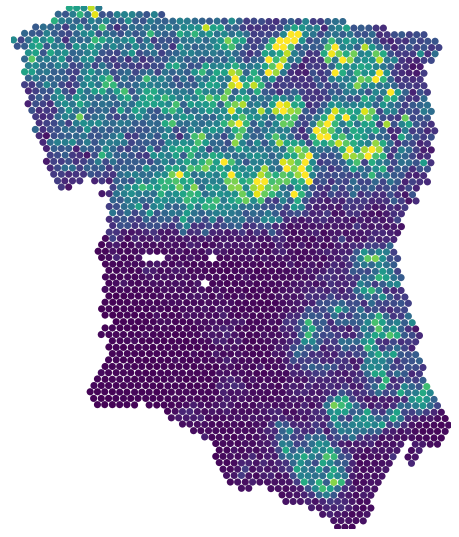
Treg composition



0 0.102

Fraction of all 30

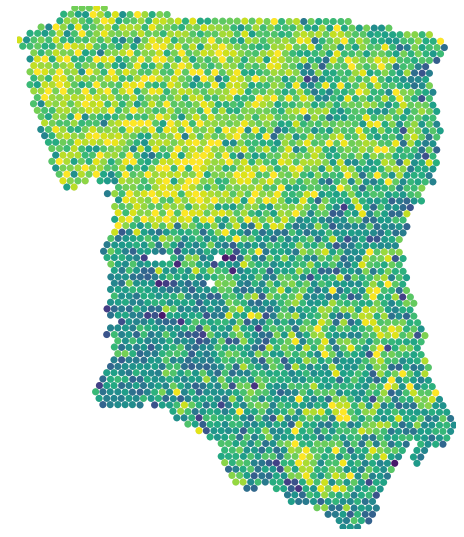
Treg estimated abundance



0 2.85

Posterior mean estimate

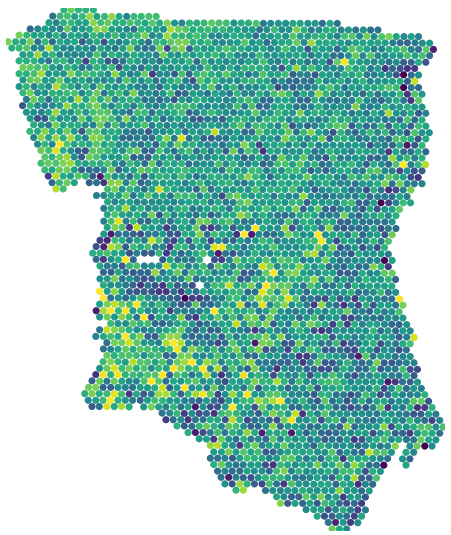
Treg Resting Vs Activated Dn



0 0.0778

UCell score

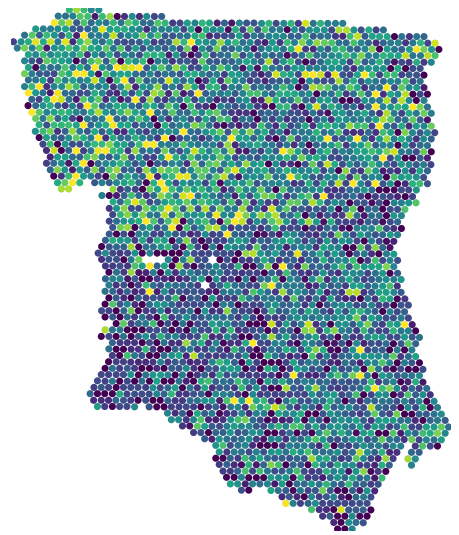
Treg Resting Vs Activated Up



0 0.0609

UCell score

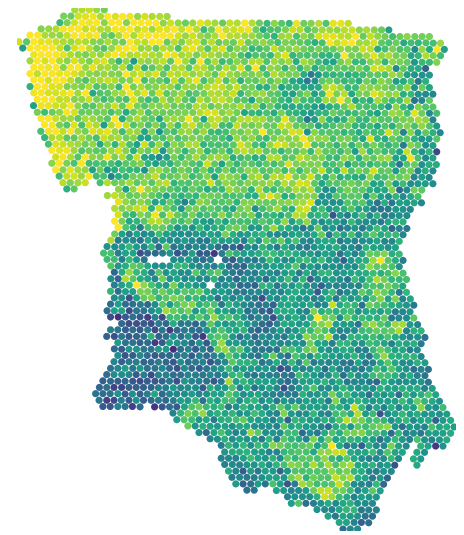
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

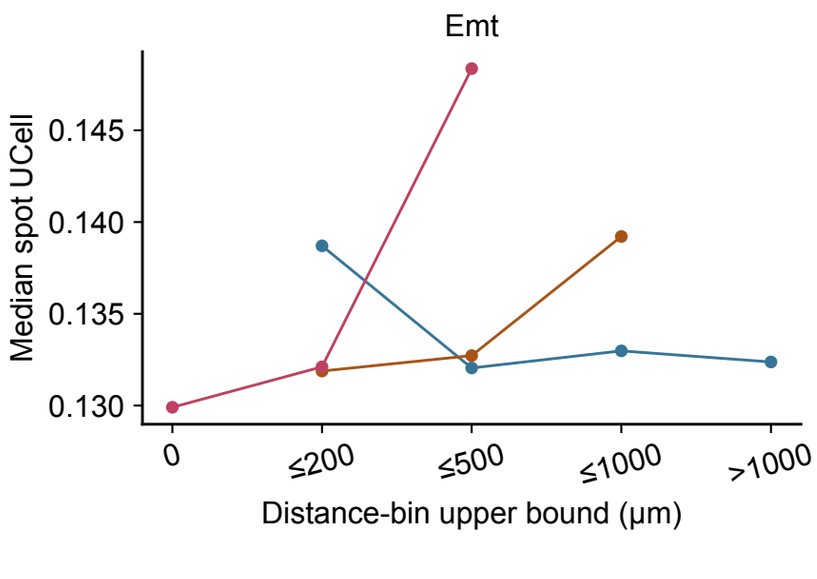


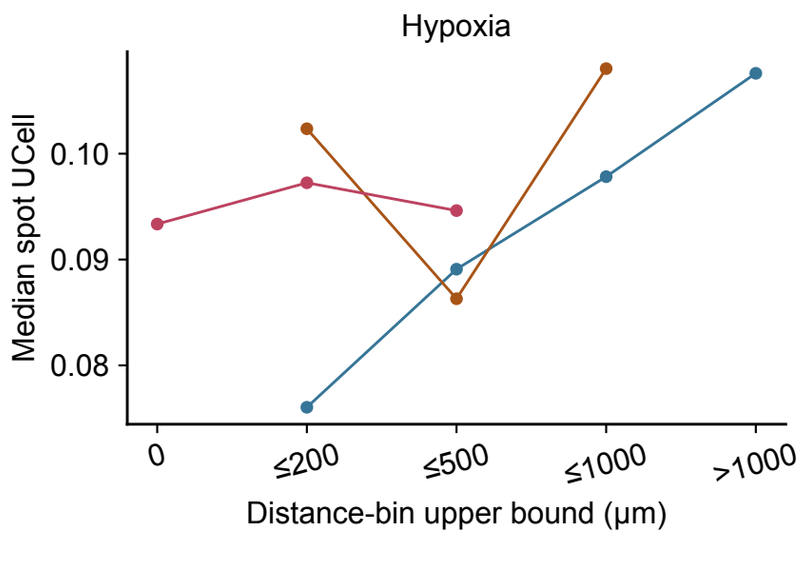
0 0.144

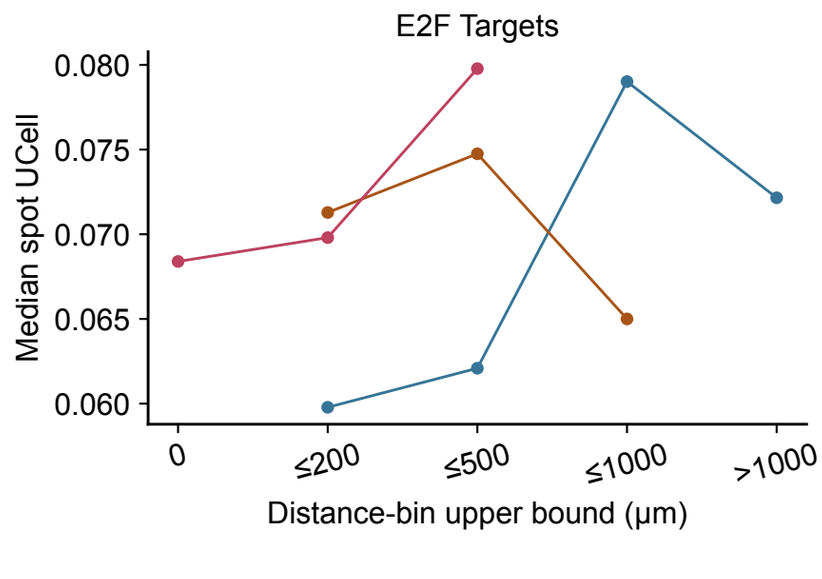
UCell score

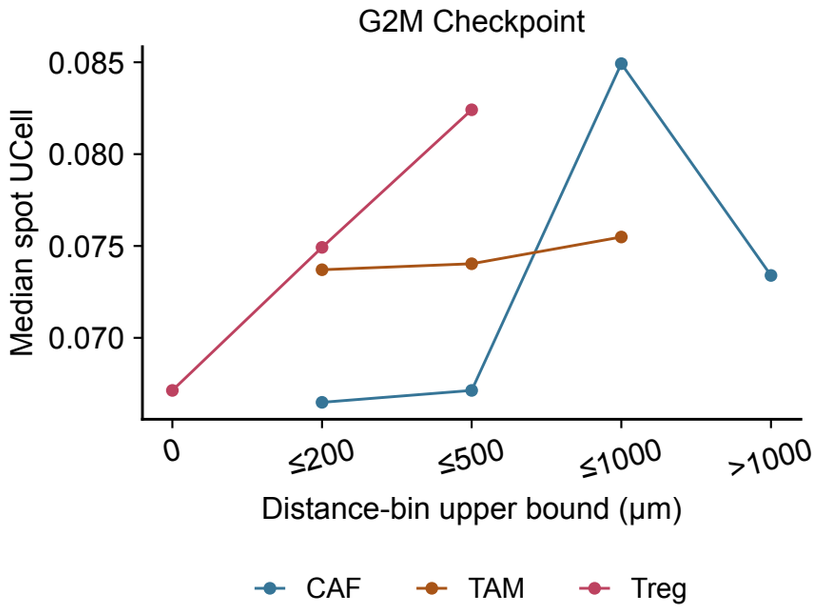
GSE238145_B18 | Programs by target distance

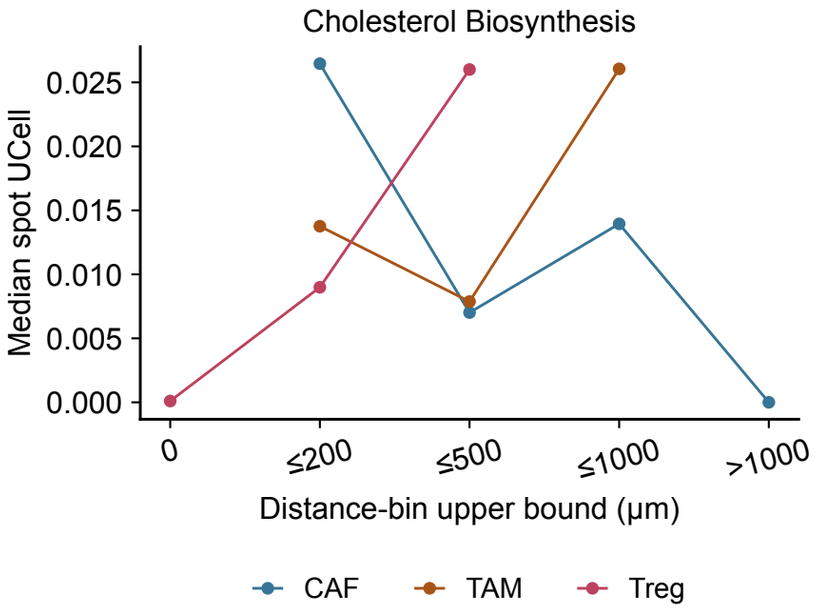
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

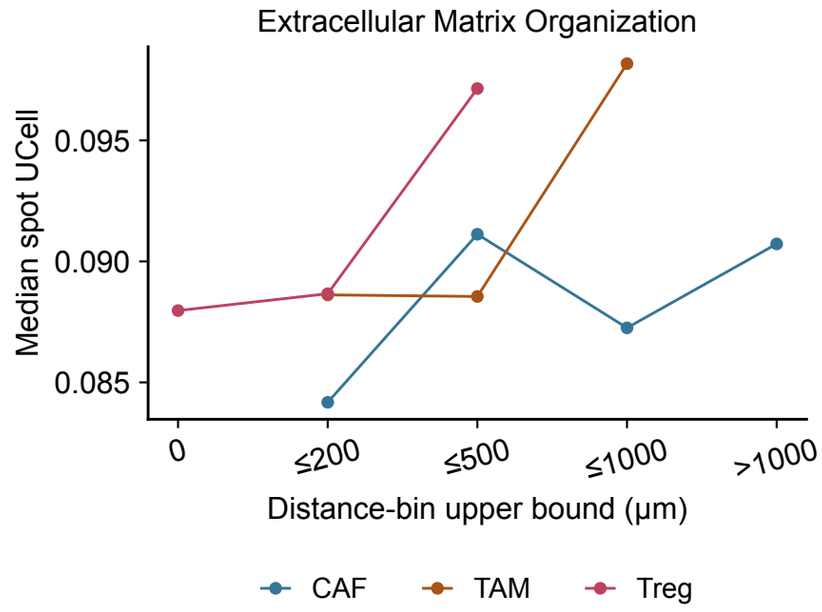






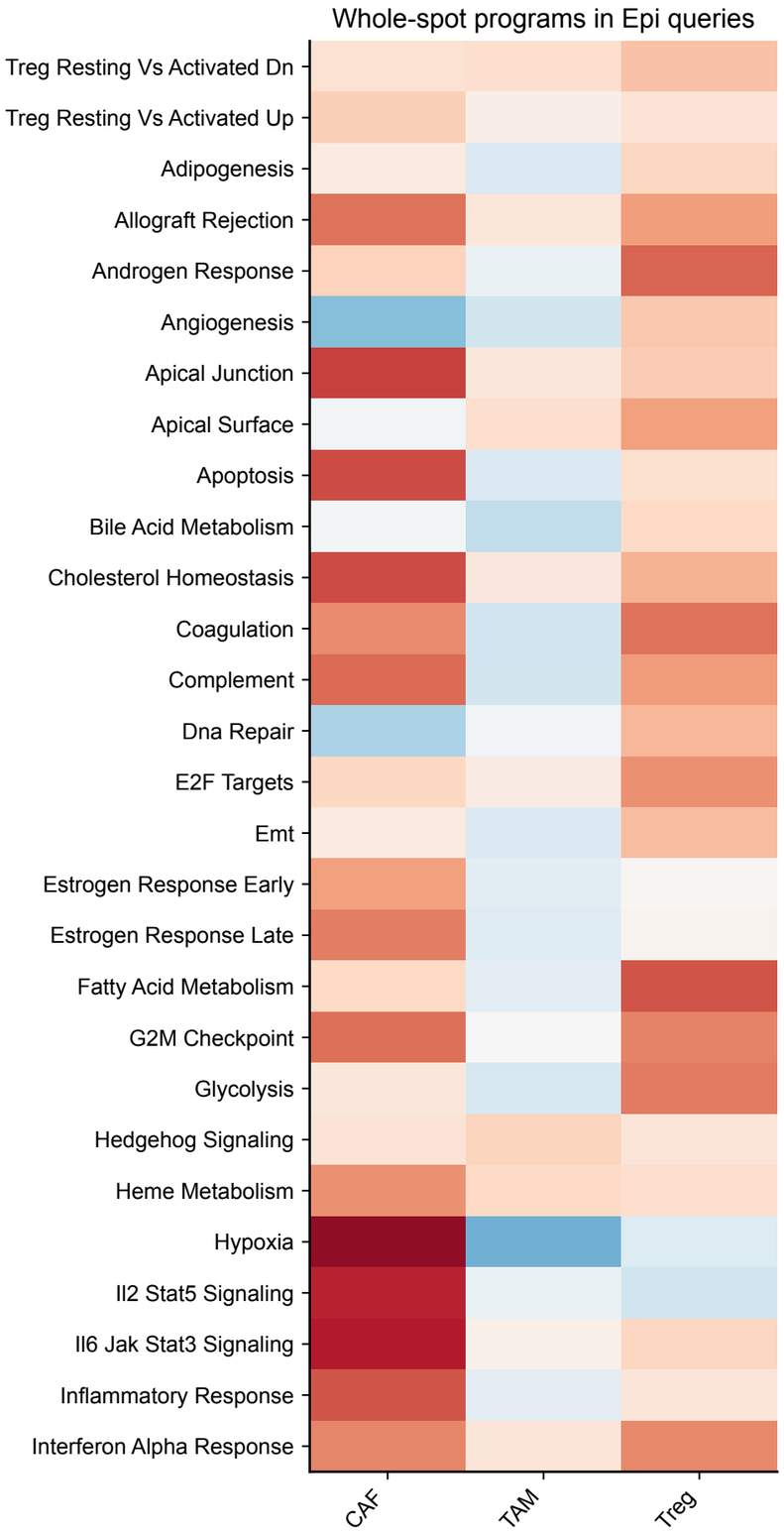


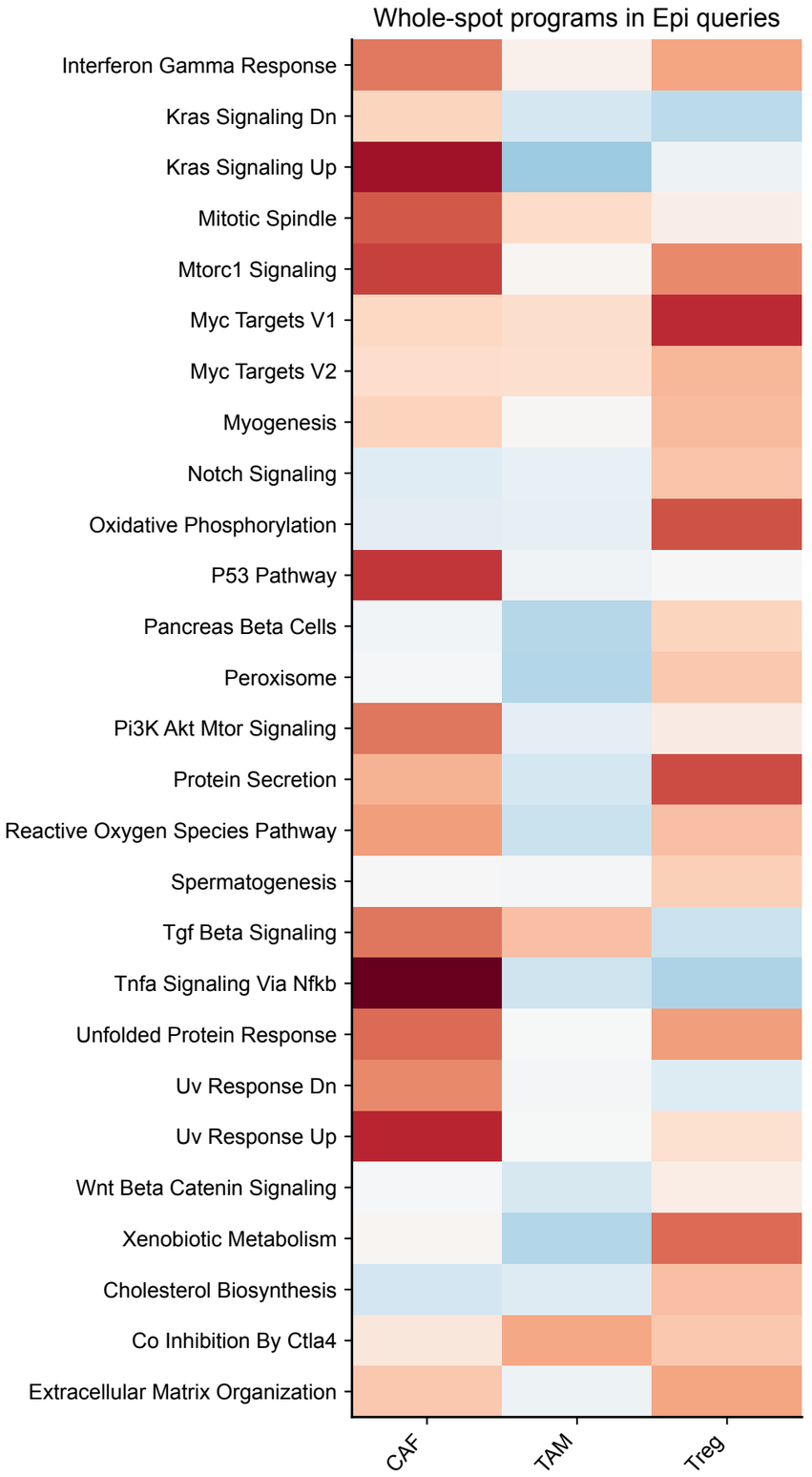




GSE238145_B18 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

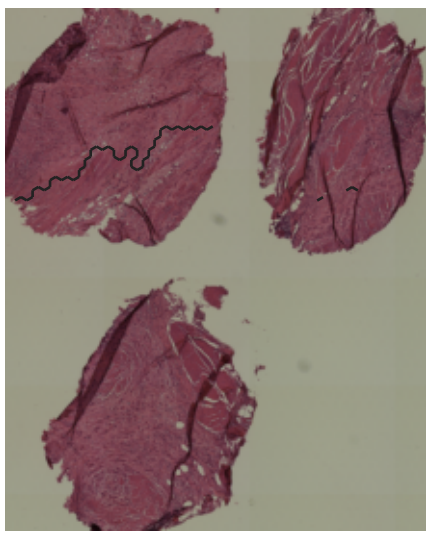




GSE238145_B5 | Tissue context

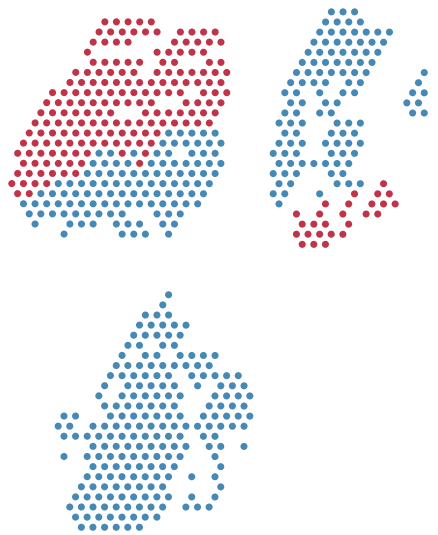
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location

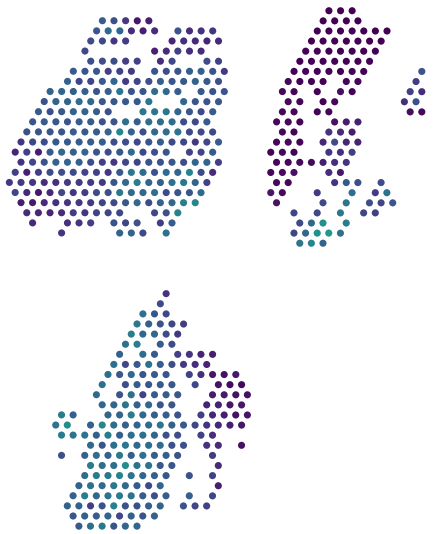


inside outside

Accepted R12 (original 0–8)

6

Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries

0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

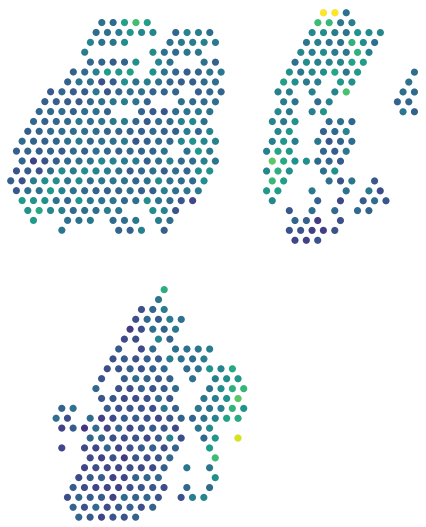
0 0.14

Matched-model AUCell

GSE238145_B5 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



0 0.418
Fraction of all 30

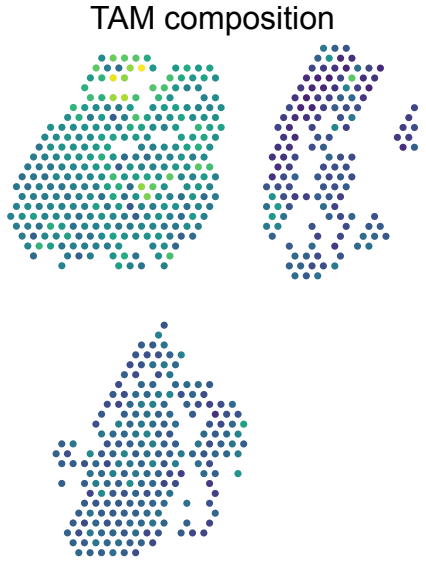
CAF P80 targets

False True

CAF distance

0 3.1e+03
Nearest spot center (μm)

TAM composition



0 0.291
Fraction of all 30

TAM P80 targets

False True

TAM distance

0 1.13e+03
Nearest spot center (μm)

Treg composition

0 0.102
Fraction of all 30

Treg P80 targets

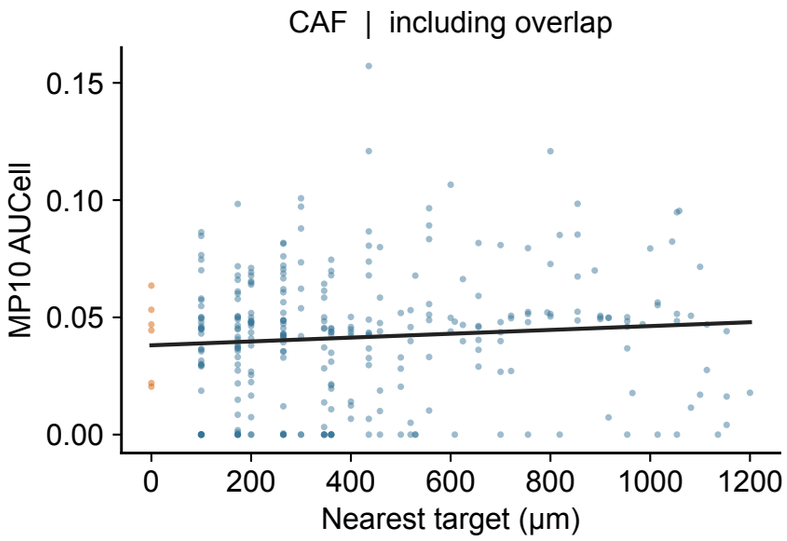
False True

Treg distance

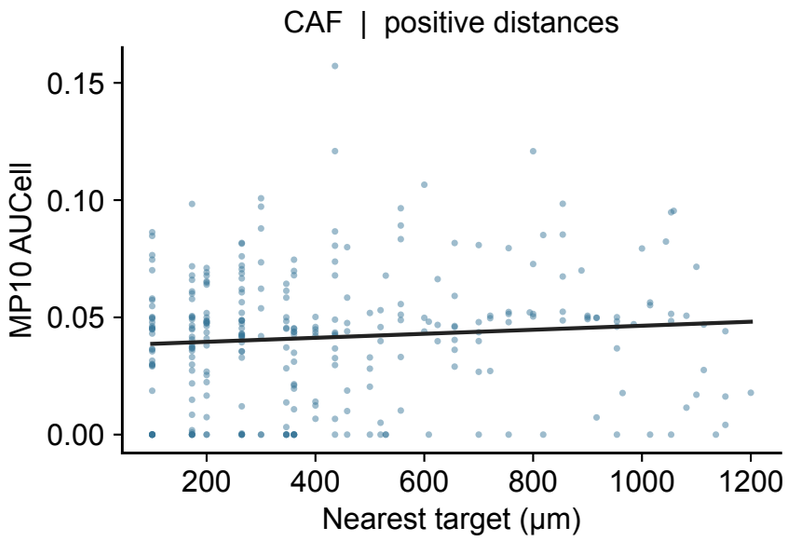
0 1.61e+03
Nearest spot center (μm)

GSE238145_B5 | MP10 versus nearest target

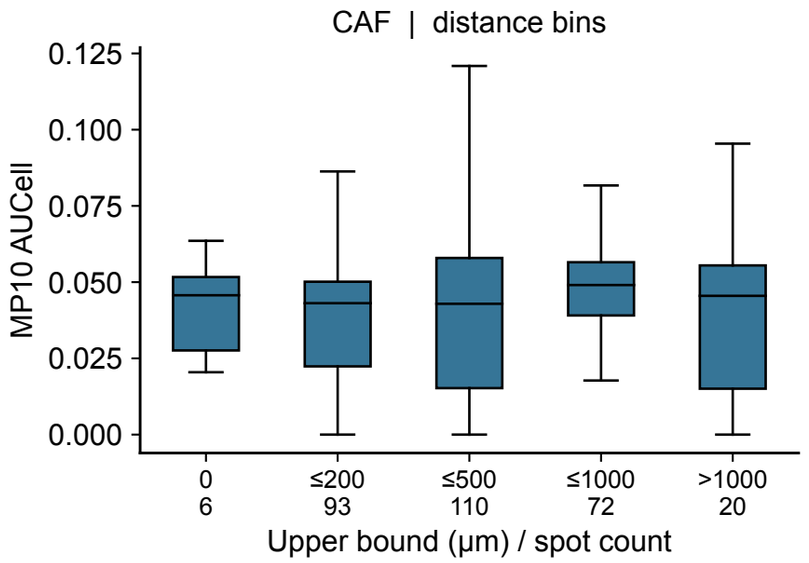
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



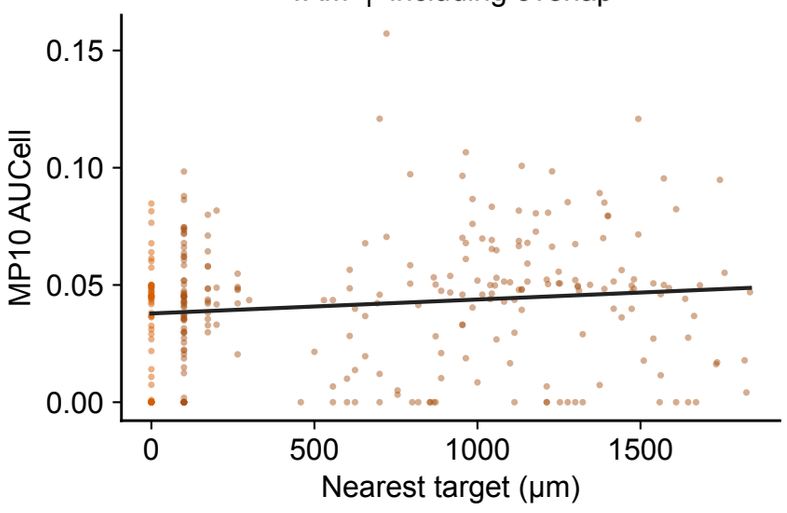
r +0.09 ρ +0.10
n 301/305 queries • 128 targets



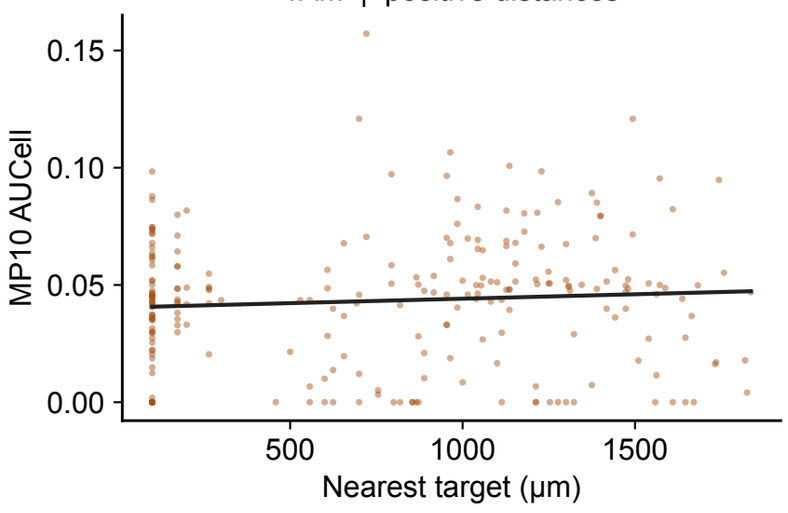
r +0.09 ρ +0.10
n 295/305 queries • 128 targets



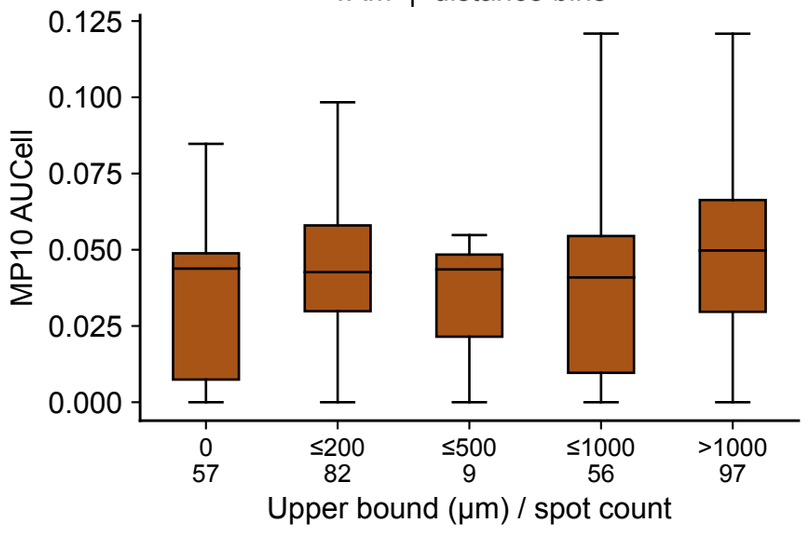
Orange = zero distance
Line = within-section linear fit



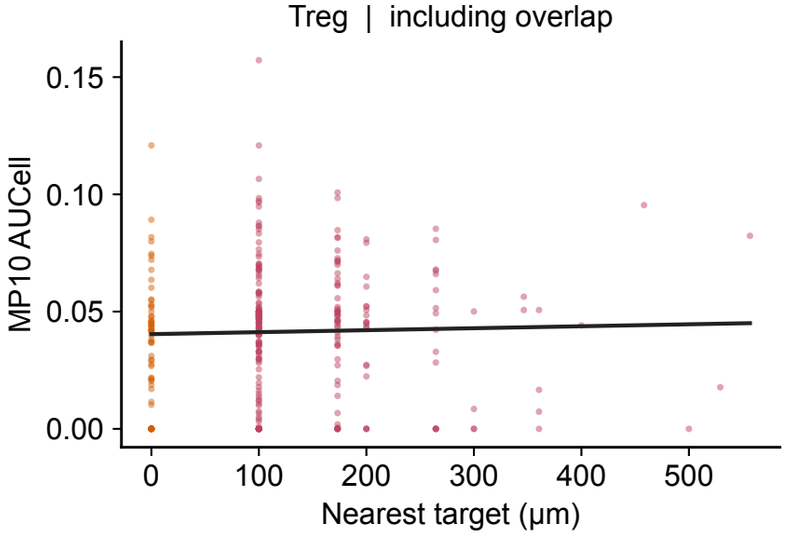
r +0.13 ρ +0.17
n 301/305 queries • 128 targets



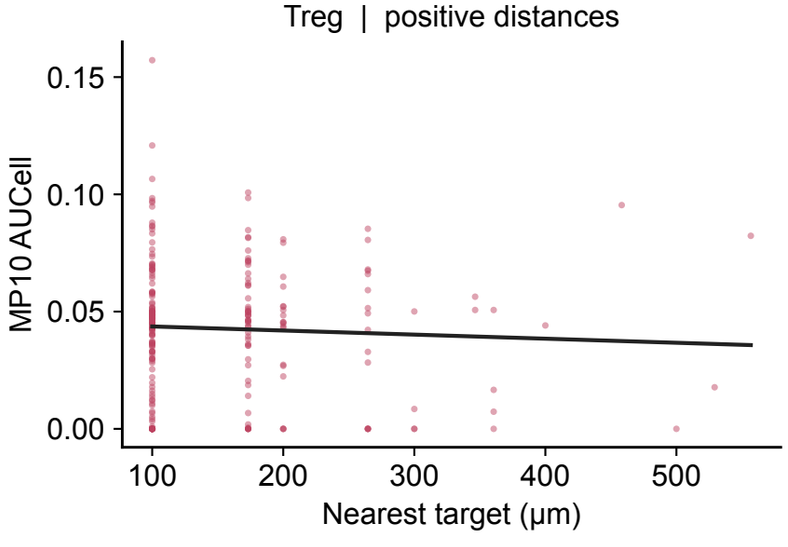
r +0.08 ρ +0.12
n 244/305 queries • 128 targets



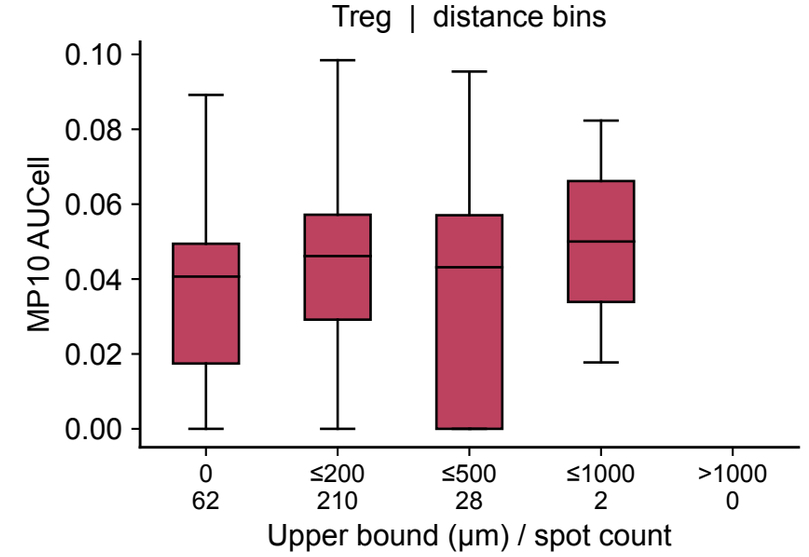
Orange = zero distance
Line = within-section linear fit



r +0.03 ρ +0.06
n 302/305 queries • 128 targets



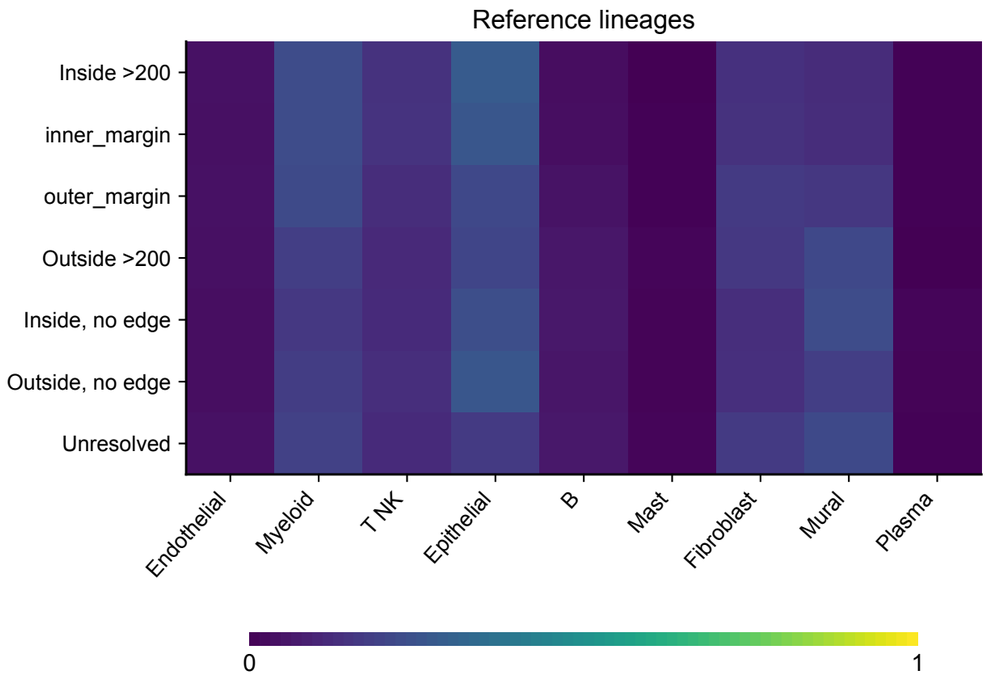
r -0.05 ρ -0.04
n 240/305 queries • 128 targets

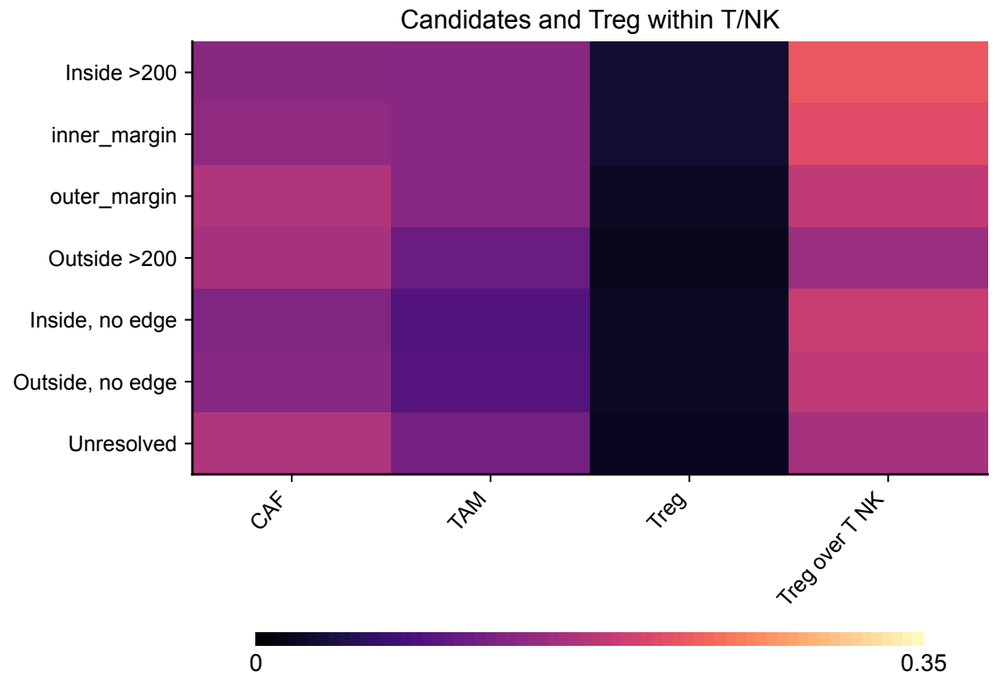


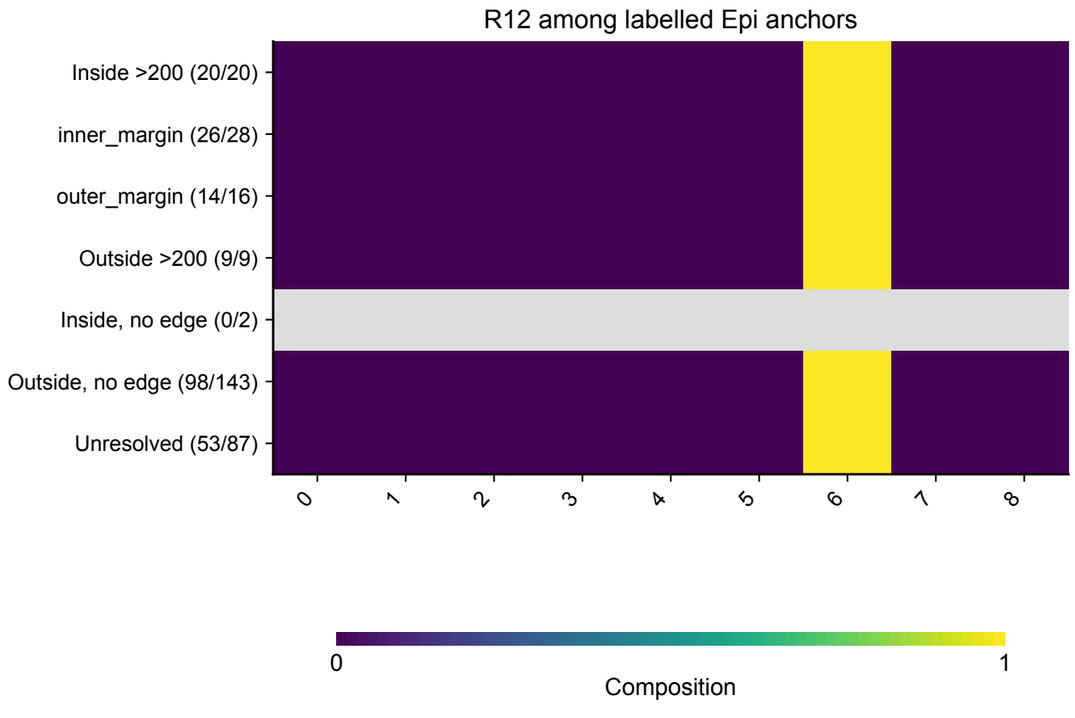
Orange = zero distance
Line = within-section linear fit

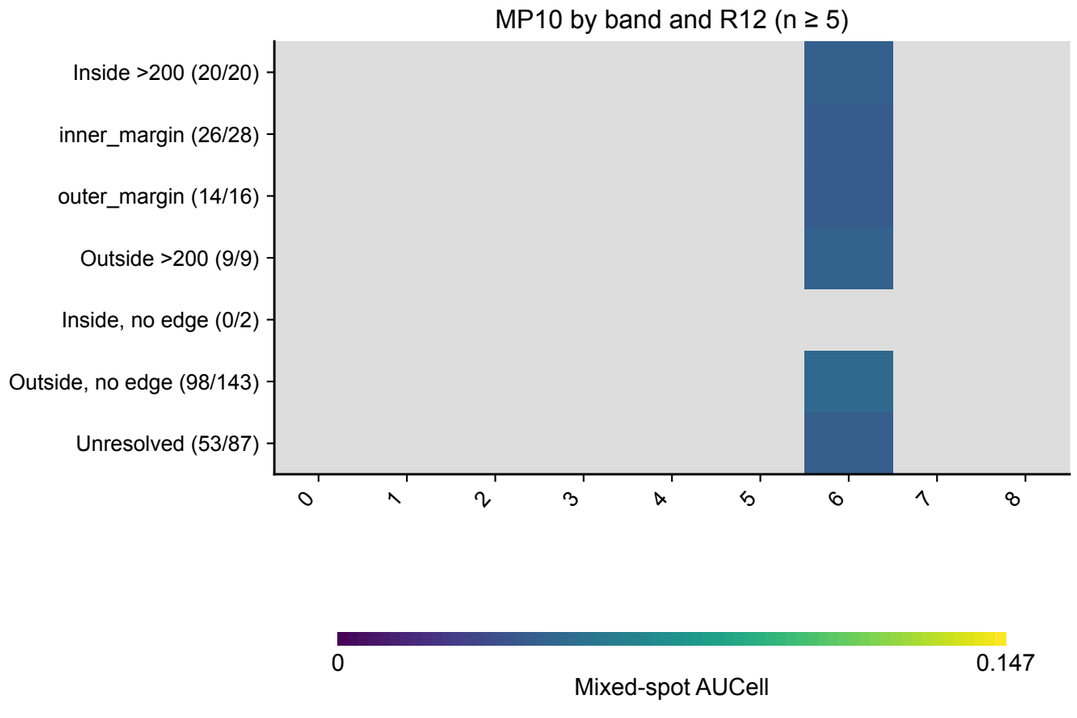
GSE238145_B5 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



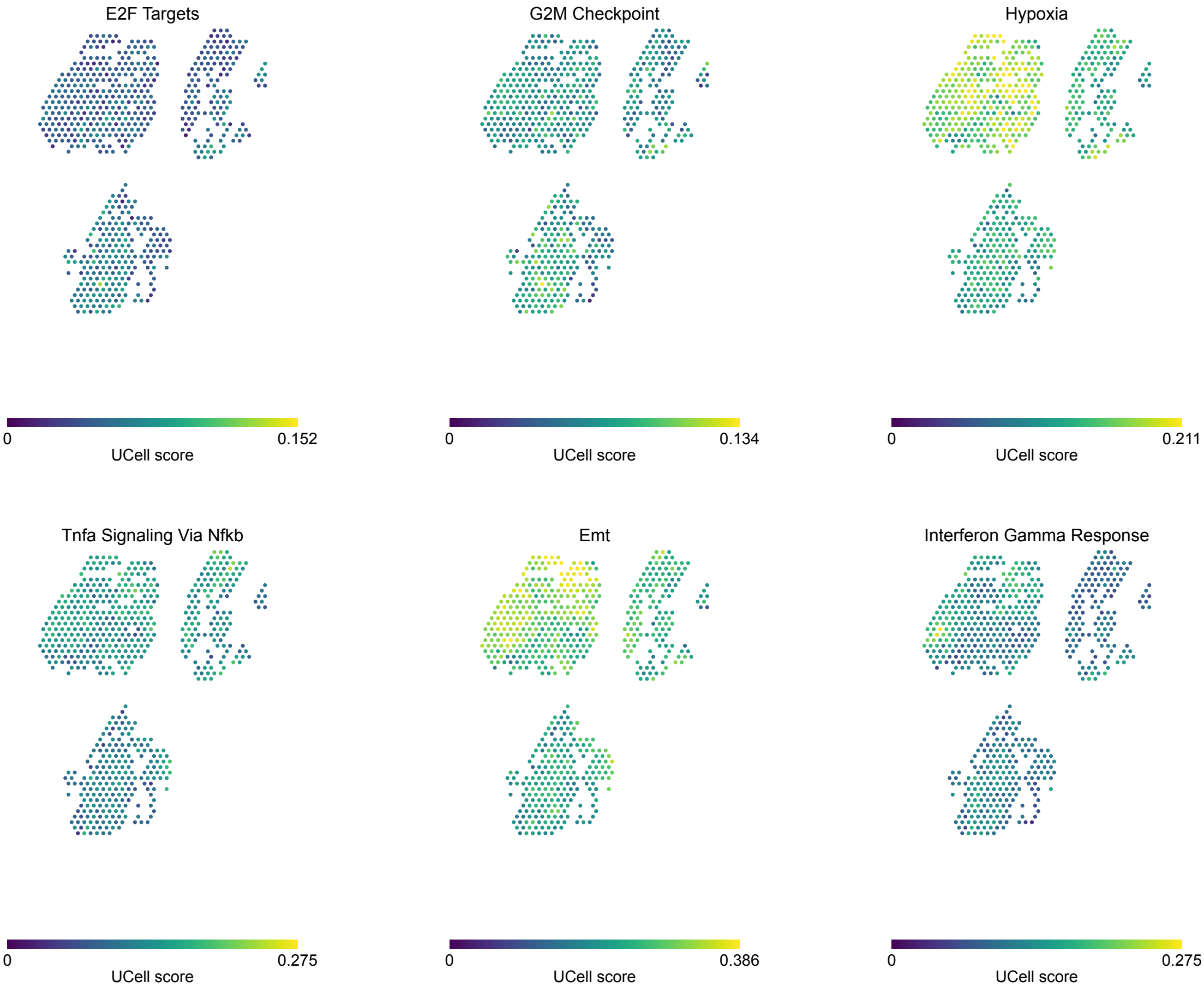






GSE238145_B5 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



GSE238145_B5 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

GSE238145_B5 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

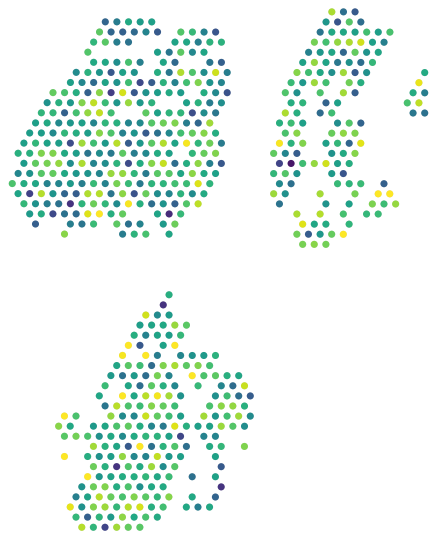
Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn



0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

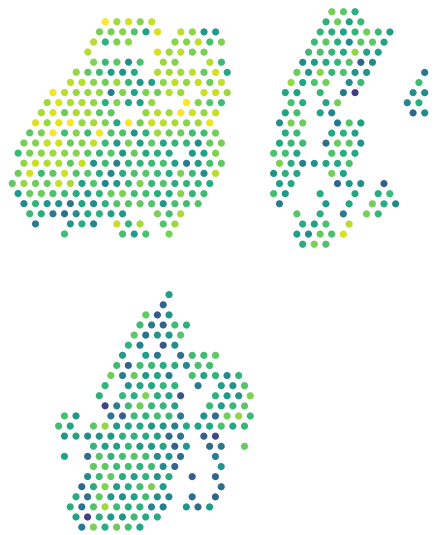
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

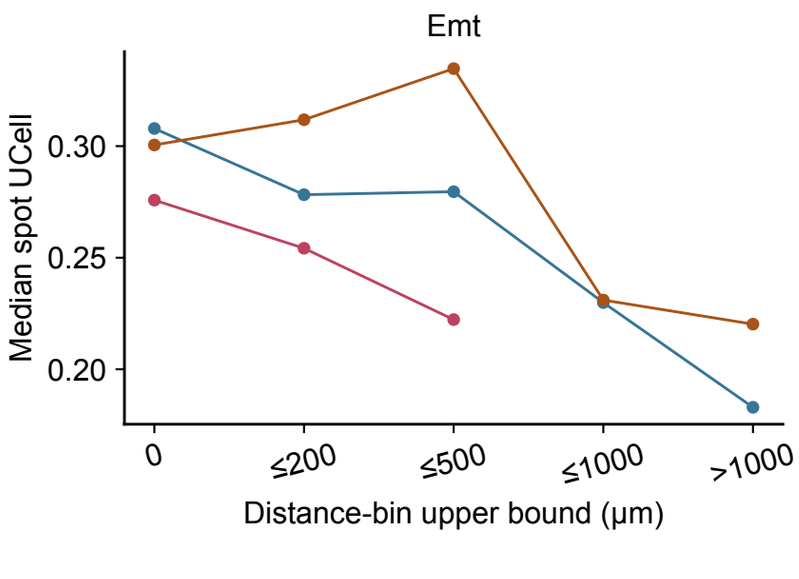


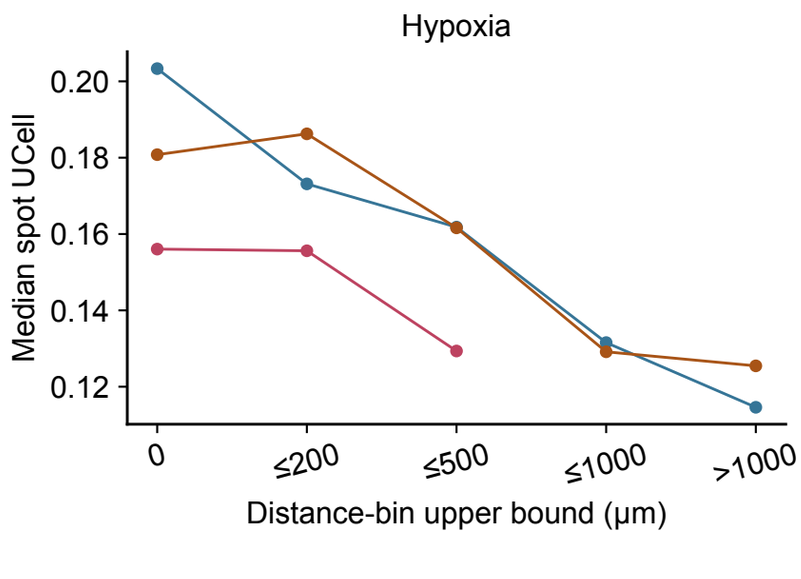
0 0.144

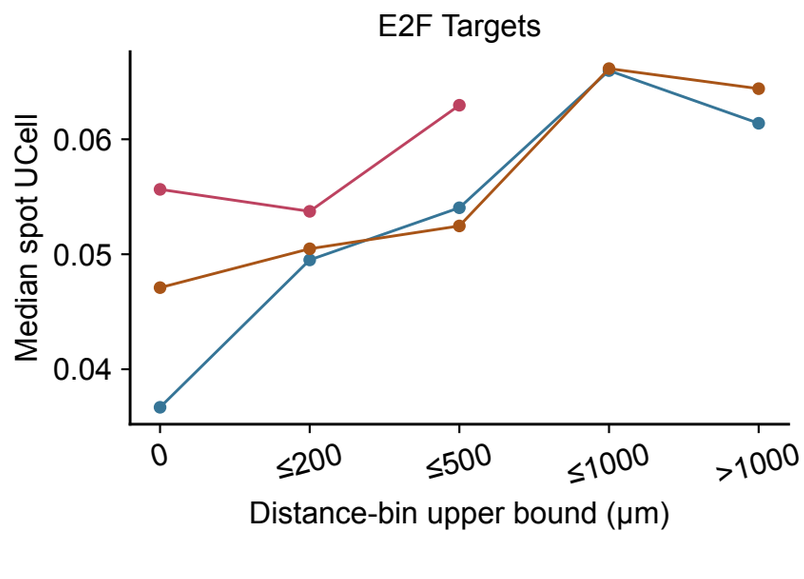
UCell score

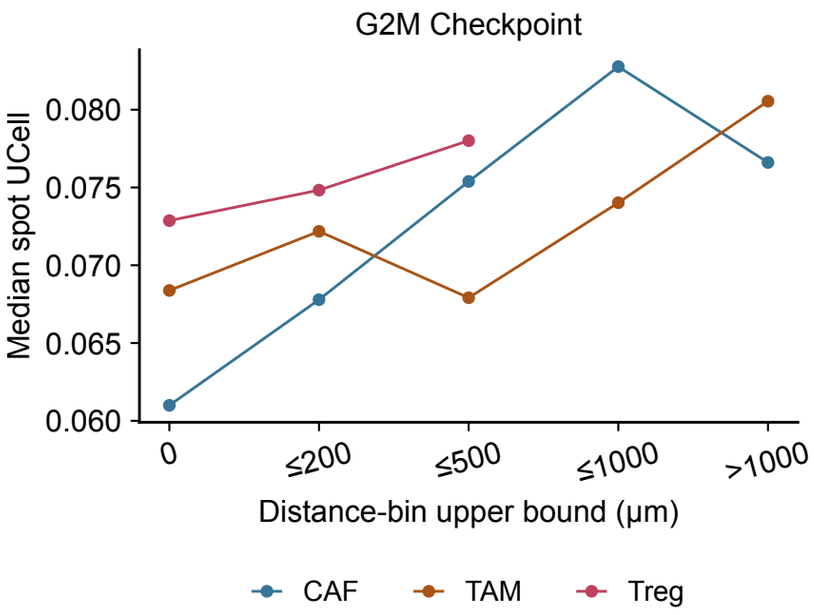
GSE238145_B5 | Programs by target distance

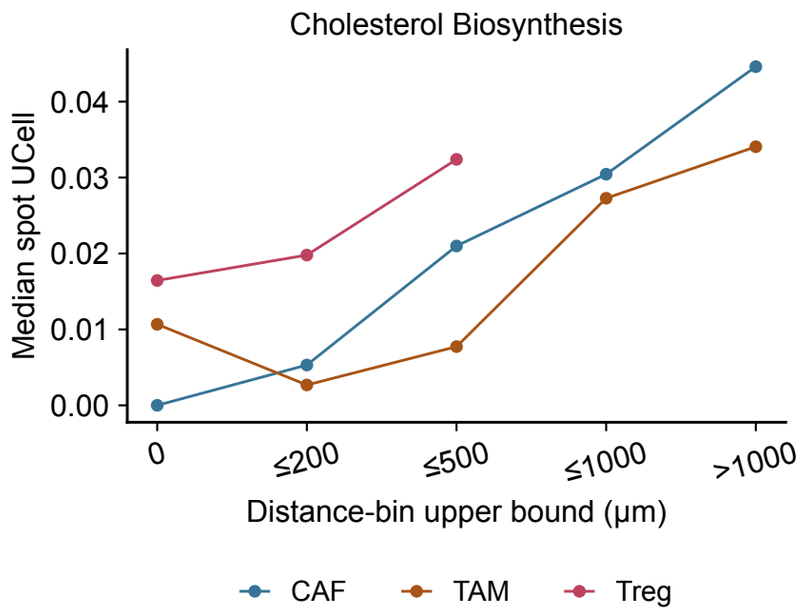
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

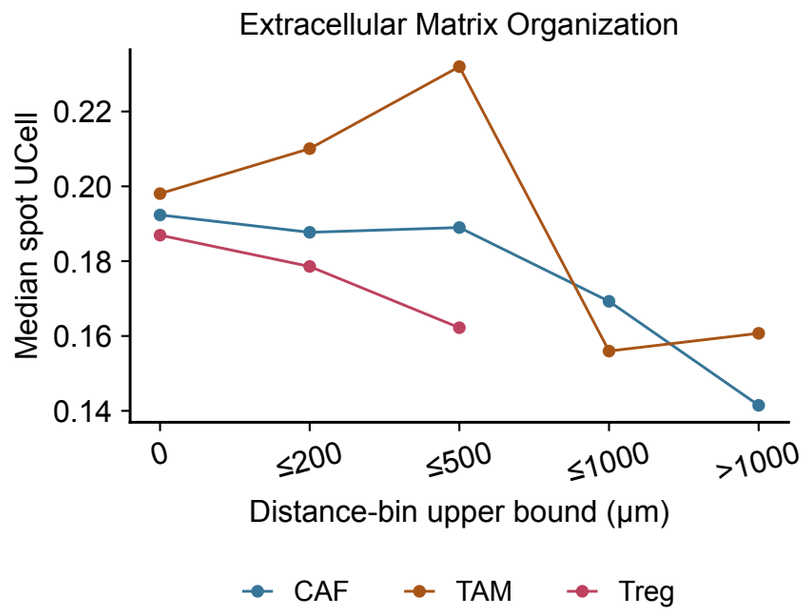






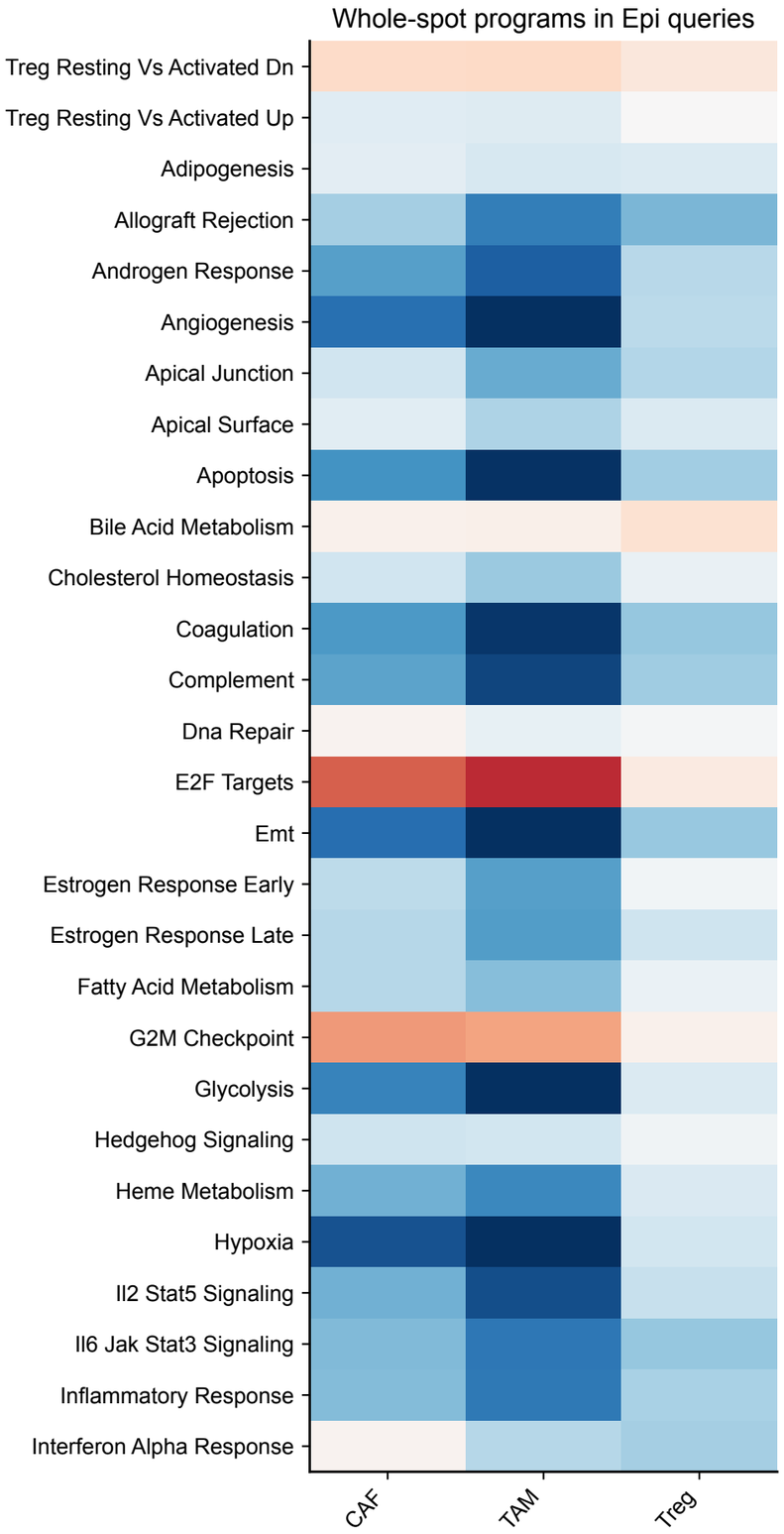


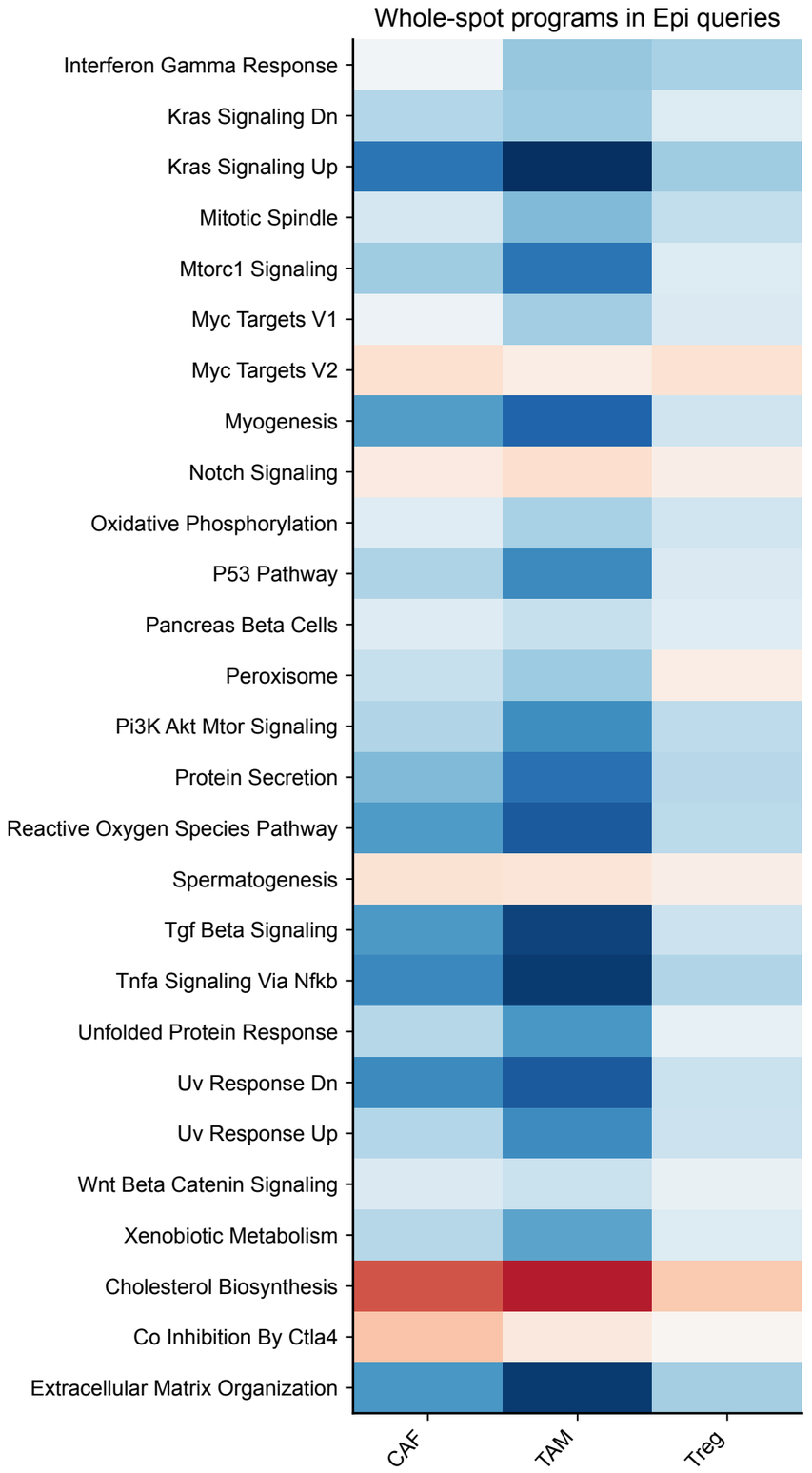




GSE238145_B5 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

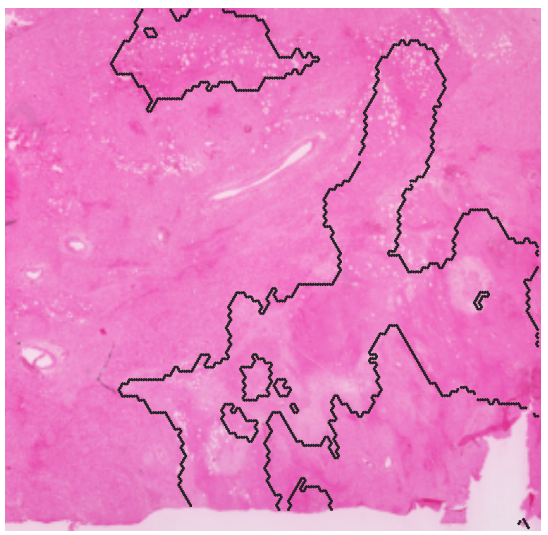




GSE238145_B9 | Tissue context

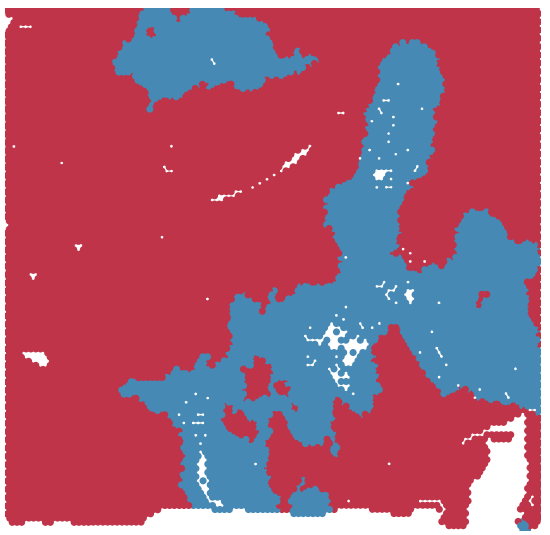
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



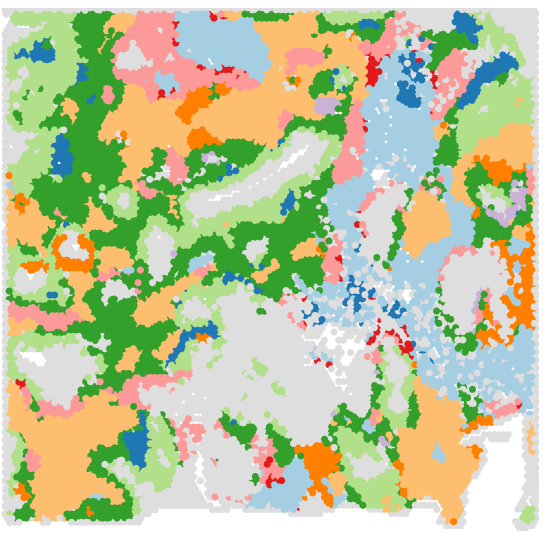
Black line: observed candidate outer edge

Final candidate location



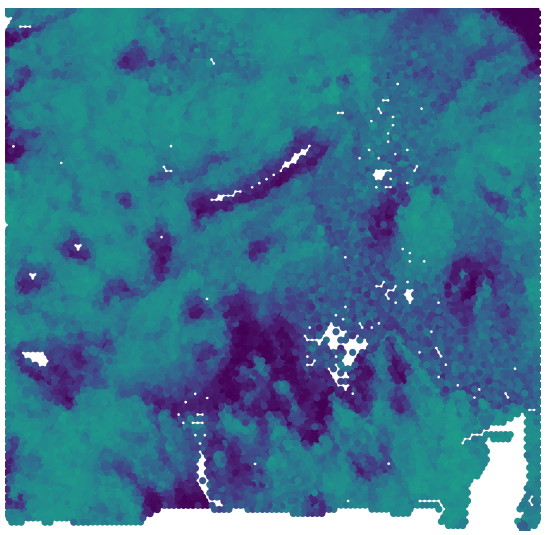
inside outside

Accepted R12 (original 0–8)



0 2 4 6 8
1 3 5 7

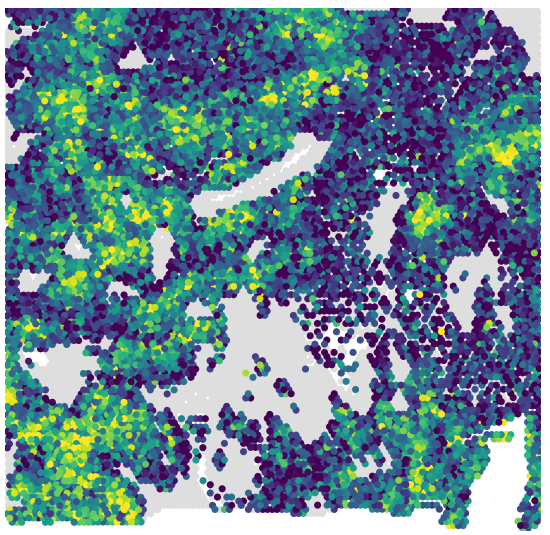
Epithelial support



0 1

Fraction of all 30

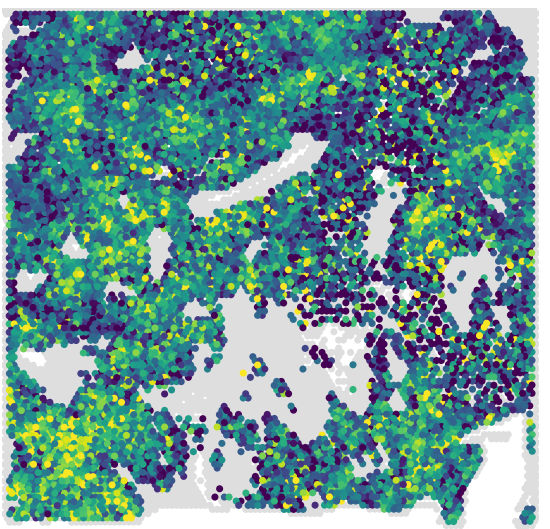
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



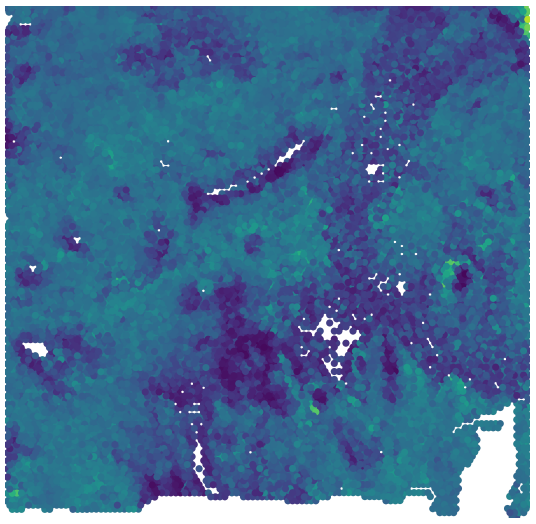
0 0.14

Matched-model AUCell

GSE238145_B9 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



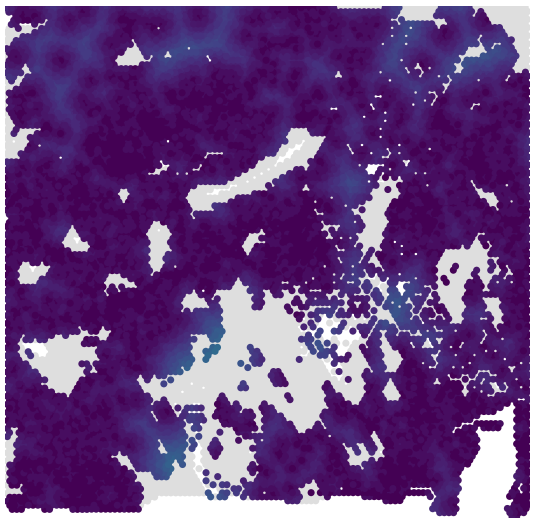
0 0.418

Fraction of all 30

CAF P80 targets

False True

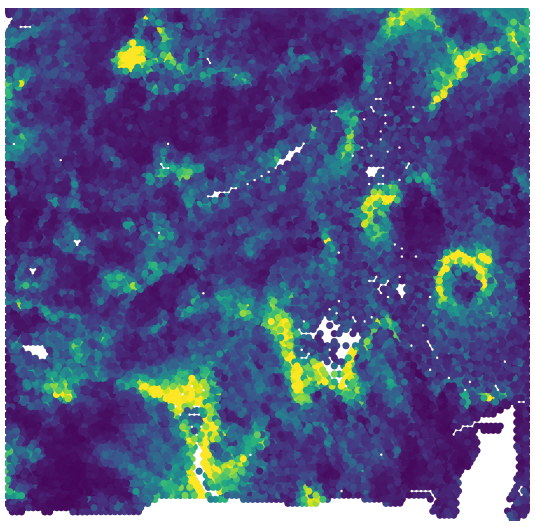
CAF distance



0 3.1e+03

Nearest spot center (µm)

TAM composition



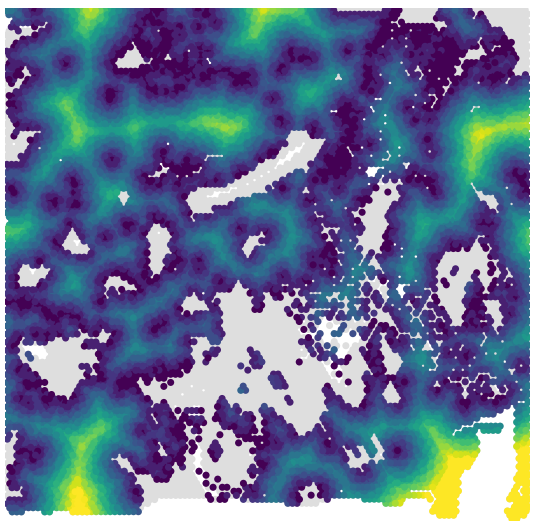
0 0.291

Fraction of all 30

TAM P80 targets

False True

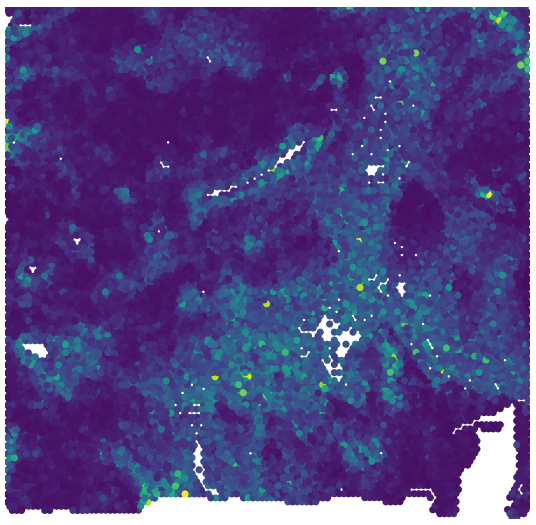
TAM distance



0 1.13e+03

Nearest spot center (µm)

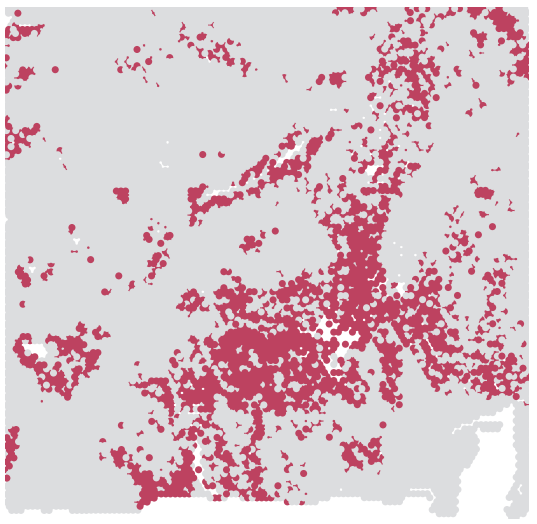
Treg composition



0 0.102

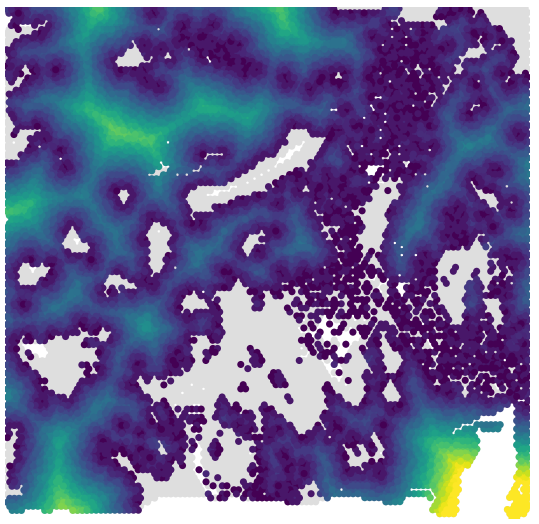
Fraction of all 30

Treg P80 targets



False True

Treg distance

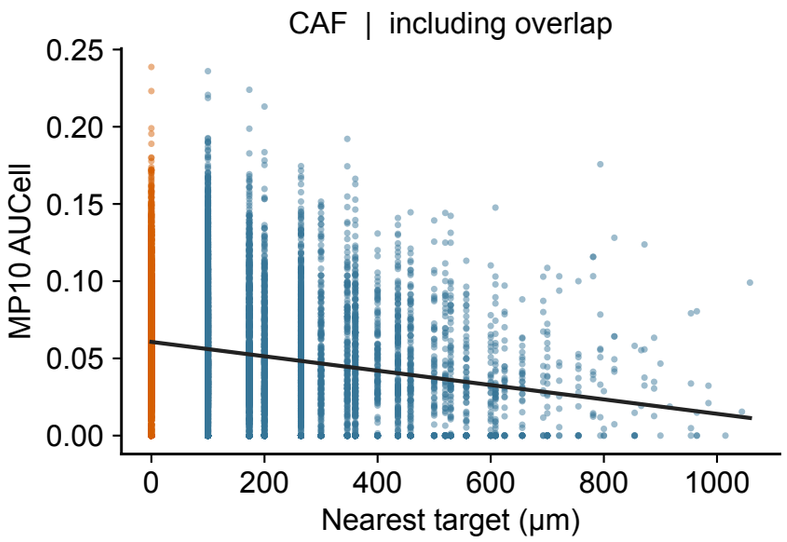


0 1.61e+03

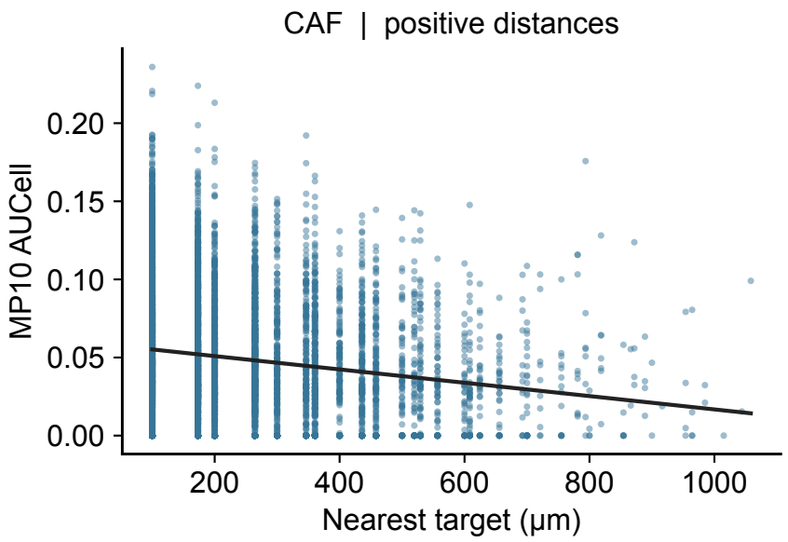
Nearest spot center (µm)

GSE238145_B9 | MP10 versus nearest target

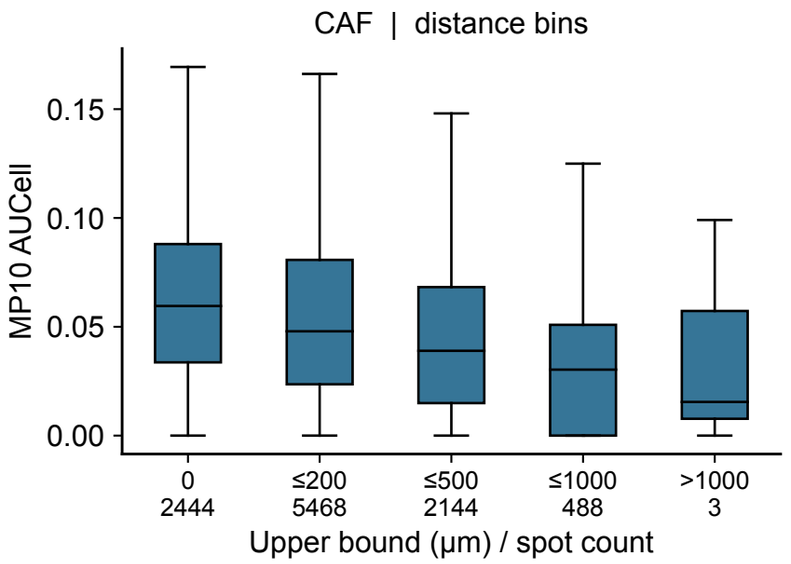
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



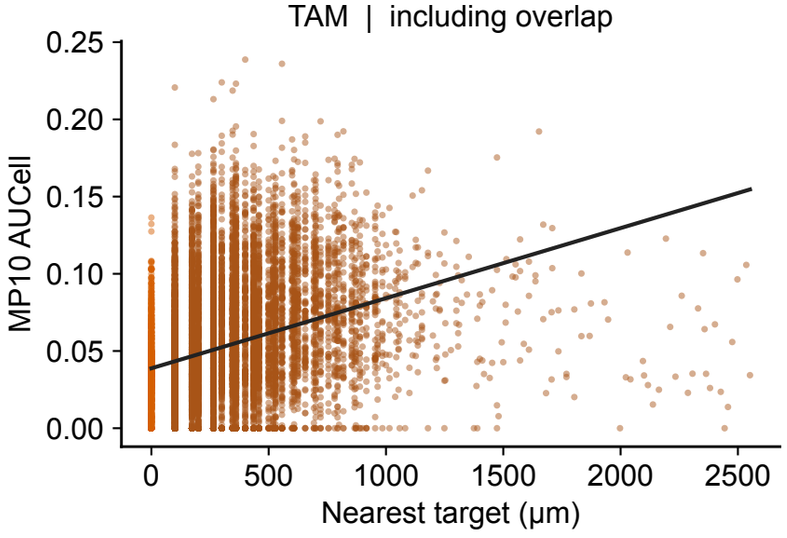
r -0.19 ρ -0.19
n 10547/10550 queries • 2658 targets



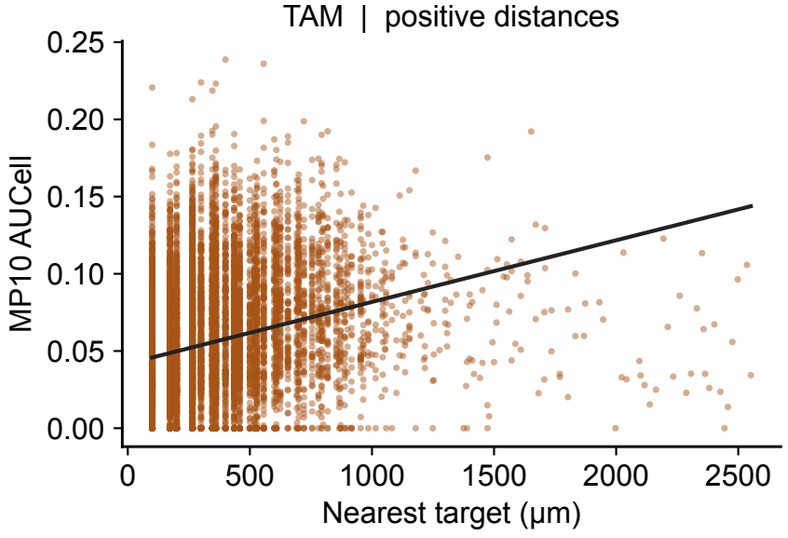
r -0.16 ρ -0.17
n 8103/10550 queries • 2658 targets



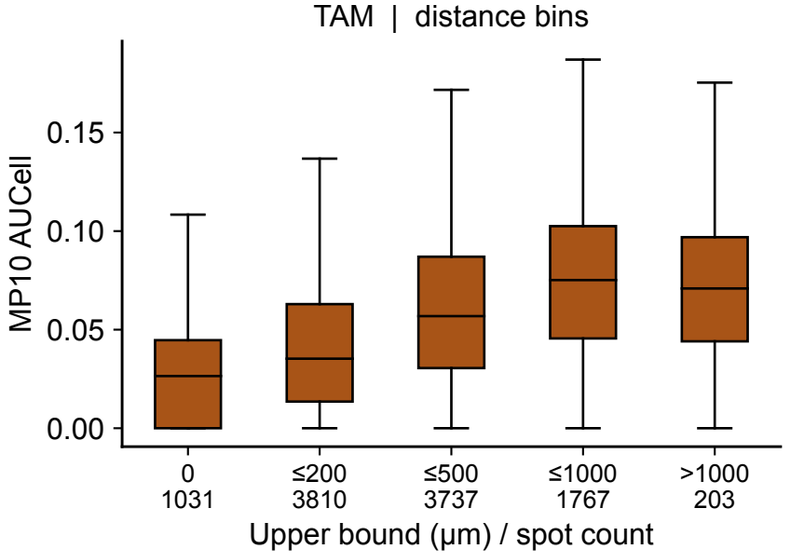
Orange = zero distance
Line = within-section linear fit



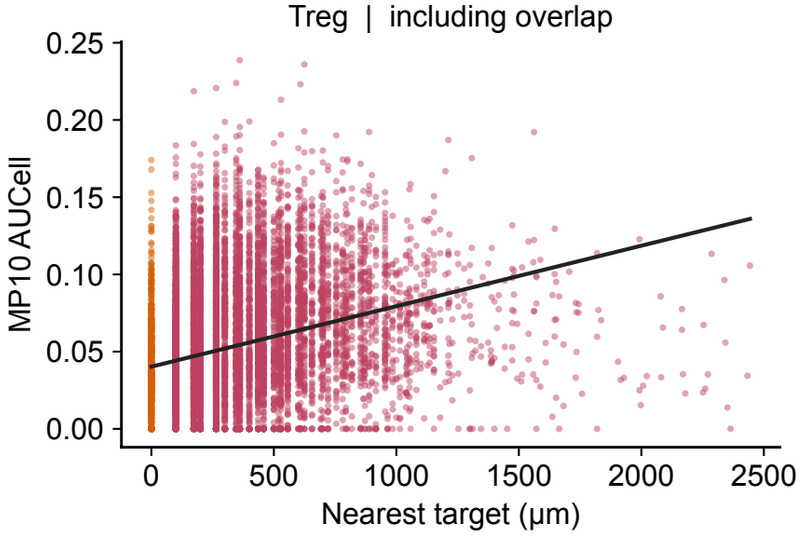
r +0.31 ρ +0.38
n 10548/10550 queries • 2658 targets



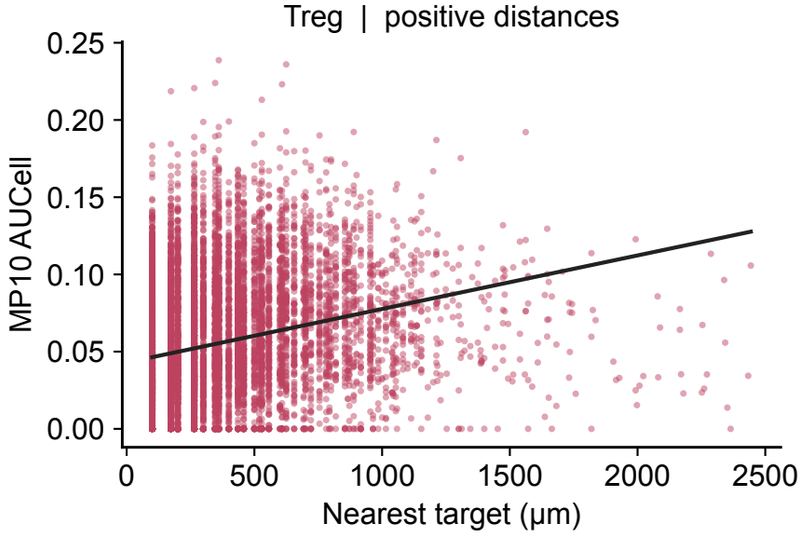
r +0.26 ρ +0.33
n 9517/10550 queries • 2658 targets



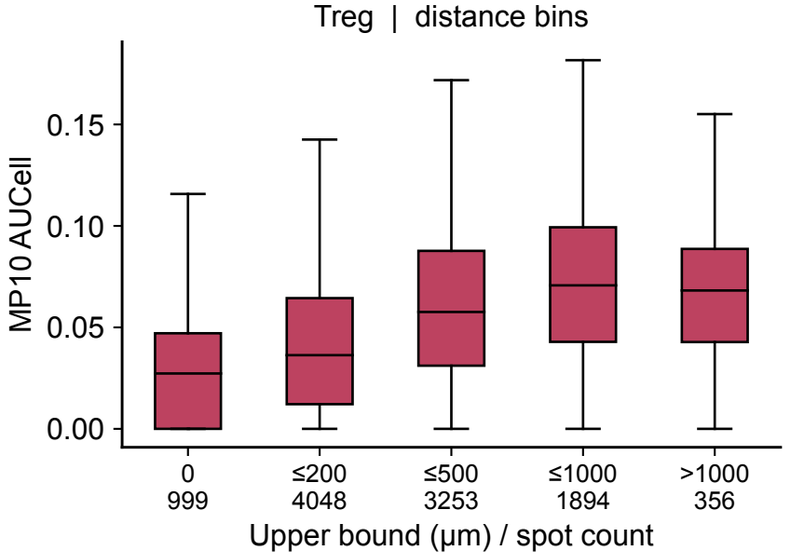
Orange = zero distance
Line = within-section linear fit



r +0.29 ρ +0.37
n 10550/10550 queries • 2658 targets



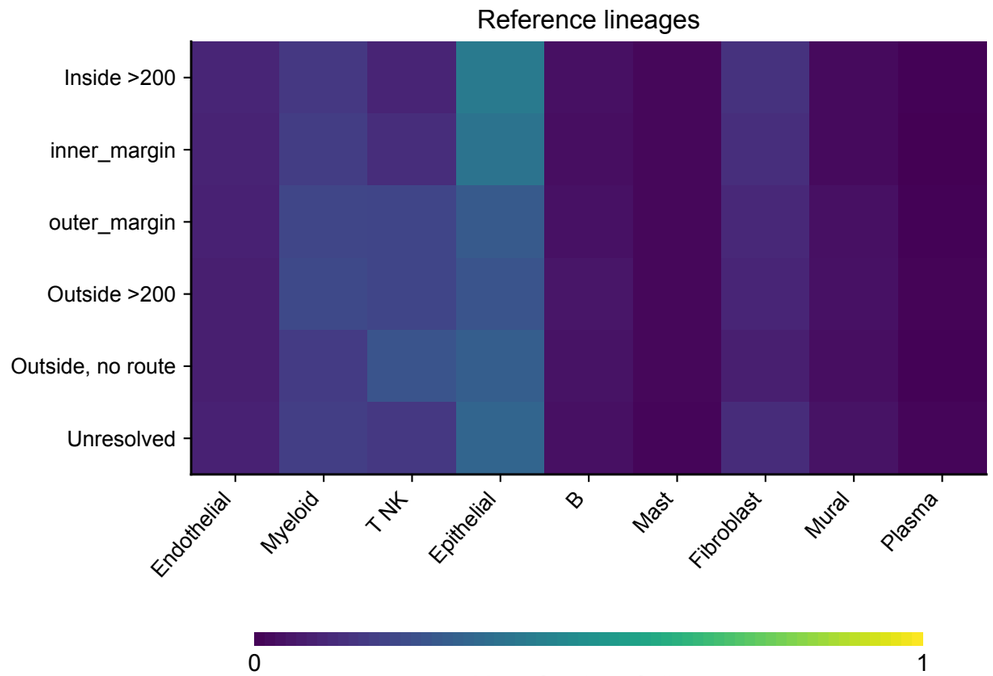
r +0.25 ρ +0.33
n 9551/10550 queries • 2658 targets

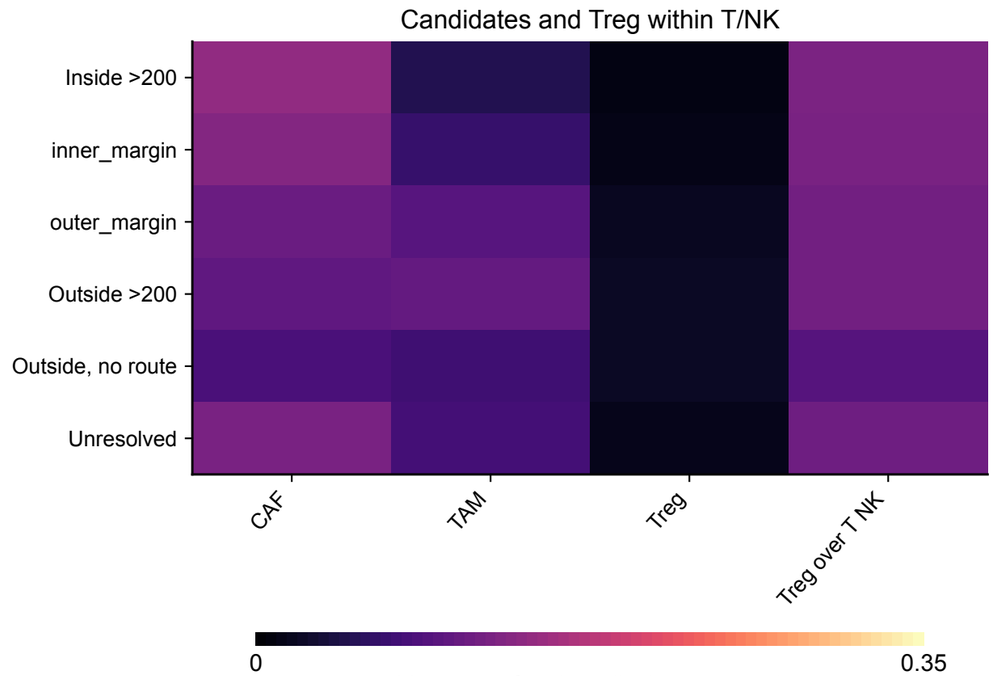


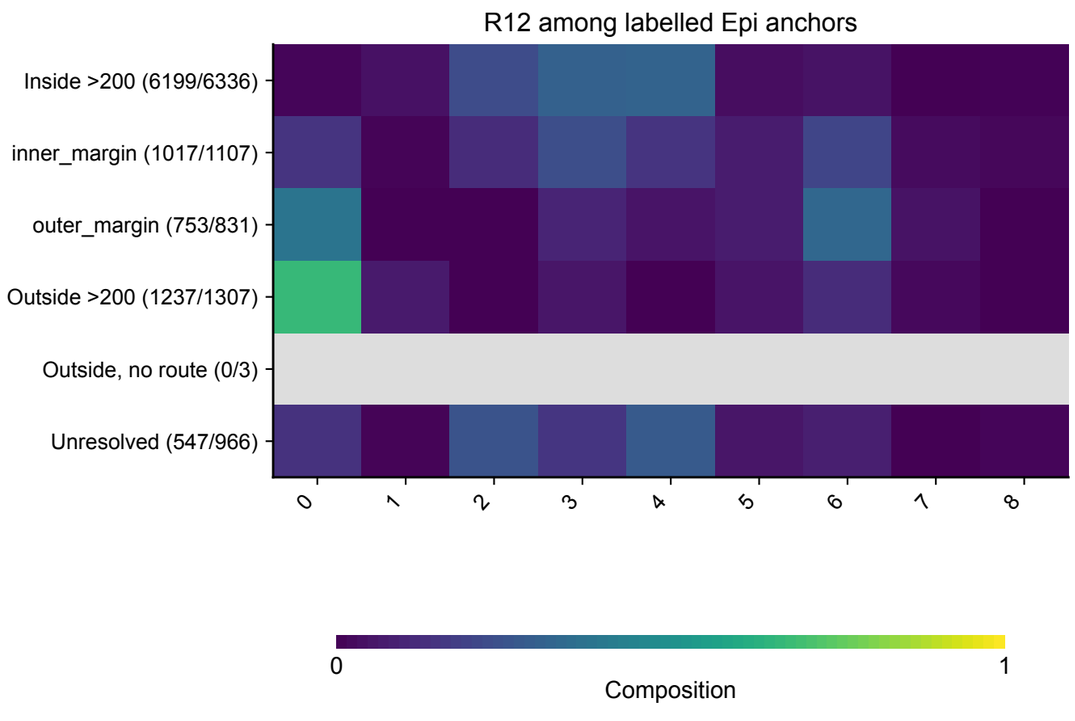
Orange = zero distance
Line = within-section linear fit

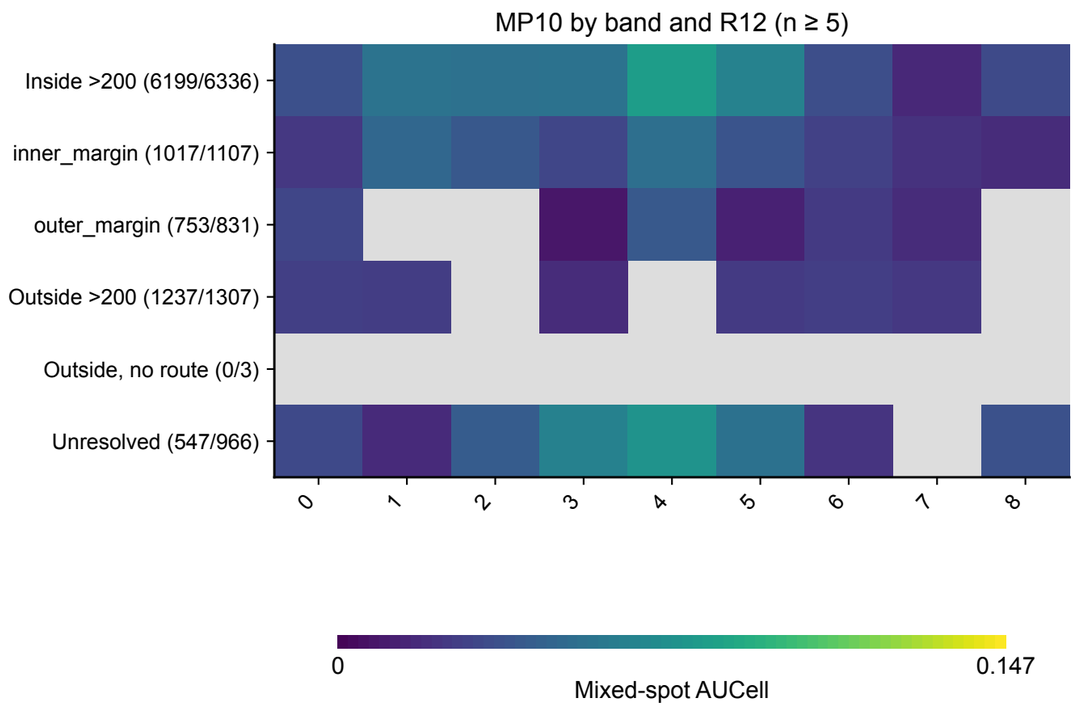
GSE238145_B9 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





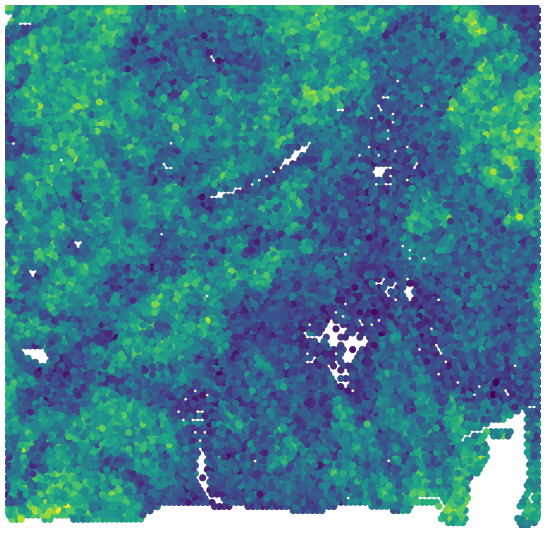




GSE238145_B9 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

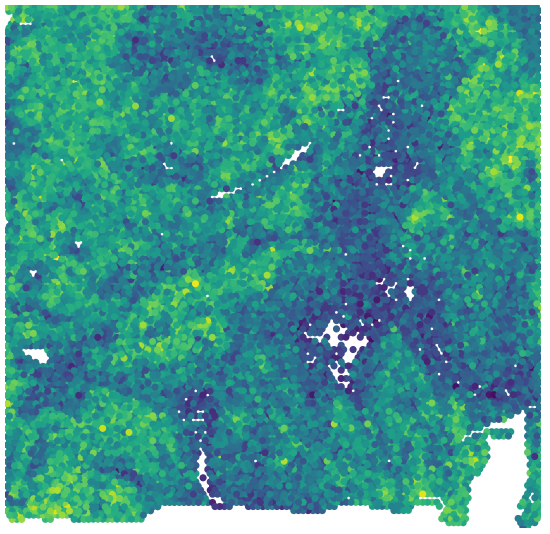
E2F Targets



0 0.152

UCell score

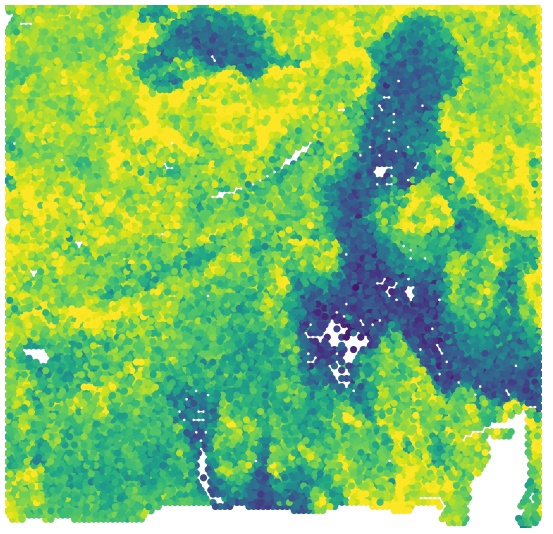
G2M Checkpoint



0 0.134

UCell score

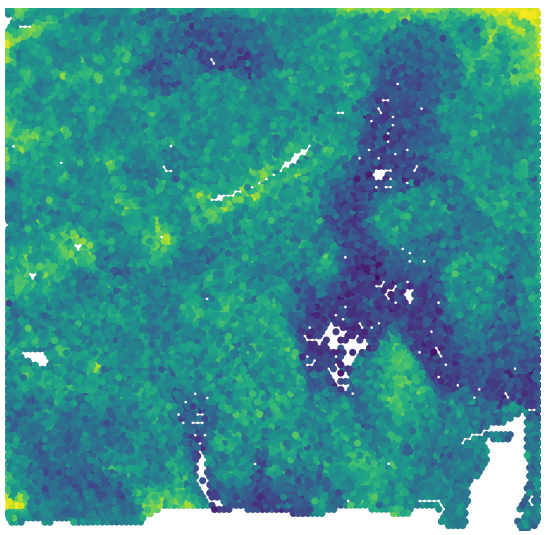
Hypoxia



0 0.211

UCell score

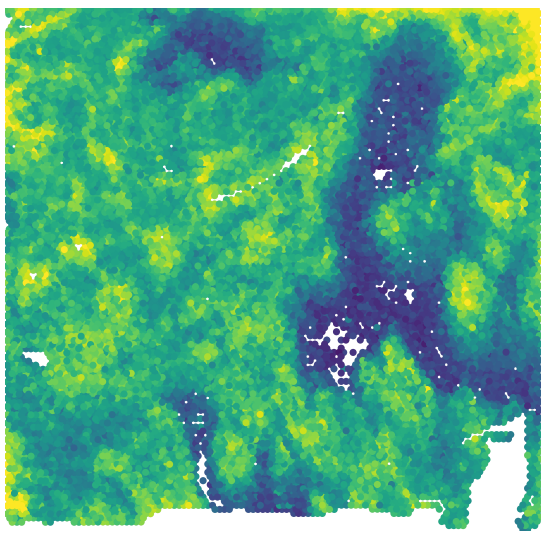
Tnfa Signaling Via Nfkb



0 0.275

UCell score

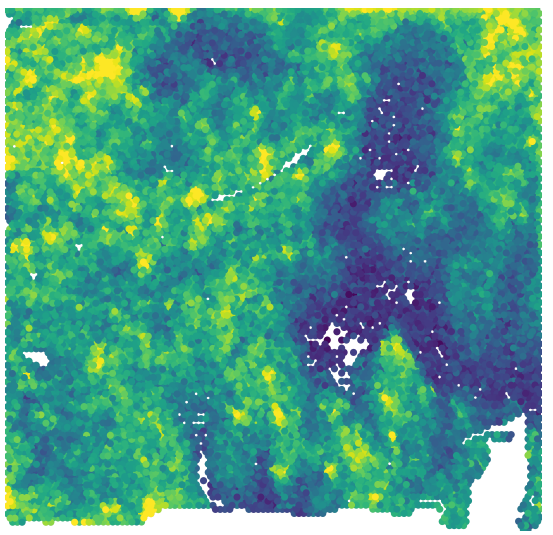
Emt



0 0.386

UCell score

Interferon Gamma Response



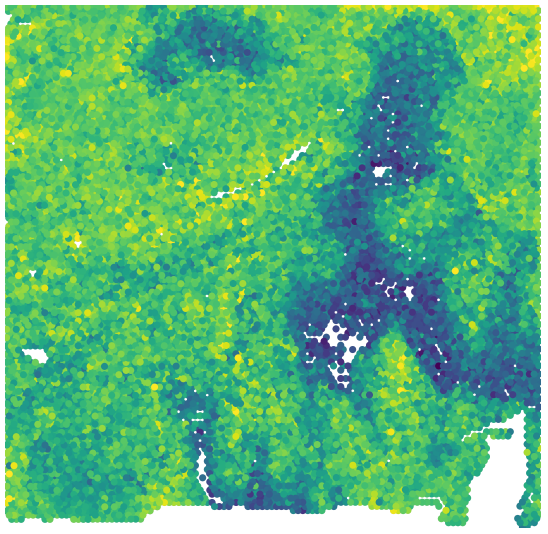
0 0.275

UCell score

GSE238145_B9 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

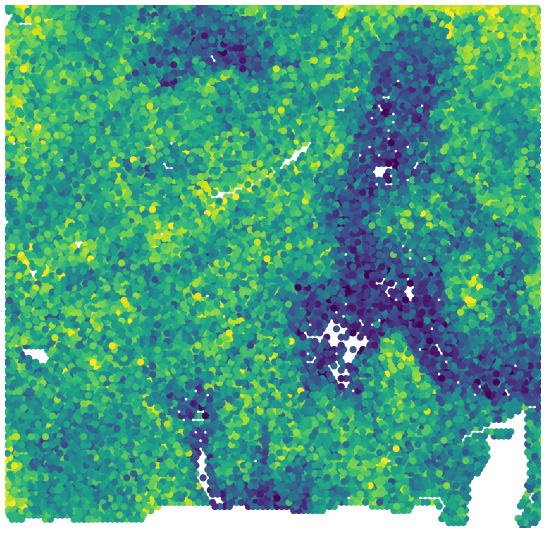
Il2 Stat5 Signaling



0 0.144

UCell score

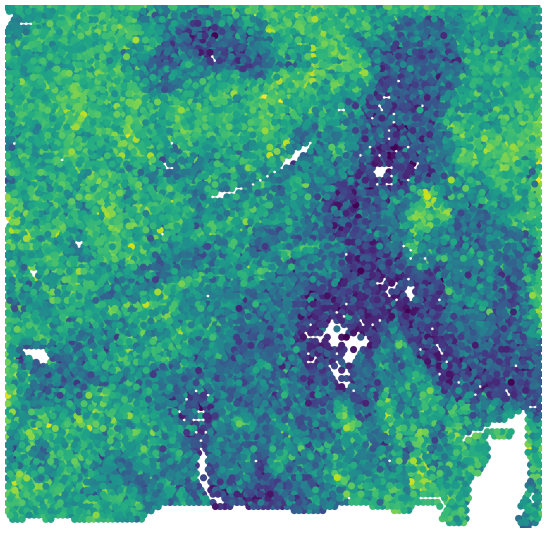
Tgf Beta Signaling



0 0.232

UCell score

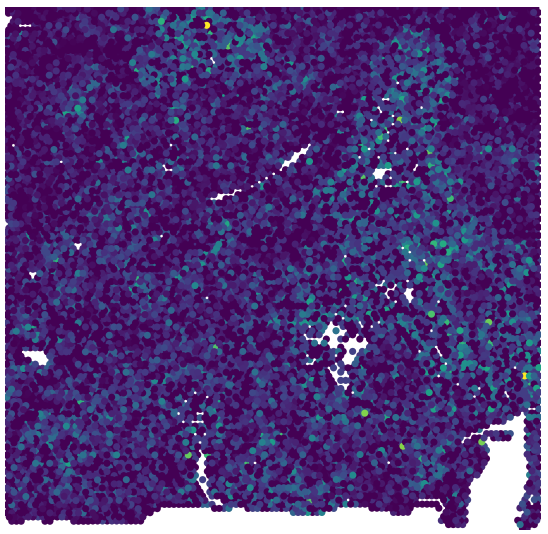
Cholesterol Homeostasis



0 0.21

UCell score

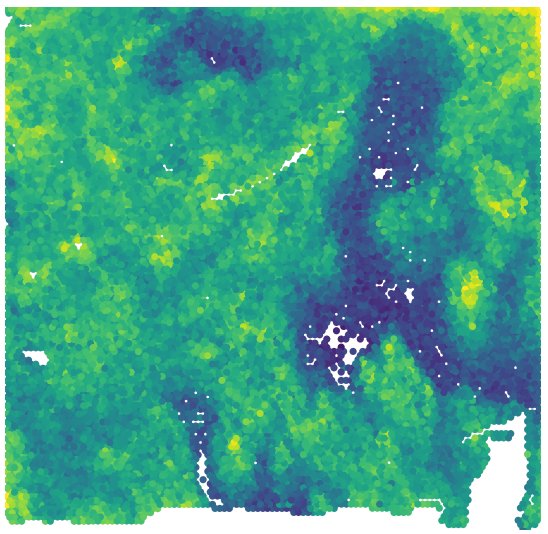
Cholesterol Biosynthesis



0 0.166

UCell score

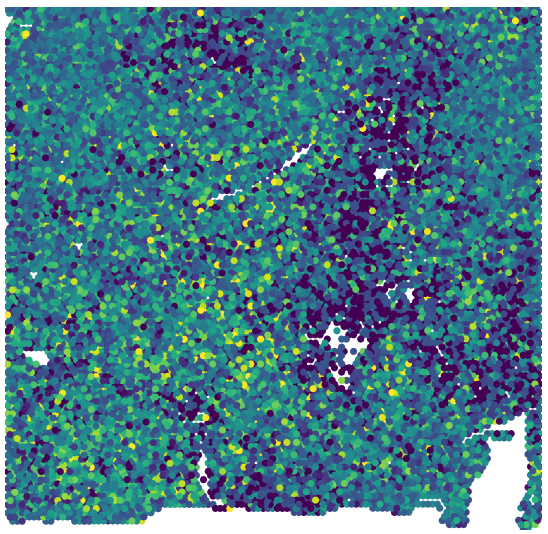
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



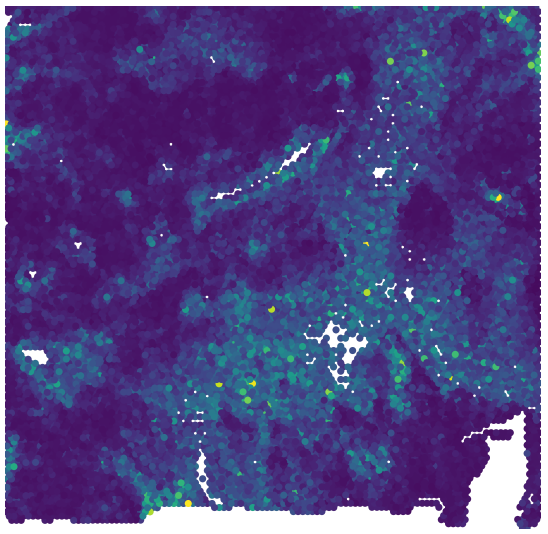
0 0.151

UCell score

GSE238145_B9 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

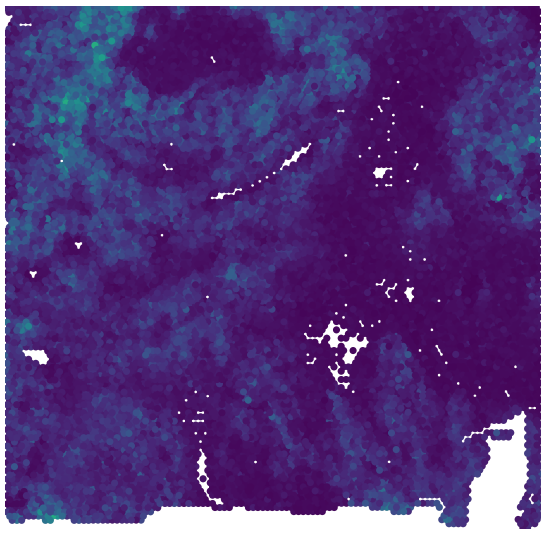
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

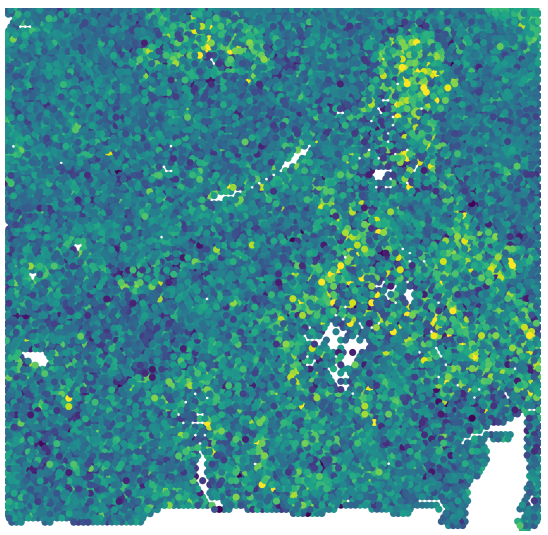
Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

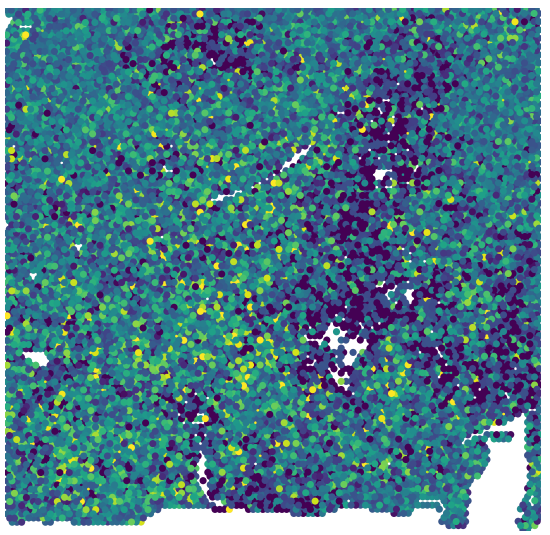
Treg Resting Vs Activated Up



0 0.0609

UCell score

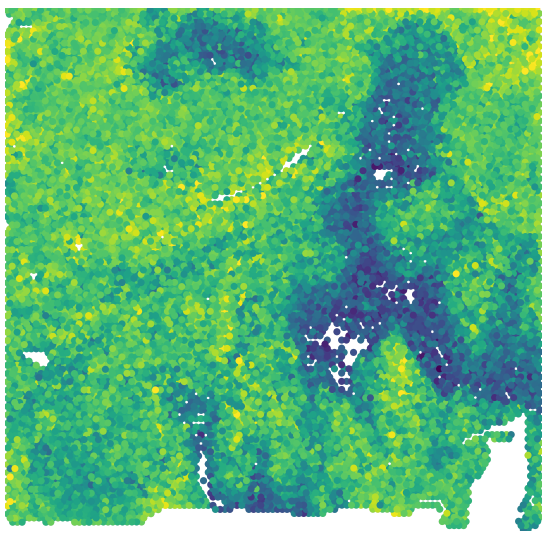
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

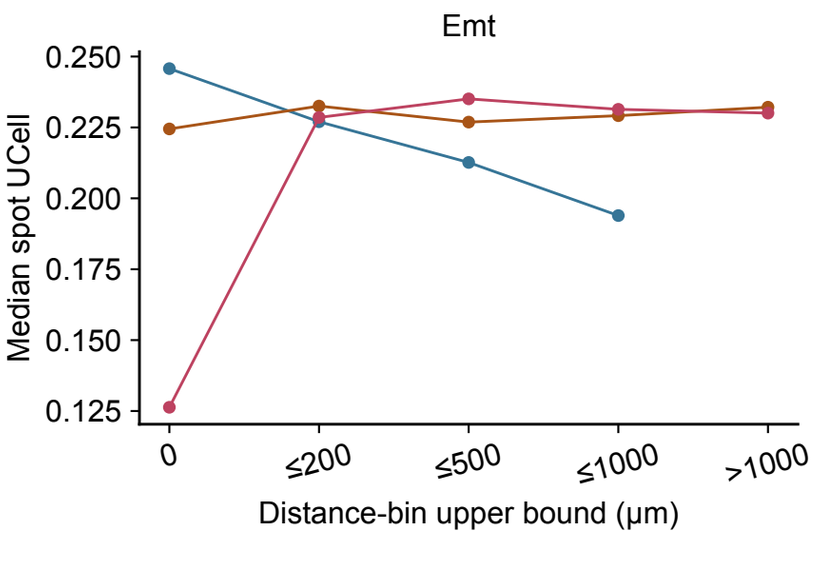


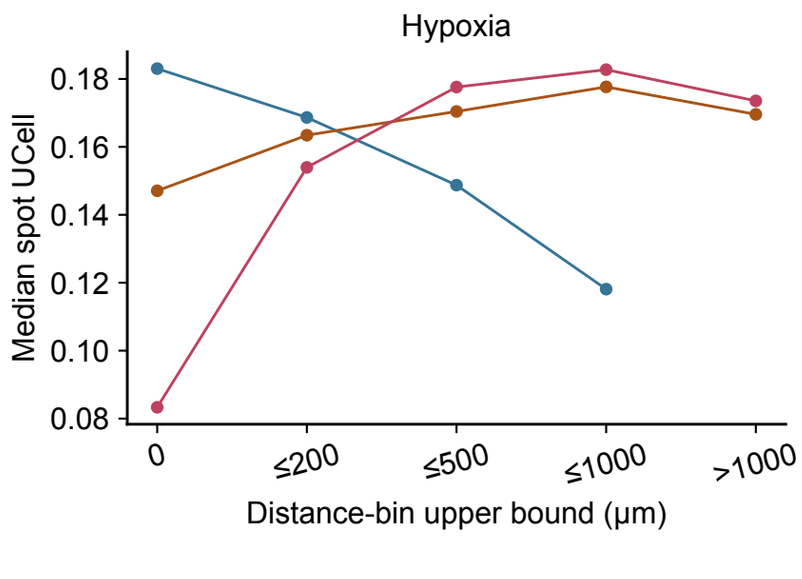
0 0.144

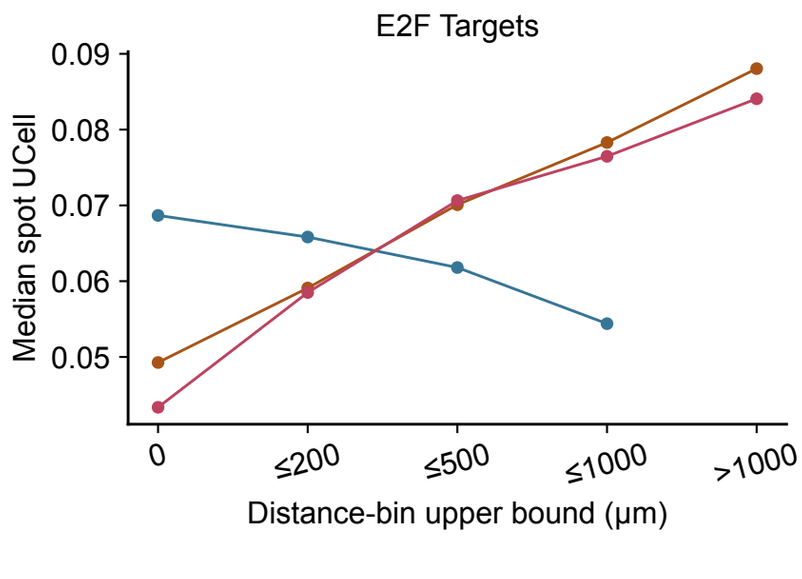
UCell score

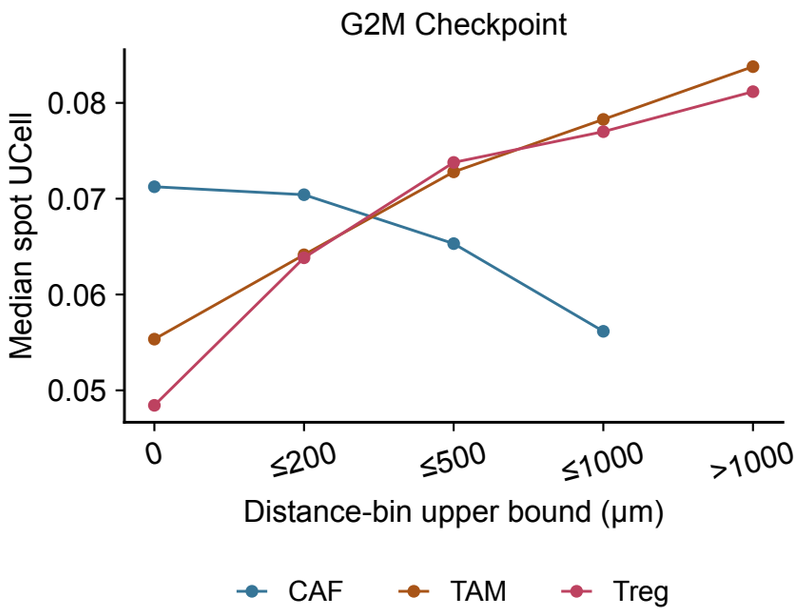
GSE238145_B9 | Programs by target distance

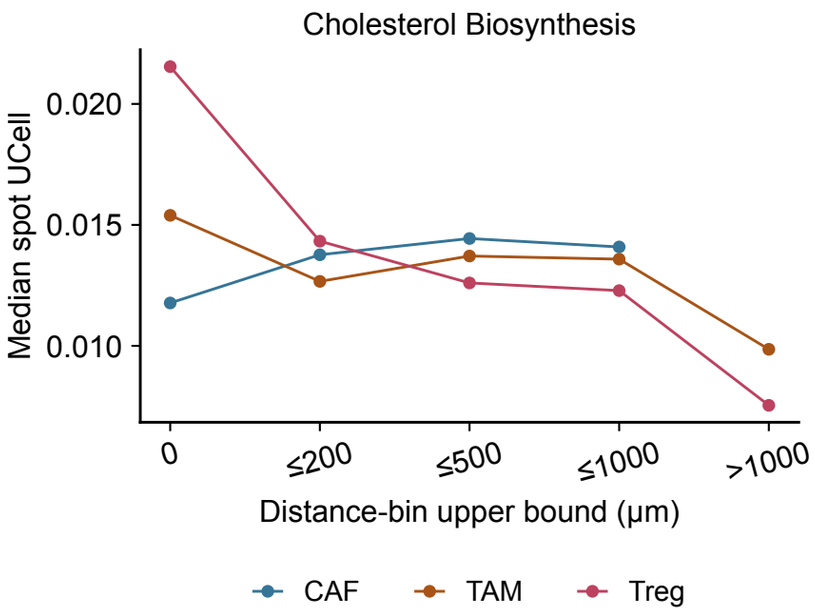
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

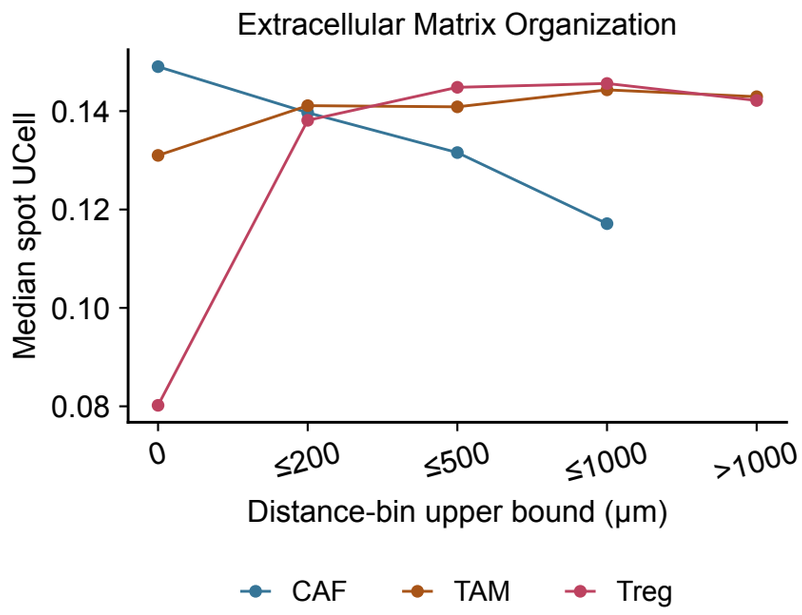












GSE238145_B9 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6